

**Modelling Two Decades of Change and Future Developments
in the Gulf of Nicoya (Costa Rica) Ecosystem
Using the Ecopath with Ecosim Toolbox**

Dissertation submitted by

Viola Alms

to the Faculty 02: Biology/Chemistry (FB02) at the University of Bremen, in
partial fulfilment of the requirements for the degree of *Doctor rerum naturalium*

Submission Date: 19th December 2024, Bremen.



Advisory Committee:

1. Reviewer: Prof. Dr. Matthias Wolff

Professor for Marine Synecology / Fisheries Biology, University of Bremen, Bremen, Germany.

Integrated Modelling, WG Resource Management, Leibniz Centre for Tropical Marine Research (ZMT), Bremen, Germany.

2. Reviewer: Dr. Marcela C. Nascimento

Ecosystem Modelling Working Group, Research Unit: Marine Evolutionary Ecology, GEOMAR Helmholtz Centre for Ocean Research Kiel, Kiel, Germany

Members of the committee:

1. Examiner: Prof. Dr. Martin Zimmer

Professor for Mangrove Ecology (University of Bremen), University of Bremen, Bremen, Germany

Working Group Leader, WG Mangrove Ecology, Leibniz Center for Tropical Marine Research (ZMT), Bremen Germany

2. Examiner: Prof. Dr. Martin Diekmann

Working Group Leader, WG Vegetation Ecology & Conservation Biology, University of Bremen, Bremen, Germany

Attending students:

1. Selma Mezger

PhD-Student (Marine Ecology), University of Bremen, Bremen, Germany

2. Siri Vietzen

Student, University of Bremen, Bremen, Germany

PhD defense: February 10th 2025, Big Seminar room, Leibniz Centre for Tropical Marine Research, Bremen, Germany

ABSTRACT

The Gulf of Nicoya is a highly productive estuary located on the Pacific coast of Costa Rica and accounts for the country's primary fish production. It is increasingly impacted by its fisheries and climate variations, urbanization, tourism developments, the input of pollutants from urban centres, and agriculture activities in the catchment areas of large rivers entering the gulf. Thus, our study intensively assesses the Gulf of Nicoya's past, present, and future states and the main driver of ecosystem change. The state of the ecosystem, food-web dynamics, detectable patterns in fishing efforts, and the external drivers' influence on the system were analyzed. Also, the economic impacts of fishing effort regulations were simulated to test and propose possible management plans. A spatial model was also built to test the effect of a temporary fishing closure on the stock recovery of target species in the Gulf of Nicoya and to propose management alternatives.

In the early 1990s, ecological and fishery data from the Gulf of Nicoya were holistically analyzed, and a trophic model was constructed using the Ecopath modelling approach. The results indicated that this tropical estuary, a hotspot for Costa Rican fisheries, was already overexploited by shrimp trawlers and longline fleets, and recent observations suggest further deterioration in this system. To evaluate the ecosystem and socio-economic changes in the Gulf of Nicoya over the last 20 years, the 1993 model was reconstructed with data from 2013 to compare both system states. Although the summary statistics of both states (i.e., 1993 and 2013) suggest that the system maintained its general functionality and even enabled the total harvest to increase by approximately 20%, a more detailed analysis of the levels of the functional groups led to the conclusion that the system was further degraded, causing a tremendous economic loss of approximately 50%. Like other coastal ecosystems, the Gulf of Nicoya is impacted by its fisheries, anthropogenic activities, and pollution from the large rivers entering the gulf. Thus, the gulf is particularly sensitive to short- and long-term changes in the climate, especially during extreme conditions in El Niño periods. For further analysis, we used a holistic approach to examine the impacts of fishing and environmental changes to identify the main drivers in the observed changes in the Gulf of Nicoya ecosystem. While the model simulations indicate that fisheries' exploitation rates mainly drive variations in the catchment, several species are also substantially affected by climate variations. This factor must be considered alongside implementing fishing regulations to form and develop appropriate management strategies. This dissertation tested the effectiveness of different management scenarios applied to the Gulf of Nicoya, including top-down and participatory fishing policies. Besides testing the current policy (which bans shrimp trawling), we used the automated fishing policy search tool of the EwE software to explore an alternative, optimized management scenario. The analysis indicates that the ban on trawling is important for the ecosystem's recovery. However, additional measures might be needed, such as reducing fishing efforts by semi-industrial purse seine fleets and artisanal longline fleets. We present two possible alternative management scenarios with different balances of ecosystem conservation and economic losses. We complement our extensive analysis by adding a spatial component using the Ecospace tool, which can simulate species' spatial distribution and fishing fleet effort over time. One obvious application is the assessment of the efficiency of existing and potential locations of MPAs to explore spatial management options and address various research questions, including the

regional consequences of climate change. The approach was used to test the effectiveness of the existing annual seasonal fishing closure in the gulf (known as “Veda” in Spanish) and to tackle further questions. For example, is excluding all fleets from the gulf necessary during fishing closure? Would a seasonal closure covering the entire area of the Gulf of Nicoya be more efficient?

This intensive assessment of the state of the Gulf of Nicoya ecosystem leads to the conclusion that a drastic decrease in fishing effort, especially of semi-industrial fleets, is needed to maintain the ecosystem’s functionality, rebuild the biomass of target species, and strengthen its resilience to environmental change. In this process, the enforcement of the fishing closure will play an important role.

Keywords: Ecosystem modelling, ecosystem-based fisheries management, Ecopath with Ecosim, Marine Protected Areas/MPAs

ZUSAMMENFASSUNG

Der Golf von Nicoya ist ein äußerst produktives Ästuar an der Pazifikküste Costa Ricas und trägt zur wichtigsten Fischproduktion des Landes bei. Er wird zunehmend durch Fischerei, Klimaveränderungen, Urbanisierung, touristische Aktivitäten, Verschmutzung aus städtischen Zentren und landwirtschaftlichen Aktivitäten in den Einzugsgebieten der großen Flüsse, die in den Golf münden, beeinträchtigt. Unsere Studie untersucht daher im Detail den vergangenen, gegenwärtigen und zukünftigen Zustand des Golfs von Nicoya sowie die Hauptfaktoren, die das Ökosystem verändern. Analysiert wurden der Zustand des Ökosystems, die Dynamik des Nahrungsnetzes, erkennbare Muster im Fischereiaufwand und der Einfluss externer Faktoren auf das System. Darüber hinaus wurden die wirtschaftlichen Auswirkungen von Fischereiaufwandsregelungen simuliert, um mögliche Managementpläne zu testen und vorzuschlagen. Darüber hinaus wurde ein räumliches Modell entwickelt, um die Auswirkungen einer saisonalen Einstellung der Fischerei auf die Erholung der Zielartenbestände im Golf von Nicoya zu testen und Managementalternativen vorzuschlagen.

In den frühen 1990er Jahren wurde eine ganzheitliche Analyse ökologischer Informationen und Fischereidaten aus dem Golf von Nicoya durchgeführt, auf deren Basis ein trophisches Modell mithilfe des Ecopath-Modellierungsansatzes erstellt wurde. Die Resultate legten nahe, dass das tropische Ästuar, ein Hotspot der costa-ricanischen Fischerei, bereits durch die Schleppnetzfisherei und Langleinenflotten überfischt wurde. Jüngste Beobachtungen deuten auf eine weitere Verschlechterung des Zustands des Ökosystems hin. Um die ökosystemischen und sozioökonomischen Veränderungen im Golf von Nicoya in den letzten 20 Jahren zu bewerten, wurde das Modell von 1993 mit Daten aus dem Jahr 2013 rekonstruiert, sodass ein Vergleich beider Systemzustände möglich ist. Die zusammenfassenden Statistiken beider Zustände (d. h. 1993 und 2013) lassen den Schluss zu, dass das System seine Gesamtfunktionalität beibehielt und sogar einen Anstieg der Fischereifänge verzeichnen konnte. Eine detailliertere Analyse der trophischen Ebenen der Funktionsgruppen führte zu dem Schluss, dass das System weiter degradiert wurde, was zu einem enormen wirtschaftlichen Verlust von etwa 50 % führte. Wie andere Küstenökosysteme wird auch der Golf von Nicoya von der Fischerei, anthropogenen Aktivitäten und der Verschmutzung durch die großen Flüsse, die in den Golf münden, beeinflusst.

Daher reagiert der Golf besonders sensibel auf kurz- und langfristige Klimaschwankungen, insbesondere während extremer Bedingungen in El-Niño-Perioden. Im Rahmen der weiteren Analyse wurde ein ganzheitlicher Ansatz verfolgt, um die Auswirkungen der Fischerei und der Umweltveränderungen zu untersuchen und die Hauptfaktoren für die beobachteten Veränderungen im Ökosystem des Golfs von Nicoya zu ermitteln. Die Modellsimulationen legen nahe, dass der Fischereiaufwand der stärkste Treiber der Schwankungen ist, wobei mehrere Arten ebenfalls erheblich von Klimaschwankungen betroffen sind. Dieser Aspekt muss neben der Implementierung von Fischereiverordnungen in die Überlegungen einbezogen werden, um adäquate Managementstrategien zu entwickeln und umzusetzen. Im Rahmen dieser Dissertation wurde die Effektivität unterschiedlicher Managementstrategien für den Golf von Nicoya evaluiert, darunter ein Top-down-Ansatz und eine partizipative Fischereipolitik. Im Rahmen der Untersuchung wurde zudem die aktuelle Politik, insbesondere das Verbot der

Garnelenschleppnetzfisherei, evaluiert. Dazu wurde das automatische Software-Tool 'Fishing Policy Search' der EwE-Software verwendet, um ein alternatives, optimiertes Managementszenario zu analysieren. Die durchgeführte Analyse belegt, dass das Verbot der Schleppnetzfisherei eine wesentliche Voraussetzung für die Regeneration des Ökosystems darstellt. Allerdings könnten zusätzliche Maßnahmen erforderlich sein, beispielsweise eine Reduzierung des Fischereiaufwands der semi-industriellen Flotten und der Langleinenflotten. Im Folgenden werden zwei alternative Bewirtschaftungsszenarien präsentiert, die sich in ihrem Gleichgewicht zwischen der Erhaltung des Ökosystems und den wirtschaftlichen Verlusten unterscheiden. Die umfassende Analyse wird durch eine räumliche Komponente ergänzt, wobei das Tool Ecospace zum Einsatz kommt. Dieses ermöglicht die Simulation der räumlichen Verteilung der Arten sowie des Fischereiaufwands im Laufe der Zeit.

Die Bewertung der Effizienz bestehender und potenzieller Standorte von Meeresschutzgebieten ermöglicht die Untersuchung räumlicher Managementoptionen sowie die Beantwortung verschiedener Forschungsfragen, darunter auch die Analyse der regionalen Folgen des Klimawandels. Der Ansatz wurde verwendet, um die Effektivität der bestehenden jährlichen saisonalen Fischereischließung im Golf (im Spanischen als "Veda" bezeichnet) zu evaluieren und weitere Fragestellungen zu beantworten. Beispielsweise wurde untersucht, ob es erforderlich ist, alle Flotten während der Fischereischließung aus dem Golf auszuschließen. Zudem wurde analysiert, ob eine saisonale Schließung, die das gesamte Gebiet des Golfs von Nicoya abdeckt, eine effizientere Maßnahme darstellen würde.

Die durchgeführte, intensive Bewertung des Zustandes des Ökosystems des Golfs von Nicoya führt zu der Schlussfolgerung, dass eine drastische Verringerung des Fischereiaufwands, insbesondere der semi-industriellen Flotten, erforderlich ist, um die Funktionsfähigkeit des Ökosystems zu erhalten, die Biomasse der Zielarten wieder aufzubauen und die Widerstandsfähigkeit des Ökosystems gegenüber Umweltveränderungen zu stärken. Die Durchsetzung des Fangverbots spielt in diesem Prozess eine wichtige Rolle.

Stichworte: Ökosystemmodellierung, Fischereimanagement, Nahrungsnetze, Ecopath with Ecosim, Meeresschutzgebiete

RESUMEN

El Golfo de Nicoya es un estuario sumamente productivo ubicado en la costa del Pacífico de Costa Rica y representa la principal producción pesquera del país. Se ve cada vez más afectada no solo por su pesca, sino también por las variaciones climáticas, la urbanización, los desarrollos turísticos, el ingreso de contaminantes de los centros urbanos y las actividades agrícolas en las zonas de captación de grandes ríos que desembocan en el golfo. Por lo tanto, nuestro estudio evalúa de manera intensiva los estados pasados, presentes y futuros del Golfo de Nicoya y el principal impulsor del cambio en los ecosistemas. Se analizaron el estado del ecosistema, la dinámica de la red alimentaria, los patrones detectables en el esfuerzo pesquero y la influencia de los impulsores externos sobre el sistema. Además, se simularon los impactos económicos de la regulación del esfuerzo pesquero para probar y proponer posibles planes de gestión. También se construyó un modelo espacial para probar el efecto de un cierre temporal de la pesca sobre la recuperación de las poblaciones de especies objetivo en el Golfo de Nicoya y proponer alternativas de gestión.

A principios de la década de 1990, se analizaron holísticamente los datos ecológicos y pesqueros del Golfo de Nicoya y se construyó un modelo trófico que utiliza el enfoque de modelado Ecopath. Los resultados arrojaron que este estuario tropical, que es un punto caliente para la pesca costarricense, ya estaba sobreexplotado por los arrastreros camaroneros y las flotas palangreras. Asimismo, recientes observaciones sugieren un mayor deterioro en este sistema. A fin de evaluar los cambios socioeconómicos y del ecosistema en el Golfo de Nicoya en los últimos 20 años, se reconstruyó el modelo de 1993 con datos de 2013 para comparar ambos estados del sistema. Si bien las estadísticas resumidas de ambos estados (es decir, 1993 y 2013) sugieren que el sistema mantuvo su funcionalidad general e incluso permitió que la cosecha total aumentara aproximadamente un 20 %, un análisis más detallado de los niveles de los grupos funcionales llevó a la conclusión de que el sistema se degradó aún más, lo que causó una enorme pérdida económica de aproximadamente el 50 %. Al igual que otros ecosistemas costeros, el Golfo de Nicoya no solo se ve afectado por su pesca, sino también por las actividades antropogénicas y la contaminación de los grandes ríos que desembocan en el golfo. Por lo tanto, el golfo es particularmente sensible a los cambios climáticos a corto y largo plazo, en especial durante las condiciones extremas en los períodos de El Niño. Para un análisis más detallado, utilizamos un enfoque holístico que examina tanto los impactos de la pesca como los cambios ambientales, e identifica los principales impulsores de los cambios observados en el ecosistema del Golfo de Nicoya. Si bien las simulaciones de modelos indican que las variaciones en la cuenca son impulsadas en especial por las tasas de explotación pesquera, varias especies también se ven sustancialmente afectadas por las variaciones climáticas. A fin de formar y desarrollar estrategias de gestión adecuadas, este factor debe considerarse junto con la implementación de la regulación pesquera. En esta disertación, probamos la efectividad de diferentes situaciones de gestión aplicados al Golfo de Nicoya, incluidas las políticas de pesca tanto de arriba hacia abajo como participativas. Además de probar la política actual (que prohíbe la pesca de arrastre de camarón), utilizamos la herramienta de búsqueda automatizada de políticas pesqueras del software EwE para explorar una situación hipotética de gestión alternativa y optimizada. El análisis indica que la prohibición de la pesca de arrastre es una medida importante para facilitar la recuperación del ecosistema. Sin embargo, podrían

necesitarse algunas medidas adicionales, como la reducción del esfuerzo pesquero de las flotas de cerco semiindustriales y de las flotas de palangre artesanales. Presentamos dos posibles situaciones de gestión alternativas, con diferentes equilibrios de conservación del ecosistema y pérdidas económicas. Complementamos nuestro extenso análisis agregando un componente espacial mediante la herramienta Ecospace, que puede utilizarse para la simulación de la distribución espacial de las especies y el esfuerzo de la flota pesquera a lo largo del tiempo. Una aplicación obvia es la evaluación de la eficiencia de las ubicaciones existentes y potenciales de las AMP para explorar opciones de gestión espacial y abordar una variedad de cuestiones de investigación, incluidas las consecuencias regionales del cambio climático. Este enfoque se utilizó para probar la eficacia de la actual veda anual de pesca estacional en el golfo (conocida como “veda”) y también abordar otras cuestiones. Por ejemplo, ¿es necesario excluir todas las flotas del golfo durante el cierre de la pesca? ¿Sería más eficiente un cierre estacional que cubra toda el área del Golfo de Nicoya?

Esta evaluación intensiva del estado del ecosistema del Golfo de Nicoya lleva a la conclusión de que se necesita disminuir drásticamente el esfuerzo pesquero, en especial de las flotas semiindustriales, con el fin de mantener la funcionalidad del ecosistema y reconstruir la biomasa de las especies objetivo y fortalecer su resiliencia al cambio ambiental. En este proceso, la aplicación del cierre de la pesca desempeñará un papel importante.

Palabras clave: modelado de ecosistemas, gestión pesquera basada en ecosistemas, Ecopath con Ecosim, Áreas Marinas Protegidas/AMP

TABLE OF CONTENTS

Abstract.....	i
Zusammenfassung.....	iii
Resumen	v
Table of Contents	vii
List of Figures.....	ix
List of Tables.....	x
Acknowledgments.....	xi
Chapter 1 General Introduction	1
1.1 Small-Scale Fisheries	2
1.2 A Case Example of Holistic Ecosystem Management	6
1.3 Scope of the Thesis.....	11
Chapter 2 The past of the Gulf of Nicoya ecosystem: Two Decades of Change	
Ecopath.....	12
Abstract.....	14
2.1 Introduction	15
2.2 Methods	16
2.3 Results	24
2.4 Discussion.....	31
Acknowledgments	36
Chapter 3 Identification of Drivers of Change of the Gulf of Nicoya Ecosystem.....	37
Abstract.....	39
3.1 Introduction	40
3.2 Materials and Methods	42
3.3 Results	47
3.4 Discussion.....	51
Acknowledgments	58
Chapter 4 Exploration of fisheries management policies in Gulf of Nicoya using ecosystem modelling	59
Abstract.....	61
4.1 Introduction	62
4.2 Materials and Methods	64
4.3 Results	73
4.4 Discussion.....	78

4.5	Conclusion	82
	Acknowledgments	83
Chapter 5	A Spatial Ecosystem Model to Assess the Efficiency of Martine Protection Plans.....	84
	Abstract.....	86
5.1	Introduction	87
5.2	Material and Methods	88
5.3	Results	97
5.4	Discussion.....	103
5.5	Conclusion	107
Chapter 6	General Discussion	108
6.1	Main Findings.....	109
6.2	Recommended Policy Plans/Management Implications	111
6.3	Critical Assessment of the Methodological Approach	113
	Bibliography	115
	Appendix	136
	Eidesstattliche Erklärung	141

LIST OF FIGURES

Figure 1.1 - Map of the total annual catches	3
Figure 1.2 - Time series of number (%) of fish stocks by status in the Gulf of Nicoya.....	4
Figure 1.3 - Maps of the study area (Gulf of Nicoya, Costa Rica)	6
Figure 2.1 - Maps of the study area (Gulf of Nicoya, Costa Rica)	17
Figure 2.2 - Semi-Industrial catch and effort (Gulf of Nicoya, Costa Rica).....	20
Figure 2.3 - Comparison of the relative yearly catches (%) by trophic level.....	24
Figure 2.4 - Trophic flow diagram of the GN ₊₂₀ model	27
Figure 2.5 - Mixed trophic impact analysis for the Gulf of Nicoya ecosystem model	28
Figure 2.6 - Keystone index of the functional groups in the Gulf of Nicoya.....	29
Figure 2.7 - The share of each target group in the total yearly monetary value (US\$).....	30
Figure 3.1 - Maps of the study area (Gulf of Nicoya, Costa Rica)	42
Figure 3.2 - Catches, fishing effort and catch per unit effort.	47
Figure 3.3 - AICc of all tested models and number of estimated parameters.	50
Figure 3.4 - Environmental time series 1997-2014	54
Figure 3.5 - Yearly catches and model predictions for the simulated EwE scenarios	55
Figure 4.1 - Maps of the study area (Gulf of Nicoya, Costa Rica)	65
Figure 4.2 - Schematic overview of the assessed management scenarios.....	69
Figure 4.3 - Relative difference in the biomass of each functional group	73
Figure 4.4 - The expected relative increase in the total biomass of all functional groups	74
Figure 4.5 - Relative differences in the catch, fishing effort and profit	75
Figure 4.6 - Results of the EwE policy search routine	76
Figure 5.1 - Maps of the study area (Gulf of Nicoya, Costa Rica)	90
Figure 5.2 - Ecospace basemap of the Gulf of Nicoya (Costa Rica).....	93
Figure 5.3 - The area of the seasonal fishing closure (veda) in the Gulf of Nicoya	96
Figure 5.4 - Relative biomass (A) and relative catch (B)	98
Figure 5.5 - Ecospace model output	99
Figure 5.6 - Ecospace model output: Distribution of the fishing effort of all fleets	100
Figure 5.7 - Relative biomass of target species.....	102
Figure 5.8 - Seasonal fishing closure. Ecospace output: Simulating biomass distribution...	103

LIST OF TABLES

Table 2.1 - Ecopath with Ecosim input/output parameters for the models GN_0 and GN_{+20} ..	21
Table 2.2 - Summary statistics of the ecosystem characteristics from the GN_0 and GN_{+20} ...	25
Table 3.1 – Hypothesis-based model fitting	45
Table 3.2 - Prey-predator interactions in the Gulf of Nicoya Model (scenario 2).....	49
Table 3.3 - Catch time series	52
Table 3.4 – Model predictions compared to reference catch data	56
Table 4.1 - Basic Ecopath with Ecosim input and output parameters	67
Table 4.2 - The Fishing Policy Search routine	70
Table 4.3 - Costs, profit and jobs per catch of all fleets	71
Table 4.4 - Mandated rebuilding of species biomass	72
Table 4.5 - Summary of the explored management scenarios and their expected outcomes ..	77
Table 5.1 - Basic Ecopath with Ecosim input and output parameters for the Gulf of Nicoya	91
Table 5.2 - Percentage of fixed and variable fishing costs	95
Table 5.3 - Scenario approach to test different fishing closure settings.....	96
Table 5.4 - Ecospace results: The relative change in catch, value, costs and effort.....	101
Table 5.5 - Ecospace results: Relative change in catches of all fleets.....	101

ACKNOWLEDGMENTS

The years of writing my PhD thesis have been an intense and versatile experience. The whole journey was accompanied by my very patient and committed supervisor Prof. Dr. Matthias Wolff. The beginning of my PhD position was marked by the research stay in San Jose, Costa Rica, where Matthias Wolff introduced me to his long-standing colleagues and partners at the CIMAR and University of Costa Rica. Especially his former colleague Prof. Dr. Alvaro Morales took the lead to supervise me at my research stay in Costa Rica. Thanks to the CIMAR, my host Alvaro Morales and his colleagues. As well as to the INCOPESCA for working with me and sharing some valuable information and data with me.

After my fruitful stay in San José, I returned to the ZMT Bremen and was lucky to have the chance to learn a lot from Matthias Wolff and the whole research group. I thank Matthias Wolff for accompanying me and my work over the years including a long period of absence due to the maternity leave with my three kids. Matthias Wolf is a very engaged supervisor, who is always available for advice. I also thank my former working group colleagues at the ZMT.

A very special thanks goes out to our Postdoc researcher Dr. Giovanni Romagnoni who joined us in 2021. He helped me to get back on track during and after my maternity leave, kept me updated and motivated throughout a time of home office, the covid pandemic and a lack of childcare. He has the ability to motivate people in a very friendly way and made mobilize all my strength, to work flexible in the late evenings and on the weekends to finish my third publication. Not to forget, that I was able to learn a lot from him about Ecopath, Ecosim and Ecospace improving the quality of my research. Giovanni was always available to help and had a good advice for every problem on hand. I thank Matthias and Giovanni for adding their valuable input and ideas to my research and helping me improve the publications through their co-authoring.

I also would like to thank my husband and my family for their support and understanding. Especially during times of paper submission and in the last weeks of my PhD I could count on your support.

Chapter 1

GENERAL INTRODUCTION

1.1 Small-Scale Fisheries

1.1.1 Small-Scale Fisheries in a Global Context

Small-scale fisheries (SSFs) support approximately 90% of all fishery jobs worldwide, with around 32 million people employed as small-scale fishers and an additional 76 million in post-harvest jobs (Bennett et al. 2020; Kelleher et al. 2012). SSFs produce more than half of all fishery landings in developing nations (Lozano and Heinen 2015), whereas 81% of the catch is used for local human consumption. Although the SSFs vary substantially by region, some characteristics are common: their use of small vessels and engines; simple or more traditional gears; small crews; their proximity to the coast, family or local ownership; and their importance for local livelihoods and subsistence (Bennett et al. 2020; Kittinger et al. 2013; Smith and Basurto 2019). The small-scale artisanal fishing sector is highly socioeconomically important for securing food production and livelihoods in coastal communities and has additional economic benefits to many countries.

There is increasing recognition of the importance of sustaining SSFs within fishers' management and development policies. Most studies of SSFs in developing countries in the last 25 years have emphasized small-scale fisherfolks' resource dependence and the open-access nature of fisheries that lead to resource degradation, poverty, and marginalization (Arthur 2020; Kalikoski et al. 2019). The interest in the 'sustainable livelihoods approach' has continuously increased. This approach aims to reduce poverty in low-income countries by understanding the strategies of artisanal fishing communities affected by variations in fishery resources (Allison and Ellis 2001). The aim is to find management directives that combine resource conservation with social and economic goals. Intensifying fishing pressures often lead to increasing competition between industrial fisheries and SSFs, which is at an increased disadvantage to SSFs (Salas et al. 2011), as well as conflicts among artisanal and industrial fishing sectors and governmental agencies. This is a common problem in different marine regions of the world. In West African waters, for example, the government responded to the depletion of marine resources with measures that mainly targeted small-scale fishers, resulting in the marginalization and increased vulnerability of SSF (Okafor-Yarwood et al. 2022). The marine sector of Bangladesh faces similar problems. Here, the number of artisanal fishers is continuously decreasing due to the increasing dominance of commercial fishers and the strong competition during peak fishing periods (Alam et al. 2021). In the Mediterranean, there is increasing competition between artisanal and industrial fishing operations and recreational fisheries, changing the role of coastal fisheries in some European countries (Lloret et al. 2018).

To add complexity to the situation, there is typically a lack of quantitative data and ecosystem information, which would help to avoid unexpected trophic dynamics in the system. Consequently, managing artisanal fisheries is challenging (Batista et al. 2014). Despite the complex situation and lack of information, countermeasures should be taken against the ongoing depletion of coastal waters, often leading to top-down regulations (Lozano and Heinen 2016). However, the top-down enforced management rules usually fail either due to noncompliance of fishers or the lack of control capacities. Besides finding the optimal balance between different management goals for successful policy-making, it can be a good approach to consider the interests and knowledge of local stakeholders (Wilson et al. 2006). Co-

management is an alternative approach to effective management. It is defined as sharing responsibility and authority between the state and resource users. It fosters stakeholder collaboration, including government agencies, non-governmental organizations, research organizations, private enterprises, and civil society (Carlsson and Berkes 2005; Evans 2010). It is suggested that co-management, which can improve social participation and compliance with the fishery's management rules of all actors (d'Armengol et al. 2018), while it reduces conflicts, fosters responsibility among resource users (Nielsen and Vedsmand 1999), and reduces management costs (Carlsson and Berkes 2005). There are several successful examples of fisheries where co-management has been practiced, mostly in coastal multispecies fisheries (d'Armengol et al. 2018), also from Latin America.

1.1.2 SSFs in the Case of Costa Rica's Fishing Hotspot

Latin American coastal waters are highly productive and support essential fisheries worldwide; thus, these waters have high economic value (Blaber 2002) and provide livelihoods for many coastal inhabitants (Figure 1.1). Coastal ecosystems often serve as important nursery, feeding, or reproduction grounds for many commercially important fish and invertebrates (Abrantes et al. 2015). Especially, mangrove forests provide organic matter and shelter and contribute to enhancing fish and invertebrate productivity; thus, mangroves strongly support the local fisheries (Wolff 2006; Hutchison et al. 2014).

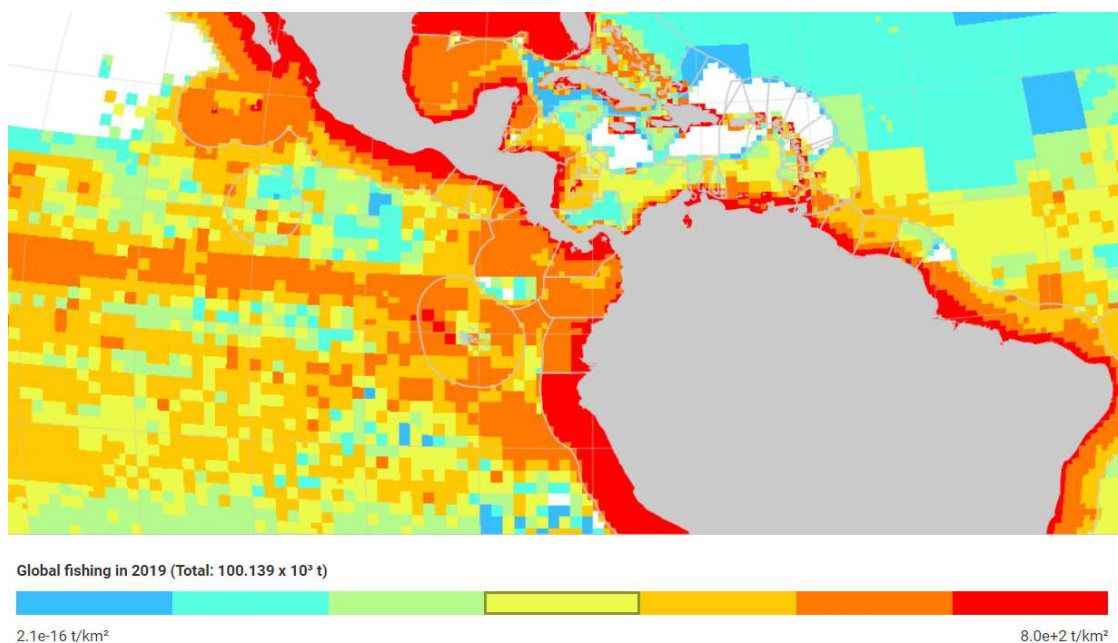


Figure 1.1 - Map of the total annual catches ($t\ km^{-2}$) in 2019 in the Eastern Pacific Ocean, Caribbean And Atlantic Ocean, displaying the density of catches sin coastal regions. From the global databases ('Sea Around Us' 2024).

The marine fishing sector in Costa Rica has steadily grown in the last decades, advancing its gear and boats, which have increased its fishing range and catch effectiveness. In Costa Rica, around 80% of landings come from artisanal fleets, most of which operate in the Pacific Ocean (Salas et al. 2011). The Gulf of Nicoya has an exceptionally high productivity, which makes it Costa Rica's principal fishing ground for the artisanal fishing fleet and where its highest landings are concentrated (Salas et al. 2011). This makes the Gulf of Nicoya one of Central America's largest and most exploited estuaries, where many stocks have already been over-

exploited and/or collapsed. According to the *Sea Around Us* statistics ('Sea Around Us' 2024), almost 90% of the fish stocks in the Gulf of Nicoya are exploited, 80% are over-exploited, and 40% have already collapsed (Figure 1.2). Thus, the Gulf of Nicoya is a suitable example of a region facing overexploitation and increasing conflicts among fishing sectors due to decreasing resources (Lozano and Heinen 2016a).

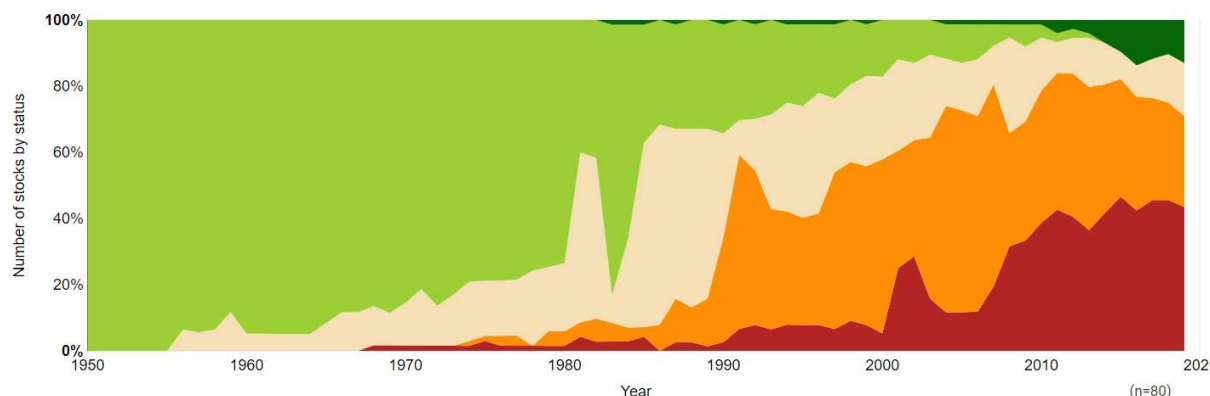


Figure 1.2 - Time series of number (%) of fish stocks by status in the Gulf of Nicoya from 1950 to 2020 (the fish stock red: collapsed, orange: is overexploited, beige: is exploited, light green: is developing or dark green: is rebuilding). The figure is derived from ('Sea Around Us' 2024)

The Gulf of Nicoya is tremendously important to artisanal fishermen, sustaining their livelihoods and maintaining food production security. SSFs are historically rooted in the communities surrounding the gulf, for which fishing is the primary source of income, making them especially vulnerable to overexploitation (Carrillo et al. 2019). In Costa Rica, the artisanal fishing sector dominates, providing livelihoods for coastal communities, and many fishermen fish for subsistence. The fisheries market is relatively small, and the commercial fisheries represent around 2-3 % of the GDP (Sabau 2017; Salas et al. 2011). However, the advanced artisanal and semi-industrial fleet focuses their production on the export market, with species such as tuna, mahi mahi (Dolphinfish), marlin, sharks, and shrimps. Small pelagic fish are sold to the commercial fisheries as bait.

To protect marine resources and to ensure a more sustainable use, some measures on the regional scale have already been implemented. The measures of input control include fishing effort limitation through granting licenses and a seasonal closure known as the “Veda” to all fishing fleets, as well as the technical limitation of gear size (mesh sizes >6.35 cm [2.5 in]). However, most of these top-down enforced management rules failed due to the fishers’ non-compliance or the lack of control capacities. These fishing regulations have been supplemented with more severe enforcement, such as the ban on shrimp trawling for the entire Gulf of Nicoya. In 2013, semi-industrial shrimp trawling was declared unconstitutional in Costa Rica, and no further licenses have been issued since 2015. The last remaining shrimp trawling license in the Gulf of Nicoya expired in 2019. This was a major step towards a more sustainable fishing industry and significantly reduced fishing efforts.

Costa Rica is a model region that can guide many other coastal areas worldwide. On the one hand, it has an immense potential for marine conservation, with an Exclusive Economic Zone (EEZ) ten times larger than its terrestrial area. On the other hand, the country has already proven its understanding of the urgency of protecting its environment, aiming to be among the global leaders in sustainability and environmental protection (UNEP, 2019). More than ever,

significant attention is being paid to global marine conservation. Like Costa Rica, many countries are interested in pushing for ocean conservation, for example, by joining the Decade of Ocean Sciences for Sustainable Development (2021–2030) proclaimed by the United Nations. The goal is to achieve a healthy, resilient, and sustainably harvested ocean for many coastal countries worldwide (Heymans et al. 2020; Ryabinin et al. 2019).

1.1.3 Description of the Study Area

The Gulf of Nicoya (10°N, 85°W) is a tidally-influenced estuary along the northwest Pacific coast of Costa Rica. It is one of the world's most productive tropical estuaries. It stretches from the northern inland (at the mouth of the Tempisque River) to the entrance of the Pacific Ocean in the south and covers an area of 1530 km². According to its shape and bathymetry, the Gulf of Nicoya can be divided into the following three regions: inner gulf, intermediate gulf, and outer gulf, named zones 201, 202, and 203, respectively, by the Costa Rican Institute of Fisheries (INCOPECA). The division is based on species composition and the fishing gear (Sánchez-Jiménez et al. 2019).

The shallow interior zone (depth < 10 m) is fringed by mangroves and mudflats and is dominated by the seasonal freshwater inflow from the Tempisque River (Proyecto Golfos 2012). The inner gulf is considered an area of high priority by the Instituto Costarricense de Pesca y Acuicultura (INCOPECA) due to its vital function as a reproduction and nursery area for commercially important fish and shrimp species (Alpízar 2011; Proyecto Golfos 2012; Vargas 2016). Sandy beaches and mangroves mostly surround the intermediate zone of the gulf. The physical parameters of the water masses are subject to intense mixing processes due to the interaction between the oceanic water and fresh water from both adjacent zones (Proyecto Golfos 2012). The region comprises reproduction areas for small pelagic fish species (Clupeiformes: Clupeidae and Engraulidae), shrimp *Litopenaeus* species (Decapoda: Penaeidae), snapper *Lutjanus* species (Perciformes: Lutjanidae), and drum *Cynoscion* species (Perciformes: Sciaenidae) and serves as a feeding ground for several commercially targeted species, such as the Mexican Barracuda (Perciformes: Sphyrinae). The outer oceanic zone extends from a minimum depth of 20 m out to 200 m at the shelf edge, is surrounded by sandy

beaches and rocky shores (Wolff et al.1998), and is mainly influenced by oceanic waters and the input from the Tarcoles River.

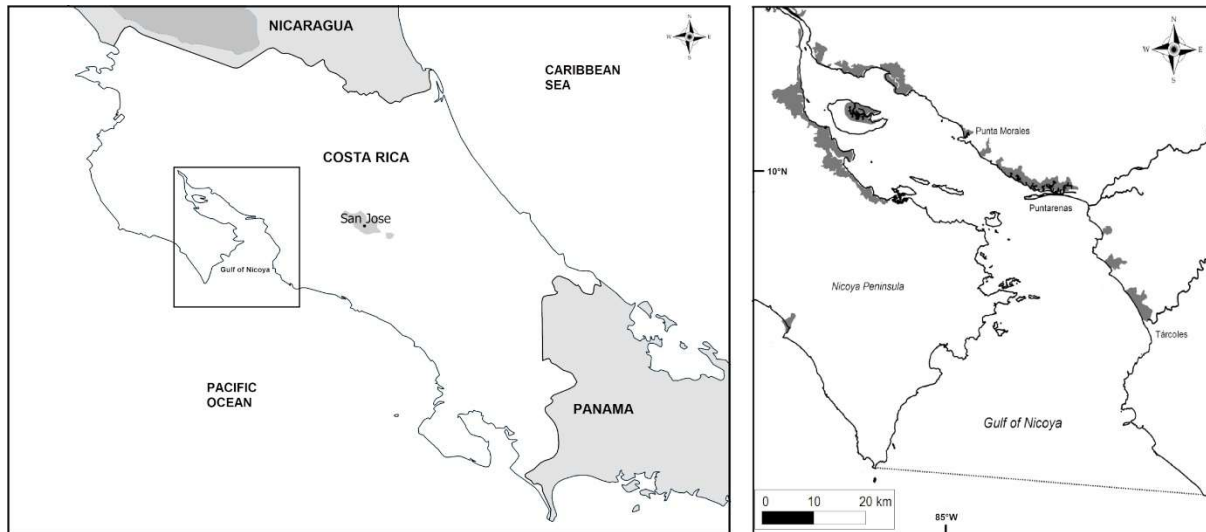


Figure 1.3 - Maps of the study area (Gulf of Nicoya, Costa Rica) including the three fishing zones (i.e., inner gulf, zone 201; intermediate gulf, zone 202; and outer gulf, zone 203).

The small-scale artisanal fishing sector is of primary economic and social importance for approximately 2000 fishermen and more than 20 coastal communities (Lozano and Heinen 2016a; Trujillo et al. 2012). Four artisanal fishing fleets operate in the gulf with different boat sizes and gear types that target various demersal and coastal species (BIOMARCC-SINAC-GIZ 2013a). Gillnets are predominantly used, followed by the longline, handline, and manual mollusk extraction combined with the manual trap fisheries targeting different crustaceans, the latter mostly carried out by women from the coastal communities. The main port of the Gulf of Nicoya, Puntarenas, is also the landing site for semi-industrial fishing fleets such as the purse seine fleet and the shrimp trawl fleet. The purse seine vessels target small pelagic fish species, such as tropical sardines and herrings (Family Clupeidae), Anchovy and Pacific anchoveta (*Anchoa spp.* and *Cetengraulis mysticetus*) in the intermediate and outer zones of the Gulf of Nicoya. Before the trawling ban, the bottom trawl fleet targeted several shrimp species, mainly in the outer gulf; however, the capture of bycatch species drastically exceeded the shrimp catches (Clarke et al. 2016).

1.2 A Case Example of Holistic Ecosystem Management

1.2.1 Ecosystem Approach to Fisheries

Changes in the biomass that occur at one trophic level are known to affect the trophic levels above and below that level, causing trophic cascades throughout the system (e.g., Pace et al. 1999; Duarte and García 2004; Andersen and Pedersen 2009). Hence, the analysis of changes in the ecosystem over time can be particularly useful for explaining changes in trophic relationships (Coll et al. 2009) and changes in the effort allocation and catches of the fishing fleets due to indirect conflicts and competition for resources (Bastardie et al. 2015).

Ecosystem food webs change over time due to changes in natural predator-prey interactions, resource limitations, anthropogenic influences, and environmental influences (Jennings and Kaiser 1998; Frederiksen et al. 2006). Fishing, as a substantial anthropogenic impact factor, greatly affects the population size of target species, their prey, and predators and may even damage the species' habitats (Frederiksen et al. 2006; Jennings and Kaiser 1998). Although overfishing is known to have caused the decline and collapse of many fish stocks worldwide, it can also alter the sensitivity of the ecosystem to natural environmental variations. It may reduce the ability to buffer the effects of extreme natural events or future climate change (Hidalgo et al. 2011). Thus, since ecosystems are driven by the combined impact of varied influencing factors, these must be identified and examined to understand the system's dynamics. In our study region, we also considered the El Niño-Southern Oscillation (ENSO) cycle that causes periodic variations in several environmental parameters, which temporarily change the coastal marine environment and its species composition (Fiedler 2002; Tam et al. 2008; Taylor et al. 2008b; Wolff et al. 2012). This phenomenon is relevant for the Eastern Tropical Pacific (ETP) region and other marine regions worldwide (Bacalso et al. 2023; Del Solar et al. 2024).

In many marine areas worldwide, ecological information is needed to achieve the goals of the 'The Decade of Ocean Science for Sustainable Development' (2021–2030) proclaimed by the United Nations (UN). To achieve a healthy, resilient, and sustainably harvested ocean (Heymans et al. 2020; Ryabinin et al. 2019), the current High Seas Treaty aims to convert 30% of the seas into protected areas by 2030 (United Nations). In many regions of the world, suitable areas for protection need to be identified. However, for the efficiency of MPAs, the right choice of location, the spatial and temporal extent, and the successful enforcement are crucial (Dimarchopoulou et al. 2024).

Therefore, the need for suitable and successful management policies, plans, and science-based decisions is greater than ever. Thus, this study presents a case example of the Ecosystem Approach to Fisheries (EAF) (FAO 2002) as a possible way to provide the necessary information for management objectives. The EAF considers the trophic interactions among functional groups within the ecosystem and includes environmental, ecological, and anthropogenic impacts on an ecosystem in the management consideration. Following the EAF, one can find optimal, sustainable exploitation rates under the constraints of biodiversity conservation and an increase in the ecosystem's health and/or in the rebuilding stocks of target species (Forrest et al. 2015). The EAF provides clear benefits compared to single-species management approaches, which are often insufficient (Cochrane 2002; FAO 2012).

In the context of the EAF, we constructed a holistic ecosystem model of the Gulf of Nicoya (Alms and Wolff 2019), developed with the open-source software Ecopath with Ecospace to study the effects of fishing across the system. Ecosystem and multispecies models are vital to approaching and exploring ecosystem-related questions. More than 450 EwE ecosystem models of aquatic systems have been published, and our procedure could be applied to many of these models. We develop and apply a set of models with the spectrum of the EwE software, starting with a mass-balanced food web model, continuing with dynamic time-series modelling, and finishing with a spatial model. In addition, we apply an automated fishing policy search tool that looks for future developments/system changes under certain management plans. We investigated different management scenarios while considering different management

objectives or a combination (Christensen and Walters 2004) by optimizing for socioeconomic and/or ecosystem conservation goals. We also holistically use this information to present a spatial ecosystem model of the Gulf of Nicoya. This model displays fishing effort and species distribution, which can be used to validate the existing MPAs' performance and to find potential areas suitable for additional protection.

By 2020, 7.74% of marine areas worldwide were conserved through protected areas and other effective area-based conservation measures (UNEP-WCMC and IUCN 2021). In 2010, the United Nations set a goal to preserve 10% of the ocean by 2020. The recent High Seas Treaty even aims to place 30% of the seas into protected areas by 2030 to conserve marine nature (Johanna J Heymans et al. 2020). In many regions of the world, Ecospace can be a valuable tool for finding suitable areas for protection.

1.2.2 Technical Approach: Ecopath with Ecosim Software

1.2.2.1 Ecopath - Static model/mass-balanced model

For the dissertation, we encompassed the full spectrum of the Ecopath with Ecosim (EwE) software, starting with a static model (Ecopath), continuing with time-series modelling and policy optimization (Ecosim), and finishing with temporal-spatial modelling (Ecospace).

The Ecopath approach has been widely used to assess and visualize structural changes in a given ecosystem (Pauly, Christensen, and Walters 2000). The open-source software Ecopath with Ecosim (EwE) versions 6.4.4 to 6.6.5 (Christensen and Walters 2004) was used for the modelling task. Ecopath with Ecosim is based on a set of linear equations for each functional group in the system through which the group's biomass production is connected to the other model groups via a prey–predator matrix. For each compartment (i), the basic equation is as follows:

$$P_i = Y_i + B_i M2_i + E_i + B a_i + P_i(1-EE)$$

where P_i is the total production rate of i ; Y_i is the total fishery catch rate; $M2_i$ is the instantaneous predation rate of group i ; B_i is the biomass; E_i is the net migration rate (emigration–immigration); $B a_i$ is the biomass accumulation rate; $P_i(1 - EE)$ represents the “other mortality” rate of i ; and EE is the ecotrophic efficiency of i and the proportion of the production that is utilized in the system (Christensen and Walters 2004; Christensen et al. 2005).

Although equation 1 allows for the balancing of the biomass production in each group with its use in the system (and the fishery), a second equation provides for estimating the energy budget of each group i as follows:

$$\text{Consumption}_i = \text{Production}_i + \text{Respiration}_i + \text{Egestion of unassimilated food}_i$$

The model was built around the main groups targeted by the fisheries. Species with similar sizes, diets, consumption, mortality, production rates, and habitats were grouped. The model comprised 22 functional groups, including three primary producer groups and ten different fish groups.

In the first article presented in this thesis (Chapter 2), two mass-balanced snapshots of the Gulf of Nicoya ecosystem have been produced, comprising the same ecosystem components at two

distinct time points (1993 and 2013) to reveal changes in the food web of the marine community (Neira, Arancibia, and Cubillos 2004). For this analysis, we determined the differences in biomass distribution, system flow structure, and system summary statistics for all system components, including primary producers, target species, and nontarget species. By applying the mixed trophic impact (MTI) analysis in Ecopath, we assessed the effect of the different fishing fleets and gear types on the marine community and the interactions among the species and the identified keystone species (Libralato et al. 2006) in both system states (i.e., 1993 and 2013). Assessing keystone species can be crucial for ecosystem management, as resource conservation plans often focus more on species of high commercial value than those of high ecological value (Navia et al. 2016). In addition, we used the Ecopath software to calculate the economic profit of the fisheries during both periods. Information regarding the resource market prices and the fishing costs (variable and fixed costs) for each of the fishing fleets was included in the Ecopath models (Alms and Wolff 2019).

1.2.2.2 Ecosim - Time-series fitting

Ecosim can incorporate time series data, which can be used to test the model performance and explore policy options for ecosystem-based fisheries management (Christensen and Steenbeek 2024). Forcing data (fishing effort or fishing mortality) is needed to trace the biomass changes of the model compartments over time, and reference data is necessary for comparison and validation of the models. In our case, fishing effort (days fishing) and time series data of the environmental parameters were used to force model group catches and biomasses over time, whereas the recorded historic landings served as (observational) reference data for the catch. During time-series fitting, a statistical measure of goodness of fit to these data is generated in Ecosim as a weighted sum of squared deviations (SS) of log biomasses from log predicted biomasses. The goodness of fit indicates how the model performs and whether it can be used for policy exploration (Christensen and Steenbeek 2024). Ecosim can be used to identify ecosystem effects of fishing and environmental factors (Christensen and Steenbeek 2024).

In the second article included in this thesis (Chapter 3), we present time-series models based on the two EwE “snapshot models” of the Gulf of Nicoya ecosystem. The Ecosim time-series fitting routine was applied (Christensen and Steenbeek 2024), using fisheries and environmental data from 1994 to 2014 to elucidate the role of the fishing effort and abiotic factors in the ecosystem. A hypothesis-based model fitting procedure was applied, for which time series of inter-annual changes in fishing effort, fleet composition, and the most relevant abiotic factors were considered. Three essential classes of ecosystem drivers were tested: anthropogenic factors (fishing, pollution), trophic control mechanisms (interactions), and the influence of the chemical and physical environment (climate). The procedure is an example of examining whether observed system changes are driven by the exploitation patterns of fisheries and/or by environmental changes. This is a central question relevant to many other ecosystems and has been applied in similar systems, e.g., the Bering Sea, the Tropics, and the Mediterranean (Coll et al. 2016; Lam et al. 2020; Mackinson et al. 2009; Punt et al. 2024). We consider identifying the main drivers behind the observed changes as crucial requisites for selecting appropriate management and conservation strategies.

Ecosim implements the ‘Fishing policy exploration’ tool in EwE 6.6.5 (Christensen and Steenbeek 2024). This optimization method can be used to search for fishing policies that

maximize particular policy goals for management. The policy exploration has two alternative objective functions, where by default, the objective function is defined based on an evaluation of the weighted policy objectives that maximize profits or net economic value (value of the catch minus costs of fishing), social benefits /jobs provided by the fishing fleets), biodiversity (biomass distribution across functional groups), ecosystem structure (Ecosystem ‘health’) (Christensen and Walters 2024).

In the third article (Chapter 4), we present the application of the policy exploration tool to find the optimal management policy for the Gulf of Nicoya. We used a scenario-based approach using the EwE simulation tool. We tested the losses and benefits of the social and economic value (profit and number of jobs) and the conservational criteria (ecosystem health and biodiversity) caused by different management scenarios after 20 years. Furthermore, the fishing policy tool was applied to explore alternative management scenarios. The automated search routine allows the assessment of appropriate management scenarios while considering different management objectives or a combination of them (Christensen and Walters 2004). The output can, for example, be optimized for socioeconomic and ecosystem conservation goals. It can distinguish between a project with long-term benefits versus a project with short-term benefits and long-term costs. Thus, the fishing policy search routine can be beneficial for different scientific questions while exploring different management strategies. One of these scenarios is a stakeholder-based policy, whereby gillnet fleet efforts are reduced by 25% and aimed at allowing high trophic level species to rebuild their stock biomass.

In an interdisciplinary collaboration with social scientists, a participatory management workshop was developed in Costa Rica (Sánchez-Jiménez et al. 2019; Sánchez Jiménez et al. 2021). In these workshops, stakeholders were invited, and ecological information about the Gulf of Nicoya ecosystem was presented using the EwE modelling software. The fishermen suggested possible policy plans based on the visualization and information about ecosystem health and future perspectives. The stakeholder-based policy was developed to encourage stakeholder cooperation and exchange and foster participatory management in the Gulf of Nicoya. In addition to finding the optimal balance between different management goals to create successful policies, it is vital to consider the interests and knowledge of local stakeholders, thereby increasing the compliance of all actors (Wilson et al. 2006). Examples show that endeavors to move from top-down management towards co-management can successfully enhance learning and knowledge among stakeholders and improve the relationship and trust between stakeholders. It may ensure the well-being of the fishers and the fishery, e.g., the coastal fisheries of Uruguay and Baja California in Mexico. (Cota-Nieto et al. 2018; Trimble and Berkes 2013).

1.2.2.3 Ecospace – Spatial model

Ecospace is a modelling tool that can be applied to a broad range of research questions that aim to incorporate a spatial component (Christensen and Steenbeek 2024). It can simulate the spatial distribution of species and fishing fleet effort over time, explore spatial management options, and address various research questions, including the regional consequences of climate change (Christensen and Walters 2004). The temporal and spatial dynamic of Ecospace models has, in many cases, contributed to Ecosystem-Based Fisheries Management (Christensen and Walters 2024), e.g., in the EEZ of the Azores, the Gulf of Mexico, and the North Sea (Brito et al. 2023;

O'Farrell et al. 2017; Romagnoni et al. 2015). Moreover, the location and efficiency of existing MPAs and optimal locations for new protected areas can be assessed.

In the fourth manuscript (Chapter 5), we present how we use the Ecospace maps to assess and display species and effort distribution in a period of exploitation by shrimp trawlers and the period after the ban of trawling. We model the effects of decreasing fishing pressure on shrimp species and several bycatch species and a long-term effect on the upper trophic levels. Additionally, we assessed the impact of the seasonal fishing closure in the Gulf of Nicoya and the consequences of changes (relaxation of regulations) in the MPA regulations due to the COVID-19 pandemic.

1.3 Scope of the Thesis

This thesis' main subject involves analyzing the past, present, and future situations/state of the Gulf of Nicoya as a fishing ground of economic relevance, a natural habitat for many marine and coastal species, and a living environment and livelihood for many coastal communities. It allows for the comparison over a long period. The analysis includes important aspects for an EAF, including food-web modelling, time-series fitting, economic validation, spatial modelling, stakeholder analysis, engagement, and fisheries management/evaluation of management policy.

This PhD thesis incorporates the practical handling of the EwE software by applying its different tools to the Gulf of Nicoya ecosystem model and interpreting the modelling results. This procedure is replicable and can be transferred to other existing or newly-built EwE models in various areas of the world. The thesis presents the wide range of ecological and economic information that can be gathered based on a solid ecosystem model using EwE as a sole instrument. It highlights that the capability and potential of EwE software go far beyond creating food-web models. The EwE software is a powerful tool that adds economic and social information to the assessment, often lacking in ecological studies. Spatial analysis and visualization can foster participatory management and stimulate stakeholder discussion.

Chapter 2

THE PAST OF THE GULF OF NICOYA ECOSYSTEM: TWO DECADES OF CHANGE ECOPATH

Chapter 2

The Gulf of Nicoya (Costa Rica) Fisheries System: Two Decades of Change.

Viola Alms and Matthias Wolff

This is the author's version of the work. Please cite the final version as:

Alms, V., and Wolff, M. (2019). The Gulf of Nicoya (Costa Rica) fisheries system: Two decades of change. *Marine and Coastal Fisheries*, 11, 139–161.

<https://doi.org/10.1002/mcf2.10050>

Submitted on 30th June 2017.

Accepted on 13th September 2018.

Published online on 25th March 2019.

Contribution statement

Project idea: **Viola Alms and Matthias Wolff**

Work design: **Viola Alms**

Acquisition and interpretation of data: **Viola Alms**

Drafted manuscript revised by Matthias Wolff.

Abstract

In the early 1990s, ecological and fishery data from the Gulf of Nicoya (Pacific Ocean) were holistically analyzed, and a trophic model was constructed using the Ecopath modeling approach. The results indicated that this tropical estuary, which is a Costa Rican fishery hot spot, was already overexploited by shrimp trawlers and the longline fleet, and recent observations suggest further deteriorations in this system. To evaluate the ecosystem and socioeconomic changes in the Gulf of Nicoya over the last 20 years, the 1993 model was reconstructed with data from 2013 to compare both system states. Although the summary statistics of both states (i.e., 1993 and 2013) suggested that the system maintained its general functionality and even enabled the total harvest to increase by approximately 20%, a more detailed analysis of the levels of the functional groups suggested that the system was further degraded as follows: (1) most shrimp, eel, and catfish species decreased their share in the catches, with severe economic losses for the fishery; (2) of the species that substantially increased their share in the catches, most were short-lived, low-value species, such as small pelagic and small demersal fishes; and (3) catches of long-lived, high-trophic-level fish, such as rays, sharks, mackerels, and barracudas, decreased. A picture emerged in which the advancing fishery of low-trophic-level species with low economic value enabled the total harvest to increase, while valuable shrimp resources and higher-trophic-level species were depleted. These developments caused a tremendous economic loss of approximately 50%.

Keywords: Ecosystem modelling, , Ecopath with Ecosim, trophic model, estuary

2.1 Introduction

Coastal waters are often highly productive and support essential fisheries worldwide; thus, these waters have a high economic value (Blaber et al., 2000). The vast majority of Costa Rican fishery catches originate from the country's relatively large Exclusive Economic Zone (10 times the terrestrial area) along the Pacific coast, a region that is known to be nutrient-rich and highly productive, and most of its target fish stocks are known to be heavily exploited (Heileman 2012; Sea Around Us Project. 2014). Costa Rica's principal fishing fleets concentrate in the Gulf of Nicoya, which accounts for a large proportion of the national catches and is among the most exploited coastal areas in Central America (Guzmán-Mora and Molina-Ureña 2008). The small-scale artisanal fishing sector is historically rooted in the communities surrounding the Gulf of Nicoya, is highly socioeconomically important for securing food production and livelihoods, and has national economic benefits. Initially, the artisanal fishing activities were confined to the inner gulf and shallow waters (Charles and Herrera-Ulloa 1994), but the fleets progressively extended to pelagic, offshore, and deepwater areas in the gulf (BIOMARCC-SINAC-GIZ 2013b).

After a period of intense exploitation by the multi-gear artisanal fleet and small industrial fleets, which peaked in the early 2000s, catches of several commercially important species began to show clear signs of decline and are likely to follow the same downward trend as the catches of many other fish stocks in the Pacific Central American Large Marine Ecosystem (Heileman 2012; Sea Around Us Project. 2014). In particular, coastal shrimp catches drastically declined, with a severe reduction of white shrimps *Litopenaeus* sp. (Trujillo et al. 2012). These developments could lead to severe ecological and socioeconomic consequences because penaeid shrimps are known to be a central food source in the Gulf of Nicoya's trophic web (Wolff et al. 1998) and are the most sought-after target group by many local coastal communities, which are highly vulnerable to fluctuations and negative trends in the fishery catches (Bakun 1999). Coastal demersal fish, which dominate the catches of many inshore artisanal fisheries in the tropical Pacific, are valuable marine resources and are also commonly overexploited (Bell et al. 2011; Trujillo et al. 2012). Similarly, in the Gulf of Nicoya, large- and medium-sized drums and croakers (Perciformes: Sciaenidae) have always constituted a substantial and profitable proportion of the artisanal catches, but currently they are caught less frequently. The decline of high-value coastal species in the Gulf of Nicoya has caused (1) the semi-industrial fleet effort to shift toward other, low-value target groups, such as small pelagic fish species; and (2) the artisanal effort to shift toward smaller-sized individuals and juveniles due to the augmented use of small illegal mesh sizes (6.35 cm [2.5 in]).

These developments have likely caused changes in the size and species composition within the functional groups. Hence, an updated analysis of the species and weight composition of catches appears to be important for the evaluation of ecological changes in the system and the associated socioeconomic consequences (Roessig et al. 2004; Ward and Myers 2005a). The depletion of coastal resources is not only associated with economic losses for the semi-industrial sector but is also likely to threaten the livelihoods of the artisanal fishing communities surrounding the Gulf of Nicoya. Hence, there is an urgent need for an ecosystem-based fishery assessment of the gulf to provide a decision-making basis for the Costa Rican management authorities. The Gulf of Nicoya has been studied by many scientists (Vargas 1995; Vargas 2016), but fishery

studies concerning commercially exploited species within the area are sparse. Since the trophic model of the Gulf of Nicoya was constructed by Wolff et al. (1998), no further attempts have been made to integrate fishery-relevant data in a holistic, ecosystem-based analysis. Since that model's construction in the early 1990s, important resources in the system have already been considered overexploited, and a further degradation of the ecosystem has been predicted (Wolff et al. 1998). We attempted to rebuild and update the model using data obtained approximately 20 years later from collaborative studies that were complemented by additional information from the literature regarding catches of target resources to analyze the ecosystem changes that occurred over the two decades between 1993 and 2013.

Changes in the biomass that occur at one trophic level are known to affect the trophic levels above and below that level, causing trophic cascades throughout the system (Pace et al. 1999; Andersen and Pedersen 2010; Duarte and García 2004). Hence, the analysis of changes in the ecosystem over time can be particularly useful for explaining changes in trophic relationships (Coll et al. 2009) and for determining the effects of certain fishing regimes on the trophic structure of the ecosystem (e.g. Neira et al., 2014; Walters et al., 1997). Thus, we used an Ecopath approach to produce mass-balanced snapshots of the Gulf of Nicoya ecosystem comprising the same ecosystem components at two distinct time points (1993 and 2013) to reveal the changes in the food web of the marine community (Christensen and Walters 2004a). For this analysis, we determined the differences in the biomass distribution, system flow structure, and system summary statistics for all system components, including primary producers, target species, and nontarget species (Gascuel et al. 2011). By applying the mixed trophic impact (MTI) analysis in Ecopath, we assessed the effect of the different fishing fleets and gear types on the marine community and the interactions among the species and the identified keystone species (Libralato et al. 2006) in both system states (i.e., 1993 and 2013). The assessment of keystone species can be crucial for ecosystem management, as resource conservation plans often tend to focus more on species of high commercial value than on species of high ecological value (Navia et al. 2016).

In addition, we used Ecopath software to calculate the economic profit of the fisheries during both periods. Information regarding the resource market prices and the costs of fishing (variable and fixed costs) for each of the fishing fleets was included in the Ecopath models.

2.2 Methods

2.2.1 General description of the Gulf of Nicoya ecosystem

The Gulf of Nicoya (10°N, 85°W) is a tidally influenced estuary along the northwest Pacific coast of Costa Rica (Figure 2.1; Wolff et al. 1998). It is one of the world's most productive tropical estuaries, and its main port, Puntarenas, is the country's principal fishing ground for the artisanal fishing fleet (J. A. Vargas 1995). The model area stretches from the mouth of the Tempisque River in the north to the entrance of the Pacific Ocean in the south and covers an area of 1,530 km². According to its shape and bathymetry, the Gulf of Nicoya can be divided into the following three regions (Figure 2.1): inner gulf (zone 201), intermediate gulf (zone 202), and outer gulf (zone 203; Vargas 2016).

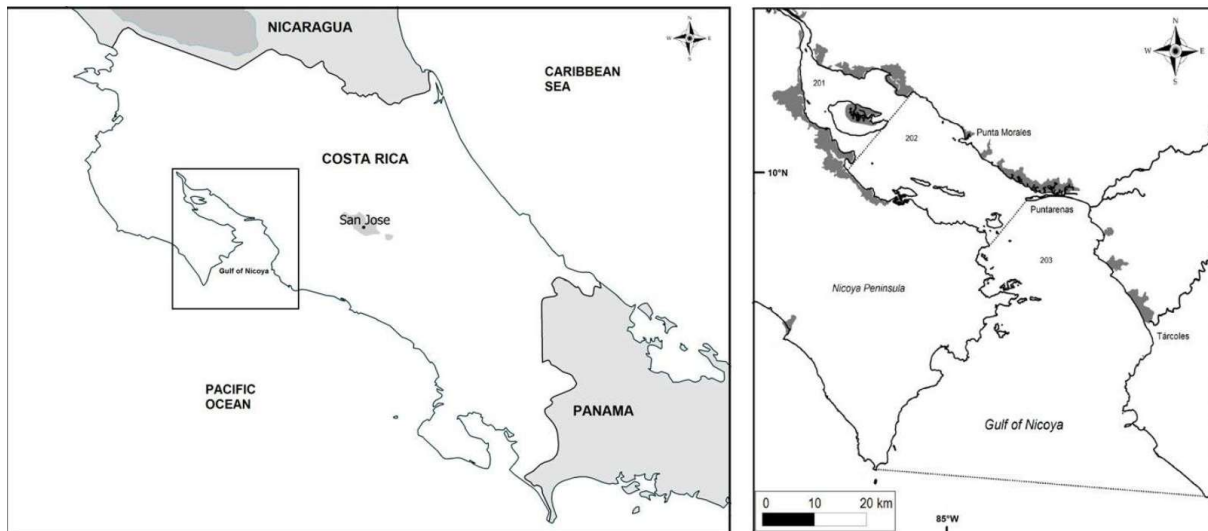


Figure 2.1 - Maps of the study area (Gulf of Nicoya, Costa Rica) including the three fishing zones (i.e., inner gulf, zone 201; intermediate gulf, zone 202; and outer gulf, zone 203).

The shallow interior zone (depth < 10 m) is fringed by mangroves and mud flats and is dominated by the seasonal freshwater inflow from the Tempisque River (Proyecto Golfos 2012b). The inner gulf is considered an area of high priority by the Instituto Costarricense de Pesca y Acuicultura (INCOPESCA) due to its important function as a reproduction and nursery area for commercially important fish and shrimp species (Alpízar, 2011; Proyecto Golfos, 2012b; Vargas, 2016). The intermediate zone of the gulf is mostly surrounded by sandy beaches and mangroves. The physical parameters of the water masses are subject to intense mixing processes due to the interaction between the oceanic water and freshwater from both adjacent zones (Proyecto Golfos 2012). The region comprises reproduction areas for small pelagic fish species (Clupeiformes: Clupeidae and Engraulidae), shrimps *Litopenaeus* spp. (Decapoda: Penaeidae), snappers *Lutjanus* spp. (Perciformes: Lutjanidae), and drums *Cynoscion* spp. (Perciformes: Sciaenidae) and serves as feeding grounds for several commercially targeted species, such as the Mexican Barracuda *Sphyraena ensis* (Perciformes: Sphyraena). The outer oceanic zone extends from a minimum depth of 20 m out to 200 m at the shelf edge, is surrounded by sandy beaches and rocky shores (Wolff et al. 1998), and is mainly influenced by oceanic waters and the input from the Tárcoles River. The eastern region of the outer gulf is a hatching and reproduction site for penaeid shrimps, sharks, and several other fish species (Proyecto Golfos, 2012a; Vargas, 2016). Similar to other tropical coastal estuarine areas, mangroves contribute to primary production in the Gulf of Nicoya by providing organic matter and playing an important role in enhancing fish and invertebrate productivity; thus, mangroves strongly support the local fisheries (Hutchison et al. 2014; Wehrmann and Cortés 2008; Wolff 2006).

2.2.2 Model Construction

2.2.2.1 Modeling approach

The previous trophic model (GN₀) that was initially developed by Wolff et al. (1998) was based on data sampled by the RV *Victor Hensen* Costa Rica expedition (1993–1994) and fisheries landings data (INCOPESCA, unpublished data). The GN₀ serves as a baseline for this study and represents the ecosystem state in 1993–1994. Based on new information obtained from collaborative studies, the literature, and the most recent fisheries data, a second model (GN₊₂₀)

was constructed to represent the state of the ecosystem in 2013. To ensure comparability, both models comprised the same functional groups. For the modeling task, the open-source software Ecopath with Ecosim (EwE) version 6.4.4 (Christensen and Walters 2004a) was used.

Ecopath with Ecosim is based on a set of linear equations for each functional group in the system through which the group's biomass production is connected to the other model groups via a prey–predator matrix. For each compartment (i), the basic equation is as follows:

$$P_i = Y_i + B_i M2_i + E_i + BA_i + P_i(1 - EE)$$

where P_i is the total production rate of i ; Y_i is the total fishery catch rate; $M2_i$ is the instantaneous predation rate of group i ; B_i is the biomass; E_i is the net migration rate (emigration–immigration); BA_i is the biomass accumulation rate; $P_i(1 - EE)$ represents the “other mortality” rate of i ; and EE is the ecotrophic efficiency of i and the proportion of the production that is utilized in the system (Christensen and Walters 2004a; Christensen et al. 2005).

Although equation (1) allows for the balancing of the biomass production in each group with its use in the system (and the fishery), a second equation allows for estimating the energy budget of each group i as follows:

$$Consumption_i = Production_i + Respiration_i + Egestion\ of\ unassimilated\ food_i$$

The model was built around the main groups targeted by the fisheries. Species with similar sizes, diets, consumption, mortality, production rates, and habitats were grouped. The model comprised a total of 22 functional groups, including 3 primary producer groups and 10 different fish groups. The functional groups and the scientific names of all species that are included in the GN models are presented in the Appendix (Table A.1).

2.2.2.2 Primary producers and zooplankton

The mangrove coverage along the Costa Rican Pacific coast was estimated to be 413 km² (Wehrtmann and Cortés 2008). In the early 1990s, the mangrove coverage surrounding the Gulf of Nicoya was estimated to be approximately 15.2 km² (Brugnoli-Olivera and Morales-Ramírez 2008). Since then, the mangrove coverage of the Gulf of Nicoya has declined by 10%, which was attributed to land use changes, such as the development of shrimp aquaculture (R. V. Li, Barquero, and Fernández 2013). The phytoplankton community is dominated by diatoms and flagellates, and the primary productivity of plankton is known to be exceptionally high (Brugnoli-Olivera and Morales-Ramírez 2008). Recent studies conducted by researchers from the University of Cadiz, Spain, provided information regarding the phytoplankton communities at different stations in the Gulf of Nicoya (Requejo 2011). Phytoplankton chlorophyll-*a* (Chl-*a*) concentrations of 1–16 mg/m³ were estimated (BIOMARCC 2013a, 2013b). For the phytoplankton biomass calculations, the following ratios were used: Chl-*a* to carbon (g) = 1:100, predominantly for diatoms; carbon to dry weight organic matter (g) = 1:2.5; and dry organic matter to wet weight organic matter (g) = 1:5 (Parsons et al. 2013; Sathyendranath et al. 2009). Regional and seasonal differences were considered; thus, a minimum biomass of 15 g/m² was measured in the inner and intermediate gulf areas, which was assumed to be most appropriate for the whole gulf area. According to previous studies, the phytoplankton primary productivity in the Gulf of Nicoya averages between 0.12 and 2.76 g C·m⁻²·d⁻¹, and the annual gross primary productivity measured in Punta Morales was 450 g C·m⁻²·year⁻¹ (Córdoba-

Muñoz 1998; Gocke et al. 2002). The phytoplankton primary productivity was assumed to be lower in other zones of the gulf according to previous samples (Wolff et al. 1998). After the RV *Victor Hensen* Costa Rica expedition in the early 1990s, the zooplankton composition was once more intensively studied in Punta Morales during 1997 (Brugnoli-Olivera et al. 2004); the dominance of copepods in the zooplankton community was confirmed, and large seasonal variations in the biomass were detected. An average annual zooplankton biomass of 4 g/m² of water column was assumed for both models based on the work of Brugnoli-Olivera et al. (2004).

2.2.2.3 *Fishing fleets and target resources*

In Costa Rican Pacific waters, the tuna fleet and the large pelagic longline fleet are the most important in terms of catch volume (FAO 2007; Trujillo et al. 2012). However, inside the Gulf of Nicoya, the small-scale artisanal fishing sector is of major economic and social importance (Lozano and Heinen 2016a; Trujillo et al. 2012). Four artisanal fishing fleets operate in the gulf with different types of gear, including gill nets, lines (bottom longlines and drifting longlines), handlines, and manual gear (traps and mollusk extraction), and target a variety of demersal and coastal species (BIOMARCC-SINAC-GIZ 2013b; 2013a). Gill nets are prominently used, mostly with small mesh sizes (< 7.62 cm [< 3 in]), which are illegal in the Gulf of Nicoya. Catches are diverse but dominated by drums (Perciformes: Sciaenidae), catfishes (Siluriformes: Ariidae), snooks (Perciformes: Centropomidae), and different penaeid shrimp species (Alpízar 2015). Additionally, two semi-industrial fishing fleets operate in the gulf: shrimp trawlers and sardine purse-seiners. The purse seine vessels target small pelagic fish species in the intermediate and outer zones of the Gulf of Nicoya, while the bottom trawl fleet targets white shrimps and several other shrimp species in the outer gulf. In this fishery, the capture of nontarget (bycatch) species drastically exceeds the shrimp catches (Figure 2.2). Due to shrinking resources in shallow waters, the shrimp trawler fleet has progressively moved toward deeper waters outside of the gulf (Wehrtmann and Nielsen-Muñoz 2009); thus, the deepwater shrimp species that are landed in Puntarenas (kolibri shrimp *Solenocera agassizii*, bigheaded shrimp *Heterocarpus vicarious*, and three-spined nylon shrimp *Heterocarpus affinis*) are presumably fished outside of the gulf and were excluded from the model (UNEP 2004). Bycatch species caught outside of the gulf are not included in the model, as the Costa Rican deepwater shrimp fishery discards almost the entire bycatch, including species of commercial value (Wehrtmann and Nielsen-Muñoz 2009). The landings of the six fishing fleets were entered separately into the Ecopath model.

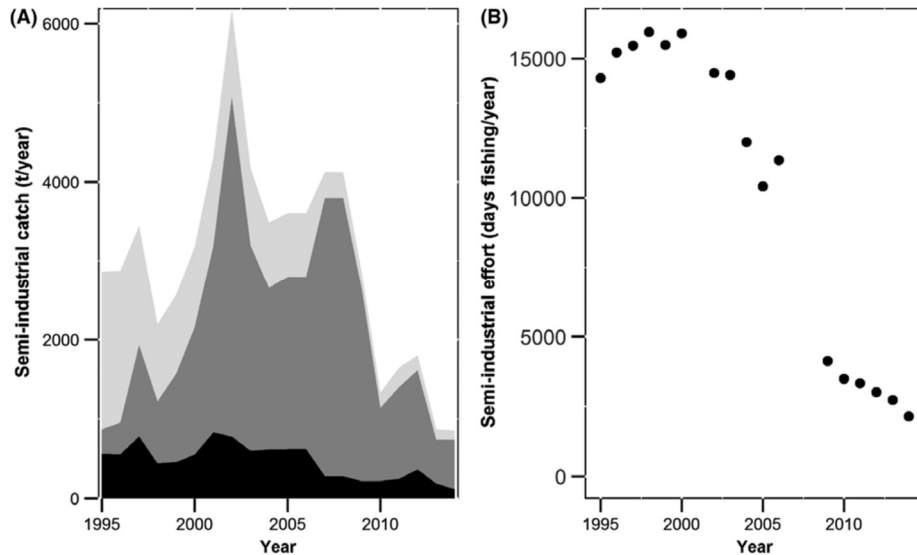


Figure 2.2 – Semi-Industrial catch and effort (Gulf of Nicoya, Costa Rica): (A) Catches (metric tons [t]/year) and (B) fishing effort (number of days fishing/year) of the semi-industrial fleet in the Gulf of Nicoya during 1994 – 2014 (in panel A, light gray = shrimp catch; dark gray = sardine catch; black = other fish catch).

All fish groups are targeted by at least one of the six identified fishing fleets. For each model group, the following basic input parameters were gathered: reported artisanal and semi-industrial catches, biomass (B), production- to-biomass ratios (P/B), consumption-to-biomass ratios (Q/B ; Table 2.1), and the diet composition of each species or group (Table A.2). Information regarding the diet composition was obtained from Wolff et al. (1998) and the recent literature (Lopez-Peralta and Arcila 2002; Olson and Watters 2003; Robertson 2009). For additional references, please refer to the Appendix (Table A.3). The artisanal landings data from each zone of the Gulf of Nicoya were obtained from the Costa Rican monthly national landing statistics (INCOPECA, unpublished data). The provided time series consists of monthly records of the main target species and groups (fish and invertebrates) that were fished and landed in the Gulf of Nicoya. The landings (kg) were separated by fishing fleet and gear. The estimated bycatch of the shrimp trawlers and discard estimates were included. The monthly records of the semi-industrial landings and the fishing effort, which was measured as the number of days fishing (as well as some data about the number of fishing trips) in the Gulf of Nicoya, were provided by INCOPECA (Departamento de Investigación y Desarrollo). Additional yearly sampling data (INCOPECA, Departamento de Investigación y Desarrollo) were obtained to estimate the relative species composition within the target groups. To enter the data into the model, the monthly landings data, discards, and shrimp trawl bycatch data were converted into metric tons per square kilometer (based on the model area of 1,530 km²) per year.

2.2.2.4 Calculations of the model parameters

The biomass estimates in the early model (GN_0) were based on fishery catches and beam and otter trawl survey data. Trends in fishery catches can serve as a first indicator of the population biomass of the target resources. For the 2013 model (GN_{+20}), the resource biomass values per unit area (metric tons/km²) were calculated from the catch data (Y) and estimated fishing mortality (F) by using the following formula proposed by Beverton and Holt (1957): $B = Y / F$:

The recent exploitation rates and F -values for most of the fish and shrimp species were obtained from INCOPECA stock assessment studies (Alpízar 2011; 2014; 2015; Alpízar et al. 2013;

Alpízar and Vásquez 2010; 2012) and from research conducted by the Universidad Nacional Costa Rica (Quiróz Aizpurúa 2014). Stock information on morays/ eels, catfishes, and flatfishes is scarce, and the biomass had to be estimated from the catch data and the previous model (Wolff et al. 1998). The values were compared to those from other studies conducted in similar areas. The biomass values for three functional groups - microphytobenthos, endobenthos, and seabirds - were estimated according to information available in the literature regarding their abundance and biomass estimations, which are based on the previous model (see Table A.3 for detailed references).

Table 2.1 - Basic Ecopath with Ecosim input and output parameters for the models GN_0 (1993) and GN_{+20} (2013) of the Gulf of Nicoya, Costa Rica. Biomass (B_i ; metric tons/km²), production-to-biomass ratio (P_i/B_i ; year⁻¹), consumption-to-biomass ratio (Q_i/B_i ; year⁻¹), ecotrophic efficiency (EE_i), production-to-consumption ratio (P_i/Q_i), and total catch (metric tons·km⁻¹·year⁻¹) are presented for each functional group. The numbers in parentheses indicate the initial input parameters prior to the model balancing routine. Input parameters that are the same between models are indicated in bold italics for the GN_{+20} model.

Group #	Functional group	Input parameters				Output parameters	
		Catch	B _i	P _i /B _i	Q _i /B _i	EE _i	P/Q
Parameters for model GN ₀ (1993)							
1	Seabirds		0.07 (0.06)	0.10	50	0.00	0.002
2	Mackerels and barracudas	0.036	0.15	0.55 (0.5)	7 (6)	0.43	0.080
3	Rays and sharks	0.048	0.15	0.60	4	0.56	0.150
4	Large drums and snooks	0.122	0.60	0.65	4.5 (5)	0.47	0.144
5	Catfishes	0.257	0.70	0.70 (0.65)	6	0.73	0.117
6	Morays and eels	0.017	0.70	0.50	5 (5.5)	0.48	0.100
7	Carangids	0.020	0.60	0.75	6	0.90	0.125
8	Snappers and grunts	0.179	0.55 (0.60)	0.90 (1.0)	5 (6)	0.93	0.180
9	Cephalopods	0.015	0.50 (0.40)	4.00	15	0.77	0.267
10	Lobster	0.001	0.25	1.00	10	0.49	0.100
11	Flatfishes	0.022	0.30	1.80	7	0.71	0.257
12	Small demersals	0.798	1.50	2.50	10	0.91	0.250
13	Small pelagics	0.560	1.90	4.00	20	0.90	0.200
14	Crabs	0.011	0.70 (0.60)	3.50	12	0.94	0.292
15	Epibenthos	0.027	10.00	4.00	20	0.47	0.200
16	Endobenthos	0.001	1.20 (1.00)	10.00 (12.00)	60	0.86	0.167
17	Shrimps	0.770	1.50	6.50 (7.00)	25	0.65	0.260
18	Zooplankton		4.00	40.00	150	0.66	0.267
19	Phytoplankton		15.00	200.00		0.21	
20	Mangroves		100	0.22		0.60	
21	Microphytobenthos		0.50	125.00		0.70	
22	Detritus		0.10			0.23	
Parameters for model GN ₊₂₀ (2013)							
1	Seabirds		0.06 (0.05)	0.09	50	0.00	0.00
2	Mackerels and barracudas	0.018	0.06	0.60 (0.55)	7	0.50	0.09
3	Rays and sharks	0.028	0.07	0.75 (0.6)	4	0.56	0.19
4	Large drums and snooks	0.132	0.35	0.70	4	0.54	0.18

5	Catfishes	0.171	0.45	0.80	6	0.56	0.13
6	Morays and eels	0.011	0.17 (0.15)	0.50	5 (5.5)	0.74	0.10
7	Carangids	0.062	0.50	0.75	6	0.99	0.13
8	Snappers and grunts	0.201	0.70	0.90	5 (5.5)	0.98	0.18
9	Cephalopods	0.010	0.30	4.00	15	0.90	0.27
10	Lobster	0.003	0.10	1.50	10	0.67	0.15
11	Flatfishes	0.020	0.20	1.80	7	0.69	0.26
12	Small demersals	0.984	1.20	2.50	10	0.82	0.25
13	Small pelagics	1.511	1.50	4.00	20	0.94	0.20
14	Crabs	0.020	0.70	3.50	12 (10)	0.77	0.29
15	Epibenthos	0.005	5.00 (6)	4.00	20	0.71	0.20
16	Endobenthos	0.000	1.00	10.00	60	0.70	0.17
17	Shrimps	0.310	0.70 (0.65)	7.50	26 (25)	0.87	0.29
18	Zooplankton		4.00	30.00	150	0.76	0.20
19	Phytoplankton		15.00	200.00		0.23	
20	Mangroves		90.00	0.22		0.39	
21	Microphytobenthos		0.50	125.00		0.39	
22	Detritus		0.10			0.18	

The P/B values, or stock turnover rates, are equivalent to the total mortality rate (Z ; Allen, 1971)) in a steady-state ecosystem. For the fish species, Z was estimated based on F and natural mortality (M) using the following relationship: $Z = F + M$.

We estimated M by using the caudal fin aspect ratio method (Pauly 1983) as follows:

$$\log_e(M) = -0.0152 - 0.279 \log_e(L_\infty) + 0.6543 \log_e(K) + 0.463 \log_e(T),$$

where L_∞ is the mean asymptotic length (cm), K is the growth coefficient (year^{-1}), and T is the mean habitat temperature ($^{\circ}\text{C}$).

For invertebrates and seabirds, the P/B values were obtained from other studies (see Table A.3). The Q/B values were obtained from physiological studies that were conducted in similar areas and ecosystems (García and Duarte 2002; Giarrizzo et al. 2013) and from the previous model. Migration rates (immigration-emigration; in metric tons·km⁻²·year⁻¹) were set to 1 because no highly migratory fish species were included in the model. Most of the seabird species included in the model are permanently resident in the Gulf of Nicoya. To account for seasonal biomass changes of the seabird species (black storm-petrel *Hydrobates melania*; certain Sulidae [blue-footed booby *Sula nebouxii*, Peruvian booby *Sula variegata*]), the biomass was calculated as a yearly average. The yearly maximum of seabird biomass was estimated at 0.09 metric tons/km².

The model was balanced based on the suggestions provided by Link (2010) while considering certain general food web model diagnostics, such as the increasing biomass from the high to low trophic levels, the increasing P/B and Q/B with decreasing trophic level, the predator-prey biomass ratios, and the analysis of vital rate ratios. The model was balanced to achieve reasonable EE values for each functional group ranging between 0 and 1 and respiration-to-biomass ratios (R/B) and P/Q values ranging mainly between 0.1 and 0.3 (Christensen et al. 2005). During the balancing process, a few minor modifications were made in the biomass

values, P/B ratios, and the diet matrix. The values that required modification are indicated in Table 2.1.

2.2.3 Network Analysis

2.2.3.1 Ecosystem changes over a period of two decades

To investigate ecosystem changes that occurred over the two-decade period of interest (1993-2013), the most important model characteristics, such as fisheries catches and model summary statistics, were directly compared. The EwE trophic flow diagram allowed for a direct comparison of the biomass distribution in the ecosystem and provided information regarding the abundance and relative importance of certain functional groups after 20 years of intensive fishing and changes in environmental conditions. The approach is particularly useful because stock assessment studies of exploited species are scarce.

To quantify the effect of the biomass changes of one functional group on the other groups in the ecosystem, the MTI matrix was used. This method was originally used to assess economic interactions (Leontief 1951) and was later used to assess the direct and indirect ecological impacts of one group/fishery on the other groups in a system (Ulanowicz and Puccia 1990). The values were estimated for each pair of functional groups in the trophic web and indicated either positive or negative effects of the impacting group on the impacted group. Negative values can be associated with top-down effects, and positive values can be associated with bottom-up effects.

Based on the MTI, the keystone index was then calculated for each group according to Libralato et al. (2006). The keystone index describes the relative impact of each group on all the other groups in the system and is the highest for those groups that have a high impact and a low biomass. The keystone species have keystone index values that are close to or greater than zero (Libralato, Christensen, and Pauly 2006).

The fishing-in-balance index (FiB) was calculated according to Pauly et al. (2000) as follows:

$$FiB = \log \left[Y_0 \times \left(\frac{1}{TE} \right)^{mTL_k} \right] - \log \left[Y_0 \times \left(\frac{1}{TE} \right)^{\frac{1}{TE_0}} \right],$$

where Y_k is the catch in year k ; mTL_k is the mean trophic level of the catch in year k ; TE is the transfer efficiency between the trophic levels ($TE = 0.1$); and the subscript 0 refers to the year used as the baseline (1993). In combination with the mTL , the indicator can be used to assess the state of the fisheries, including masking factors, such as the geographic expansion or technological improvements in the fleet (Pauly, Christensen, and Walters 2000).

2.2.3.2 Economic analysis of the fishery

Information regarding the current market prices in Costa Rican colones per kilogram in the early 1990s and 2013 was obtained from INCOPESCA. The prices were entered separately for the artisanal and semi-industrial fleets. The average market price that was directly paid to the fishermen at the landing site was calculated for each functional group. Costa Rican colones were converted to U.S. dollars (US\$) by using the respective purchasing power parity (PPP) conversion factor per international US according to the World Development Indicators (Prince and Fantom 2014). The PPP is based on the relative cost of living and inflation rates in the

country and reflects differences in the price levels of goods (e.g., Hoyt and Iñíguez 2008) . A PPP conversion factor of 364 was used for 2013, and a PPP conversion factor of 100 was used for 1993. The converted prices were entered into the models as off-vessel prices (US\$ per kg) and were used to calculate the value of the total catches of each functional group and the profit of the fishing fleets. The costs of each fishing fleet were entered as a percentage of the fishery's total value in the given year, leading to a percentage of the fleet's total profit (Christensen and Walters 2004). A distinction was made between (1) fixed costs, which accrue without any fishing activity taking place and are mainly composed of the loans for the vessels and other equipment; and (2) variable costs, such as fuel and ice. The variable costs are fleet specific and directly relate to the fishing effort of the fleet. The relative percentages of the fixed costs and effort-related costs were estimated for each of the artisanal and semi-industrial fleets in the Gulf of Nicoya separately according to an economic study by Coyle et al. (2015) on the Costa Rican fishing fleet.

2.3 Results

2.3.1 Network Analysis

The basic Ecopath input and output parameters for both balanced trophic models of the Gulf of Nicoya (GN₀ and GN₊₂₀) are presented in Table 2.1. The EE values estimated by the EwE software varied from 0.00 to 0.99 for different functional groups. The EE value is the fraction of production that is used in the system (Christensen et al. 2005). A high EE value (close to 1) indicates that a large fraction of the production is either passed up the food chain or exported out of the system by the fisheries, as is the case for shrimps, small pelagic fish species, and other mid-trophic-level fish.

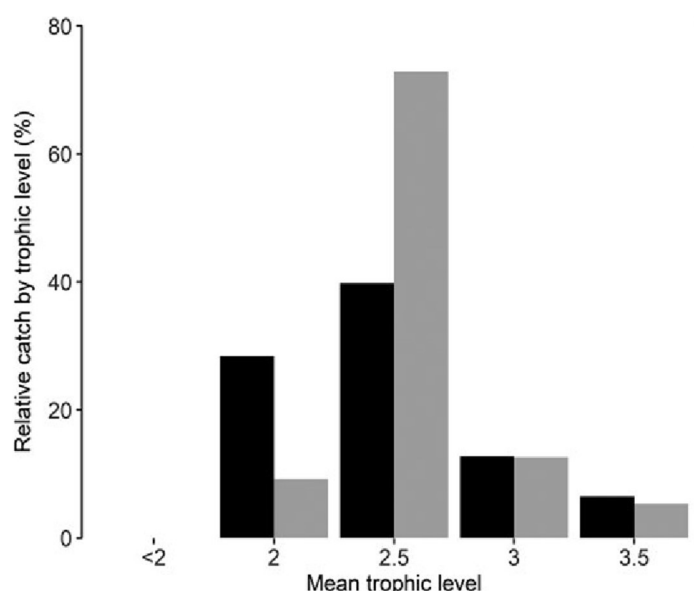


Figure 2.3 - Comparison of the relative yearly catches (%) by trophic level in the Gulf of Nicoya during 1993 (black bars) and 2013 (gray bars).

The diet matrix is presented in Table A.2. The summary statistics (Table 2.2) that were calculated by the EwE software for both models were directly compared. The gross efficiency of the fishery and the overall catch levels increased by 21.1% and 20.8%, respectively, between

1993 and 2013, whereas the profit from the fishery decreased by 48% due to the profound changes in the fishery targets. The model output indicated that catches at the lowest and highest trophic levels decreased, while catches at trophic level 2.5 strongly increased compared to those in 1993 and represented more than 70% of all catches in the Gulf of Nicoya in 2013 (Figure 2.3). The mTL of the catch remained at a low value of approximately 2.8 (Table 2.2).

The total system biomass and size in terms of total flows (throughput) of the system declined by 12.2% and 0.5%, respectively, over the 20-year period. The functional groups that decreased or increased in biomass in 2013 compared to 1993 are shown in the trophic flow diagram (Figure 2.4). The functional groups are ordered according to their trophic level (1 to 4). The sizes of the nodes and lines in Figure 2.4 are proportional to the biomass and the flow between the functional groups, respectively. Mangroves, which hold the largest share of the biomass in the system, declined severely. The biomass comparison showed a decline in benthic invertebrates and shrimps, which are the most prominent groups in the Gulf of Nicoya food web and function as an important link between the trophic levels. Certain species commonly caught as bycatch of the shrimp trawl fisheries, such as catfishes, morays, and eels, declined over the two-decade period of interest. The top predators – rays, sharks, mackerels, and barracudas – also declined. Important trophic flows link phytoplankton with zooplankton, benthos, and small pelagic fishes. Benthos, shrimps, and small pelagics play an important role in the system as a connection between the primary producers and the upper trophic levels.

Table 2.2 - Summary statistics of the ecosystem characteristics from the GN_0 and GN_{+20} models of the Gulf of Nicoya. The fishing-in-balance index (FiB) was not a direct model output; rather, it was calculated based on the trophic levels and catches in the model.

Parameter	GN_0 (1993)	GN_{+20} (2013)	Units	Change (%)
Total system throughput	6,892.33	6,856.97	Metric tons $\cdot$ km ⁻² $\cdot$ year ⁻¹	0.52
Mean trophic level of the catch	2.797	2.796		0.04
Gross efficiency catch/net primary production	0.00093	0.00113		21.10
Total primary production/total respiration	6.968	6.844	Metric tons/km ²	1.77
Total biomass excluding detritus	140.67	123.56		12.16
Total catch	2.886	3.488	Metric tons $\cdot$ km ⁻² $\cdot$ year ⁻¹	20.86
FiB	0.05	0.1		
Total market value	15,986.3	7,920.26	US\$	50.46
Total costs	10,230.65	4,929.92	US\$	51.81
Profit	5,755.66	2,990.33	US\$	48.05

The MTI plot (Figure 2.5) suggested that the artisanal gill-net fleet, including the use of illegal mesh sizes, had the strongest negative impact on different species in the ecosystem, particularly on predatory species (e.g., large drums, snooks, mackerels, barracudas, and catfishes). Removal of the top predators by the fishing fleet can lead to positive effects on the prey species (e.g., the gill-net fleet removes predators of lobsters). The semi-industrial shrimp fleet negatively impacted its target species along with a variety of bycatch species, whereas the selective small pelagic fishery affected only its target species.

The bottom-up effects of the prey on the predators were clearly visible. Zooplankton had a negative impact on phytoplankton (grazing), epibenthos had a negative impact on macrophytes, and cephalopods had a negative impact on flatfish. Certain species compete for the same resources, such as catfish and morays/eels as well as carangids (jacks and pompanos) and snappers/grunts.

Compared to 1993, the negative impact of the fishing fleets on sharks/rays, mackerels/barracudas, and small pelagics was greater in 2013. The shrimp trawlers increasingly impacted other nontarget species. The impact of the developing trap fishery on lobsters was evident. Furthermore, competition between the demersal fish groups increased, as indicated by the negative interactions with each other on the MTI plot. Demersal groups competed for the same prey, such as epibenthos, which substantially declined in biomass.

The EwE keystone analysis (Figure 2.6) of both years (1993 and 2013) indicated the high relative importance of phytoplankton and zooplankton in the ecosystem and the links to the higher trophic levels, such as benthos, shrimps, and small pelagic fishes. The 2013 model suggested that certain groups, such as snappers/grunts and small demersal fishes, increased in importance within the Gulf of Nicoya ecosystem.

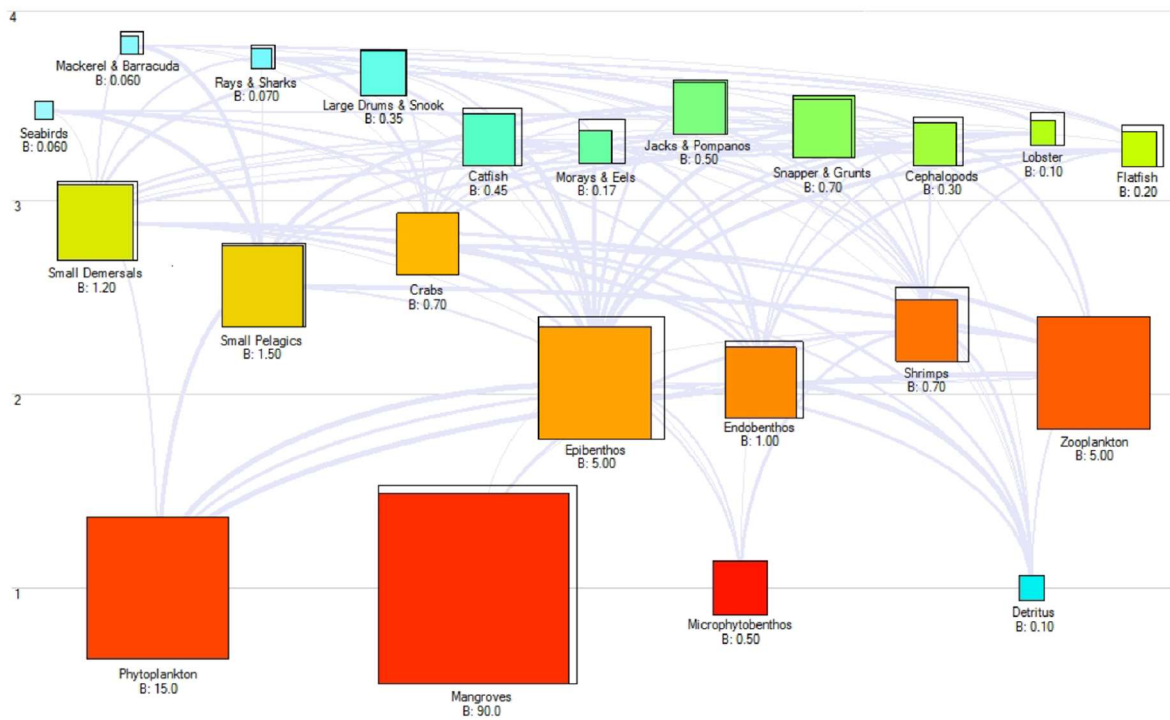


Figure 2.4 - Trophic flow diagram of the GN₊₂₀ model (2013) of the Gulf of Nicoya ecosystem and the relative changes over the two decades since 1993. Each square represents a functional group, and the size is proportional to the group biomass (B; metric tons [t]/km²). The biomasses in 2013 (GN₊₂₀) are represented by the filled colored squares, and the biomasses in 1993 are represented by the open squares. The trophic flows are indicated by the gray lines, which are proportional in size to the magnitude of flows. The y-axis indicates the trophic level of the group.

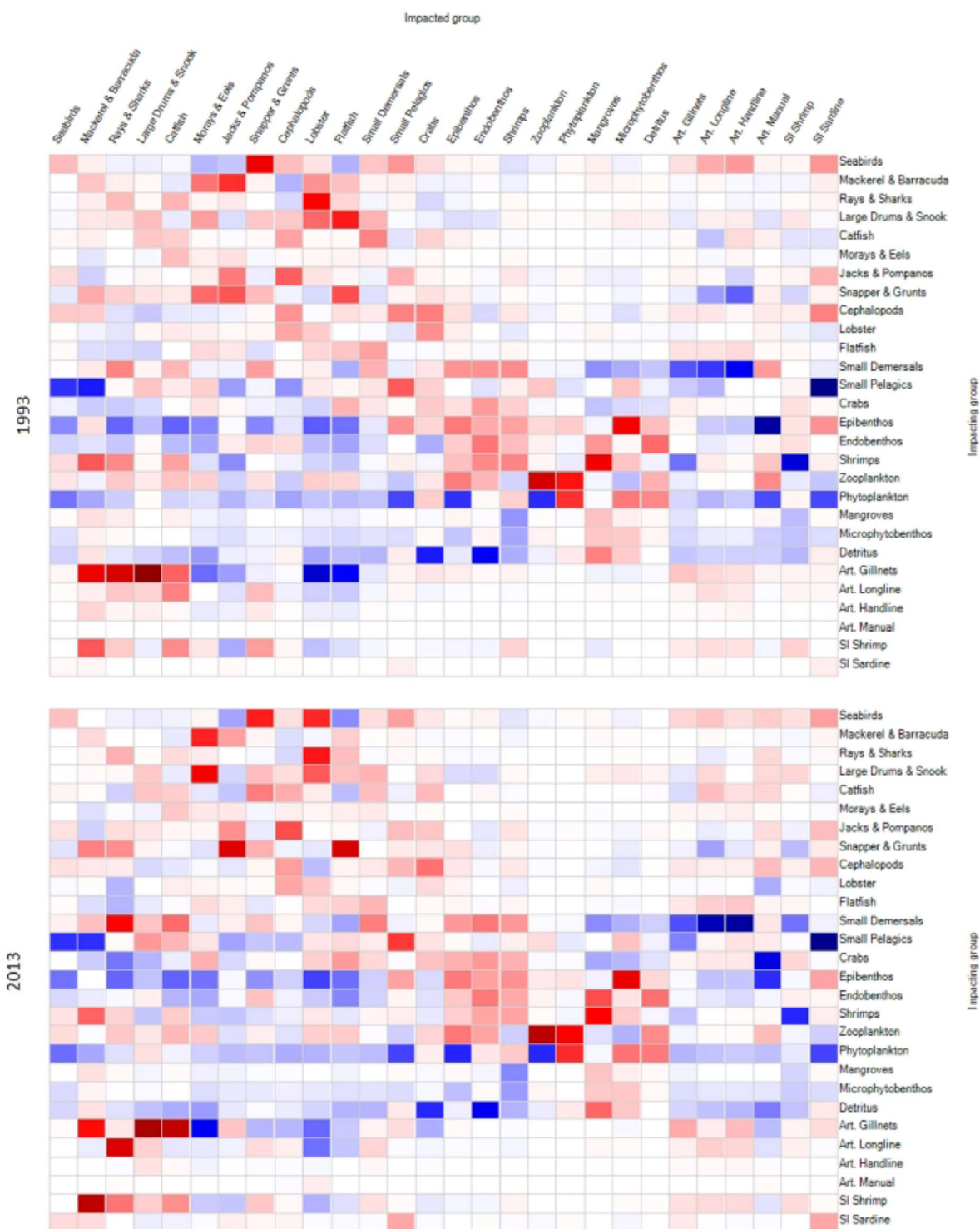


Figure 2.5 - Mixed trophic impact analysis for the Gulf of Nicoya ecosystem model in (A) 1993 (GN_0) and (B) 2013 (GN_{+20}), indicating the impact of one group on the other functional groups in the ecosystem. The color of the squares indicates the proportional impact (blue = positive; red = negative) of the functional group or fishing fleet listed at the right (impacting group) on the group or fleet listed along the top (impacted group; fleet types: Art. = artisanal, SI = semi-industrial).

2.3.2 Economic Analysis

The economic value of the yearly catches of all fishing fleets that operated in the Gulf of Nicoya declined by 50% in 2013 compared to that in the early 1990s (Figure 2.7). In 1993, shrimps accounted for the largest percentage (72%) of the total yearly value of artisanal and semi-industrial fleets. In 2013, shrimps still accounted for the largest share of the overall value; however, their contribution severely declined (by 39%). In both years, small demersal fish species were the second most important target group in terms of the yearly catch value. Small pelagic species only contributed a small share (7%) to the total value of the catch despite their large contribution to the catch biomass (20%).

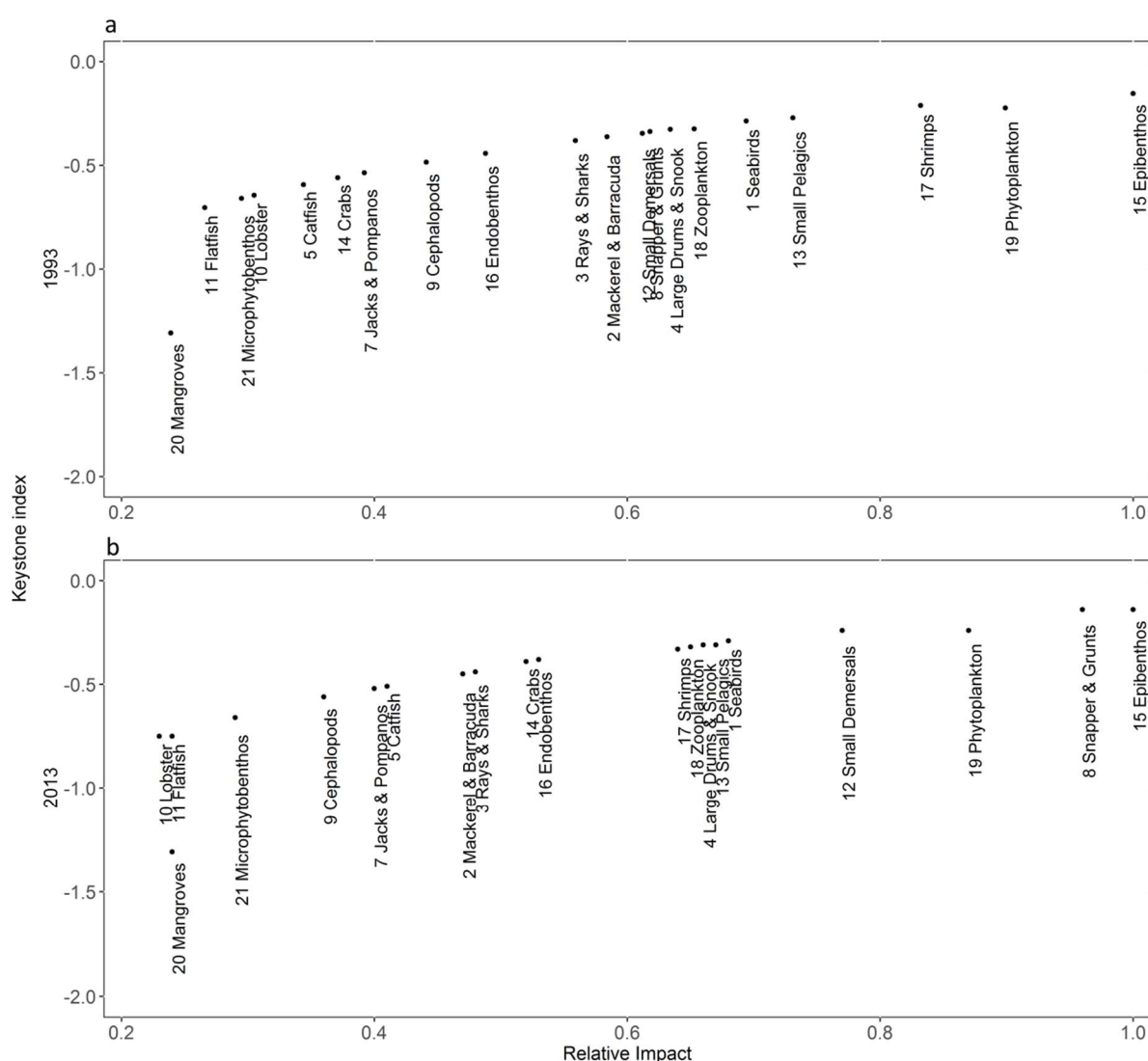
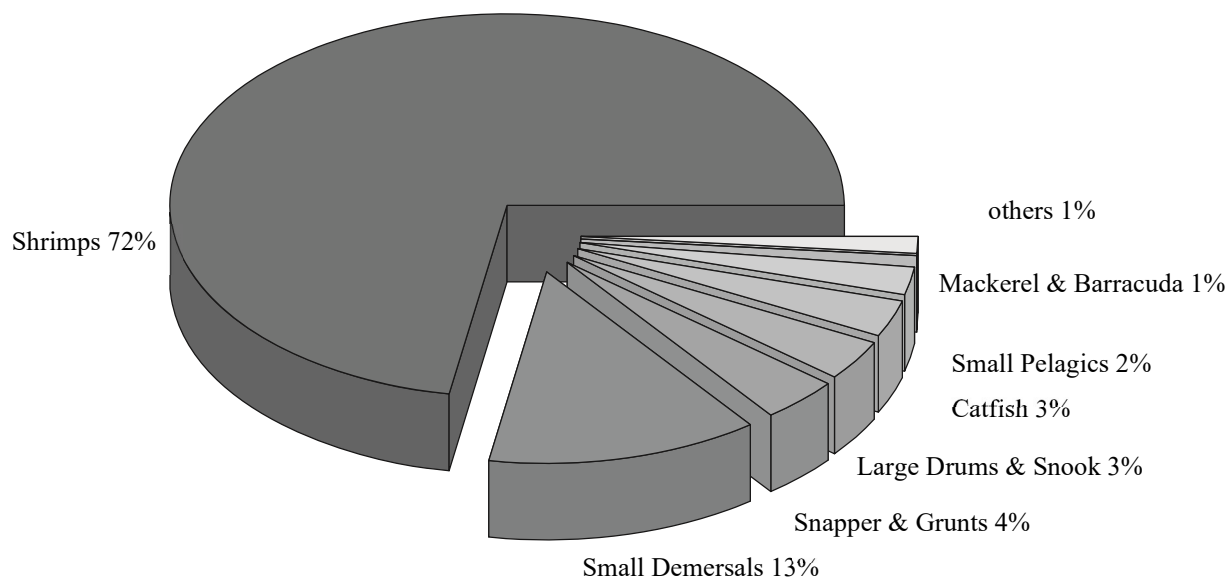


Figure 2.6 - Keystone index (keystoneness) of the functional groups in the Gulf of Nicoya during (A) 1993 and (B) 2013. Keystone species with high importance for the ecosystem are indicated by keystone index values close to zero (on the y-axis) and a relative impact close to 1 (on the x-axis). Functional groups are defined in Appendix Table A.1.

Relative value of the catch by functional group (1993)

15,986 US\$



Relative value of the catch by functional group (2013)

7,920 US\$

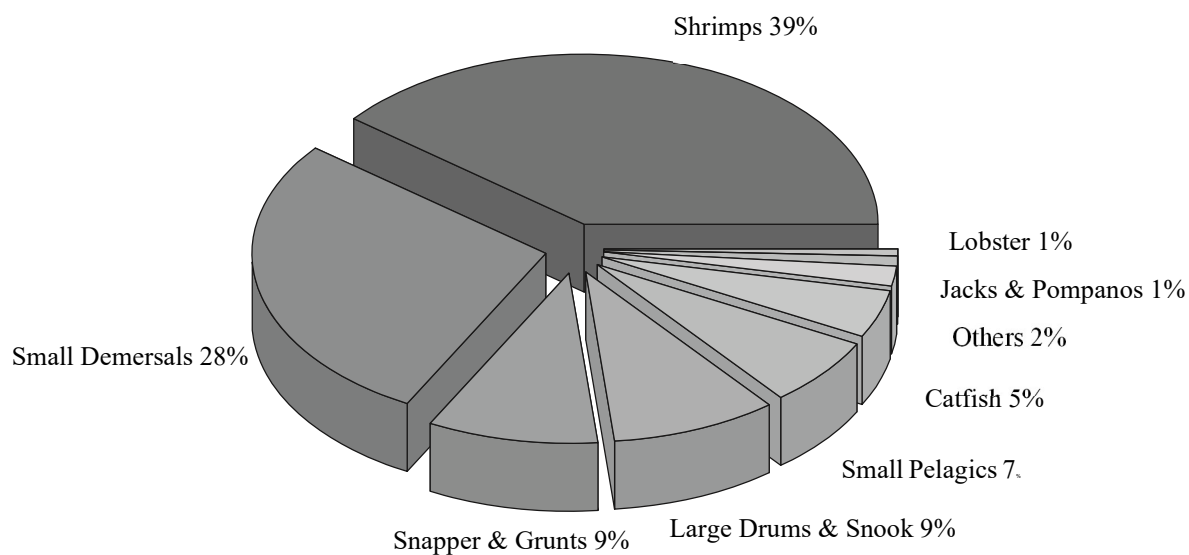


Figure 2.7 - The share of each target group in the total yearly monetary value (US\$) of the fisheries in the Gulf of Nicoya, calculated from the average market prices and yearly catches (Instituto Costarricense de Pesca y Acuicultura) in 1993 and 2013.

2.4 Discussion

2.4.1 Ecosystem Changes

The Gulf of Nicoya has been intensively exploited for many decades by both artisanal multi-gear fisheries and sardine purse seiners in the gulf and by shrimp trawlers in the outer zone (García Lozano 2014; Salas et al. 2011). This study demonstrated an increase in total fishery catches in the Gulf of Nicoya over two decades from 1993 to 2013, despite former descriptions of an overfishing state in the gulf during 1993 (Wolff et al. 1998). However, the overall increase in the catches was due to shifts in important target species in the fishing sector, which affected the trophic structure of the ecosystem.

One important change was the increase in semi-industrial sardine catches, which was consistent with the severe decrease in shrimp catches. However, the semi-industrial and artisanal catches of small pelagic fish appeared to have already peaked in the early 2000s, and a declining trend since 2008 can also be observed in those catches (BIOMARCC 2013a, 2013b; FAO 2017). The model revealed that the EE of small pelagic fishes had already been reached by 2013, as almost 100% of their production was either consumed by predators or exported out of the system by the fisheries. Thus, it is likely that the maximum sustainable catch and EE of small pelagics were already exceeded in 2008 (see Figure 2.2). Hence, a further increase in the exploitation rate for small pelagic species would not be advisable.

Two simultaneous changes in the fishing pattern occurred in the Gulf of Nicoya: (1) catches of small and low-trophic-level species increased; and (2) the spatial scale of the fishing fleet increased. When shrimp catches dropped, the low-trophic-level fish species were increasingly targeted, which might have had severe impacts on the gulf ecosystem. The gulf is an extremely productive, phytoplankton-dominated estuary (Gocke et al. 2002); thus, phytoplankton production is an important factor in the system, supporting the high fishery catches in the gulf. The results of the Ecopath keystone analysis indicated the high relative importance of phytoplankton, zooplankton, and the links to the higher trophic levels (benthos, shrimps, and small pelagics), suggesting a bottom-up-driven system. In a comparison of 33 Ecopath models (Libralato et al. 2006), marine mammals or sharks were often ranked the highest in “keystoneness” (e.g., Azores Archipelago: Guénette et al. 2001); a Galapagos reef: (Okey 2004). However, primary producers, zooplankton, small pelagics, and benthic organisms have been frequently observed to be among the most important groups, particularly in coastal or semi-enclosed ecosystems, including Chesapeake Bay (Baird and Ulanowicz 1989), the Gulf of Maine (Heymans 2001), and the Gulf of Thailand (Christensen 1998). These previous results are consistent with our findings for the Gulf of Nicoya.

The model shows a further decline in the biomass of shrimps, which also constitute one of the keystone species in the marine community. Keystoneness is often associated with a structuring role of the species in the food web (Libralato et al. 2006), highlighting the importance of shrimps as a link between the low and high trophic levels. The high EE of shrimps shows that almost all the biomass production is used in the system. Thus, their decline directly affects their main predators (i.e., flatfishes, lobsters, and jacks/pompanos), as shown by the MTI. These predator species all declined in biomass.

The consequences might be more severe due to the parallel decline in the biomass of benthic invertebrates, which occurred over the 20-year period; benthic invertebrates negatively impacted the same predator groups. In the context of declining benthic biomass, the impact of the artisanal fleet should not be disregarded. The heavy exploitation of bivalves by the coastal communities (yearly average = 20 metric tons in the last 5 years) has most likely caused the severe decline in epibenthos biomass. Furthermore, over the two decades of interest, shrimp trawlers in the Gulf of Nicoya might have caused the observed decrease in biomass of slow-growing benthic invertebrates by gradually degrading the habitat. In addition to the negative effects of the fisheries, these organisms are negatively affected by the declines in mangrove habitat.

Furthermore, the ecosystem model indicates the presence of increasing fishing pressure on small pelagic fishes, which is another important link between the trophic levels and are also identified as keystone species. In the Gulf of Nicoya, which is an ecosystem with only a few planktivorous fish species, small pelagics occupy the key position in the ecosystem as foraging fish, and a further reduction in this group's biomass might lead to trophic cascades throughout the food web (Duarte and García 2004). Shrimp trawlers are not only known to affect the target species but also affect bycatch species and often cause disturbances to marine benthic habitats (Thrush and Dayton 2002).

Shrimp trawling is likely the most significant source of fishing mortality in Costa Rica's marine ecosystem (Trujillo et al. 2012). A fraction of the bycatch is potentially discarded; however, the full extent of the discards is unclear and has been rather underestimated in our models.

The increasing bycatch of the shrimp trawlers is most likely the reason for the dramatic drop in the biomass of eels. The same trend of declining eel populations could be observed along the Pacific coast of South America. Populations of the Pacific Bearded Brotnia *Brotnia clarkae* rapidly declined by up to 30% in South America due to increased fishing pressure (Lea, Béarez, and McCosker 2010). The Conehead Eel *Cynoponticus coniceps* is also known to be frequently caught as bycatch of the shrimp trawlers (Lea et al., 2010). Another reason for the decline in biomass could be the habitat degradation and especially the reduction of mangrove areas.

Several other taxa, such as catfishes, rays, sharks, mackerels, and barracudas, have been affected by the shrimp trawlers due to the increasing fishing mortality, increasing catches of juvenile fish, and habitat destruction that interferes with the reproduction and nursery areas of these species inside the Gulf of Nicoya (Trujillo et al. 2012). These developments were also observed from the MTI analysis, which showed an increasing fishing impact on these species in 2013 compared to the early 1990s.

Over the two decades, an increase in crab and lobster catches was observed. This increase might be due to the development of the trap fisheries, which target crustaceans mainly for the export market. Furthermore, previous studies have shown that the removal of demersal predatory fish species can lead to an increase in benthic macroinvertebrates; thus, the substantial decrease in predator biomass in the Gulf of Nicoya could have caused trophic cascading effects (Baum and Worm 2009).

Shifts in the species composition in marine ecosystems over time are often due to anthropogenic influences, such as fishing, pollution, and environmental factors (e.g. Last et al., 2011). In the

Gulf of Nicoya, changes in the fishing pattern occurred due to spatial expansion of the fishery toward the outer gulf to target more oceanic species. Similar trends could be observed in other coastal areas of Latin America (Wehrtmann and Nielsen-Muñoz 2009). However, the decreasing shrimp catches indicate that the sparse coastal shrimp resources cannot be substituted with the deepwater shrimp fisheries, as these deepwater species are more vulnerable to overexploitation due to their low capacity for recovery (Morato et al. 2006; Wehrtmann and Nielsen-Muñoz 2009). Catches of certain high trophic-level, large pelagic predators also increased due to the offshore expansion of the fishing grounds. Not only did the species composition in the catches change but also the catches of small and juvenile sciaenids increased. A large fraction of small sciaenids is removed from the gulf at a low weight and size before they attain the weight at first maturity (Alpízar and Vásquez 2012).

Both trends – fishing more heavily on low-trophic-level species and the expansion of the spatial scale to areas where large fish are still more abundant—are common phenomena in tropical fisheries, along with improvements in the gear and modernization of the semi-industrial fleet (Corrales and Vindas 2014). Such a gear improvement occurred in the early 2000s, when many illegal gill nets with smaller mesh sizes were introduced into the Gulf of Nicoya, potentially increasing the yield of smaller and juvenile fish, and increasing the success of catching low-value, small coastal fish.

Catches of the lowest and highest trophic levels declined due to the decline of shrimps and large predators, and catches are now dominated by mid-trophic-level species (trophic level = 2.5; i.e., small pelagic and small demersal fishes). However, we found no overall decline in the mTL, an indicator of increasing negative impacts on the ecosystem. One has to consider that the model does not account for the possible size-class and age-class reductions of the species within the model compartments. Furthermore, the mTL of the catch is not a sufficient measure to characterize the fishing impact on the ecosystem, as trends might be masked due to the development of the fishing sector. Hence, one should consider the FiB as an additional indicator (Pauly, Christensen, and Walters 2000). In 1993, the FiB was close to zero (0.05; i.e., close to a balanced state); in 2013, the FiB was 0.1. The FiB index increases as fishery catches increase more rapidly than assumed according to the trophic level declines (Kleisner et al. 2011). It has been shown that an increase in the FiB ($\text{FiB} > 0$) is often due to geographic expansion of the fishing fleets (Christensen et al. 2011; Kleisner et al. 2011). Therefore, the values explain the constant mTL despite the strong ecological impact of the fishing fleet.

The fishing fleets that operate in the Gulf of Nicoya differ considerably in their impact on the ecosystem due to their gear selectivity. The MTI analysis reflected the selectivity of the different fishing gears. The small pelagic fish fleet operates selectively and almost exclusively catches the target species. In contrast, the shrimp trawlers have an estimated shrimp-to-bycatch ratio of at least 1:7 (Trujillo et al. 2012), and the species landed and discarded are diverse. Among the artisanal fleets, the use of gill nets has the most negative impact on the functional groups in the ecosystem. Demersal species, such as large drums, snooks, and catfishes, are particularly affected. In 1993, rays and sharks were negatively affected by the gill-net fleet; however, this effect decreased greatly in 2013, likely due to the declining biomass of those functional groups and their overall decreasing catches. Another impact factor of the fishing fleet on the ecosystem is the use of small mesh sizes. It has been suggested that since the early 2000s,

at least 80% of gill nets in use consist of mesh sizes smaller than 7.62 cm (3 in; Alpízar and Vásquez 2012), which catch almost all the white shrimp and 36% of the major fish species (Alpízar 2015).

The gross efficiency of the fisheries describes the ratio between the catch and total primary production; in the Gulf of Nicoya, the gross efficiency value for 2013 (0.00113) exceeded the weighted global average of 0.0002 (Christensen et al. 2005) but was comparable to values reported for other coastal systems (e.g. Ullah et al. 2012). Generally, higher values tend to occur in ecosystems where most fisheries' target species are of lower trophic levels and only a few large predatory species are caught (Godinot and Allain 2003); such is the case in the Gulf of Nicoya, where shrimps and small pelagic fishes represent the bulk of the catches. The gross efficiency of the fishery increased in 2013 compared to that in 1993, most likely due to the gear improvement and the increasing catch volumes of small pelagic fishes, small demersal fishes, and invertebrate species.

Besides the fisheries, environmental factors may have caused changes in the distribution and production of certain species over time. In the Gulf of Nicoya, several environmental factors can be considered to affect shrimp catches. Substantial changes in oceanographic factors, such as sea surface temperature and salinity, could not be observed. However, the 10% loss of mangrove biomass in the Gulf of Nicoya, as an important nursing ground for coastal shrimp species, might have negatively affected shrimp production and enhanced the decline of shrimp biomass. Large-scale environmental processes, such as the El Niño – Southern Oscillation, are also known to affect environmental conditions in the study region; however, these periodical variations are unlikely to have caused the severe decline in shrimp biomass over two decades. To analyze which environmental factors influence the ecosystem, further study is needed whereby environmental information is added to the model to simulate the effects on the ecosystem.

2.4.2 Economics

Although the overall catches increased between 1993 and 2013, the total value and profit of the catches decreased substantially (by 50%) over the two decades of interest (Figure 2.7). The heavy loss of profits was due to changes in the catches toward low-value species, such as small pelagic and small demersal species. In 2013, shrimps accounted for over 28% of the total value of the catch and therefore constituted only half as much as that in 1993. However, snappers, grunts, and small pelagic fishes increased in importance.

The observed species shifts within certain functional groups did not affect the general model behavior but appeared to have strong impacts on the economy of the fishery. This was the case for the drums and croakers, with only a few species of commercial interest, among which prices vary considerably. Catches of five of the seven commercially important species have declined over the last 10 years – in some cases, by as much as 70%. Finally, changes in the size composition of the catch at the species level must be considered. Currently, a large fraction of juvenile fish, particularly drums and croakers, is believed to be caught in the Gulf of Nicoya (Alpízar 2014). These changes of ecological and socioeconomic concern were not fully revealed by our EwE model. The analysis of the relative age-class distribution indicated increasing catches of small-sized species, which were below the mean weight at first maturity.

The different fleets have different profit margins according to their vessel (engine) sizes and the type of gear in use due to the variance in daily expenses (Bolaños 2005; Gasalla et al. 2010). In some cases, artisanal fishing fleets can operate more profitably than semi-industrial fishing fleets due to the gear used and the lower operational and fixed costs (Gasalla et al. 2010; Lery et al. 1999; Tietze 2001). Similarly, it is assumed that the artisanal fleet in the Gulf of Nicoya could develop an advantage over the semi-industrial fleet because of the low costs of artisanal fishing and the benefits of using more selective gear. Furthermore, some artisanal fishery-caught products, such as snappers, large drums, snooks, and sharks, achieve higher market prices (INCOPECA, unpublished market price statistics). However, with further declining inshore resources, the catches presumably fail to provide a sufficient profit, and the artisanal fleet may easily be outcompeted by the semi-industrial fleet in terms of efficiency and profit maximization due to the semi-industrial fleet's larger fishing vessels, greater fishing radius, and advanced ability to adapt its gear and the targeted species (Morato et al. 2006). Both the deepwater shrimp fisheries outside the Gulf of Nicoya and the semi-industrial longline fleet within the gulf have further increased (Arauz 2000; Wehrtmann and Nielsen-Muñoz 2009). The semi-industrial longline fleet was able to increase catches of large pelagic fish, which are a profitable target group (e.g., market prices for marlins and tunas increased by 20-35%; INCOPECA, unpublished data). The ratio of coastal to deepwater catches shifted from 3:2 to 1:4 and mitigated the decline in coastal catches (Mug-Villanueva 2002). Eventually, the artisanal fishermen will suffer direct negative effects from the lack of coastal resources, and the socioeconomic losses for the whole country are high due to the severe decrease in Costa Rican shrimp and fish exports since the early 2000s (Cornick, Jimenez, and Román 2014).

2.4.3 Management Considerations and Outlook

The results of this study confirm the urgency for the enforcement of regulations to protect the marine coastal resources in the Gulf of Nicoya. A minimum mesh size of 7.62 cm (3 in) for gill nets was implemented but with little success. One important step would be to regulate and enforce the minimum mesh sizes for the gill-net fleet to counteract the trend of increasing catches of juvenile fish, such as sciaenids. The specimens are not only removed from the gulf before they reach the age at first maturity and before they can reproduce, but they also generate low market values for the fishermen. Another measure is the temporal and spatial closures of certain areas in the Gulf of Nicoya. To protect the country's large marine environment, certain conservation measures of this type have already been taken. The Gulf of Nicoya was declared a "Marine Area of Multiple Use" (Area Marina de Uso Multiple [AMUM]) by the Costa Rican government in 1995, with the aim of protecting and conserving marine and coastal resources through integrated management (Melissa 2012). To protect the marine resource in the AMUM, a "veda" (an annual 3-month period) was implemented during which the Gulf of Nicoya is closed to fisheries. The temporal closure is supposed to overlap with the reproduction peak of commercially important species to support their recovery (Coyle et al. 2015). The closure can be a good way to reduce fishing effort in general and to support the reproduction of declining species, such as shrimps, sharks, sciaenids, and others.

Due to a lack of efficient control, difficulties in the execution of the fishing ban have occurred (Coyle et al. 2015). During the seasonal closure, a reduction in fishing effort and catches in zones 201 and 202 can be observed; however, catches in the subsequent month are accordingly

higher. Moreover, the subsidies given to the fishermen during the temporal closure are hardly sufficient to secure their livelihoods. Consequently, opportunities for alternative livelihood sources for fishermen need to be assessed. Another aim is to entirely ban shrimp trawlers from the coast of Costa Rica by not renewing expired shrimp trawling licenses in the future. This would be an efficient measure to gradually remove trawling. Regulations are needed to resolve the conflict between the artisanal and semi-industrial fishing fleets in terms of the competition for the same limited resources. Furthermore, measures should be taken to protect the keystone species in the ecosystem since the observed decreases in the abundance of key functional groups might have significant implications for ecosystem functioning, and a further decline should be avoided through management. An important step could be the validity control of fishing licenses to address the problem of illegal fishing. The management plans might have led to a small reduction in fishing effort; however, due to the lack of enforcement, the measures have remained primarily ineffective. Thus, it is unlikely that these measures have caused considerable changes in the ecosystem over recent decades.

The two-snapshot approach used in this study is instrumental for the assessment of the status of the ecosystem and the major changes that occurred over the two decades of interest. Furthermore, the ecosystem-based analysis highlights the importance of certain keystone species that should be considered in management plans due to their high ecological value for the Gulf of Nicoya system (Navia et al. 2016). The models can provide the basis for further simulations to anticipate possible future trends in development (Neira et al. 2014), including fishing effort and environmental trends.

Acknowledgments

This study was financed through the Project MONICOYA (“Modelling two decades of change of the Nicoya Gulf [Costa Rica] ecosystem: implications for future management”) of the Leibniz Centre for Tropical Marine Research (ZMT), in which V.A. was able to conduct her doctoral research. The work was also supported by the Bremen International Graduate School for Marine Sciences (GLOMAR). This research did not receive any specific grants from funding agencies in the public, commercial, or not-for-profit sectors. The facilities in Costa Rica and some of the data describing the Gulf of Nicoya were provided by the Centro de Investigación en Ciencias del Mar y Limnología (CIMAR) and the Universidad de Costa Rica. The research project was strongly supported by our main project partner in Costa Rica, Alvaro Morales Ramírez, who provided the infrastructure and guidance in Costa Rica and helped to develop the project collaboration between ZMT and CIMAR. Several other colleagues from CIMAR and the Universidad de Costa Rica supported the project, including Jorge Cortés Núñez, José A. Vargas Zamora, Helena Molina Ureña, and Catalina Benavides Varela. Our colleague, Astrid Sánchez Jiménez, supported the execution of the stakeholder workshop in Puntarenas, Costa Rica. Berny Marín Alpízar (INCOPESCA) supported the study with valuable advice and guidance concerning the marine resources in the Gulf of Nicoya. Isabel Seguro Requejo and Sara Soria Píriz (Universidad de Cádiz, Spain) supported the project with information about the phytoplankton distribution in the Gulf of Nicoya. There is no conflict of interest declared in this article.

Chapter 3

**IDENTIFICATION OF DRIVERS OF CHANGE
OF THE GULF OF NICOYA ECOSYSTEM**

Chapter 3

Identification of Drivers of Change of the Gulf of Nicoya Ecosystem (Costa Rica)

Viola Alms and Matthias Wolff

This is the author's version of the work. Please cite the final version as:

Alms, V., and Wolff, M. (2020). Identification of Drivers of Change of the Gulf of Nicoya Ecosystem (Costa Rica). *Frontiers in Marine Science*, 7, 707

<https://doi.org/10.3389/fmars.2020.00707>

Submitted on 21th March 2020.

Accepted on 3rd August 2020.

Published on 28th August 2020.

Contribution statement

Project idea: **Viola Alms** and Matthias Wolff

Design of the work: **Viola Alms**

Acquisition and interpretation of data: **Viola Alms**

Drafted manuscript revised by Matthias Wolff.

Abstract

The Gulf of Nicoya is a large tropical estuary located on the Pacific coast of Costa Rica and accounts for the country's main fish production. It is increasingly impacted not only by its fisheries, but also by urbanization, tourism developments, the input of pollutants from urban centers and agriculture activities in the catchment areas of large rivers entering the gulf. Similar to other coastal ecosystems, the gulf is particularly sensitive to short- and long-term changes in the climate, such as the precipitation rate, Sea Surface Temperature, wind speed and current regimes. While the gulf has been studied for decades, until now no attempt has been made to combine the analysis of the impacts of fishing and environmental changes on the ecosystem. By following a holistic approach that uses fisheries and environmental time series data from collaborative research and simulation models, this study aims at identifying the main drivers of the observed changes in the Gulf of Nicoya ecosystem. While the model simulations indicate that variations in the catch of some target species (e.g., cephalopods and shrimps) are mainly driven by the fisheries' exploitation rates, several other species (e.g., corvina, snook, small pelagics, and crabs) are also substantially affected by climate variations, particularly during El Niño periods of high Sea Surface Temperature extremes and increased precipitation. During these periods, phytoplankton productivity and zooplankton biomass decreases with bottom-up effects on the entire food chain. Indicators of the environmental state of the system thus have to be considered along with regulations of the fishing pressure to form the basis for the development of appropriate management strategies.

Keywords: Ecosystem modeling, fisheries management, climate variability, ecopath with ecosim, trophic interaction

3.1 Introduction

Ecosystem food webs change over time either due to changes in natural predator-prey interactions, resource limitations and anthropogenic influences, or due to environmental influences that cause changes in the distribution, growth performance or survival rate of the species. Fishing as a major anthropogenic impact factor additionally greatly affects the population size of target species, their prey and predators and may even damage the species habitats (Frederiksen et al. 2006; Jennings and Kaiser 1998). Although overfishing is known to have caused the decline and collapse of many fish stocks around the world, it can also alter the sensitivity of the ecosystem to natural environmental variations and reduce the ability to buffer the effects of extreme natural events or future climate change (Hidalgo et al. 2011). At the same time, the limits overfishing of sustainability critical threshold for fishing can be affected by environmental changes (Neira et al. 2009). Fishing induced changes in the biomass that occur at one trophic level affect the trophic levels above and below that level, causing trophic cascades throughout the system (e.g., Andersen and Pedersen, 2010; Duarte and García, 2004). Ecosystems are driven by the combined effect of varied influencing factors, that have to be identified and examined to understand the dynamics of the system. To assess the role of small-scale and industrial fisheries within the ecosystem multiple factors need to be evaluated. The historical development of the operating fishing fleets, the different gear types and target species, as well as the temporal distribution of the fishing effort are all important factors that need to be considered for a holistic assessment of the Gulf of Nicoya ecosystem and its changes over time. We assume that the historical and ongoing removal of large numbers of small pelagic species and the existence of a variety of small demersal species should have most likely shaped the ecosystem over the last decades (Alms and Wolff 2019). This process, together with the fisheries-induced substantial reduction of shrimp biomass over the last two decades, might be particularly relevant since it may have greatly reduced system biomass and productivity in low and mid trophic levels, reducing the energy basis for higher trophic levels of the gulf (Duarte and García 2004). Our previous research on the Gulf of Nicoya ecosystem (Wolff et al., 1998; Alms and Wolff, 2019) allowed us to describe the energy flow structure and biomass changes that have occurred in the system over a period of two decades. We attributed an important role in causing these changes to fisheries and were able to show that phytoplankton and zooplankton largely control higher trophic levels groups, such as benthos, shrimps, and small pelagic fishes, suggesting that bottom-up control mechanisms are crucial to the system.

Since in such environments; changes in abiotic factors due to climate variations may directly affect the primary productivity, we further examine these bottom-up effects in this study. Additionally, we aim at exploring the possibility of mixed trophic interactions or wasp-waist control mechanisms within intermediate trophic level species that exercise both bottom-up control on their predators and top-down control on their prey (Cury et al., 2000).

According to the Food and Agriculture Organization (FAO), most inter-annual fluctuations in fish production in the Eastern Central Pacific Ocean (Fishing Area 77) result from abundance changes in mid-trophic level pelagic fish. These fish often assume a central role in marine ecosystems and influence both higher and lower trophic levels (Frederiksen et al. 2006). Duarte and García (2004), however, point out that this “wasp-waist” role of small pelagic fish that has been proven for temperate and subtropical upwelling regions may not remain valid for tropical

and therefore less productive upwelling areas, a statement we would like to analyze for the Gulf of Nicoya.

Previous studies prove that the El Niño-Southern Oscillation (ENSO) cycle occurring in the Eastern Tropical Pacific Ocean (ETP) causes periodic variations in several environmental parameters, which temporarily change the coastal marine environment and its species composition (Fiedler 2002; Tam et al. 2008; Taylor, Wolff, et al. 2008; Wolff et al. 2012). In the southern Humboldt and southern Benguela upwelling systems, environmental variations and trophic interactions were found to be more important than the fisheries in causing variations in species abundance (Shannon et al. 2008). Atmospheric pressure and Sea Surface Temperature (SST) are the main indicators of El Niño events (NOAA, 2020), although rain fall, wind and current patterns may also change dramatically (Wolff 2010). On average, the atmospheric pressure is low in the Western Tropical Pacific and relatively high in the Eastern Tropical Pacific. Here, sea level Sea Surface Temperature (SST) greatly rises during El Niño events, concurrently with changes in other climatic parameters, such as wind direction, wind speed and precipitation rates (Ji et al., 2015). In the entire ETP region, climate impacts of the ENSO cycle are perceptible and in Costa Rica, strong effects of the ENSO cycle have been observed in the Gulf of Nicoya as well as in the Golfo Dulce, the second important gulf system of the Costa Rican Pacific coast (Quesada-Alpízar and Morales-Ramírez 2004).

During the particularly strong El Niño phase in 1997/98, as well as during the ENSO phase in 2009, changes in the structure of the phytoplankton and zooplankton communities have been observed (Brugnoli-Olivera and Morales-Ramírez 2008). The phytoplankton and primary productivity decreased in these years and lower densities of diatoms and dinoflagellates were reported in the Gulf of Nicoya (Brugnoli-Olivera and Morales-Ramírez 2008; Vargas et al. 2014). This reduction of low trophic level biomass and productivity is then usually followed by a reduction in biomass of plankton, fish and seabirds, leading to severe consequences for the regional fishing sector and coastal communities (Lizano and Alfaro, 2004; Wolff et al., 2012) as their catch of pelagic species, coastal demersal species, shrimp, squid and other marine organisms (FAO) is affected. For an effective implementation of an ecosystem-based fisheries management in the region it is crucial to understand the response of the ecosystem to different fisheries in place and to the environmental drivers as a whole. This is based on the understanding of the biomass flows through the food web, starting at the lowest trophic level (Brugnoli-Olivera and Morales-Ramírez, 2008; Tam et al., 2008).

Accordingly, we used a routine adapted to specific time series using fisheries and environmental data for a period from 1995 to 2014 to elucidate the role of the fishing effort and abiotic factors in explaining observed changes. For this purpose, we used the open-source software Ecopath with Ecosim (EwE) and applied a hypothesis-based model fitting procedure, for which we considered time series of both inter-annual changes in fishing effort and fleet composition as well as the most relevant environmental factors (Sea Level Pressure, Sea Surface Temperature, Chlorophyll-a concentrations, wind direction and speed, precipitation rates, particulate inorganic and organic carbon concentrations and photosynthetically active radiation). A central question examined in our study is whether the observed system changes were driven by the exploitation patterns of fisheries or/and by environmental changes. The basis for this analysis was provided by two EwE “snapshot models” of the Gulf of Nicoya ecosystem for the years

1995 (Wolff et al., 1998) and 2015 (Alms and Wolff, 2019). We consider the identification of main drivers behind the observed changes as crucial requisites for the selection of appropriate management and conservation strategies of the gulf. Therefore, on the basis of the present research, future measures can be taken to sustainably improve the habitat of the Gulf of Nicoya.

3.2 Materials and Methods

3.2.1 Description of the Study Site

The Gulf of Nicoya (10°N , 85°W) is an estuary influenced by the tides on the northwest Pacific coast of Costa Rica (Figure 3.1). The mangrove-fringed gulf extends from the mouth of the Tempisque River to the Pacific Ocean, with an area of 1530 km^2 . It is of considerable ecological importance due to its function as a feeding, reproduction and nursing site for many species, such as shrimps, sharks, small pelagic fish and seabirds. Due to its high productivity, the estuary is the country's principal fishing ground for the artisanal fishing fleet on the one hand (Vargas, 1995) and for the semi-industrial fishing fleets on the other hand. In the inner shallow part of the Gulf of Nicoya, the small-scale artisanal fishing sector is of major economic and social importance (Trujillo et al., 2012; Lozano and Heinen, 2016). More than 20 coastal communities around the Gulf of Nicoya depend on the artisanal fishing sector, with rare options for alternative livelihoods (Proyecto Golfos, 2013).

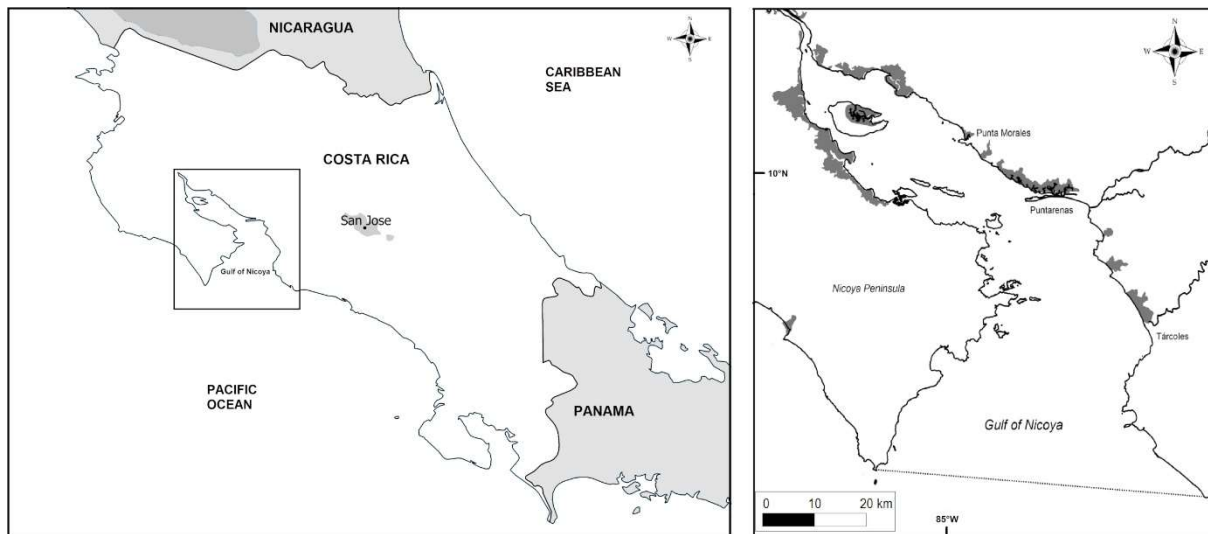


Figure 3.1 - Maps of the study area (Gulf of Nicoya, Costa Rica) , modified from Alms and Wolff (2019).

3.2.2 Fisheries and Time-Series Data Used

The Gulf of Nicoya is the country's main fishing ground. The present research is based on fisheries data of the artisanal and semi-industrial fishing sector.

Four artisanal fishing fleets with approximately 2000 licensed fishermen operate in the gulf with different boat sizes and gear types that target a variety of demersal and coastal species (BIOMARCC-SINAC-GIZ, 2013). Gillnets are prominently used and its usage further increased in the 2010s (Sánchez-Jiménez et al. 2019). Fishermen use small motorized boats (pangas). Catches are diverse but dominated by drums (Perciformes: Sciaenidae), catfishes (Siluriformes: Ariidae), snooks (Perciformes: Centropomidae), and different penaeid shrimp

species. The line fleet uses bottom longlines and drifting longlines and operate bigger motorized boats (lanchas). Target species are sharks, barracuda, comber and others. Smaller non-motorized boats fish for drums, croakers and snook, using handlines in the inner shallow part of the gulf. Manual mollusk extraction is mostly carried out by the women of the coastal communities, combined with the manual trap fisheries targeting different crustaceans. The artisanal fleet provides around 55% of the total catch in the Gulf of Nicoya (in 2013).

Additionally, two semi-industrial fishing fleets operate in the gulf: shrimp trawlers and sardine purse-seiners. The purse seine vessels target small pelagic fish species in the intermediate and outer zones of the Gulf of Nicoya. The bottom trawl fleet targets several shrimp species, mostly in the outer gulf, however the capture of bycatch species drastically exceeds the shrimp catches. Artisanal fisheries landing data of the Gulf of Nicoya on the one hand was obtained from the Costa Rican national landing statistics registered by the Instituto Costarricense de Pesca y Acuicultura (INCOPECA). The time series data collected from 1994 to 2014 consists of monthly records of the main target species and groups (fish and invertebrates) fished and landed inside the Gulf of Nicoya, measured in kg. The catches were organized by fishing fleet and gear. The monthly catch data was then converted into tons per square kilometer of the entire area (1530 km²) per year. Fishing effort time series data were added to the model to realistically simulate changes in the fishing pressure over the last 20 years (1994-2014). Fishing effort data (days fishing and number of fishing trips) per fleet is available for the years prior to 2005. Thereafter, only combined fishing effort was available for both the artisanal and semi-industrial fleets. We then divided this effort for the artisanal and semi-industrial fleets using percentages estimated from previous years and current information about recent developments in the fishing fleets and fishing zones (INCOPECA). Biomass estimates of all target groups were entered for the beginning and for the endpoint of the simulations (1993 and 2013) from the EWE reference models presented in Alms and Wolff (2019).

3.2.3 Environmental Forcing Functions

In order to account for possible environmental impacts on the different model compartments, time series data derived from satellite images were used for the simulations. The data was taken from the Sea-viewing Wide Field-of-view Sensor (SeaWiFS) project and the Moderate Resolution Imaging Spectroradiometer (MODIS) provided by the NASA (NASA Ocean Color). As a first approach to find significant environmental drivers, the most common marine climate factors were tested, i.e., daytime and night-time Sea Surface Temperature (°C), sea surface salinity (ppt), zonal wind speed (m s⁻¹) and direction, tropical easterly trade wind speed (m s⁻¹), and precipitation rates (mm). Additionally, specific ENSO related parameters were used, i.e., Eastern Equatorial Pacific Sea Level Pressure, SST in El Niño regions 1 + 2 and 3, 3.4 and 4 (NOAA, 2020), as well as proxies for primary productivity, such as Chlorophyll-*a* (Chl_a) concentrations (mg m⁻³), photosynthetically active radiation (Einstein m⁻² day⁻¹), dissolved organic matter (mg m⁻³), particulate organic matter (mg m⁻³) and high colored dissolved organic matter. The time series data were standardized by scaling the values to the (0.5,1.5) range, according to the following formula:

$$x' = (max' - min') \div (max - min) \cdot (x - max) + max$$

whereas each time series value x was scaled according to the new maximum: $\max' = 1.5$ and new minimum value $\min' = 0.5$.

3.2.4 Hypothesis Testing and Model Selection

The EwE time series fitting routine was used to identify the model that matches the variation in the abundance of the functional groups with the historical catch time series data of the Gulf of Nicoya. For the choice of model procedure forcing data is needed to trace the biomass changes of the model compartments over time and reference data is necessary for comparison and validation of the models. In conclusion, three essential classes of ecosystem drivers were tested: anthropogenic factors (fishing, pollution), trophic control mechanisms (interactions) and the influence of the chemical and physical environment (climate).

Fishing effort (days fishing) and environmental time series data of the environmental parameters mentioned above were used to force model group catches and biomasses over time, whereas the recorded historic landings served as (observational) reference data for the catch. The effect of the forcing data on the goodness of model fit could then be observed in the resulting biomass and catch estimates. We used the predicted variations of environmental factors in the ETP related to future climate change as given by the IPCC (IPCC, 2007) and selected the predictions for South and Central America (Latin America) for two emission scenarios, namely the representative concentration pathways RCP4.5 (radiative forcing of 4.5 W m^{-2} by 2100) and RCP8.5 (radiative forcing $> 8.5 \text{ W m}^{-2}$ by 2100).

The vulnerability settings in the model represent the assumed trophic control mechanisms between functional groups (later trophic effects). While vulnerabilities can be set between 1 and 100, 2 is used as the default setting. At a low vulnerability setting (close to 1) an increasing predator biomass does not cause a substantial increase in the predation mortality of the prey (bottom–up control). On the contrary, at high vulnerability values (up to 100) an increasing predator biomass causes a similar increase in the predation mortality of the prey (top–down control). In the time series routine, the EwE software modifies the vulnerability parameter of the predator groups from its default 2 to find the setting which reduces the sum of square difference between the predicted and the observed data (Scott et al., 2016). The Stepwise Fitting Procedure for automated fitting as described by Scott et al. (2016) was used, allowing to test a high number of settings to find the best fit model. The routine tested for the most sensitive predator and predator-prey interactions. The search for vulnerability parameters set for the predator columns of the model resulted in better outcomes than by each prey/predator cell, thus only these results will be presented in the following. For the model fitting procedure 22 time series were used as a reference, thus a maximum of $22-1 = 21$ vulnerability parameters were estimated. In the same manner, primary productivity (PP) anomalies can be estimated by the software. The software searches for an anomaly PP function to simulate potential bottom–up control mechanisms on the ecosystem. The time series data contained 19 years, thus $2-19$ spline points could be calculated (Scott et al., 2016). For the model time series fitting procedure, the best practice is to use statistical hypotheses testing (S. Mackinson et al. 2009; Scott et al. 2016) that is based on ecological knowledge.

Table 3.1 summarizes the scenarios selected. In general terms it was hypothesized that temporal changes in primary productivity due to regional oscillations in environmental conditions on one

hand, and fishery induced changes in the trophic control of the food web on the other hand (through changes in the relative biomass contribution of the different fishing target species) both might have contributed to the observed temporal variation in the volume and composition of the catch.

Table 3.1 – Hypothesis-based model fitting. Selection of five model scenarios, including a combination of time series data [Yearly Biomass (t/km²), fishing effort (days fishing), trophic effects (vulnerability setting by predator, between 0 and 100)], primary production anomalies, time series data of several climatic and physical factors forcing primary productivity.

Hypothesis	Scenario	Biomass	Fishing effort	Trophic effects	PP anomalies	Time series data
1 Changes in fishing effort alone explains observed biomass and catch changes over time	Fishing only, default vulnerability, v = 2	X	X			
2 Fishing changes the biomass proportions of model compartments causing trophic effects and biomass changes in model groups	Fishing + vulnerability search	X	X	X		
3 Changes in fishing effort and parallel changes in primary productivity cause observed changes in catches and biomass of compartments	Fishing, default vulnerability + Primary production anomaly	X	X		X	
4 In addition to the above listed (3) trophic control mechanisms also drive observed changes	Fishing + vulnerability search + Primary production anomaly	X	X	X	X	
5 In addition to the above listed (3) environmental parameters that explain additional inter-annual fluctuations and long-term trends in catches and biomass	Fishing + vulnerability search + Environmental time series	X	X	X		X

The most representative model was selected based on the calculated sum of squares and the Akaike information criterion as an output of the Ecosim routine (AIC). The AIC was calculated as follows:

$$AIC = n \times \log\left(\frac{\min SS}{n}\right) + 2k \quad (1)$$

Here, n is the number of observations (data points) being fitted and k is the number of parameters estimated ($k_{\max}$ = number of time series - 1), SS is the sum of square residuals and minSS is the calculated sum of squares.

To prevent overfitting and to select the most parsimonious model for the given data, the AIC is a good indicator to use, as it includes penalties for increasing the number of adjusted parameters. The AIC assumes that all data points are independent. Since this is not always the case with fisheries data, a conservative approach was chosen and the number of parameters (k - 1) was based on the number of time series instead of single data points. For small sample sizes it is recommended to use the AIC corrected for small sample sizes (AICc) and since the AICc converges to AIC as n increases, the AICc can be employed regardless of the sample size (Burnham and Anderson, 2004). The AICc is derived from the AIC as follows:

$$AICc = AIC + \left(2k \frac{k + 1}{n - k - 1}\right)$$

According to the protocol created by Burnham and Anderson, 2004, several additional measures associated with the AIC were calculated to improve model comparison:

The relative difference (Δ_i) in AICc compared to the best model (AICc_{min})

$$\Delta_i = AICc_i - AICc_{\min}$$

the relative likelihood of the model having the best fit

$$relative\ likelihood = \exp\left(-\frac{\Delta_i}{2}\right)$$

and the Akaike weights (W_i):

$$w_i = \frac{\exp\left(-\frac{\Delta_i}{2}\right)}{\sum \exp\left(-\frac{\Delta_i}{2}\right)}$$

where w_i is a measure of the strength of evidence for each model, indicating the probability that the model is the most accurate one among the set of possible models. The models were ranked by AICc and their relative difference was compared to the minimum AIC. In general, the model was selected that provided the lowest (most negative) AICc value, and models with fewer parameters were favored. However, to select the most representative model that conveniently explains variations in the data, several approaches can be followed. In this case, all models with an AIC difference $\Delta_i < 2$ compared to the best model were considered. The favored models were selected based on the SS decreases, the number of estimated variables and the ability of the model to explain variations in the historical catch data. In order to compare the trends of the observed and predicted values, the data were visualized (see Figure 3.5). To validate the selection of the model, the model's functional groups and their data availability (catch and effort data) were taken into account.

3.2.5 Uncertainty and Sensitivity Analysis

In order to examine the sensitivity of the results with regard to uncertainty of the input parameter values, Monte Carlo simulations were run for each scenario. Each trial (number of trials = 100) represents an Ecosim run with a randomly selected set of Ecopath parameters (B, P/B, EE, BA) for each group, within a distribution centered around the original values. The coefficient of variation of each parameter was selected according to the certainty of the input data set in the pedigree, defined by the quality of the data from estimates from low to high precision sampling (Heymans et al., 2016). The predicted biomass trajectories and the sum of squares of the best iteration runs were compared to the initial input data to detect changes in model behavior due to changes in the input data (Travers et al. 2010). These steps ensure an analysis of the data that is considerably less sensitive to errors.

3.3 Results

3.3.1 Fisheries Data

Trends in the fishing effort and catch per unit effort of the artisanal and semi-industrial fleets are displayed in Figure 3.2. It can be observed that the semi-industrial fishing effort increased until 2000, followed by a slight decline in the early 2000s and a further drastic decline after 2006 and 2007, resulting in an effort reduction of 65% in total. The artisanal fishing effort remained high until 2005, with some variations, and then declined drastically. Recent levels of fishing effort are reduced by 55% compared to 2005; also due to the seasonal fishing ban “veda” that lasts 3 months. A peak of artisanal catches can be observed in the late 1990s, followed by decreasing catches.

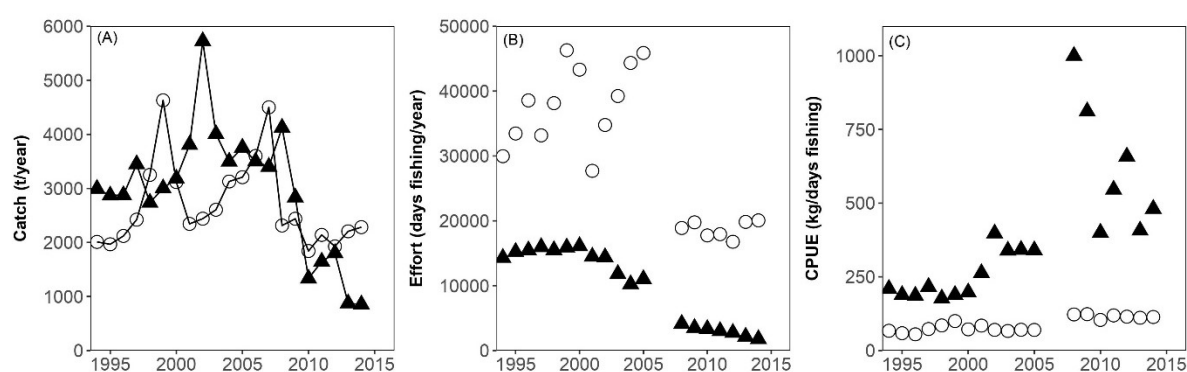


Figure 3.2 – Catches, fishing effort and catch per unit effort. (A) Semi-industrial (filled triangles) and artisanal (circles) catches (t/year), (B) semi-industrial (filled triangles) and artisanal (circles) fishing effort (days fishing/year), (C) semi-industrial (filled triangles) and artisanal (circles) catch per unit effort (CPUE) (kg/day fishing) for the years 1994-2014 (missing effort data 2006 and 2007) in the Gulf of Nicoya (Costa Rica).

3.3.2 Ecosim Simulations

The EwE time series simulations of the last two decades indicated that the fishing effort explains a large part of the variability in the catch and biomass data of some functional groups (Figure 3.2). The semi-industrial fleet’s effort improves the model of the target species shrimp and sardine, as well as of small demersal fish and crabs as bycatch species of the shrimp trawl fleet.

Since not all trends in the catch data can be explained by variations in the fishing effort of different fleets, the possible role of trophic effects was taken into account and the vulnerability settings of 11 different predator groups were changed from its default to simulate top-down and bottom-up effects (Table 3.2). This allowed for further improvement of the model fit and a reduction of the SS.

The assessment of the prey-predator interactions in the model of the Gulf of Nicoya provides important information on the food web structure of the system. The most representative vulnerability settings are displayed for each predator group, whereas bottom-up (B), top-down (T) interactions are indicated in Table 3.2. The top predatory fish species in the system, mackerel and barracuda as well as rays, sharks and cephalopods, are bottom-up controlled as well as the lower trophic level species, such as benthos, shrimps, and crabs. According to the modeling results, only few species exert top-down control on their prey, such as jacks and

pompanos, lobster, seabirds and zooplankton. Mid-trophic-level species, small pelagics and small demersals are assumed to exert mixed control on their prey and predators.

Furthermore, some of the functional groups seemed to be further affected by inter-annual production anomalies. The consideration of these anomalies (scenarios 3 and 4) further improved the model (Figure 3.3). The best model was obtained by adjusting the trophic interactions and adding environmental forcing data (Scenario 5) related to changes in primary productivity. Moreover, adding Chl_a time series to the model, a proxy for the available amount of phytoplankton in the system, improved the model substantially. Together with changes in the vulnerability settings of 10 groups, the model (sum of squares) improved by as much as 69% (and showed the lowest AICc). Table 3.3 gives an additional overview of the factors that improve the model of each functional group.

The variations of the physical environment in the Gulf of Nicoya linked to the ENSO cycle were represented in the model by the Eastern Equatorial Sea Level Pressure (SLP) time series, leading to substantial improvement of the model. Changing the vulnerability setting of once again 10 groups resulted in the most representative model and a 60% reduction of the SS (as well as low AICc values) (Figure 3.3).

Figure 3.4 elucidates the relation between phytoplankton productivity (Chl_a as proxy), the SLP and the ENSO cycle. This shows that there is a negative relationship between the East Equatorial Sea Level Pressure and the El Niño phenomenon. El Niño years are associated with negative anomalies in Sea Level Pressure (SLP), e.g., in the years 2002 and 2009. A pronounced positive peak in the SLP in 1999 can be associated with a strong La Niña event, the counterpart of El Niño. Positive SST anomalies occur during episodes of El Niño (e.g., 2002, 2009). Maximum Chl_a concentrations coincide with La Niña episodes (2005, 2012).

In Figure 3.5 the two Ecosim model scenarios and the baseline scenario are presented together with the catch reference time series-data for the 10 selected functional groups. The baseline scenario includes solely the fishing effort and the two best fitted scenarios include either Sea Level Pressure or Chl_a concentrations. Both time series can be used separately to force phytoplankton productivity, improving the overall model fit. For several functional groups the model's predictions of the catch are relatively close to the observational data. Especially for important target groups (shrimps and small pelagics) the model can predict changes in the catches. For some other groups, such as rays and sharks, catches for some years are above or below the predicted values. The results show that some functional groups (e.g., lobster, small demersals, small pelagics, and shrimps) are more affected by the environmental factors than others and can be predicted best by scenarios including phytoplankton forcing. Table 3.4 provides the related sum of square (residual sum of squares, regression sum of squares and total sum of squares) values for each target group and scenario presented in Figure 3.5.

Table 3.2 - Prey-predator interactions in the Gulf of Nicoya Model (scenario 2) . The vulnerability settings for each predator group are displayed, resulting in the lowest model sum of square calculated by the fitting procedure. Trophic interactions are labeled B: bottom-up interaction with $v < 2$, T: top-down interaction with $v > 2$, and D: default setting.

Predator prey	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 Seabirds																		
2 Mackerel and barracuda																		
3 Rays and sharks			B															
4 Large drums and snook																		
5 Catfish			B			D												
6 Morays and eels		B	B	D				D										
7 Jacks and pompanos		B	B			D	T	D										
8 Snapper and grunts	T			D														
9 Cephalopods	T		B	D	D	D	T	D	B	T								
10 Lobster		B	B	D														
11 Flatfish		B	B	D				D										
12 Small demersals	T	B	B	D	D	D			B		D							
13 Small pelagics	T	B	B				T	D	B									
14 Crabs	T	B	B	D	D	D	T	D	B	T								
15 Epibenthos	T		B	D	D	D	T	D	B	T	D	D		B			B	
16 Endobenthos	T		B		D	D				T	D	D		B			B	
17 Shrimps			B	D	D	D	T	D	B		D	D		B				
18 Zooplankton							T		B			D	D	B	B	B	B	T
19 Phytoplankton												D	D		B	B		T
20 Mangroves															B	B	B	
21 Microphytobenthos												D	D		B	B	B	
22 Detritus				D	D	D		D				D		B	B	B	B	T

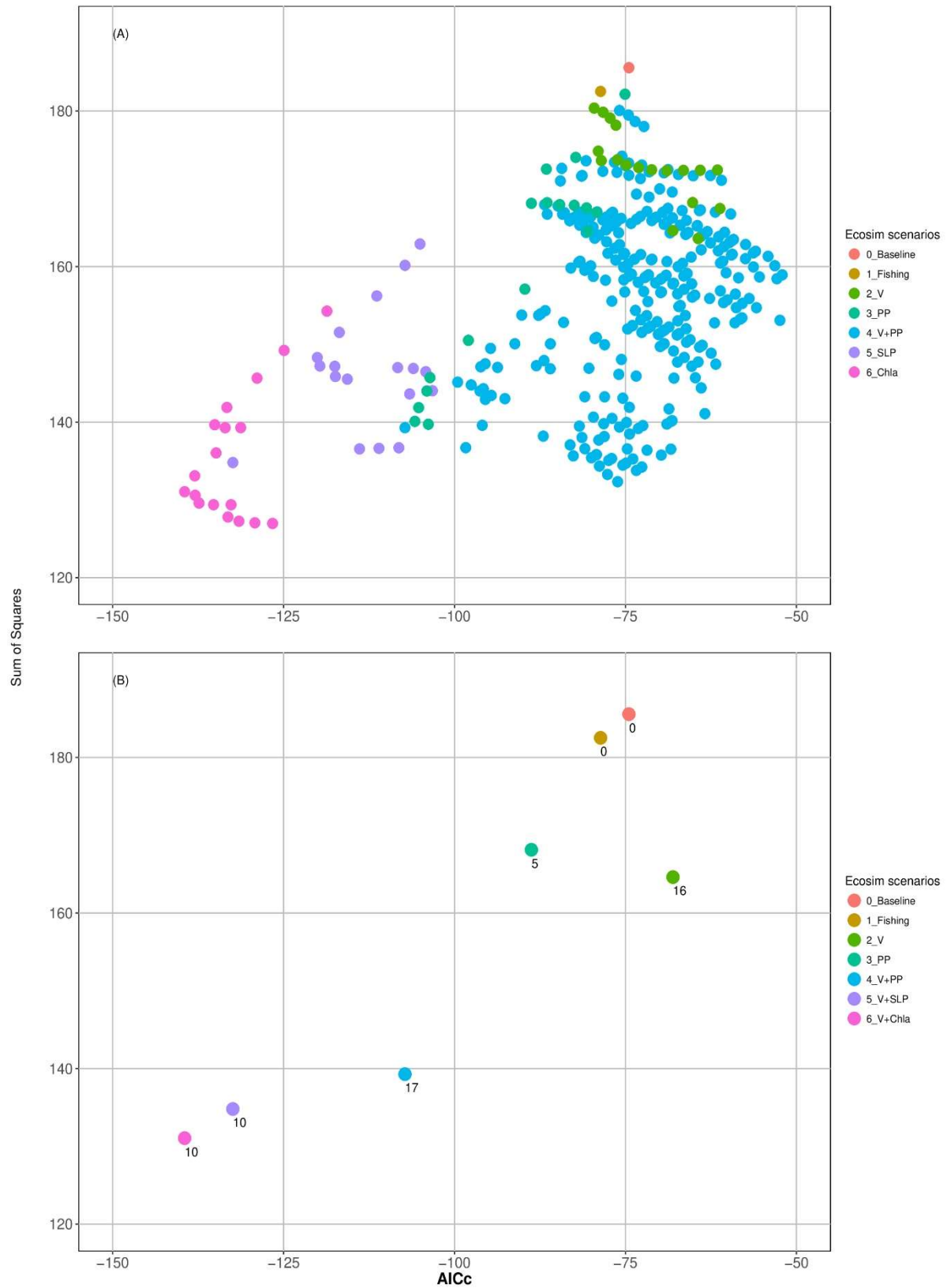


Figure 3.3 – AICc of all tested models and number of estimated parameters. **(A)** Sum of squares and the corrected Akaike information criterion (AICc) of all tested models of each scenario and **(B)** number of estimated parameters (vulnerabilities and or spline points of primary production anomalies) of the 'best models' of each scenario.

3.4 Discussion

3.4.1 Combined Ecosystem Effect

The aim of the present study was to identify factors influencing the changes observed in the Gulf of Nicoya ecosystem over the past two decades, and to understand how these factors control the energy flows in the system (bottom-up, top-down or mixed/wasp waist control). We hypothesized that a combination of several drivers might affect the ecosystem and may decrease its natural resilience toward future environmental changes (Côté and Darling 2010). As it has been shown in previous studies, ecosystem parameters calculated from Ecopath models can be used as indicators of the status of the ecosystem (Heymans et al. 2014; Wolff et al. 2012). Accordingly, fisheries and/or climate impacts on the ecosystem should result in changes in the sensitive system state indicators.

The officially registered overall fishing effort in the Gulf of Nicoya declined drastically since the early 2000s. On the one hand, the reduction is due to the implementation of the seasonal fishing ban “veda” in the inner Gulf of Nicoya, which has directly reduced the yearly effort since 2006 by approximately 60%. On the other hand, the shrimp fishing fleet reduced its capacities due to a lack of resources and the expiration of licenses. In 2013 shrimp trawling was declared unconstitutional in Costa Rica and no further licenses were said to be granted in the future. The last remaining shrimp licenses expired in 2018 and 2019.

Although the overall fishing effort decreased, the CPUE of the artisanal and semi-industrial fleet increased compared to the early 1990s, despite the heavy exploitation of the resources. However, gear improvements changed the relation between the CPUE and the actual abundance of species and tend to mask the trends of the species exploitation status (Arreguín-Sánchez and Calderón-Aguilera 2002; Ward and Myers 2005b). Illegal fishing gear improvements, i.e., toward Taiwanese lines and smaller gillnet mesh sizes might have allowed increasing catches per unit effort. These changes in the fishing fleet have most likely increased the overall efficiency of the fishing gear and the catchability of certain target resources.

We come to the conclusion (Table 3.3) that both the semi-industrial and artisanal (gillnet) fleet negatively influence the distribution of some target species (e.g., cephalopods, rays, and sharks). However, the negative impact on the major target species is dominated by the shrimp trawl fleet, which negatively influences its target species as well as its bycatch species and by the semi-industrial sardine fleet. The artisanal handline fleet does not appear to have a relevant influence on any target species. The manual mollusk extraction is a very selective harvesting method, however, the removal of piangua (*Anadara tuberculosa*, Arcoida: *Arcida* spp.) drastically increased in the last decade and exceeded sustainable levels.

Table 3.3 - Catch time series : factors [fleet effort, trophic effects, primary production anomalies: Sea Level Pressure (SLP) and Chlorophyll a] that improve the model (reduce the sum of squares) of each functional group and the total improvement of the sum of squares for all groups for each factor analyzed.

Functional group	Fishing effort	Trophic effects	PP anomaly
Cephalopods	Gillnets	X	
Corvina and snook		X	SLP
Crabs	Manual, SI shrimp		SLP
Eel			SLP
Jacks and pompano			Chl _a
Lobster	Art. manual		Chl _a
Mackerel and barracuda			Chl _a
Rays and sharks	Longline	X	Chl _a
Shrimps	SI shrimp, Art. gillnets	X	
Small demersals	SI shrimp		Chl _a
Small pelagics	SI sardine	x	Chl _a
Total reduction model SS	-24.91	-18.52	-69.30 (-60.56)

The evaluation of the trophic effects of the most representative models helped to investigate the most relevant control mechanisms between predators and their prey (Table 3.2). As seen in Table 3.3, the top predatory fish species in the system as well as cephalopods are bottom-up controlled by their prey. The same force applies to lower trophic level groups, such as benthos, shrimps and crabs. According to our analysis, only few species exert top-down control on their prey, such as jacks and pompanos, lobster, seabirds and zooplankton. It seems that mid-trophic-level species exert wasp-waist control on their prey and predators. This is in agreement with findings of Griffiths et al. (2013) and Roux et al. (2013). As has been found in these studies, the mid-trophic levels are only represented by few species, such as plankton feeding small pelagic fish.

Our analysis confirms these findings for the Gulf of Nicoya ecosystem and highlights the importance of the mid-trophic level species, which are heavily exploited by both semi-industrial fisheries (small pelagics) and artisanal fleets (small demersals). The observed decline of these species did not only have ecological, economic and social consequences (Alms and Wolff 2019; Queiros et al. 2019), but may also lead to unfavorable and irreversible shifts in the ecosystem. This has been observed in other ecosystems, namely through the expansion of jellyfish (Roux et al., 2013) or may lead to the decrease of predatory species, such as fish, seabirds, and mammals (e.g., Crawford and Shelton, 1978; Pichegru et al., 2010)

The modeling approach we applied in our study clearly showed that the variations in the catch cannot solely be explained by variations in fishing pressure. Similar results have been found in other systems, such as the Southern Benguela system (Shannon et al. 2008). Adding

environmental data related to the primary productivity of the system to the simulations considerably increased the reliability of the model.

Especially primary producers are strongly influenced by the environmental conditions through changes in nutrient or light availability (through wind, current, or upwelling pattern changes) and growth conditions (if in situ temperatures changes) (Hays et al. 2005). Since changes in phytoplankton productivity are closely related to zooplankton abundance (Runge, 1988, FAO), e.g., bottom-up force through the entire food chain can be the consequence. For the analysis of the effects of inter-annual environmental variations on the ecosystems in the ETP region, the El Niño-Southern Oscillation (ENSO) cycle needs to be considered, which is known to be a major factor of climate variability in the Eastern Tropical Pacific with global impact. It is anticipated that ENSO will potentially modify in frequency and intensity due to global climate change (Cai et al. 2014; Collins et al. 2010).

In our study, the Chl_a concentration appeared to be the most relevant of tested environmental drivers. It is a proxy for the amount of plankton in the water column, which directly affects the bottom of the food chain. Increasing phytoplankton availability enhances zooplankton growth, resulting in higher availability of prey for low trophic level species foraging on zooplankton. The associated increased production will then cascade through the entire food web.

On the contrary, a reduced primary production will lead to a lack of foraging success of species in higher trophic levels. Changes in the vertical and horizontal distribution of the plankton could also cause changes in the spatial distribution of foraging fish with effect on their predators and fishery through an alteration in catchability. Our results indicate that particularly the lower trophic level groups, such as small demersal- and small pelagic fish benefit from increased phytoplankton availability. Since catches of predatory fish species, such as rays, sharks, mackerel and barracuda, as well as the catches of lobster also positively benefit from high Chl_a concentrations, this could be due to a bottom-up effect due to increased prey availability of these groups. Plankton feeding fish are known to aggregate closer to the coast due to food concentration in these areas. Foraging fish follow their prey and likewise occur closer to the coast. Thus, the catchability of certain fish species increases for the coastal fisheries.

Furthermore, other abiotic factors of the marine environment (e.g., wind direction and speed, light availability) can directly influence higher trophic levels of the system, for instance by affecting larval dispersal and the productivity of certain species (Runge, 1988).

The environmental indicator that led to the most representative model was the East Equatorial Sea Level Pressure (SLP), which is negatively related to the El Niño phenomenon (Figure 3.4). At high SLP (La Niña events), a lower SST can be observed (station ElNiño4) as well as stronger winds and upwelling, and lower precipitation rates. As we expected a negative impact of El Niño on the catches, the East Equatorial SLP as the inverse of SST anomalies proved to be a good measure. In the Eastern Pacific, the ENSO cycle is known to alter the oceanographic condition in the area and to affect a variety of marine communities (National Oceanic and Atmospheric Administration, 2020b). As it has been observed in the upwelling regions of the Humboldt Current system, small pelagic fish biomass decreases during strong El Niño years and the total fisheries production declined up to 50% (Taylor et al., 2008; Wolff et al., 2003, 2012).

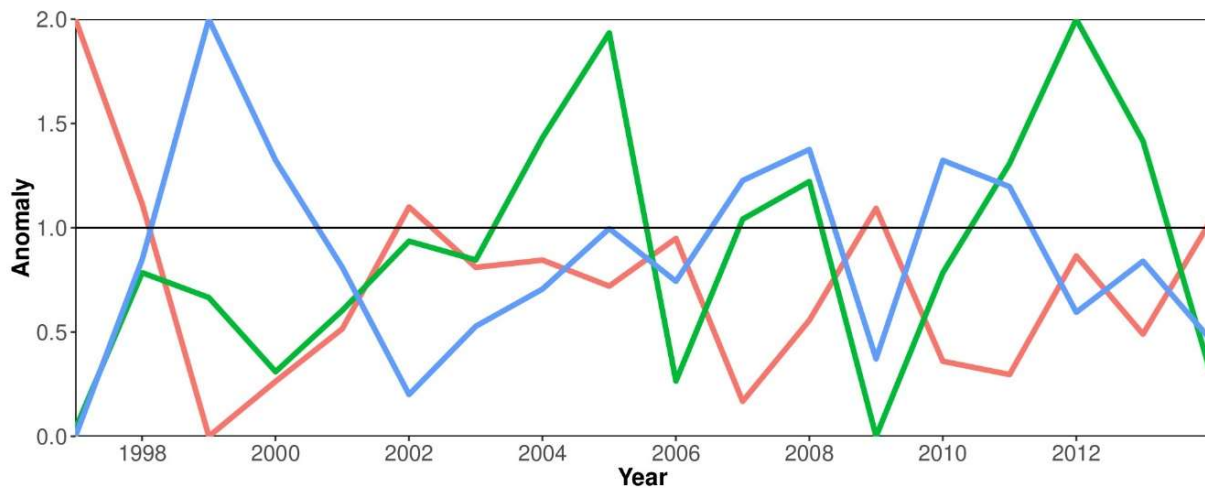


Figure 3.4 - Environmental time series 1997-2014: anomalies of Sea Surface Temperature (SST) (red line), Chlorophyll a concentration (green line) and the Sea Level Pressure (blue line) in the Equatorial Eastern Pacific Ocean associated with the El Niño-Southern Oscillation cycle.

Our study reveals that several species in the Gulf of Nicoya are influenced by these large-scale environmental conditions associated with the ENSO cycle. We specifically observed that variations in the catches of functional groups (large drums and snooks, crabs and morays and eels) are related to the ENSO cycle.

At higher temperatures during El Niño events some species in the Gulf of Nicoya migrate further out of the gulf into cooler oceanic waters while others, i.e., small pelagic fish (sardine), aggregate close to the beaches at the Pacific coast of Costa Rica. Similar aggregation patterns of small pelagic fish can be observed in other regions (Gutiérrez et al. 2007). A shallow and more pronounced thermocline can therefore limit the vertical distribution of certain pelagic species and may increase their catchability in the upper water column (Bigelow and Schroeder 2002) (Bigelow et al., 2002). Warm water masses can also cause other species to move offshore into deeper and cooler waters, reducing their catchability.

According to the FAO, most of the inter-annual fluctuations in total fish production in the Eastern Central Pacific (Fishing Area 77) are due to changes in the abundance of small pelagic fish (FAO), that play a central role in this and many other marine ecosystems (Frederiksen et al., 2006; Shannon et al., 2008). ENSO related changes in water properties (e.g., heavy rains washing nutrients into the gulf or calm periods leading to enhanced water stratification and oxygen deficiency) might also increase the frequency of toxic algae blooms that have previously been observed in the Gulf of Nicoya, and eventually cause fish mass mortalities in the Gulf of Nicoya (Vargas-Montero and Freer 2004).

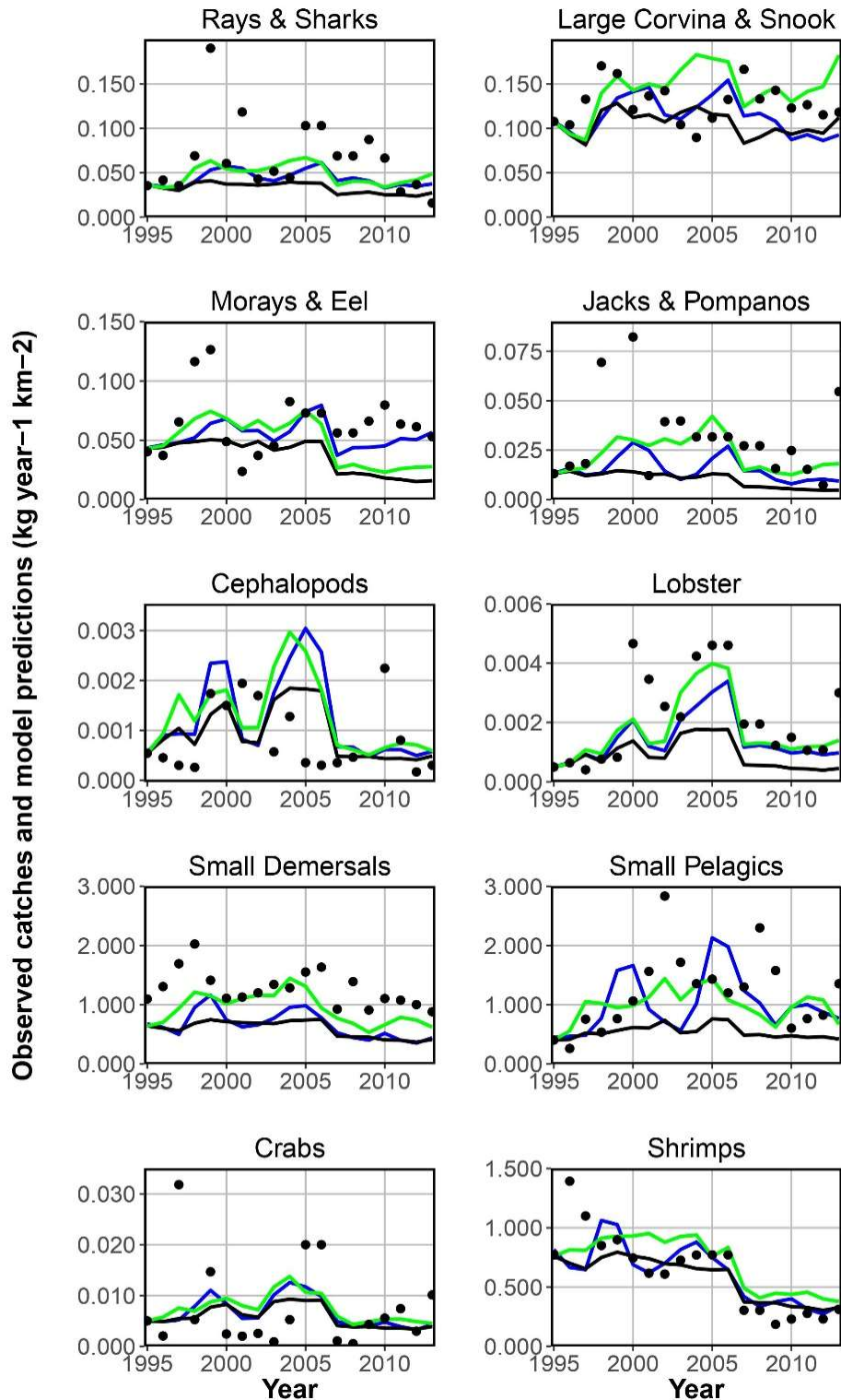


Figure 3.5 - Yearly catches (filled circles) and model predictions for the simulated EwE scenarios : fishing effort (black line); fishing effort, trophic effects and Sea Level Pressure forcing phytoplankton productivity (blue line); fishing effort, trophic effects and Chlorophyll-a concentration forcing phytoplankton productivity (green line).

The findings of our study suggest that there are multiple factors that impact the ecosystem, besides the exploitation by the semi-industrial fishing fleets. Environmental forcing plays an important role in the explanation of species biomass variabilities, especially in relation to the ENSO cycle, together with bottom-up effects, due to fluctuations in primary productivity. Furthermore, mid-trophic level species, such as small pelagic fish, have a pivotal role in the food web of the Gulf of Nicoya.

Table 3.4 – Model predictions compared to reference catch data: Calculated sum of squares: residual sum of squares (SSE), regression sum of squares (SSR), and total sum of squares (TSS) for the model's predictions of the selected scenarios and important target groups compared to the reference catch data. The lowest SSE, SSR and TSS values, indicating the model scenario with the best model fit, are marked in bold. (1) Fishing effort only; (2) fishing effort, trophic effects and Sea Level Pressure (SLP) forcing phytoplankton productivity; (3) fishing effort, trophic effects, and Chlorophyll-a concentrations forcing phytoplankton productivity.

	Fishing effort			Fishing effort, trophic effect and SLP			Fishing effort, trophic effects and Chla		
Species	SSE	SSR	TSS	SSE	SSR	TSS	SSE	SSR	TSS
Cephalopods	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Crabs	0.0012	0.0001	0.0013	0.0012	0.0001	0.0013	0.012	0.0001	0.0013
Jacks & pompanos	0.0213	0.0115	0.0328	0.0166	0.0077	0.0243	0.0115	0.0041	0.0156
Large corvina & snook	0.0226	0.0132	0.0358	0.0185	0.0124	0.0309	0.0297	0.0182	0.0479
Lobster	0.0001	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Morays & eel	0.0278	0.0175	0.0453	0.0136	0.0039	0.0174	0.0180	0.0107	0.0287
Rays & sharks	0.0481	0.0225	0.0706	0.0330	0.0119	0.0449	0.0307	0.0091	0.0398
Shrimps	0.8307	0.6720	1.5027	0.9318	1.058	1.9904	0.9409	1.0539	1.9948
Small demersals	9.8339	9.0453	18.8792	8.1687	7.9805	16.1492	3.5483	3.4970	7.0453
Small pelagics	14.6412	8.3203	22.9615	11.6363	5.0199	16.6562	6.9506	2.1752	9.1258

3.4.2 Uncertainty and Sensitivity Analysis

Although we reported a substantial reduction of the officially reported fishing effort in the gulf, it can be assumed that unreported catches, fishing without licenses and fishing during the closed season have occurred during the observational period. These unreported fishing activities cannot be quantified and are thus not part of the study.

The Ecopath models have been carefully validated with a prescribed balancing routine of Link (2010). The balancing process of the models has been described in Alms and Wolff (2019). We adhered to the suggestions of best-practice in Ecopath with Ecosim food-web models for ecosystem-based management by Heymans et al. (2016). Adding data to the model such as model drivers and reference data sets using the Ecosim tools helped to further validate the quality and accuracy of the created static Ecopath model. The model was additionally calibrated with historical fishing datasets and the comparison of the model predictions to reference data allowed for the validation of the model performance (Johanna Jacomina Heymans et al. 2016). The calculated EwE Pedigree Index of 0.29, based on the confidence intervals of all input parameters, indicated that the model's level of precision is in the intermediate to the upper range, compared to other published Ecopath models worldwide (Morissette 2007). We believe that the AICc is a good measure for validation of the selected model. We followed the recommendation to adapt the AIC for small sample sizes (Burnham and Anderson 2004).

The aim was to provide a holistic modeling approach, in the sense looking at combined effects influencing the species biomass variations. We covered trophic effects, the effects of fishing, the most prominent climate variables, and physical and chemical factors (also as proxy for pollution). However, the selection of factors was limited by the availability of time-series data and not every single possible parameter could be tested. Some factors, such as human disturbances, sedimentation, solid waste pollution and agricultural pollution, could not be

assessed. For future research it would be useful to incorporate social-ecological interactions into the model.

3.4.3 Management Outlook and Implications

The present study leads to the conclusion that fisheries management policies entirely based on fisheries information only, with no consideration of the influence of environmental factors is not sufficient for the development of a holistic management strategy. Both fishing regulations and environmental monitoring needs to go hand in hand. We found that the shrimp trawl fleet predominately affects the distribution of important target species. Thus, the advised fishing ban and the expiration of licenses concerning the trawler can be seen as a promising management measure. The sardine purse-seine fleet needs to be seen critically as we could highlight the importance of small pelagic fish to the ecosystem and their susceptibility to climate change. The impact of the artisanal fleet is minor compared to the semi-industrial fleet. However, the gillnet fleet using small mesh sizes can be potentially harmful for sustainable fisheries. To reduce illegal fishing activities stronger control mechanisms and rigor enforcement are needed. Furthermore, the manual mollusk extraction needs to be limited, as yearly around 8 million individuals are harvested from the Gulf of Nicoya (SINAC 2019).

Environmental monitoring needs to be considered and combined with fisheries assessment studies, due to the considerable influence of these factors on the entire trophic system as well as their social-economic impacts. A regular monitoring program for the Gulf of Nicoya is necessary to track changes in primary production, and phytoplankton community composition in the Gulf of Nicoya, that may be driven by both increased coastal pollution and environmental change. To apply appropriate management measures, a forecasting approach is needed (Clark et al. 2001) that takes into account knowledge of the current and predicted ecosystem structure and functioning (Valette-Silver and Scavia 2003).

Our study may be used as a reference to further assess the regional impacts of future climate variations and extreme events in an ecosystem already altered by the fisheries. In the Gulf of Nicoya, like in other tropical environments, we need to consider the effects of increasing temperatures, as stated by Cheung et al. (2013) and Palomares and Pauly (2019). In contrast to colder regions, no species adapted for warm regions will be available to replace species that are displaced by the high temperature thus, there will most likely be a temperature-driven loss of species in these areas.

Of particular relevance for future management scenarios are regional changes that have been predicted by the Intergovernmental Panel on Climate Change (IPCC) until the year 2100. According to the IPCC the Gross Primary Productivity is expected to decrease, mainly due to an increase in Colored Dissolved Organic Matter (CDOM), which absorbs a fraction of the Photosynthetically Active Radiation (PAR) (Thrane et al. 2014; Wolf and Heuschele 2018). One can thus expect that Chl_a concentrations decrease in the Gulf of Nicoya and that phytoplankton composition, quantity and distribution will change. In addition, mangrove coverage is expected to further decrease due to sea level rise and changes in salinity (Ward et al., 2016). While mangroves only account for an estimate of 1% of the gulf's primary production (Wolff et al. 1998), they contribute to the system's productivity in the Gulf of Nicoya by providing organic matter and protected areas, enhancing fish and invertebrate

productivity (Wolff et al. 1998). Considering the predicted decrease in primary productivity, it seems necessary for a sustainable ecosystem management to adapt the volume of fishery catches to potentially lower overall resource productivity.

Furthermore, there is a high uncertainty concerning the future dynamics of the ENSO cycle. The frequency and intensity of El Niño events is likely to be altered by climate change, however it is unknown to what extent (e.g. Cai et al. 2014; Collins et al. 2010). As our simulations suggest, catch and biomass fluctuations in the Gulf of Nicoya are linked to ENSO related variations in the Equatorial Pacific Sea Level Pressure, and a constant monitoring of the regional ENSO impact is thus imperative.

As we have suggested previously (Alms and Wolff, 2019), it is crucial to think of measures to sustain primary productivity and to protect keystone groups, benthos and shrimps in the Gulf of Nicoya, but also adequately manage the mid-trophic level species such as small pelagic and small demersal fish. It is known that in the gulf, like in other coastal areas and estuaries with high abiotic variability (Heymans et al. 2014), these groups are central for ecosystem functioning and need to be prioritized in future management plans. In addition to adequate fisheries regulation, possible measures for improving the overall management of the Gulf of Nicoya could be the improvement of wastewater management and treatment and solid waste disposal plans. For the conservation of the coastal marine environment the further protection and restoration of mangroves is an important issue that needs to be further pursued (SINAC 2019).

Furthermore, our models can be used to directly engage stakeholders in the management plans, as it has been attempted by Sánchez-Jiménez et al. (2019). The EwE software was used as an interactive communication tool with artisanal fishers in the Gulf of Nicoya, to facilitate discussions on changes in the fishing ecosystem over time.

Acknowledgments

The facilities in Costa Rica and some of the data about the Gulf of Nicoya were provided by the Centro de Investigación en Ciencias del Mar y Limnología (CIMAR) and the Universidad de Costa Rica. The research project was strongly supported by our main project partner Alvaro Morales Ramírez in Costa Rica, who provided the infrastructure and guidance in Costa Rica and helped to develop the project collaboration between the Leibniz Centre for Tropical Marine Research and the CIMAR. Several other colleagues from the CIMAR and the University of Costa Rica supported the project, Jorge Cortés Núñez, José A. Vargas Zamora, Helena Molina Ureña, and Catalina Benavides Varela. Berny Marín Alpízar (INCOPESCA) supported the study with valuable advice and guidance concerning the marine resources in the Gulf of Nicoya. Moritz Stäbler (ZMT) supported the time series modeling task with his experience and advice on the Ecopath with Ecosim modeling approach.

Chapter 4

EXPLORATION OF FISHERIES MANAGEMENT POLICIES IN GULF OF NICOYA USING ECOSYSTEM MODELLING

Chapter 4

Exploration of fisheries management policies in the Gulf of Nicoya (Costa Rica) using ecosystem modelling

Viola Alms, Giovanni Romagnoni, and Matthias Wolff

This is the author's version of the work. Please cite the final version as:

Alms, V., Romagnoni, G., and Wolff, M. (2022). Exploration of fisheries management policies in the Gulf of Nicoya (Costa Rica) using ecosystem modelling. *Ocean & Coastal Management*, 230, 106349. <https://doi.org/10.1016/j.ocecoaman.2022.106349>

<https://doi.org/10.1016/j.ocecoaman.2022.106349>

Submitted on 24th February 2022.

Accepted on 28th August 2022.

Published online on 14th September 2022.

Contribution statement

Project idea: **Viola Alms**

Design of the work: **Viola Alms**

Acquisition or interpretation of data: **Viola Alms** supported by Giovanni Romagnoni

Drafted manuscript revised by Giovanni Romagnoni and Matthias Wolff.

Abstract

The Gulf of Nicoya is a highly productive estuary located at the Pacific coast of Costa Rica. Previous studies have used trophic models to examine changes in the biomass of key species and in the food web in the last 25 years, revealing an overfished, degraded system with decreasing biomass of valuable target species. The ecosystem degradation was mainly driven by intensive fishing, while climate variations affected resource productivity additionally. This study tested the effectiveness of alternative scenarios including combinations of the current top-down fishing policy and the ban on shrimp trawling, together with a participatory management scenario developed in a previously performed stakeholder workshop. In parallel, the automated fishing policy search of the Ecopath with Ecosim (EwE) software was used to explore an optimized alternative management scenario. The analysis indicates that the ban on trawling is an important measure to allow for the recovery of certain target species, such as shrimps, demersal fish and their predators. However, this ban would not suffice to substantially rebuild the biomass of all key species in the system. Thus, two possible alternative management scenarios are proposed: in the first one, the economic losses are minimized, ecosystem health increases by 10% (by rebuilding target species biomass) and employment provided by fishing decreases (-15%). In the second scenario, higher economic losses are accepted (mainly for the semi-industrial fisheries sector) which allows for a higher increase in ecosystem health and biodiversity. Both scenarios call for additional reductions in fishing efforts, mainly by the semi-industrial purse-seine fleet and the artisanal longline fleet. This study exemplifies how holistic ecosystem models can be used for management advice, future policymaking and how stakeholders can be engaged in this process.

Keywords: Trophic model, fishing policy, fisheries management, Ecopath with Ecosim, participatory management

4.1 Introduction

The marine fishing sector in Costa Rica has steadily grown in the last decades, advancing its gear and boats, increasing the fishing range and its catch effectivity. These development lead to overexploitation and also increased conflicts among fishing sectors due to decreasing resources (Lozano and Heinen 2016b). Especially the small-scale fisheries is the primary source of income and food for many coastal communities, and at the same time the most vulnerable to overexploitation (Carrillo et al. 2019). The situation calls for an effective management policy and suitable marine conservation strategies.

Costa Rica has already proven to understand the urgency of protecting its environment. The country aims to be among the global leaders in sustainability and has successfully established a functioning environmental protection system, fights climate change with an ambitious decarbonization plan (UNEP, 2019). One quarter of the terrestrial area is under some kind of protection and Costa Rica has managed to successfully promote reforestation. Although Costa Rica pushed for ocean conservation, in the past it has played a subordinate role in this process, as less than 1% of its marine area is protected. This is especially striking as the potential of marine conservation is immense, with Costa Rica having an Exclusive Economic Zone (EEZ), ten times larger than its terrestrial area. Now more than ever before significant attention is paid to marine conservation. It is not without reason that the United Nations proclaimed a Decade of Ocean Science for Sustainable Development (2021-2030). The Ocean Decade is linked to the 2030 Agenda and facilitates a variety of marine conservation strategies, like the goal to achieve a healthy, resilient and sustainably harvested ocean (Johanna J Heymans et al. 2020; Ryabinin et al. 2019). In this context, a promising example of an international cooperation to promote the conservation and sustainable use of marine resources is the Eastern Tropical Pacific Marine Corridor (Corredor Mariano del Pacífico Este Tropical - CMAR 2019). This large Marine Protected Area (MPA) is supposed to protect marine biodiversity with special attention on migratory species, for an area across the EEZ boundaries of Ecuador, Costa Rica, Colombia and Panama (Enright et al. 2021), which makes it a significant large-scale initiative.

Also, on the regional scale, some measures have been taken in the past to protect the Gulf of Nicoya ecosystem, Costa Rica's main fishing ground. The goals of these policy measures were to ensure a more sustainable use of harvested resources and to protect the existence of the small-scale fisheries, that is traditionally rooted in the coastal communities around the estuary and is of high economic and social importance.

The measures of input control include fishing effort limitation through granting licenses and a seasonal closure (in Spanish: "*veda*") to all fishing fleets, as well as the technical limitation of gear size (mesh sizes > 6.35 cm [2.5 in]). However, most of these top-down enforced management rules failed either due to noncompliance of fishers or the lack of control capacities. Furthermore, Costa Rica is supporting Marine Areas of Responsible Fishing (AMPRs), areas of exclusive harvesting for small-scale fishing communities fostering co-management.

However, to turn the tide the fishing regulations had to be supplemented with a more drastical enforcement, the ban on shrimp trawling for the entire Gulf of Nicoya. In 2013 semi-industrial shrimp trawling was declared unconstitutional in Costa Rica and no further licenses were issued since 2015. The last remaining shrimp trawling license in the Gulf of Nicoya expired in 2019.

This was a major step towards a more sustainable fishery and a reduction in fishing effort. Not only shrimp species benefit from the ban but also a variety of bycatch species, as well as the bottom flora and fauna.

To evaluate the effectiveness of these efforts, the first research question of the present study is to assess whether the current ban on trawling is the optimal management solution and whether it is sufficient to achieve a recovery of a variety of species and the restoration of the of the Gulf of Nicoya marine ecosystem.

The assessment of the ecosystem benefits, was based on the principles of the Ecosystem Approach to Fisheries (EAF) (FAO 2002). The EAF considers the trophic interactions among functional groups within the ecosystem, and includes environmental, ecological, and anthropogenic impacts on an ecosystem in the management consideration. Following the EAF approach, one can find optimal sustainable rates of exploitation under the condition of biodiversity conservation, increase in ecosystem health and/or rebuilding stocks of target species (Forrest et al. 2015). It provides clear benefits compared to single-species management approaches, that are often insufficient (Cochrane 2002; FAO 2012). In the context of the EAF approach, we used a holistic ecosystem model of the Gulf of Nicoya (Alms and Wolff 2019), developed with the open-source software Ecopath with Ecosim (EwE). Ecosim simulations were used to depict effects of changes in the fishing effort levels on the species/group biomasses and ecosystem health over time.

In the first step to assess the effectivity of the different management options, we tested the success of the policy that is currently in place (ban of trawling), which was afterwards compared to other hypothetical management plans, using a scenario-based approach. For this approach we developed a set of hypothecial scenarios. One of these scenarios is a stakeholder-based policy based on the reduction of gillnet fleet efforts by 25% aimed at allowing high trophic level species to rebuild their stock biomass. It has been suggested by the fishermen themselves during participatory management workshops based on the visualization and information about the ecosystem health and future perspectives using the EwE modelling software (Sánchez Jiménez et al. 2021). The stakeholder-based policy was especially developed to foster participatory management in the gulf of Nicoya. It is known, that besides finding the optimal balance between different management goals for successful policymaking, it is important to consider the interests and knowledge of local stakeholders, thereby increasing the compliance of all actors (Wilson et al. 2006). In fact, according to the UN Sustainable Development Goals (SDGs), fisheries play a pivotal role in food security but also on job provision for coastal communities (Food security and decent work conditions; SDGs 2 and 8). Especially tropical artisanal coastal fisheries face a variety of challenges that threaten future sustainability and at the same time, the management of artisanal fisheries is particularly difficult (Batista et al. 2014). As a further scenario, the combination of the artisanal gillnet effort together with the ban of shrimp trawling was selected. The scenario deliberately affects both the artisanal and the semi-industrial sector, to consider the conflicts in the use of coastal resources as well as competition between artisanal and semi-industrial fishing.

Our second research question was to assess, whether an alternative fishing policy can be found that is more effective than the current policy or the stakeholder-based policy. Thus, in the second step, we used an automated modelling routine to explore other random management

scenarios. For the analysis, the fishing policy optimization routine embedded in the EwE software was applied. This routine maximizes defined management goals based on weighted policy objectives. This approach allows for the definition of specific management objectives that are realistic trade-offs between ecosystem and social-economic benefits. Ecosystem benefits include ecosystem health, a balanced biodiversity and stable biomass levels of the functional group in the system, whereas social and economic benefits include the employment through fishing and related processes, as well as fishing itself as a source of income (FAO 2020). The routine is a powerful tool to find a successful fishing policy, that should at best entail benefits for sustainability, society, equity of costs and benefits, and decrease of conflicts e.g., from competition between artisanal and semi-industrial fleets (FAO 2020; Hilborn and Walters 2013).

The method allows for the calculation of profit from fishing for each fishing fleet in a given scenario. It is evident that all measures taken towards a healthier ecosystem and the recovery of species, will entail short-term economic losses and social constraints in the Gulf of Nicoya, e.g. Ovando et al. (2016). However, there will also be consequences if no actions are taken, due to the further depletion of valuable resources by continued extensive fishing. This has been observable by the decrease of valuable shrimps as well as by increasing catches of low value (and low-trophic-level) species, that have already caused a tremendous economic loss of 50% of the profit (Alms and Wolff 2019).

Our study is an example of how holistic ecosystem models can be used for policymaking and how the models can be useful for participatory management, to facilitate the cooperation between stakeholders and to increase compliance.

4.2 Materials and Methods

4.2.1 Study area and fishing fleets

The Gulf of Nicoya (10°N, 85°W) is a tropical estuary at the northwest Pacific coast of Costa Rica (Figure 4.1). The model area stretches from the northern inland (mouth of the Tempisque River) to the entrance of the Pacific Ocean in the south and covers an area of 1,530 km². The inner part of the Gulf is characterized by shallow waters and tidally influenced mud- and sandflats. The outer oceanic zone extends to a maximum depth of 200 m at the shelf edge (Wolff et al. 1998). The Gulf of Nicoya has an exceptionally high productivity, which makes it Costa Rica's principal fishing ground for the artisanal fishing fleet, which provides around 55% of the total catch in the Gulf of Nicoya (in 2013). The small-scale artisanal fishing sector is of major economic and social importance for approximately 2000 fishermen and more than 20 coastal communities (Lozano and Heinen 2016b; Trujillo et al. 2012).

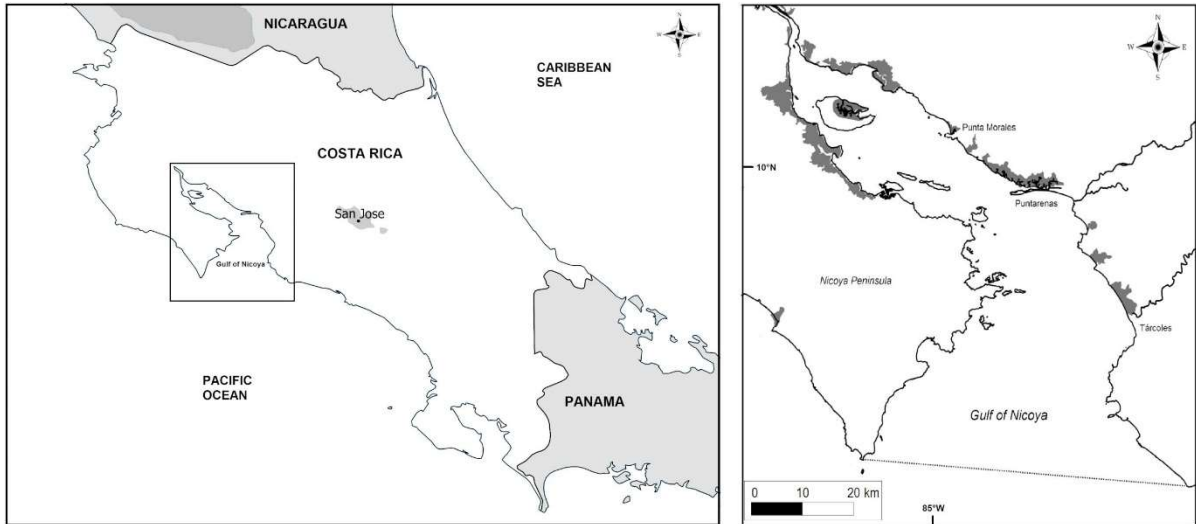


Figure 4.1 - Maps of the study area (Gulf of Nicoya, Costa Rica) , from Alms and Wolff (2020).

Four artisanal fishing fleets operate in the gulf with different boat sizes and gear types that target a variety of demersal and coastal species (BIOMARCC-SINAC-GIZ 2013a). Gillnets are predominantly used, followed by the longline, handline and manual mollusk extraction, which is mostly carried out by women from the coastal communities, in combination with the manual trap fisheries targeting different crustaceans.

The main port of the Gulf of Nicoya, Puntarenas, is also the shrimp trawlers landing site for the semi-industrial fishing fleets - the purse-seine fleet and the shrimp trawl fleet. The purse-seine vessels target small pelagic fish species (*Anchoa* spp., *Cetengraulis mysticetus* and *Opisthonema* spp.) in the intermediate and outer zones of the Gulf of Nicoya. The bottom trawl fleet used to target several shrimp species, mostly in the outer gulf, however the capture of bycatch species drastically exceeded the shrimp catches (Clarke et al. 2016). As a result, shrimp trawling has been banned from the Gulf of Nicoya since 2019. For further descriptions of the ecosystem please refer to (Alms and Wolff 2019; 2020).

4.2.2 Model description

For our study we used the balanced trophic model of the Gulf of Nicoya (Table 4.1), constructed by Alms and Wolff (2019) based on collaborative studies and the most recent fisheries data available (INCOPESCA, unpublished data). It represents the state of the ecosystem in the year 2013. The model comprised a total of 22 functional groups, including 3 primary producer groups and 10 different fish groups and was balanced and fitted to time-series data (historic fisheries catches and effort data).

For the modelling task, the open-source software Ecopath with Ecosim (EwE) version 6.4.4 (Christensen and Walters 2004) was used. Ecopath with Ecosim is based on a set of linear equations for each functional group in the system through which the group's biomass production is connected to other model groups via a prey–predator matrix. For each compartment (i), the basic equation is as follows (1):

$$P_i = Y_i + B_i M2_i + E_i + B A_i + P_i (1 - EE),$$

where P_i is the total production rate of i ; Y_i is the total fishery catch rate; $M2_i$ is the instantaneous predation rate of group i ; B_i is the biomass; E_i is the net migration rate (emigration–immigration); BA_i is the biomass accumulation rate; $P_i(1-EE)$ represents the “other mortality” rate of i ; and EE is the ecotrophic efficiency of i and the proportion of the production that is utilized in the system (Christensen et al. 2005; Christensen and Walters 2004a). Although equation (1) allows balancing of the biomass production in each group with its use in the system (and the fishery), with a second equation estimating the energy budget of each group i is possible as follows (2):

$$Consumption_i = Production_i + Respiration_i + Egestion\ of\ unassimilated\ food_i$$

The model was built around main groups targeted by the fisheries. Species with similar sizes, diets, consumption, mortality, production rates, and habitats were grouped.

Table 4.1 - Basic Ecopath with Ecosim input and output parameters for the Gulf of Nicoya (Costa Rica) model 2013 (Alms and Wolff 2019). Biomass (B_i ; metric tons/km²), production-to-biomass ratio (P_i/B_i ; year⁻¹), consumption-to-biomass ratio (Q_i/B_i ; year⁻¹), ecotrophic efficiency (EE_i), production-to-consumption ratio (P_i/Q_i), and total catch (metric tons·km⁻²·year⁻¹) are presented for each functional group.

Functional group		Input parameters				Output parameters	
		Catch	B_i (t/km ²)	P_i/B_i (/year)	Q_i/B_i (/year)	EE_i	P/Q (/year)
1	Seabirds		0,06	0,09	50	0	0
2	Mackerel & Barracuda	0,017	0,06	0,6	7	0,61	0,09
3	Rays & Sharks	0,016	0,07	0,6	3	0,71	0,2
4	Large Drums & Snook	0,118	0,35	0,7	4	0,54	0,18
5	Catfish	0,153	0,45	0,8	6	0,55	0,13
6	Morays & Eels	0,010	0,17	0,5	5	0,77	0,1
7	Jacks & Pompanos	0,055	0,5	0,75	6	0,8	0,13
8	Snapper & Grunts	0,132	0,45	1,3	5	0,97	0,26
9	Cephalopods	0,000	0,3	4	15	0,81	0,27
10	Lobster	0,003	0,1	1,5	10	0,58	0,15
11	Flatfish	0,010	0,2	1,8	7	0,53	0,26
12	Small Demersals	0,882	1,2	2,5	10	0,81	0,25
13	Small Pelagics	1,355	1,5	4	20	0,9	0,2
14	Crabs	0,010	0,7	3,5	12	0,7	0,29
15	Epibenthos	0,005	5	4	20	0,67	0,2
16	Endobenthos		1	10	60	0,69	0,17
17	Shrimps	0,310	0,7	7,5	26	0,84	0,29
18	Zooplankton		5	30	150	0,71	0,2
19	Phytoplankton		15	200		0,23	
20	Mangroves		90	0,22		0,39	
21	Microphytobenthos		0,5	125		0,39	
22	Detritus		0,1			0,05	

The model has previously been balanced based on the suggestions provided by Link (2010), as described in Alms and Wolff (2019). The EwE modelling routines are particularly sensitive to the selected vulnerability settings; thus, we ran the analysis with the optimized model fit to time-series data. We used the Stepwise Fitting Procedure for automated fitting of Ecopath with Ecosim models (Scott et al. 2016) as described in Alms and Wolff (2020). Compared to the

model parameterization of Alms and Wolff (2019), successive reanalysis suggested some minor changes. In particular, due to the high predation rates of the abundant small demersal fish species, we opted to slightly reduce the vulnerability of shrimps to the predation of small demersal (from 3 to 1). This modification allowed more realistic model behavior while maintaining the model performance (i.e., the sum of squares representing the fit to time series data did not change substantially).

4.2.3 Management scenarios and simulations

To find the optimal management policy for the Gulf of Nicoya, we used a scenario-based approach. We tested the losses and benefits of the social and economic value (profit, number of jobs) and of the conservational criteria (ecosystem health, biodiversity) caused by each of the different management scenarios (Figure 4.2), based on the fishing effort control of each fishing fleet. We also tested scenarios that would result in a future increase or decrease in the total biomass of the ecosystem, as well as for the biomass of each functional group after 20 years of a given management scenario, using the EwE simulation tool. We aimed for a medium-term strategy, according to the suggestion of the FAO management cycle or EAF (Bianchi 2008); thus, we projected the management scenario 10 years into the future.

We assessed the effectiveness of the current management plan, which successively banned shrimp trawlers from the Gulf of Nicoya until 2019 by not renewing the corresponding fishing licenses. (scenario 1, Figure 4.2) Furthermore, the benefits of a 25% effort reduction of the artisanal gill net fleet were evaluated, based on suggestions from a collaborative stakeholder workshop in Puntarenas, Costa Rica (Sánchez-Jiménez et al. 2019), that included the perception of the fishermen themselves. Additionally, a combination of the existing ban on trawling and the gillnet effort reduction was considered (scenarios 2 and 3, respectively Figure 4.2).

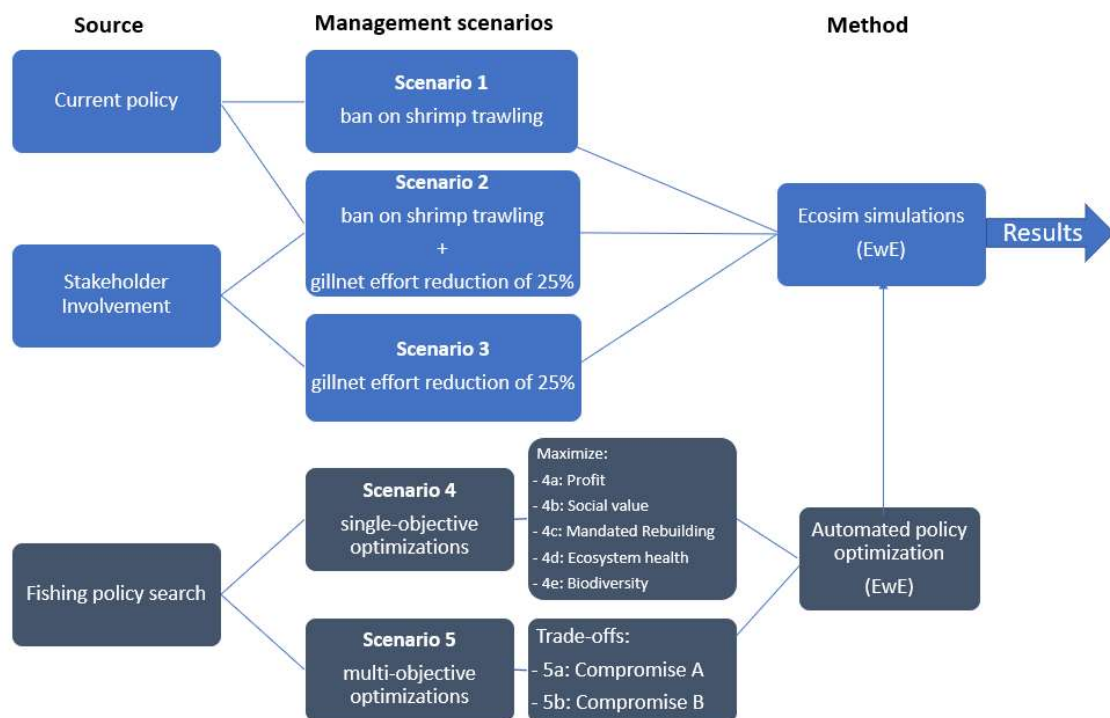


Figure 4.2 - Schematic overview of the assessed management scenarios for the Gulf of Nicoya (Costa Rica).

Furthermore, the Fishing Policy Search tool of the EwE package was used to explore alternative management scenarios. The automated search routine allows the assessment of appropriate management scenarios while considering different management objectives or a combination of them (Christensen and Walters 2004b). The output can be optimized for socio-economic goals and ecosystem conservation goals.

The fishing policy search routine uses a nonlinear optimization procedure known as the Davidson-Fletcher-Powell (DFP) method, repetitively changing the relative fishing rates in multiple model runs, to improve an objective function (Walters et al. 2002). Subsequently, the EwE simulations were used to analyze the given output (effort rates) of the search routine. The annual economic discount rate was kept at 4% for all simulations. The rate describes the costs and benefits that arise at different times and implicates at which rate society is willing to trade off present for future benefits (NOAA 1999). It can be distinguished between a project with long-term benefits versus a project with short-term benefits and long-term costs.

For the single-objective optimization runs the model optimizes the fishing effort of the fishing fleets in order to maximize only one of the management goals (Table 4.2), for which a weight of 1 ($W=1$) was assigned, whereas the other objectives were assigned with a weight of 0 ($W=0$). In each run, one of the objectives was maximized and the effects on all other objectives were studied (Table 4.2).

In the next step, we assessed multi-objective optimizations, a more realistic approach to include the interests of different stakeholders, by testing a combination of several management objectives. The goal was to identify a trade-off, which balances ecosystem conservation goals, social value (employment provided by fishing) and profits from fishing, and is not detrimental

for one single sector. For the following analysis, a combination of different weights was assigned to different management objectives (Table 4.2). Following the approach suggested by Christensen and Walters (2004b), we modified the weight of each objective until we found a compromise that considers all objectives in a suitable manner. We selected two compromise-scenarios for further analysis. Firstly, we present Compromise A, which implies less extreme effort reductions for one single fleet, affecting both the artisanal and semi-industrial fishing sector. Secondly, Compromise B was selected, a trade-off oriented towards ecosystem health and increasing biodiversity, with severe effort reductions for the semi-industrial fleets. The fishing effort of the selected scenarios was applied to the EwE model to simulate future changes in the ecosystem. The policy search considers the following management goals.

Table 4.2 – *The Fishing Policy Search routine : Relative weights for the single-objective and multi-objective optimization runs of the policy search routine (EwE) for the assessed management scenarios for the Gulf of Nicoya. Single objective: One objective is maximized at a time, assigned a value of 1 (all others assigned a value of 0). Multi-objective: All goals that are considered for maximization are assigned a value >1.*

Objective	Single-objective optimizations					Multi-objective optimizations	
	Profit	Social value	Mandated rebuilding	Eco-system	Bio-diversity	Compromise A	Compromise B
Economic value (profit)	1	0	0	0	0	1	1
Social value	0	1	0	0	0	1	1
Mandated rebuilding	0	0	1	0	0	0	0
Ecosystem structure	0	0	0	1	0	5	10
Biodiversity	0	0	0	0	1	5	10

4.2.3.1 The economic value (profit from fishing) and social value (employment)

The selling prices at the landing site were entered into the model as off-vessel prices (US\$ per kg) for each functional group and were used to calculate the revenue from fishing (INCOPECA, unpublished market price statistics). The costs of each fishing fleet were entered as a percentage of the fishery's total value in the given year (Table 4.3). We distinguished between fixed costs, which accrue without any fishing activity taking place; and variable costs (e.g., fuel and ice) which are directly related to the fishing effort of the fleet. The relative percentages of fixed costs and effort-related costs were entered for each of the artisanal and semi-industrial fleets in the Gulf of Nicoya separately. These values are based on an economic study of the Costa Rican fishing fleet by Coyle et al., (2015), who calculated the expenses and profits per kg catch in the Gulf of Nicoya. Additionally, economic information provided by the project "Biodiversidad Marino Costera en Costa Rica, Desarrollo de Capacidades y Adaptación al Cambio Climático" were used (BIOMARCC-SINAC-GIZ 2013b) as well as information on the value chain of the fisheries in the Gulf of Nicoya (Proyecto Golfos 2012b). These values were then compared to other similar studies (Lercari and Arreguín-Sánchez 2008; Gasalla et al. 2010). The profit is defined as the *value of the catch (catch price, by species) - the cost of fishing (fixed + variable costs)* or, more formally,

$$\pi_f = \sum_k^K p_{k,f} * H_{k,f} - (FC_f + E_f * VC_f)$$

Where π_f is the profit for fleet f , determined as the sum of harvest H for all species k in the portfolio of caught species K in fleet f , multiplied by the fleet- and species- specific price $p_{k,f}$ minus the fixed cost FC and variable cost VC multiplied by effort E . The social value for each fishing fleet was calculated as the fleet specific number of jobs per catch value. The social employment weights are displayed in Table 4.3.

Table 4.3 – Costs, profit and jobs per catch of all fleets : Percentage of fixed and variable cost for each fishing fleet and the fleet specific number of jobs per catch value in the Gulf of Nicoya (Costa Rica).

Fishing fleet	Fixed cost (%)	Variable cost (%)	Profit (%)	Jobs/catch
Artisanal Gillnets	20	60	20	3.5
Artisanal Longline	23	65	12	3
Artisanal Handline	20	55	25	4
Artisanal manual collection/traps	20	62	18	2
Semi-industrial shrimp trawl	35	50	15	1
Semi-industrial sardine purse-seine	25	65	10	1.5

4.2.3.2 Ecosystem conservation

The first of the three ecosystem conservation goals (4c) is the mandated rebuilding of species biomass, which allows for the assessment of the policy measures needed for preserving or rebuilding the population of certain functional groups in the ecosystem. A threshold biomass (relative to the biomass in Ecopath) was entered for each functional group, and the policy routine optimizes the fleet effort structure to most effectively reach these biomass levels for as many functional groups as possible. As threshold biomasses we chose to use the baseline biomass values of the ecosystem state in 1993 (Alms and Wolff 2019). These values originate from our baseline model of the Gulf of Nicoya representing the ecosystem state in 1993. Due to our modelling results, we believe that at that time the ecosystem was in a healthier state, with higher biomass levels across the ecosystem, and a higher biodiversity compared to 2013 after two decades of further degradation. As ecological weighting factors for the conservation of each functional group (structured relative weight) we used the reciprocal P/B ratios, which put a higher weight on long-lived, slow-growing groups. This approach allows for the maximization of the trophic transfer efficiency, increasing biodiversity and the maintenance of ecosystem stability (Okey and Wright 2004; Walters et al. 2002) (Table 4.4).

Table 4.4 -Mandated rebuilding of species biomass: (scenario 4c) threshold biomasses defined as desired level of restoration for each functional group. We chose the biomass values of the Gulf of Nicoya baseline model from 1993 (Alms and Wolff 2019), representing a healthier state of the ecosystem. The structured relative weight was calculated as the reciprocal P/B ratios of each functional group, representing the relative importance of each group for the system.

	Functional group	Mandated rel. B	Structure rel. W
1	Seabirds	1	11.2
2	Mackerel & Barracuda	2	1.6
3	Rays & Sharks	2	1.6
4	Large Drums & Snook	2	1.4
5	Catfish	1.5	5
6	Morays & Eels	4	2
7	Jacks & Pompanos	1.2	2
8	Snapper & Grunts	1	0.8
9	Cephalopods	1.6	6
10	Lobster	2.5	5
11	Flatfish	1.5	5
12	Small Demersals	1.3	3
13	Small Pelagics	1.3	3
14	Crabs	1	0.2
15	Epibenthos	2	0.2
16	Endobenthos	1	0
17	Shrimps	2.2	2
18	Zooplankton	0	0
19	Phytoplankton	0	0
20	Mangroves	1	4.6
21	Microphytobenthos	0	0
22	Detritus	0	0

The second ecosystem conservation objective is the ecosystem structure (health). To classify the ecosystem health of the model, different network parameters can be used, such as the maturity of the ecosystem, number of trophic flows and age structure, average trophic level of the catch.

In order to maximize ecosystem health, the automated policy search uses an ecosystem maturity index (Odum 1969), calculating the longevity-weighted summed biomass over all functional groups (Christensen and Walters 2004b). The group-specific biomass/production ratio indicates the longevity of functional groups and can thus be used as a proxy for ecosystem health (W. W. L. Cheung and Sumaila 2008).

The third objective is the ecosystem biodiversity. As a measure of biodiversity, a modified version of Kempton's Q75 index (Kempton 2002) is used to calculate an index of biomass diversity (Christensen and Walters 2004b).

The goal is to achieve increasing diversity, especially in higher trophic levels (Cury et al. 2011; Geers et al. 2016). Often, increasing biodiversity is accompanied by increasing ecosystem maturity and a healthy ecosystem (Cury et al. 2011).

4.3 Results

4.3.1 Ecosim simulations (Scenarios 1 – 3)

4.3.1.1 Scenario 1: Existing ban on trawling from the entire Gulf of Nicoya.

Model simulations of the future effects of the existing ban on shrimp trawling indicate that it will most likely lead to the desired long-term recovery of shrimp biomass, with a projected increase of 6,5% (until 2033). The policy measure also entails increasing biomass of demersal fish species and related predatory species (Figure 4.3), as well as an overall increase in the total ecosystem biomass (Figure 4.4). However, it might not be sufficient to substantially rebuild the biomass of all key species in the ecosystem (Figure 4.3). The elimination of the shrimp trawling industry from the Gulf of Nicoya leads to a substantial loss of profit for the fisheries. The value of the catch, landed by the semi-industrial fleet decreases by around 85% (Figure 4.5). However, it can be observed that artisanal catches could increase slightly (gillnet, handline, longline fleet) with the ban on trawling (Figure 4.5).

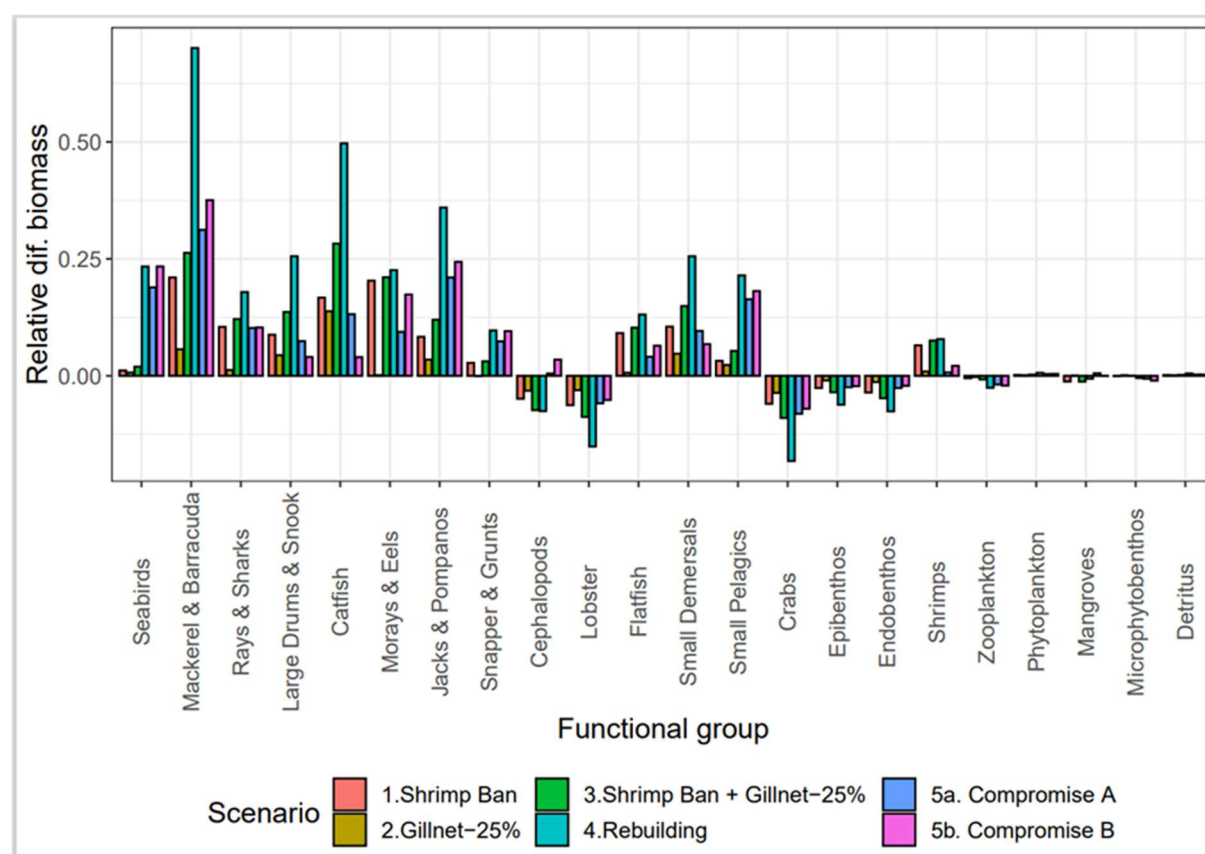


Figure 4.3 - Relative difference in the biomass of each functional group in the Gulf of Nicoya (Costa Rica) after 20 years of a selected management scenario compared to the baseline year.

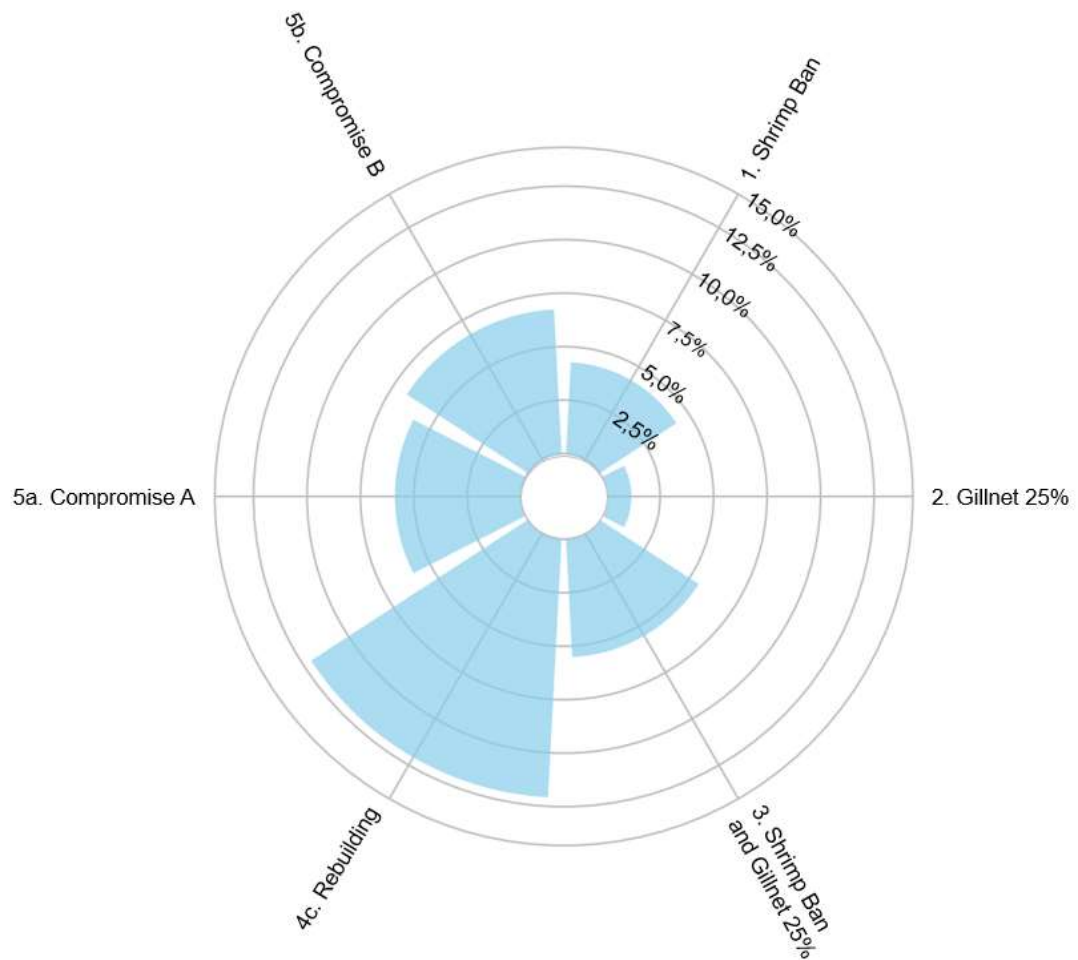


Figure 4.4 - The expected relative increase [%] in the total biomass of all functional groups (sum of biomass) in the Gulf of Nicoya (Costa Rica), presented for each of the selected management scenarios after 20 years (2033) compared to the baseline year (2013).

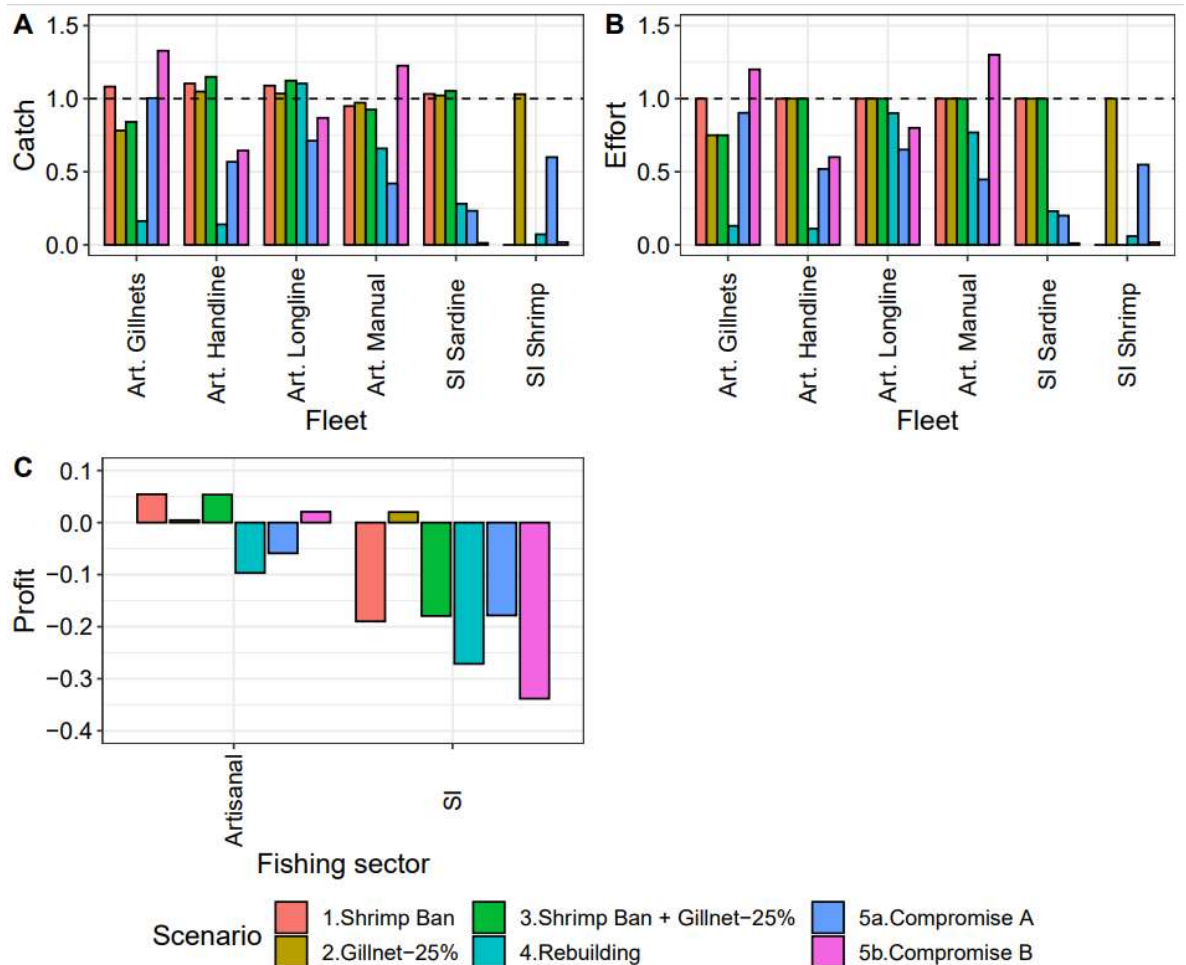


Figure 4.5 – Relative differences in the catch, fishing effort and profit: Relative difference in the catch (A) and fishing effort (B) of each fishing fleet compared to the baseline value (dashed line) and relative difference in the profit of the artisanal and semi-industrial fishing sector (C) in the Gulf of Nicoya (Costa Rica) after 20 years of a selected management scenario compared to the baseline model without any policy restrictions.

4.3.1.2 Scenario 2: Effort reduction of the gillnet fleet by 25%

The stakeholder-proposed effort reduction for the gillnet fleet (-25%) implies benefits for several species, such as flatfish, small demersal fish as well as mackerel and barracuda (Figure 4.3). However, the benefits for most species are only minor and the overall ecosystem biomass cannot be increased as effectively, compared to the other management scenarios. At the same time, the effort reduction causes a significant loss of profit (more than 20%) and jobs to the artisanal fishing sector (Figure 4.5a).

4.3.1.3 Scenario 3: Combination of scenarios 1 and 2

A combined approach, that includes adding the gillnet effort reduction to the existing ban of trawling is a powerful measure, causing an apparent increase in the biomass of several target species. The biomass of shrimp, small demersals, catfish and top predators can be sustainably increased (up to 25%) over the long term (Figure 4.3), and the ecosystem biomass rises by more than 10% (Figure 4.4). In this case, the loss of profits and employment affects both the semi-industrial and the artisanal fleet.

4.3.2 Fishing policy search (Scenarios 4 – 5)

4.3.2.1 Single-objective optimizations

Results of the single-objective optimization search are presented in Figure 4.5a, which displays the relative changes of each objective for each run optimizing one of the management goals. The benefits for these goals are not balanced and the maximization of economic profit is not compatible with any conservation goals, whereas the social value can be combined with biodiversity. While shrimp trawling is linked to high profits, it is not favorable in the scenarios promoting ecosystem health and biodiversity. The relative changes in fleet effort for each management scenario suggested by the fishing policy search routine are presented in Figure 4.5b. Severe effort changes are suggested for single-objective scenarios. To maximize social value, for instance, an 8 times increase of effort of the manual collection and the sardine purse-seine fleet and a 6 times increase of shrimp trawling effort is suggested. The handline fleet and manual collection are in better agreement with biodiversity. All scenarios, except for insufficient gillnet effort reduction, cause a severe loss of profit to the semi-industrial fishing sector. At the same time, it provides an opportunity for the artisanal fleet to increase catch and profits. To maximize the ecosystem health, strict effort limitations for all fleets are necessary, the same holds for the mandated rebuilding of all functional groups to a biomass level from 25 years ago, causing high social and economic costs. The rebuilding process would entail severe profit declines for all fleets, except the longline fleet.

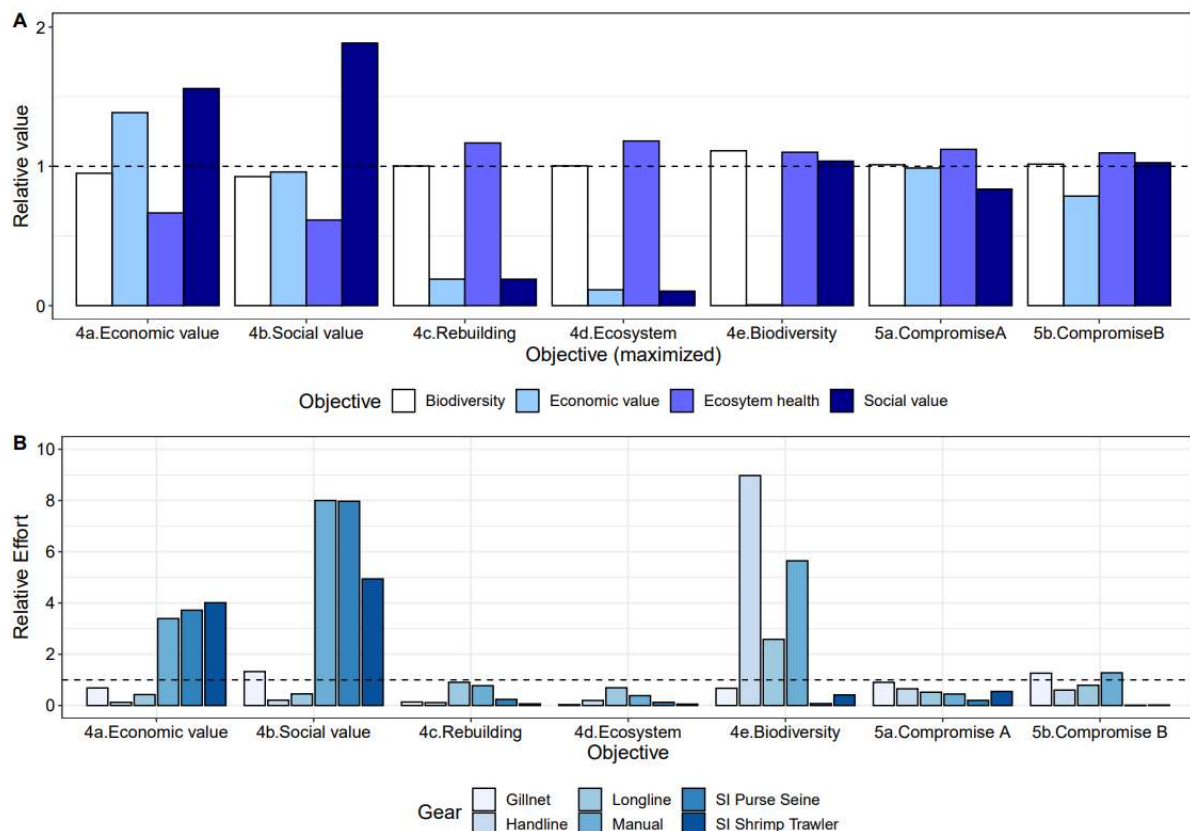


Figure 4.6 - Results of the EwE policy search routine : A: Changes in the relative values of each of the four management objectives (fill), as a result of maximizing one of the management objectives 4 (single-objective optimization) or several objectives 5 (multi-objective compromises). B: Suggested relative fishing effort for each fleet (fill) in the Gulf of Nicoya (Costa Rica) by the policy search, in order to maximize different management goals (scenarios).

4.3.2.2 Multi-objective optimizations

The two compromises are considered trade-offs between all optimization goals, including both socio-economic and ecosystem conservation aspects. Both compromises manage to minimize losses to the social value and/or profit, while improving biodiversity and ecosystem health in the Gulf of Nicoya. Compromise A and B are compared in the summary table of all scenarios (Table 4.6)

Table 4.5 - Summary of the explored management scenarios and their expected outcomes. The expected increase/decrease in fishing effort and the related changes in profit for scenarios 1-5. And the changes in social value, ecosystem health and biodiversity calculated by the Policy search routine, only for scenarios 4 and 5. The colors indicate the expected increase/decrease of relative values. Red: decrease >10%, Yellow: increase/decrease <10%, Green: increase >10%.

	Scenarios	Effort changes						Expected results			
		Gillnet	Handline	Longline	Manual	Shrimp trawl	Purse-Seine	Profit	Social value	Ecosystem	Biodiversity
1.	Ban of shrimp trawling	Red	Yellow	Yellow	Yellow	Yellow	Yellow	Red			
2.	25% effort reduction gill nets	Red	Yellow	Yellow	Yellow	Yellow	Yellow	Yellow			
3.	Ban of shrimp trawling and 25% effort reduction gill nets	Red	Yellow	Yellow	Yellow	Yellow	Red	Red			
4.	Single-objective optimization										
	Net economic value (profit)	Red	Red	Red	Green	Green	Green	Green	Green	Red	Yellow
	Social value (employment)	Green	Red	Red	Green	Green	Green	Yellow	Green	Red	Yellow
	Mandated rebuilding	Red	Red	Red	Red	Red	Red	Red	Red	Green	Yellow
	Ecosystem structure	Red	Red	Red	Red	Red	Red	Red	Yellow	Green	Yellow
	Biodiversity	Red	Green	Green	Green	Red	Red	Red	Yellow	Yellow	Green
5.	Multi-objective optimizations										
	Compromise A	Yellow	Red	Red	Red	Red	Red	Yellow	Red	Green	Yellow
	Compromise B	Red	Red	Red	Green	Red	Red	Red	Yellow	Green	Green

4.3.2.3 Compromise A

Compromise A focuses on increasing ecosystem health (+10%), as well as increasing biodiversity, with marginal over-all losses of profits (loss of profit to the semi-industrial fleet), but causing substantial social cost (-15%) (Figure 4.6a). The measures of effort control are balanced among the artisanal and semi-industrial fishing sector (Figure 4.6b). In this compromise, we allow the purse-seiners to continue with only 20% of their original effort. Shrimp trawlers would be allowed to operate with up to 50% of their original effort. It is important to note that this is the outcome of a theoretical modelling routine. The shrimp trawling has already been banned and our modelling analysis shows that it is a successful and efficient policy measure. The compromise assumes effort reduction for each artisanal fleet (decrease between 10-55% each).

4.3.2.4 *Compromise B*

A second compromise (B) increases future ecosystem health and biodiversity without severe social costs and loss of livelihoods for the artisanal fishermen. It entails profit losses of 20% mostly at the expenses of the semi-industrial fleet (Figure 4.6a). Compromise B maintains the ban on shrimp trawling, combined with a ban on purse-seine fishing in the entire gulf, only maintaining artisanal fishing activities. An additional reduction of longline fishing effort is suggested to further promote a recovery of predatory species (Figure 4.6b). Compromise B can obtain the second highest increase in overall ecosystem biomass by 6% (Figure 4.4) and the recovery of a majority of functional groups, including shrimp, small pelagics, large drums and snook as well as other predatory species. Benthic species are not showing a positive effect in any tested scenario.

4.4 Discussion

4.4.1 Complexity of policy management approach

In many marine systems there is a variety of stakeholders who are linked to fisheries along the economic supply chain (Jacinto and Pomeroy 2011), which make fisheries management scientifically and socially complex (Hare 2020). To face the challenge of overfishing in many marine areas, according to (Beddington et al. 2005) integrated management framework are required that include co-management and collective decision-making are needed. To provide these information in a holistic manner, we followed the Ecosystem Approach to Fisheries (EAF) by defining management strategies that consider the ecosystem as a whole, with its main species and the trophic interactions between them, as well as the effects of fishing (FAO 2012). We aimed to find a balance between ecosystem and biodiversity conservation by rebuilding biomass of key species on the one hand and socio-economic aspects and stakeholder interests on the other hand.

In the Gulf of Nicoya, the Costa Rican fisheries hotspot, a complex range of stakeholders are involved. On the one hand, there is the semi-industrial fleet that operates in the gulf, fishing with purse-seine nets which target small pelagic fish as well as the shrimp trawlers that targeted different inshore and offshore shrimp species (until 2019). On the other hand, artisanal fishing fleets operate in the gulf, using a diverse set of gears (handline, longline, gillnets, traps, manual bivalve extraction) and catching a variety of species.

Women of the fishing communities contribute to small-scale artisanal fishing, mainly the collection of bivalves, and most importantly to a variety of pre- and post-harvest activities, that continue to be poorly recognized (Cruz et al. 2022). Their contribution needs to be especially considered in the development of policy plans.

4.4.2 Efficiency of the ban on trawling (current policy)

The first step of our assessment of the management scenarios tested the current policy, the ban on shrimp trawling. We can state from our modelling results, that a severe reduction of shrimp trawling is essential for the recovery of the Gulf of Nicoya ecosystem. All tested management scenarios that include the goals of increasing ecosystem health and biodiversity, suggest a

reduction of trawling by 50-100%. Modelling results imply that the existing shrimp trawling ban will have several positive effects on the recent and future ecosystem health.

As a result of the ban on trawling, firstly, we would expect an increase of shrimp biomass. The speed of recovery strongly depends on the current status of the stock. From examples of the past, we know that it can be a difficult and protracted process. In the case of the Indonesian shrimp fishery, for example, the over-fished shrimp stocks showed a recovery within a 7-year period after the ban (Chong et al. 1987). In Hong Kong waters first signs of recovery were visible 3,5 years after the trawl ban, however the biomass remained unchanged for the next 5 years after the ban. Furthermore, the ban on trawling will have a positive effect on the bycatch species, ocean bottom flora and fauna, and demersal bottom-associated species. The recovery of bycatch populations, such as small demersal fish species, can be faster than the increase of shrimp biomass, as has been shown in an embayment in Kenya, (Munga et al. 2012) and in (estuarine) coastal waters of Hong Kong, China (Mak et al. 2021). We expect a recovery of vulnerable bycatch species, such as eel (*Cynoponticus coniceps*, *Brotula clarkae*) and flatfish, both frequently caught as bycatch by the shrimp-trawling fishery (Herrera et al. 2016). Increasing abundance of lower trophic level species will most likely be followed by the increase in biomass of longer-lived predatory species, like drums and snook, mackerel, barracuda and sharks. Especially demersal sharks and rays are a vulnerable bycatch species in the trawl fisheries of the Gulf of Nicoya (Clarke et al. 2016).

Despite the benefits of the ban on trawling, we see from our modelling results that this recent policy is presumably not sufficient to achieve a full recovery of biomass levels over all functional groups. The ecosystem has strongly degraded over the last 30 years, and more severe measures are necessary to turn the tide. Given the fact that the catch of the artisanal fleet would likely increase in the future, due to the loss of competition from the shrimp trawlers, we believe that some additional restrictions are necessary. One also has to keep in mind that a full recovery of all species in the system is not very likely to be achieved simultaneously due to the variety of trophic-interactions and other complex external influences (Froese et al. 2016; Steven Mackinson et al. 2009).

4.4.3 Stakeholder engagement in the decision-making process

Furthermore, there is often a gap between stakeholder interests and ecological knowledge (Bethel et al. 2014; Evans 2010), that can be bridged by involving stakeholders in the decision-making process. In our study we were able to show how the Gulf of Nicoya EwE model can be used to initiate a discussion with engaged stakeholders about appropriate management of target species in the ecosystem. As shown in several workshops (Sánchez-Jiménez et al. 2019), the fishermen have fundamental knowledge about past and present resource availability in their fishing areas. With this knowledge, EwE simulations could be used to present and visualize possible scenarios of the future development of resource availability and the consequences and benefits of effort control. One outcome of these workshops was that according to the fishermen, gillnet effort should be reduced (by 25%) to rebuild the stock biomass of key resources. However, our study revealed that a gillnet effort reduction as the only measure (without a concurrent trawling ban) is not powerful enough in terms of ecosystem conservation but causes a significant loss of profit to the artisanal fishermen.

We rather recommend to strengthen the enforcement and control mechanisms of illegal mesh sizes (smaller than 6.35 cm [2.5 in]), to let juvenile target species reach the age of maturity and to allow population sizes to increase. This is also suggested by Fonseca et al. (2005), who state that mesh sizes larger than 60-80 mm have a better compatibility with sustainable management strategies. We suggest that stakeholder engagement can be particularly intensified in the entire EwE modelling processes starting with the data collection, continuing with the model development and model validation and lastly end with management simulations. As it has been shown in several cases, stakeholder engagement can be used for management strategy evaluation (Fulton, Jones, et al. 2011) and is a key aspect in providing good governance, to dissolve top-down hierarchies and to improve the compliance of all actors (Pita et al. 2010; Soma 2003).

4.4.4 Policy Search – is there another more effective fishing policy?

We analyzed the results of the fishing policy search (EwE) to suggest an optimized policy plan. The results highlight the discrepancy between socio-economic and conservational goals and the challenge of finding the right balance for policy-making. Increasing profits come with increasing semi-industrial fishing efforts at the expense of ecosystem health. Effort reduction of the artisanal fleet in favor of ecosystem health entails the loss of jobs for the fishermen in the communities around the Gulf of Nicoya.

We ran an additional scenario, the mandated rebuilding of species, to present the measures that would be necessary to restore biomass levels of functional groups to the known state of 1993 (baseline model by Alms and Wolff (2019); Wolff et al. (1998)), disregarding the feasibility and social-economic aspects. In this scenario the extent of ecosystem degradation is highlighted. Necessary measures would involve a ban on trawling, and a reduction of gillnet, handline and purse-seine effort to a minimum. The scenario would only allow the longline fleet to operate in the Gulf of Nicoya (-10%), together with manual collection (-20%), leading to an 80% loss of profit and social value.

Since this scenario is not viable, we looked for compromises from both multi-objective optimizations runs, that are trade-offs between the four management objectives. One compromise (A) is oriented slightly more towards maintaining social-economic benefits (limiting effort reduction for each fleet), while the other compromise (B) focuses on ecosystem health and biodiversity conservation. After the assessment of our modelling results, we conclude that compromise B has the best long-term outcomes. This scenario suggests keeping the existing ban on shrimp trawling, supplemented by the ban on purse-seine fishing. This scenario would allow only artisanal fishing activities in the Gulf of Nicoya. All artisanal fleets can maintain their recent effort, except for a 20% reduction of longline effort, which is an additional measure to allow for long-term recovery of the top predators, such as rays and sharks. One major advantage of this policy option is that the obstacle of implementing the ban on shrimp trawling has already been overcome. Compromise B will surely lead to a substantial loss of profit to the fishing industry compared to prior to the trawling ban. However, one needs to keep in mind that the shrimp industry had collapsed already due to the overfishing of inshore shrimp species and deep-water shrimp, followed by declining resources (-50% biomass) (Alms and Wolff 2019). Decreasing biomass of target resources resulted in lower CPUE, which reduced the profit and income of the fishing vessels due to a high level of effort and costs

relative to the value of the harvested catch (FAO 2020; Froese et al. 2018). This tremendous loss of profit to the fisheries (-50%) occurred between 1993 and 2013 (Alms and Wolff 2019). While the sardine purse-seine fleet would surely suffer from the suggested ban, it might still be early enough to prevent a similar collapse of the small pelagic fish population in and around the Gulf of Nicoya. In the long run, after a recovery of the small pelagic target species, the purse-seiners could gradually be allowed back to operate in the outer parts of the Gulf of Nicoya. Thus, this short-term burden could lead to a long-term increase in profits (Brown et al. 2015; Froese et al. 2018). It is known that the semi-industrial fishing fleets compete with artisanal fishing practice (Stiles et al. 2010) and the industrial purse-seine fisheries may impact upon artisanal and subsistence fishers (Leroy et al. 2016). Both semi-industrial fleets remove great amounts of biomass from the mid-trophic levels. In the case of the Gulf of Nicoya, these groups are small pelagic fish and several shrimp species, which have been identified as keystone species and important links between trophic levels (Alms and Wolff 2019). The collapse of more key target fish populations needs to be prevented, to sustain fishing resources for the artisanal fleet, as fishing provides basic income and food security, with rare options for alternative livelihoods available (Proyecto Golfos 2013). The compromise B favors the artisanal sector and most likely even allows for the artisanal fleets to increase catches and/or CPUE. This can be an advantage for its implementation.

The alternative compromise (A) reduces the impact on the industrial fleet, increasingly impacting the artisanal fleet. In this compromise, we allow the purse-seiners to continue with only 20% of its original effort. The shrimp trawlers would be allowed to operate at 50% of its original effort. However, as the shrimp trawling has already been banned and the modelling analysis shows the positive effect it has, we strongly recommend maintaining the ban. The overall biomass cannot be increased as effectively as in compromise B. The compromise puts effort reductions on each artisanal fleet (decrease between 10-55% each). Critical aspects of compromise A are the social consequences and the loss of jobs, which also affect artisanal fishermen, who would have to either find alternative occupation or rely on state subsidies, which are often given due to a lack of alternative employment. In this case fishermen are encouraged to stay in the fishing industry, despite the lack of resources (Sumaila et al. 2010).

Ideally, policies that facilitate alternative income opportunities can be provided, similar to what has been done in other coastal regions of the world. Examples of alternative occupation options established for fishers being left out after fleets reduction include aquaculture (Ghana, Anning et al. 2012), the farming of shrimps (Brazil, Cavalli et al. 2008), or the engagement in touristic activities, such as recreational fishing trips/sea trips offered by the fishermen (Porter and Orams 2014). In Costa Rica artisanal fishermen collecting bivalves have been alternatively employed for the restoration of mangroves, not only providing a livelihood but ecosystem services to fish, shrimp, and society. Same actions have been taken in other marine areas like India, Malaysia, Nigeria and Panama (BDCP 2007; Porter and Orams 2014; Suman 2019; Woods et al. 2022).

According to Johnson et al. (2021), the collapse of ecosystem services, such as marine fisheries, could result in a dramatic decline of the global GDP in the next decade. At the same time, the preservation of the ecosystem health is a major factor for economic growth. Thus, short-term economic losses due to fishing effort reductions to allow for ecosystem recovery might be preferable. Eventually, a successful management plan will pay off in terms of higher catch

rates, increase in (financially) valuable species and higher catches as well as less effort for artisanal fishermen.

As it has been stated by Coglan and Pascoe (2014), fishermen have to handle a high amount of uncertainty concerning their future catch rates, prices and costs. With predictive models we have the right tools at hand to develop reliable fisheries management plans that minimize these livelihood-threatening uncertainties. Ideally, these processes are fulfilled in cooperation with relevant stakeholders. For a suitable management strategy that regulates the fishing effort for each fleet, one has to consider that the ecosystem will be additionally affected by climate variations and other external factors, such as pollution or change in the land use (FAO 2020). As it has been shown by Alms and Wolff (2020), particularly during El Niño periods with extreme Sea Surface Temperatures and increased precipitation, decreasing phytoplankton productivity and zooplankton biomass are expected to occur with bottom-up effects on the entire food chain. Thus, a stringent effort control, that incorporates a buffer for external effects, will help the ecosystem to recover and to increase its resilience towards climate change in the future.

4.5 Conclusion

The here presented modelling routine is a useful tool to explore management options and inform stakeholders about the benefits of specific policy measures, and the results of our assessment can serve as the basis for management advice. However, modelling results always include some uncertainties and as it is highlighted by Christensen and Walters (2004b), models cannot fully represent natural dynamics. According to Plagányi and Butterworth (2004) the limitation of the EwE approach lies in the quality and quantity of available data. The model is balanced and fit to time-series to reduce uncertainty, but the Ecosim modelling tool as well as the policy search are sensitive to the parameter settings used. Thus, it is recommended to confirm the results through further analysis, before the model results are directly applied to policymaking. The EwE plug-in Ecosampler could be used to measure the impact of the input parameter sensitivity on the modelling results e.g. for the use of time-series simulation using Ecosim (Steenbeek et al. 2018). Other options to validate the results, could be stock assessment or bioeconomic and socio-economic analysis.

Following the EAF guidelines, after the implementation of the selected policy measures a monitoring system is needed, followed by a phase of assessing the data and possibly revising the policy plan (FAO 2012). Furthermore, the ban on trawling is a rare and valuable opportunity for a "before - after" comparison to assess the recovery in ecosystem structure and function after the removal of a major distraction (Tao et al. 2021).

The goals of the Ocean Decade are numerous and several next steps of producing scientific knowledge for our study area in Costa Rica and Latin America are relevant. One of the goals is a "mapped and protected marine ecosystem". With that, the location of possible MPAs and/or predestined areas of seasonal closure (veda), can be assessed through a spatial model using the Ecospace tool, the spatial-temporal component of Ecopath with Ecosim. It could be assessed how the "veda" can be used best for conservation and whether it is necessary to exclude all

fishing fleets from the region, depending also on other policy measures implemented in the Gulf of Nicoya.

Acknowledgments

The facilities in Costa Rica and some of the data about the Gulf of Nicoya were provided by the Centro de Investigación en Ciencias del Mar y Limnología (CIMAR) and the Universidad de Costa Rica. The research project was strongly supported by our main project partner Alvaro Morales Ramírez in Costa Rica, who provided the infrastructure and guidance in Costa Rica and helped to develop the project collaboration between the Leibniz Centre for Tropical Marine Research and the CIMAR. INCOPESCA supported the study by providing fisheries statistics and guidance concerning the marine resources in the Gulf of Nicoya.

Chapter 5

**A SPATIAL ECOSYSTEM MODEL TO ASSESS THE EFFICIENCY
OF MARTINE PROTECTION PLANS**

Chapter 5

A Spatial Ecosystem Model to assess the Efficiency of Marine Protection Plans

Viola Alms, Giovanni Romagnoni, and Matthias Wolff

This is a draft manuscript in preparation for submission.

Contribution statement

Project idea: **Viola Alms**

Design of the work: **Viola Alms** supported by Giovanni Romagnoni

Acquisition or interpretation of data: **Viola Alms**

Drafted manuscript revised by Giovanni Romagnoni and Matthias Wolff.

Abstract

The Gulf of Nicoya, located at the Pacific coast of Costa Rica, is a highly productive estuary and the country's primary fishing region. Previous assessment using holistic trophic models to examine the ecosystem's status revealed an overfished and degraded system, with decreasing biomass of valuable target species. The ecosystem degradation was mainly driven by intensive fishing, especially by the semi-industrial shrimp trawling fleet that operated in the gulf until 2019. Besides a trawling ban, an annual temporary closed season, which existed for many years was implemented as another countermeasure against depletion. However, the regulations governing this fishing closure have recently been relaxed.

This study tested both the effects of removing trawling on the ecosystem and the desired benefits, the temporary Marine Protected Area (MPA) has on the target species. The goal is to search for conservation measures that are needed to ensure the efficiency of the fisheries closure.

To achieve this, the study utilized Ecospace, a spatial modelling tool that is part of the Ecopath with Ecosim (EwE) software suite. With Ecospace, species biomass and fishing effort can be analyzed over space and time, with results visualized through maps.

Findings from our previous study identified trawling ban from the gulf, as an important measure, allowing various species to recover. In our model, we observed significant recovery of bycatch species and upper-trophic-level species. However, rebuilding the shrimp species biomass appears to require either more time or more protective measures.

The 3 months closed season is an important conservation measure for many species in the Gulf of Nicoya; however, our model simulations suggest room for improvement. Extending the fishing closure to encompass the entire gulf and apply it to all fishing fleets would significantly increase its efficiency. Moreover, including the artisanal manual collection into the closure should be considered, as in our simulations all its target species show a declining biomass over time, until 2039.

This study shows how spatial models can be used to simulate and visualise processes of spatio-temporal change in the ecosystem and how existing fishing closures can be checked for its efficiency.

Keywords: MPA, fishing closure, spatial modelling, Ecopath with Ecosim, shrimp trawling

5.1 Introduction

The Gulf of Nicoya is a productive estuary and the hotspot for the Costa Rican fisheries, which has been subject to overexploitation for many decades, especially caused by the semi-industrial shrimp trawlers. Since the early 1990s, the ecosystem continuously degraded, with many declines in target species catch, causing a substantial economic loss (Alms and Wolff 2019). Like other coastal ecosystems (Blaber 2002; Del Solar et al. 2024; Trégarot et al. 2024), the Gulf of Nicoya is not only impacted by its fisheries, but also by anthropogenic activities, pollution and climate change (Vargas 2016). To address these concerns, Costa Rican authorities have implemented conservation measures to regulate fishing, such as spatial and temporary restrictions in specific areas of the Gulf of Nicoya and a ban on bottom trawling since 2019.

Holistic ecosystem modelling emerged as a key tool to evaluate the effectiveness of these conservational measures and to provide a valuable basis for management advice (Fulton et al. 2011). This is especially crucial in data-poor regions, where the lack of integrated ecosystem-based information limits the development of policies and marine management plans (Feary et al. 2014). Globally, ecological information is needed to achieve the goals proclaimed by the United Nations (UN). ‘The Decade of Ocean Science for Sustainable Development’ (2021–2030) aims to achieve a healthy, resilient and sustainably harvested ocean for many coastal countries worldwide (Heymans et al. 2020; Ryabinin et al. 2019). While the UN aimed to preserve 10% of the ocean by 2020, the current High Seas Treaty aims to convert 30% of the seas into protected areas by 2030 (United Nations). To achieve these goals, suitable areas for protection need to be identified in many regions of the world. However, the efficiency of MPAs significantly relies on the right choice of location, the spatial and temporal extent as well as successful enforcement (Dimarchopoulou et al. 2024).

Ecosystem food webs are dynamic, due to changes in natural predator-prey interactions, resource limitations and anthropogenic influences, or due to environmental influences (Frederiksen et al. 2006; Maureaud et al. 2017; Pinnegar et al. 2000). Also, ecosystems are spatially heterogeneous, with species abundance and productivity varying across regions (Hervann et al. 2020). Here, Ecospace emerges as a useful tool. It is a spatial modelling tool embedded in the Ecopath with Ecosim software suite. It can be used to explore spatial management options and to address various research question, including the regional consequences of climate change (Christensen and Walters 2004a). It can valuably incorporate temporal and spatial factors in the analysis as well as defining fisheries restricted areas. Though the tool is less widely used than other EwE software components, its importance is increasingly recognized, especially in terms of the recent UN endeavors (Mutsert et al. 2024).

Prior studies using Ecopath with Ecosim (Alms et al. 2022), suggested a ban on shrimp trawling as a necessary measure to allow ecosystem recovery. Simulations showed that the ban could lead to the desired long-term recovery of shrimp biomass, along with increases in biomasses of bycatch species and predatory species, as well as an overall increase in the total ecosystem biomass. After decades of intensive exploitation of shrimp species and associated bycatch, the shrimp trawlers were effectively banned from the Gulf of Nicoya since 2019. In this study, we hereby evaluate the local consequences of this ban on species distribution and fishing efforts in the gulf using the Ecospace model. While shrimp stock recovery depends on the stock status, it

can be a challenging and a lengthy process (e.g. Chong et al. 1987; Munga et al. 2012). We also simulate the changes in the distribution of the bycatch species, which tend to recover faster than the shrimps itself (Munga et al. 2012), and look at the bottom-up effects on the predatory species. The restoration of shrimp biomass and small demersal fish biomass (as bycatch species) will likely cascade through the system, and positively affect the upper trophic levels. Thus, we model the recovery phase after a prolonged intensive trawling using the Ecospace software.

Our first goal is to simulate the efficiency of the existing seasonal fishing closure in certain areas of the Gulf of Nicoya. The closure, known as ‘veda’, is an annual three-monthly closure on all fishing fleets, in the inner and intermediate Gulf of Nicoya, except for manual collection, which has been active for decades. The first closure was established in 1985 for trammel nets only, then further extended, including trawling, purse seine and gillnets. The regulations have undergone further adjustment over the years, since the three-month closure (from May to July) was established in 1990. Since then, the fishing closure occurs every year to reduce fishing effort and allow certain species to reproduce, grow and recover from the intense fishing pressure. The closure aligns with spawning seasons of commercially important species, such as snapper (*Lutjanus* spp.), Sciaenidae (*Cynoscion* spp.), jacks and pompanos, catfish (*Bagre* spp.), small demersals (Mullidae), small pelagics (*Anchoa* spp., *Cetengraulis* spp., among others), snook (*Centropomus*), mackerel and barracuda (*Scomberomorus sierra*, *Sphyraena ensis*), camarón blanco (*Penaeus*) and crabs (*Callinectes*). We aimed to simulate the consequences of the fishing closure for the spatial distribution of species and fishing fleet effort. Most small artisanal fishermen are unable to fish during the closed season, while the advanced fleet is believed to partly shift to the outer gulf. An additional aspect of our analysis is the assessment of the importance of certain regulations of the fishing closure. In recent years, some changes were made to the regulations of the fishing closure. Since the Covid-19 pandemic, the fisheries institute *Instituto Costarricense de Pesca y Acuicultura* (INCOPESCA) has eased the regulations, allowing semi-industrial sardine vessels to fish for two of the three months in the closed area. It remains to be verified whether these changes affect the potential success of the fishing closure in any way. Our research addresses the following key questions about the closure’s regulation: Is it necessary to close the whole gulf for all fleets? Is it advisable that the manual collection of molluscs and crustaceans is exempt from the ban? Is there a need for temporal or spatial extension of the fishing closure? These research questions are tackled by the modelling analysis.

5.2 Material and Methods

5.2.1 Study area

The Gulf of Nicoya (10°N, 85°W) is a tidally influenced estuary on the northwest Pacific coast of Costa Rica (Figure 5.1). The model area spans 1530 km², from the northern inland (mouth of the Tempisque River) to the Pacific Ocean entrance in the south. According to its shape and bathymetry, the Gulf of Nicoya is divided into three regions (Figure 5.1): inner gulf (zone 201), intermediate gulf (zone 202) and outer gulf zone 203 (Vargas 2016). The division is based on the species composition and fishing gears used, as defined by the Costa Rican institute of fisheries (INCOPESCA) (Sánchez-Jiménez et al. 2019).

The shallow interior zone (depth < 10 m) of the Gulf is characterized by shallow waters and tidally influenced mud- and sandflats. It is fringed by mangroves and dominated by the seasonal freshwater inflow from the Tempisque River (Proyecto Golfos 2012). The inner gulf is considered a high-priority area due to its importance as a reproduction and nursery ground for commercially important fish and shrimp species (Alpízar 2011; Proyecto Golfos 2012; Vargas 2016). The intermediate zone of the gulf is mostly surrounded by sandy beaches and mangroves. Here, the physical parameters of the water masses vary due to intense mixing between the oceanic water and freshwater from both adjacent zones (Proyecto Golfos 2012b). This region comprises reproduction areas for small pelagic fish species (Clupeiformes: Clupeidae and Engraulidae), shrimps *Litopenaeus* spp. (Decapoda: Penaeidae), snappers *Lutjanus* spp. (Perciformes: Lutjanidae) and drums *Cynoscion* spp. (Perciformes: Sciaenidae). It also serves as feeding grounds for several commercially targeted species, such as the Mexican Barracuda *Sphyrna ensis* (Perciformes: Sphyrna). The outer oceanic zone extends from a minimum depth of 20 m to 200 m at the shelf edge, surrounded by sandy beaches and rocky shores (Wolff et al. 1998). This zone is mainly influenced by oceanic waters and the Tárcoles River input. The eastern region of the outer gulf is a hatching and reproduction site for penaeid shrimps, sharks and several other fish species (Proyecto Golfos 2012; Vargas 2016). Like in other tropical coastal estuarine areas, mangroves in the Gulf of Nicoya contribute to primary production, by providing organic matter and play an important role in enhancing fish and invertebrate productivity; thus, significantly supporting the local fisheries (Clarke et al. 2016; Hutchison et al. 2014; Wehrtmann and Cortés 2008; Wolff 2006). Due to its high productivity, the estuary is the country's principal fishing ground for both the artisanal fishing fleet (Vargas, 1995) and the semi-industrial fishing fleets, thereby, providing approximately 55% of the total catch in the Gulf of Nicoya (in 2013). The small-scale artisanal fishing sector is socially and economically important for approximately 2000 fishermen and over 20 coastal communities (Lozano and Heinen 2016a; Trujillo et al. 2012). Four artisanal fishing fleets operate in the gulf using different boat sizes and gear types to target various demersal and coastal species (BIOMARCC-SINAC-GIZ 2013a). Gillnets are predominantly used, followed by the longline, handline and manual mollusk extraction, which is mostly conducted by women from the coastal communities, synergizing with the manual trap fisheries targeting different crustaceans.

The main port of the Gulf of Nicoya, Puntarenas, is also the shrimp trawlers landing site for the semi-industrial fishing fleets, the purse-seine fleet and the shrimp trawl fleet. The purse-seine vessels target small pelagic fish species (*Anchoa* spp., *Cetengraulis mysticetus* and *Opisthonema* spp.) in the intermediate and outer zones of the Gulf of Nicoya. The bottom trawl fleet mostly targeted several shrimp species in the outer gulf; but unfortunately, the capture of bycatch species drastically exceeded the shrimp catches (Clarke et al. 2016).

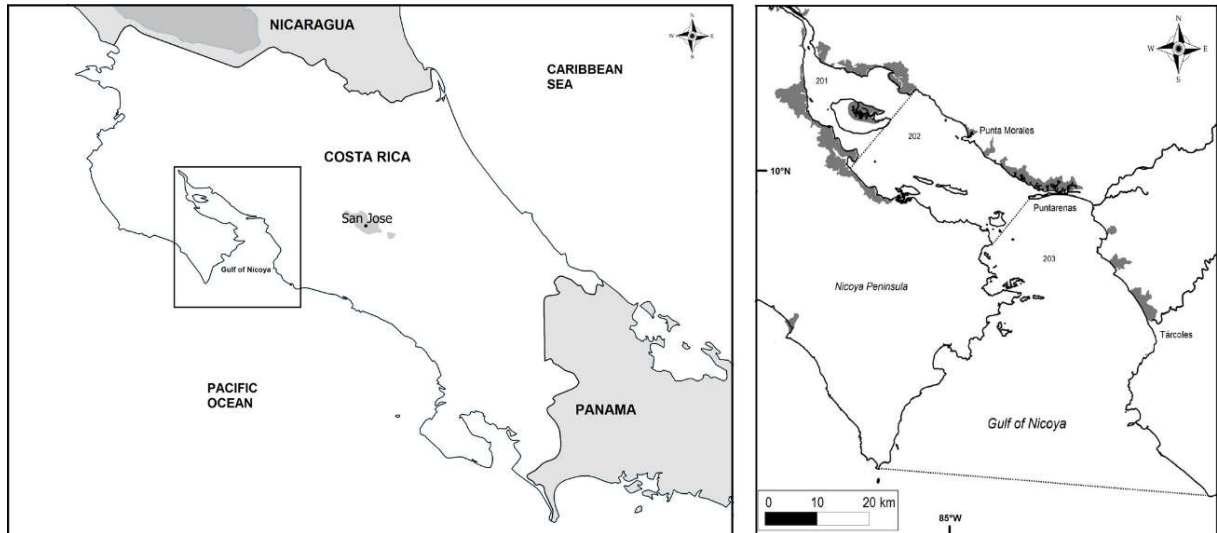


Figure 5.1 - Maps of the study area (Gulf of Nicoya, Costa Rica) , from Alms and Wolff (2020).

5.2.2 Modelling background

5.2.2.1 Food-web Model – Ecopath

For the modeling task, the open-source software Ecopath with Ecosim (EwE) versions 6.6.5 (Christensen and Walters 2004) was used. The Ecopath approach has been widely used to assess and visualize structural changes within ecosystems over time (Pauly, Christensen, and Walters 2000). Ecopath with Ecosim operates on a set of linear equations for each functional group in the ecosystem, connecting the group’s biomass production to other groups through a prey–predator matrix. For each compartment (i), the basic equation is as follows:

$$P_i = Y_i + B_i M_{2i} + E_i + B a_i + P_i(1 - EE)$$

where P_i is the total production rate of i ; Y_i is the total fishery catch rate; M_{2i} is the instantaneous predation rate of group i ; B_i is the biomass; E_i is the net migration rate (emigration–immigration); $B a_i$ is the biomass accumulation rate; $P_i(1 - EE)$ represents the “other mortality” rate of i ; and EE is the ecotrophic efficiency of i and the proportion of the production that is utilized in the system (Christensen and Walters 2004; Christensen et al. 2005).

While equation 1 allows for the balancing of the biomass production in each group with its use in the system (and the fishery), a second equation allows for estimating the energy budget of each group i as follows:

$$Consumption_i = Production_i + Respiration_i + (Egestion\ of\ unassimilated\ food)_i$$

The model was built around the main groups targeted by the fisheries, while grouping species with similar sizes, diets, consumption, mortality, production rates and habitats. In total, the model comprised 22 functional groups, including 3 primary producer groups and 10 different fish groups.

The balanced trophic model of the Gulf of Nicoya by Alms and Wolff (2019) was used as a basis for our further modelling approach. This EwE model of the Gulf of Nicoya was built around the main groups targeted by the fisheries. Species with similar sizes, diets, consumption, mortality, production rates and habitats were grouped, representing the state of the ecosystem in 2013. To construct the spatial model (*EwE Ecospace*) and better capture spatial patterns of

interest, we made modifications, including dividing the shrimp functional group into two categories: coastal shrimp (*Penaeus* spp., *Farfantepenaeus* spp., *Trachypenaeus* sp.) and deep-water shrimp (*Solenocera* sp., *Heterocarpus* sp.), according to their preferred depth ranges. To split the functional group according to the selected species, we used INCOPESCA landings data and additional information from Wehrtmann and Nielsen-(Muñoz 2009). The P/B and Q/B were adjusted for each shrimp group separately, considering their physiology and using information from other studies in similar areas and ecosystems (García and Duarte 2002; Giarrizzo et al. 2013; Wehrtmann and Nielsen-Muñoz 2009).

Our updated model comprises of 23 functional groups, including 3 primary producer groups and 10 different fish groups (Table 5.1). The model was balanced and fitted to time-series data, including historic fisheries catches, recent monitoring data and effort data. For further details please refer to Alms and Wolff (2020).

Table 5.1 - Basic Ecopath with Ecosim input and output parameters for the Gulf of Nicoya (Costa Rica) model 2013 (Alms and Wolff 2019). Biomass (B_i ; metric tons/km²), production-to-biomass ratio (P_i/B_i ; year⁻¹), consumption-to-biomass ratio (Q_i/B_i ; year⁻¹), ecotrophic efficiency (EE_i), production-to-consumption ratio (P_i/Q_i) and total catch (metric tons·km⁻²·year⁻¹) are presented for each functional group.

Functional group		Input parameters			Output parameters	
		B_i (t/km ²)	P_i/B_i (/year)	Q_i/B_i (/year)	EE_i	P/Q (/year)
1	Seabirds	0,06	0,09	50	0	0,002
2	Mackerel & Barracuda	0,06	0,5	7	0,733	0,071
3	Rays & Sharks	0,07	0,5	3	0,856	0,167
4	Large Drums & Snook	0,35	0,5	4	0,921	0,125
5	Catfish	0,43	0,47	6	0,977	0,078
6	Morays & Eels	0,17	0,5	5	0,769	0,100
7	Jacks & Pompanos	0,50	0,75	6	0,796	0,125
8	Snapper & Grunts	0,45	1,3	5	0,936	0,260
9	Cephalopods	0,30	4	15	0,801	0,267
10	Lobster	0,10	1,2	10	0,801	0,120
11	Flatfish	0,20	1,3	7	0,735	0,186
12	Small Demersals	1,20	3	10	0,681	0,300
13	Small Pelagics	1,50	4	20	0,896	0,200
14	Crabs	0,70	3,3	12	0,785	0,275
15	Epibenthos	5,00	3,8	20	0,706	0,190
16	Endobenthos	1,00	9	60	0,871	0,150
17	Shrimps	0,50	5	25	0,989	0,200
18	Shrimps_DeepSea	0,22	3,5	13	0,658	0,269
19	Zooplankton	5,50	30	150	0,684	0,200
20	Phytoplankton	15,00	200		0,248	
21	Mangroves	90,00	0,22		0,328	
22	Microphytobenthos	0,50	125		0,386	
23	Detritus	0,10			0,055	

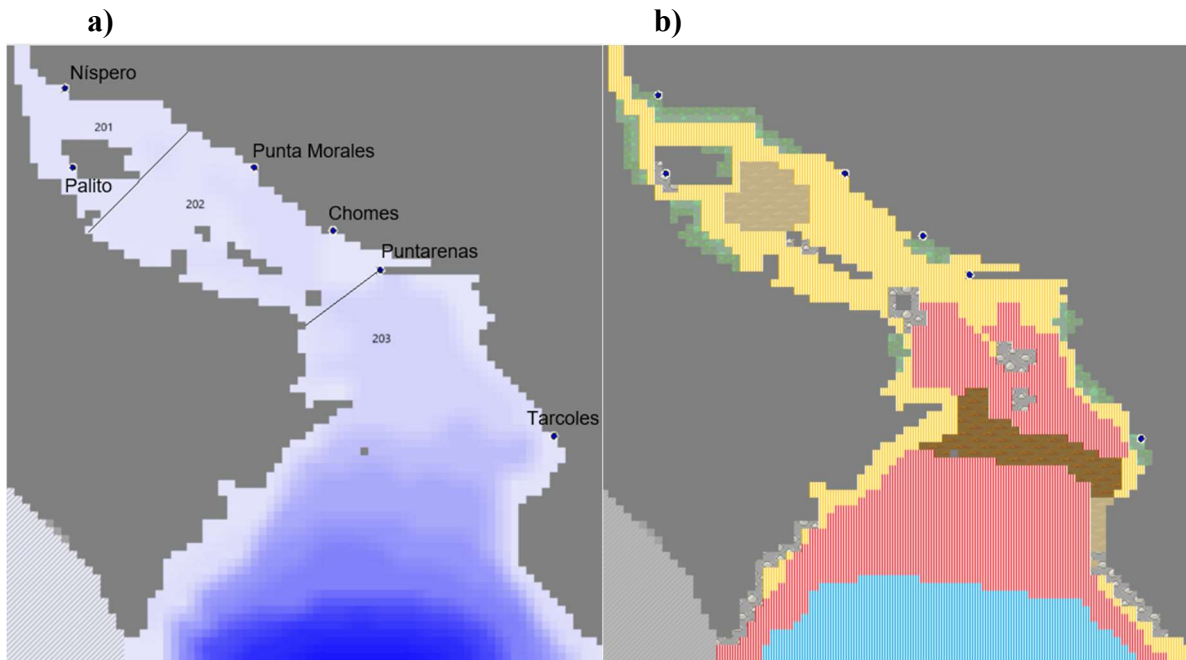
5.2.2.2 The Spatial Model – Ecospace

Ecospace can be used for the simulation of the spatial distribution of species and fishing fleet effort over time. It is particularly useful for exploring spatial management options and addressing various research questions, such as the regional consequences of climate change (Christensen and Walters 2004a; Mutsert et al. 2024). Ecospace models are spatially explicit, time dynamic models based on the *Ecopath* mass-balance and *Ecosim* time dynamic routines (Christensen and Walters 2024). A set of differential equations is defined based on the biomass components of change for consumer functional groups, expressed as:

$$\frac{dB_i}{dt} = g_i \cdot \sum_{j=1}^n Q_{ji} - \sum_{j=1}^n Q_{ij} + I_i - (F_{it} + e_i + M0_{it}) \cdot B_{it}$$

where B_{it} is the biomass of i at time t , g_i is the growth efficiency, I_i is the immigration rate; F_{it} is the mortality rate due to harvesting (fishing mortality); e_i is the emigration rate; and $M0_i$ the other mortality. The terms Q_{ji} and Q_{ij} represent the consumption due to predation by j on i , and by i on j , respectively. For primary producers, the consumption rate term is replaced by a production rate function $f(B_{it})$ represents primary production rate as a function of the group biomass (Christensen and Steenbeek 2024).

The Ecospace model is defined as a basemap based on a two-dimensional grid, where each cell is designated as land or water, and each cell can have unique characteristics (Christensen and Steenbeek 2024). For our basemap, an area of 1530 km² was defined using an 80x76 grid, with 1 km² grid cells. To facilitate regional analysis, the Gulf of Nicoya was divided into three regions, as defined by the fishing authorities: the inner gulf, intermediate gulf and outer gulf (zones 201, 202 and 203). The map also includes the primary port known as Puntarenas, which serves the semi-industrial and longline fleet, along with smaller ports mainly used by the artisanal fleet (Figure 5.2).



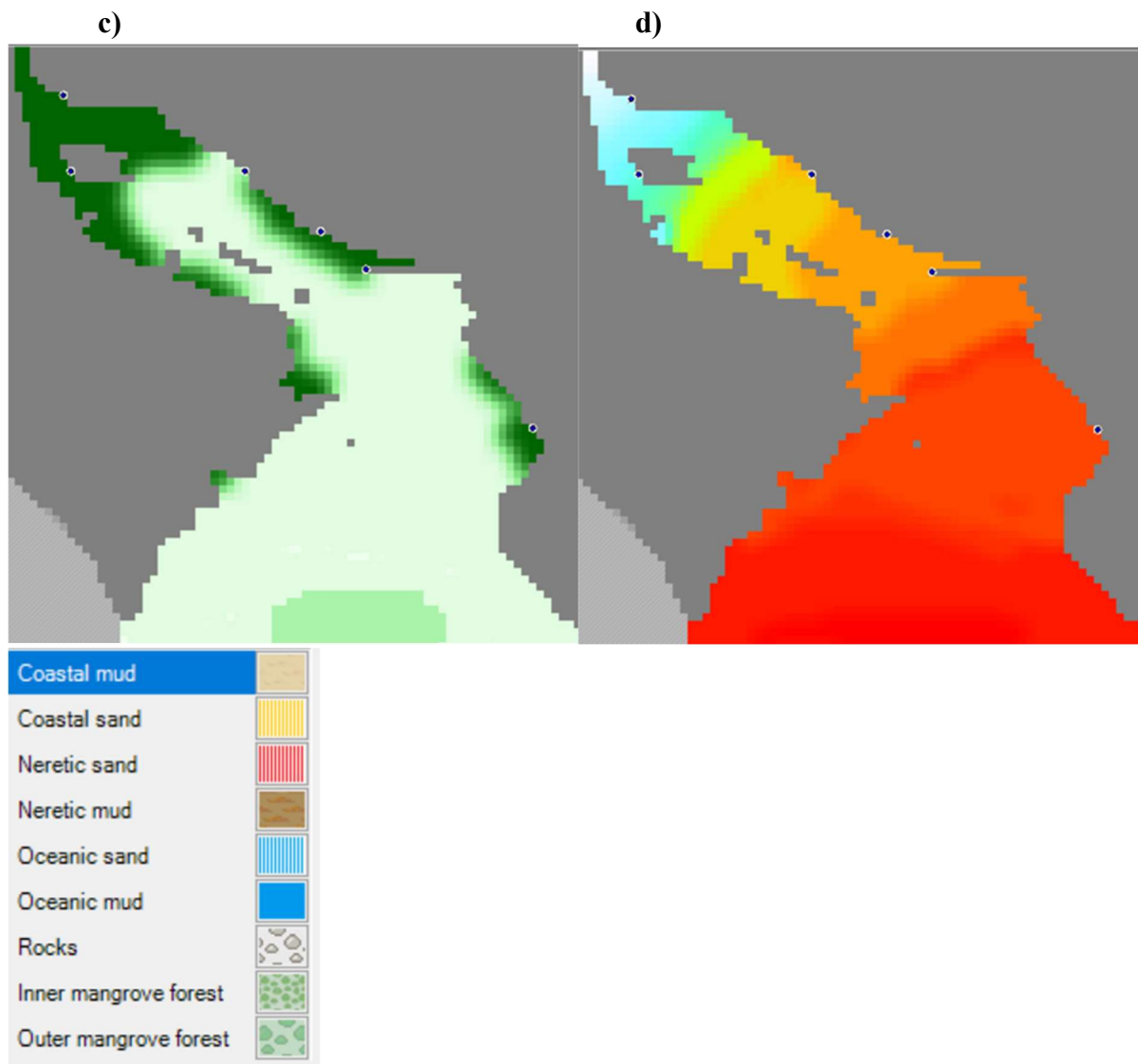


Figure 5.2 - Ecospace basemap of the Gulf of Nicoya (Costa Rica) with the main ports: **a)** displaying the bathymetry from shallow waters in light blue to deeper waters in dark blue (5-200m) and the three zones (201, 202 and 203). **b)** the distribution of habitat types (see legend), **c)** primary productivity as relative value low (light green) to high (dark green) **d)** Salinity from low values in light blue to higher values in red (8-34 ppm).

In the basemap, we defined 9 different habitat types, differentiated by a combination of bottom type (sand, mud, rocks) and depth zones (coastal, neritic, oceanic). Additionally, we determined the mangrove forest habitat, mainly located in the inner and intermediate gulf, mainly consisting of *Rhizophora* sp., *Pelliciera* sp. and *Avicennia* sp. We distinguish between the inner and outer mangrove forest. The outer mangrove area is a feeding area for different species, whereas the dense inner mangrove forest is more challenging for certain species to enter (small pelagics, flatfish). It serves as an area of shelter for shrimps, lobster, crabs, bivalves. The distribution of these habitat types is presented in Figure 5.2b.

Further maps of the spatial distribution of different parameters were created for the Gulf of Nicoya. The water depth reaches from 5 m in the inner gulf close to the mouth of the river Tempisque up to 200 m in the outer oceanic part of the gulf (Figure 5.2a). The primary productivity is highest around the mouth of the river Tempisque and in coastal areas close to other river inputs (Figure 5.2c). Salinity was defined as values between 8–34 ppm displayed in Figure 5.2d. The Gulf of Nicoya consists of brackish water at the mouth of the Tempisque river

in the inner Gulf of and of oceanic water in the outer gulf. The spatial environmental information (depth, salinity and primary productivity) is based on a GIS database conducted by the CIMAR/University of San José, Costa Rica (unpublished data).

To model the distribution of species, defining the spatial distribution within the ecosystem of each functional group is necessary. The model operates based on the habitat capacity model (Christensen and Steenbeek 2024; Mutsert et al. 2024), where species distribution is driven by species preferences for certain factors, such as the habitat type, abiotic environmental factors (e.g. salinity or depth) and the prey and predator distribution. This can either be done by selecting suitable and unsuitable habitat types for each species and/or by defining the spatial distribution through environmental response functions. According to Villy Christensen and Carl J. Walters (2024), the environmental response function is often more suitable to describe the distribution of the species than the defined habitats only. As environmental responses, we defined the optimum depth ranges for certain species (small demersal fish, pelagic fish, shrimps and lobster), the abundance related to the distance from the coast (0–19 km) (seabirds) and salinity preferences (cephalopods). These information were taken from different studies (Blaber 2008; Briones-Fourzán and Lozano-Álvarez 2013; García and Duarte 2002; Guzman et al. 2008; Hegde et al., 2016; Hunnam 2021; Schmidt and Dias 2012; Wehrtmann and Nielsen-Muñoz 2009). Ecospace models describe the biomass of each functional group, within each cell in the grid over time by considering trophic interactions, fishing and species' movement (Christensen and Walters 2024). The model calculations are based on several input parameters, such as the habitat preference for each functional group and the dispersal rates, indicating the average annual movement distances of the functional groups across the ecosystem (Christensen et al., 2008). The Dispersal rate is entered as the absolute value of average individual annual movement distances due to random movements (Christensen and Steenbeek 2024). Based on Christensen and Steenbeek (2024) suggestions, the range of values should be set between 3–300 km year⁻¹. We assigned low values (3–10 km year⁻¹) to mainly non-dispersing species (benthos, lobster, crabs), medium values (30–60 km year⁻¹) to demersal groups (flatfish, small demersals, morays & eels) and high values (up to 300 km year⁻¹) to pelagic groups and plankton (small pelagics, mackerel & barracuda, rays & sharks, plankton). Some functional groups have a low general dispersal rate, such as sessile species (crabs and benthos in general) compared to high dispersal rates of pelagic fish species and plankton. The dispersal rate is entered as a base rate in the suitable habitat and with adjustments applied when species enter unfavourable habitats. We assume that for all species, the vulnerability to predation is higher in unsuitable habitats, especially for benthos species, like crabs and lobster. Some fisheries related parameters are needed to define the distribution of fleet effort throughout the model area. Basic fishing fleet related parameters have been previously defined in the Ecopath and Ecosim model, such as the landings, discards and market prices for each functional group. In Ecospace, we additionally defined the habitat fishery, identifying which fleet operates in specific habitat types. Another important information needed is the definition of costs for each fishing fleet, the percentage distribution of effort related costs and fixed costs (Table 5.2). This information is needed to determine the fishing costs related to the distance covered by the fleet from port to its fishing region. Finally, the effective power set for each fishing fleet is among the key parameters governing the effort distribution in Ecospace (Mutsert et al. 2024; Romagnoni et al. 2015). This scaling factor can be adjusted when a functional group only occur in a small part

of the model area and serves as a target species for a fleet. In our model, we retained the default setting of 2 for the fleets, except for manual collection (1.5), where the distribution of the bivalves is mainly located in the mangrove forests.

Table 5.2 - Percentage of fixed and variable fishing costs (effort related and sailing related costs) for each fishing fleet and the fleet specific number of jobs per catch value in the Gulf of Nicoya (Costa Rica).

Fleet name	Fixed cost (%)	Effort related cost (%)	Sailing related cost (%)	Profit (%)
Art. Gillnets	20	5	55	20
Art. Longline	23	5	60	12
Art. Handline	20	20	35	25
Art. Manual	20	5	57	18
SI Shrimp	35	3	47	15
SI Sardine	25	5	60	10

5.2.2.3 Model Simulation of Conservation Plans

We assessed the effects of the decline and elimination of shrimp-trawling effort on species in the ecosystem using a set of indicators. Coastal and deep-sea shrimp stock biomass was defined as one indicator, while the spatial distribution and relative biomass of bycatch and predator species were compared between a period of active shrimp trawling to 20 years after the last trawler had left the gulf. The goal was to explore whether shrimp and bycatch species would show signs of recovery, potentially followed by increased biomass at higher trophic level (predator) species. We also compared the fishing effort distribution of all fleets between 2013 to 2039.

Ecospace defines fisheries restricted areas (MPAs) that block out fishing from specific Ecospace cells in the basemap for a given fleet during a selected time period (Steenbeek and Coll 2024). We used this tool to test the effect of the seasonal closure (veda) in the Gulf of Nicoya. The fishing closure is a traditional closure of the inner and intermediate gulf (called zones 201 and 202, Figure 5.3), where closure is for all fleets (except for artisanal manual collection) over a period of 3 months. We tested the effects of the original fishing closure on the effort and species biomass distribution over time, which can be quantified and visually displayed in the Ecospace output maps. We then adopted a scenario-based approach to test and compare the results of different fishing closure settings, where we implemented changes in the size, duration of the closure and the number of fleets involved (Table 5.3). Shrimp trawlers are not subject to the ban because they usually and only fish in the outer gulf (called zone 203, Figure 5.3) and artisanal manual collection is not part of the closure neither.

We defined 6 scenarios to test the effects of different regulations of the fishing closure. The first scenario- traditional fishing closure- existed since 1990s. The inner and intermediate gulf if closed for all fleets, except for manual collection, for three months. In the next scenario (2), we test the effects of including manual collection to the closure, since its target species are intensively exploited. In the third scenario, we model the effect of a spatial extension of the fishing closure to the entire gulf. What would be the benefits for the target species to close the all zones) for fishing for 3 months? In the next scenario (4), we simulate the effects of a fishing closure that lasts for 4 months to ascertain the effects of a temporal extension. Scenario 5 tests how the recent changes in the regulations of the fishing closure effect the effort and catch distribution in the gulf. Recently, in response to the constraints of the COVID-19 pandemic and

the need for small pelagic fish as bait for commercial fisheries, INCOPESCA adjusted the regulations. Under the revised rules, the semi-industrial sardine fleet is allowed to fish 2 of 3 closed months.

The criteria to assess the effects of the different fishing closure setting (scenario 1-6) are measured as overall relative effort and catch by fleet, the relative biomass of the functional groups as well as spatial fishing effort distribution of the fleets and spatial biomass distribution of all functional groups. We also compare the long-term trends of catches for the fishing fleets (2013-2039).

Table 5.3 - Scenario approach to test different fishing closure settings in the Gulf of Nicoya (Costa Rica) . Artisanal Gillnets: GN, Artisanal Longline: LG, Artisanal Handline: HL, Artisanal Manual Collection: MC, Semi-industrial Sardine Purse-Seine: SP, Semi-industrial Shrimp trawling: ST.

Scenario		Spatial scale (zones)	Temporal scale	Fleets banned	Fleets allowed
1	Fishing closure	201,202	3 months	GN, LG, HL, SP	MC
2	Manual collection not allowed	201,202	3 months	All fleets	None
3	Spatial extension	201,202,203	3 months	All fleets	None
4	Temporal extension	201,202	4 months	GN, LG, HL, SP	MC
5	New regulations	201,202	3 months	GN, LG, HL, ST	MC, SP
6	No closure	No closures			



Figure 5.3 - The area of the seasonal fishing closure (veda) in the Gulf of Nicoya (Costa Rica) (black stipes), in blue the fishing regions. Shades of blue represent inner gulf, called zone 201 (light), central gulf, called zone 202 (intermediate) and outer gulf, called zone 203 (intense) areas. Black dots represent the main harbors.

5.2.3 Model Validation

According to (Christensen and Walters 2024), the fishing effort distribution predicted by Ecospace should be compared to observed spatial effort patterns (VMS and/or AIS data), to validate and modify input parameters. In our study area, spatial data are rare, but we utilized

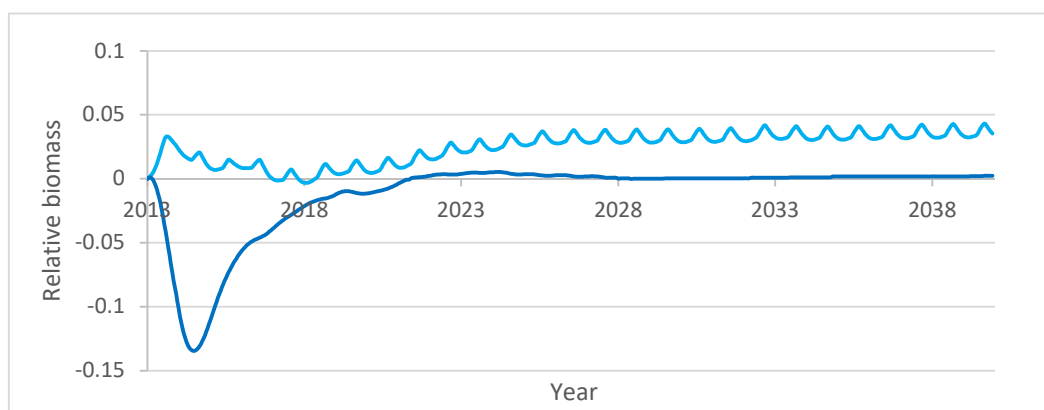
the limited available resources to validate our model performance. We used global fishing watch (GFW) data, based on data from the vessel monitoring system (VMS) provided by the Costa Rican Fisheries and Aquaculture Institute (INCOPECA) to visually validate the model performance of fishing effort distribution. Furthermore, we used monitoring data from fishermen (2013) for a plausibility check of fleet and biomass distribution.

Additionally, we conducted a sensitivity analysis to test the reliability of the Ecospace model. This included performing a robustness test following the guidelines outlined by (Romagnoni et al. 2015, 2019). We tested different dispersal rate settings and evaluated the model sensitivity to alternative values using fictional MPAs in the model. The dispersal rate is a critical parameter, as the model performance is vulnerable to its setting. Accurate estimates are needed for model performance. Information on specific dispersal rates for species/functional group can be taken from ecological studies. However, such data are not always available, necessitating the use of multiple methods to estimate these parameters (Christensen and Walters 2024). A common assumption is to base the parameter on species lifestyle, coupled with sensitivity of the model to this parameter.

5.3 Results

5.3.1 An Ecosystem Without Shrimp Trawlers

We compared a set of previously defined indicators, (see Section 2.2: Modelling background), between a period of intensive trawling (2013) to a projected time, 20 years after the last trawler left the gulf (2039). The Ecospace output results indicate that shrimp biomass of both shrimp groups (coastal and deep-sea) slowly increased after trawling until 2039 (Figure 5.4a). The increase of the coastal shrimp biomass is faster than the deep-sea shrimp biomass. The oscillation of the graph is due to the seasonal fishing closure



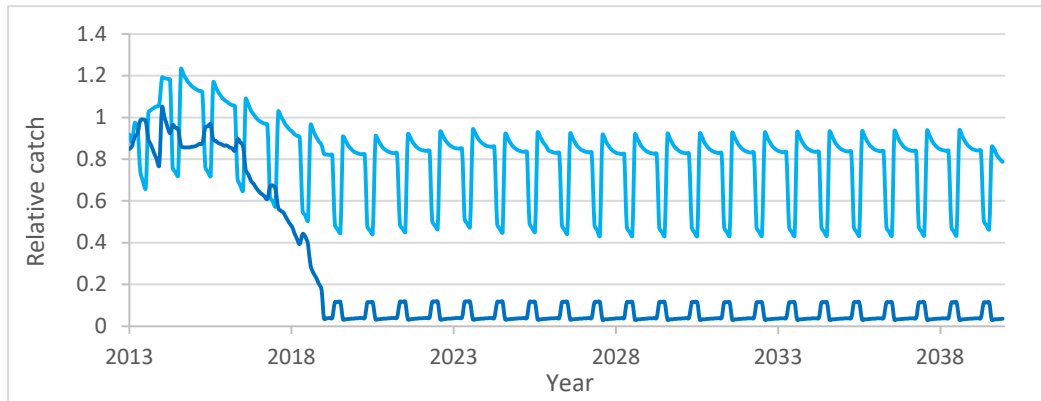


Figure 5.4 - Relative biomass (**A**) and relative catch (**B**) of the coastal and deep-sea shrimp species in the Gulf of Nicoya over time (2013-2039), coastal shrimps: light blue, dark blue: deep sea shrimps

In the beginning of the modelling period (2013), the number of coastal shrimps landed by the artisanal fleet was similar to the number of landings by the semi-industrial trawlers. After the ban, the artisanal fleet slightly increased its catches; however, fishing mortality of coastal shrimps declined by around 40% (Figure 5.4b). In 2013, the deep-water shrimps were almost solely caught by the shrimp trawlers; thus, the fishing mortality declined drastically by 2039 (Figure 5.4b).

Besides the direct target species, the model indicates a recovery for several other species in the ecosystem. Bycatch species, such as Catfish, Morays & Eels and Small Demersals (Figure 5.5 and Figure A.4 in the Appendix), starts to recover. The biomass distribution of the predatory species, like Large Drums & Snook and Mackerel & Barracuda also increased in our model prediction by 2039. Rays & Shark distribution shifts more towards the inner Gulf of Nicoya, the same can be observed for Cephalopods (Figure 5.5 and Figure A.4 in the Appendix).

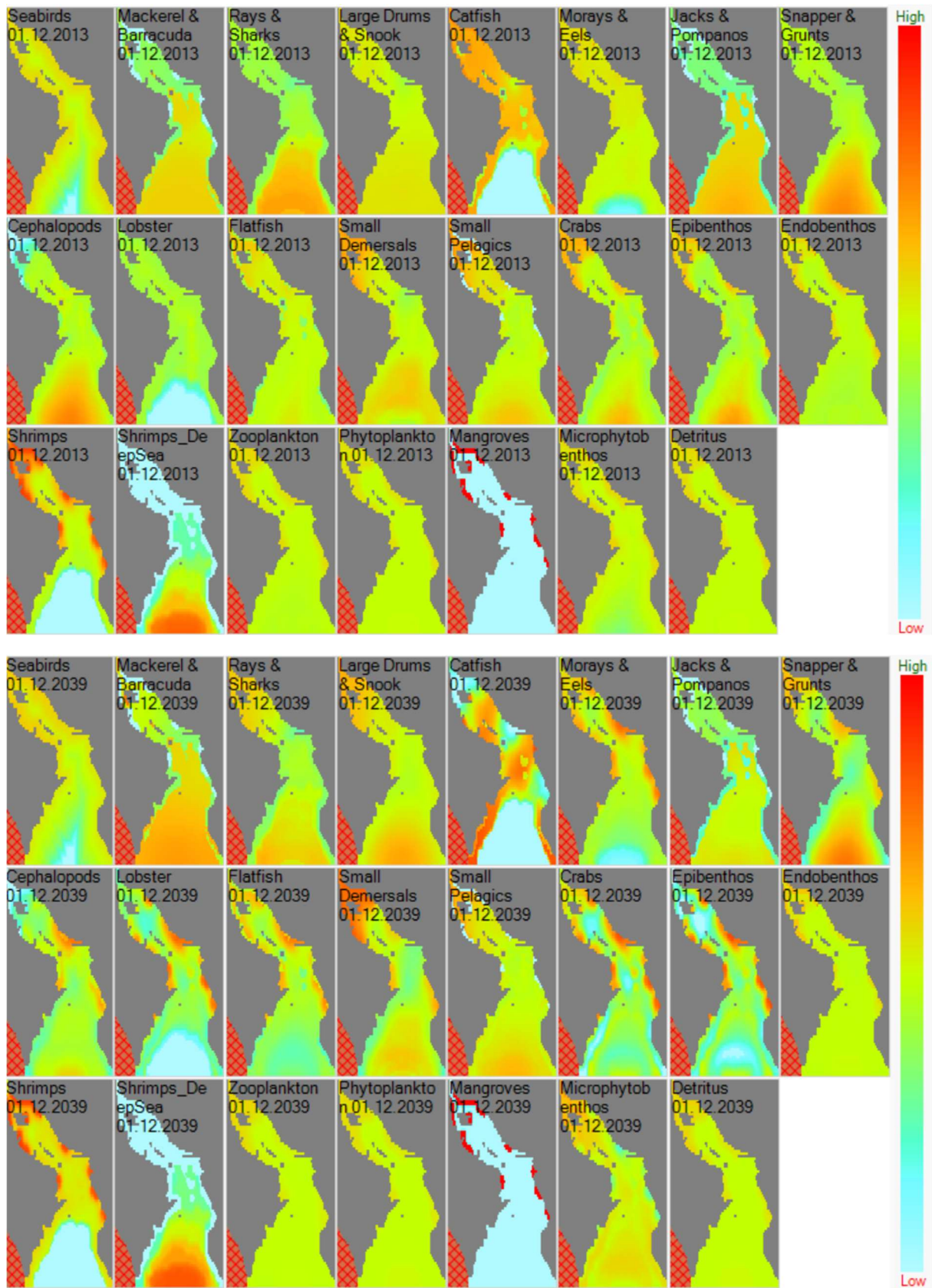


Figure 5.5 - Ecospace model output : Distribution of the biomass of all functional groups in the Gulf of Nicoya ecosystem (Costa Rica) in 2013 compared to 2039. From blue (green) over green to high (red) values.

After the ban on trawling, the fishing effort distribution of the remaining fishing fleets also changed in some areas. The distribution of the fishermen and women collecting mollusks and crustaceans (manual collection) drifted from some areas and intensified in other coastal areas. The entire west coast of the Gulf of Nicoya is considerably exploited (Figure 5.6). A shift of effort can be also observed for the handline fleet. Effort is concentrated closer to the coast and ports, especially in the inner gulf (Figure 5.6).

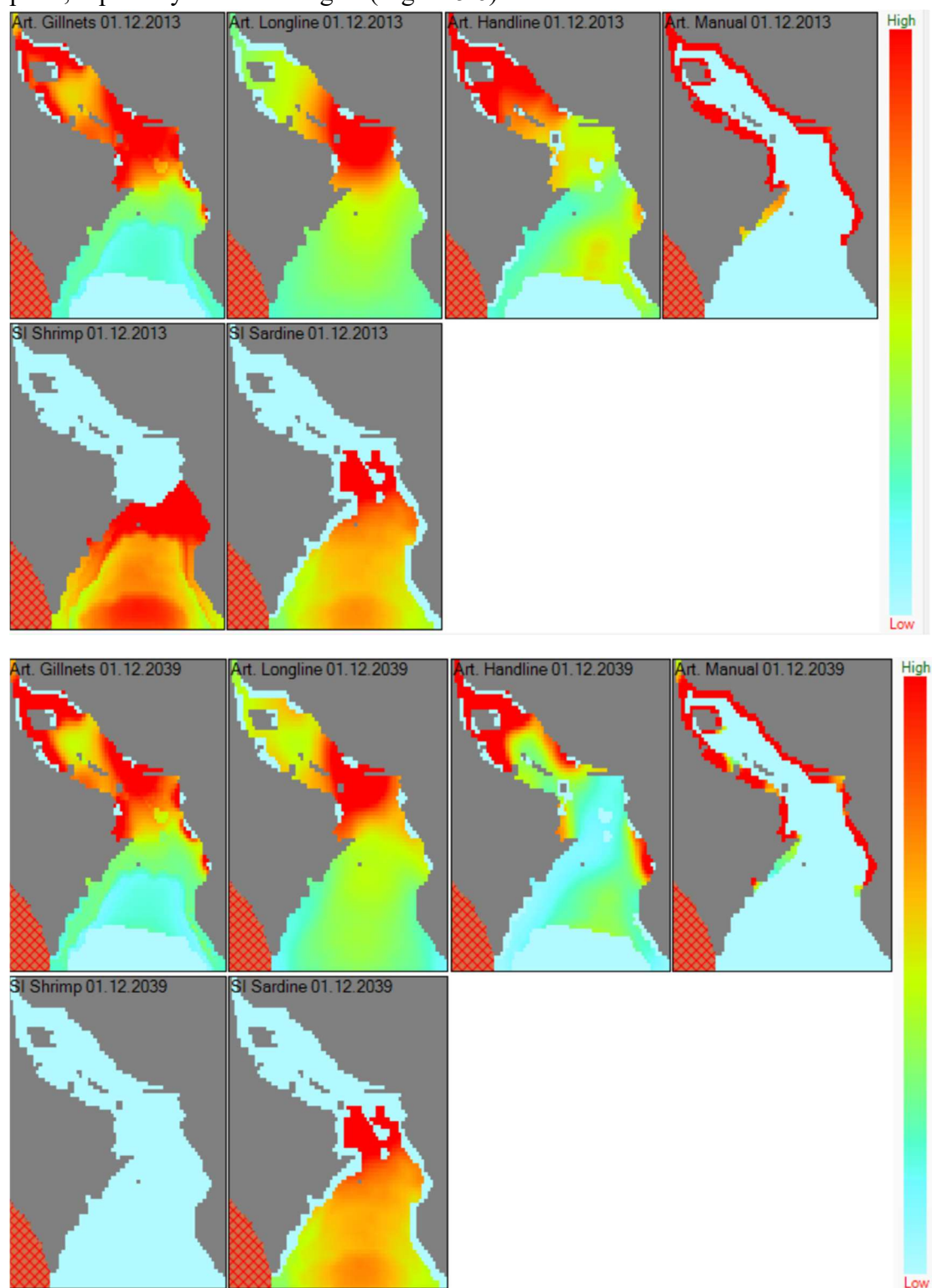


Figure 5.6 - Ecospace model output: Distribution of the fishing effort of all fleets in the Gulf of Nicoya (Costa Rica) a) in 2013 and b) in 2039

The catches of all remaining fleets increased after 20 years of trawling ban (Table 5.4). The artisanal gillnet fleet, which targets similar species as the shrimp trawlers, increased its catches by 20%. The artisanal handline fleet could even increase catches by 70%. The longline and purse-seine fleet increased catches by around 10%. The increasing catches go along with marginal effort increases (by 1%). Thus, from the modelling result, all fleets increased the catch value. As the effort only increased slightly along with a minimum increase in costs, the overall profit of the fleets increased. The manual collection is an exception. It can be observed that the manual collection sector increased its fishing effort by 10% and over tripled the catch.

Table 5.4 - Ecospace results: The relative change in catch, value, costs and effort of all remaining fleets after the ban of trawling in the Gulf of Nicoya (Costa Rica) in 2039 compared to the initial values in 2013.

Fleet name	Catch (E/S)	Value (E/S)	Cost (E/S)	Effort (E/S)
Art. Gillnets	1.224244	1.2611	1.129135	1.030662
Art. Longline	1.106568	1.116496	1.03619	1.010802
Art. Handline	1.726026	1.679562	0.8768923	1.035697
Art. Manual	3.269391	3.223467	0.8716791	1.116019
SI Sardine	1.125936	1.125936	1.064756	1.0567

5.3.2 Optimization of the Seasonal Fishing Closure

In the next step, we evaluated the efficiency of the seasonal fishing closure in the Gulf of Nicoya, under its original version, the revised regulation introduced by INCOPESCA and some alternative settings to potentially improve the success of the fishing closure (Scenario 1-6, Table 5.5). Our modelling result revealed that the fishing closure can be an efficient measure to reduce the overall annual effort; however, its regulations that lead to the best performance remains unverified. Our modelling results indicate that the original fishing closure (Scenario 1) reduces the overall catch by 1.7% compared to a scenario without any restrictions (Scenario 6). The recent changes in regulation of the closure, allowing the sardines to fish 2 of the 3 months. This new fishing closure (Scenario 5) is less efficient with a lower annual catch reduction than the original fishing closure. A temporal extension of the seasonal closure from 3 to 4 months (Scenario 4) would lead to an extra reduction of 1.4% of the catch compared to the original fishing closure.

Table 5.5 - Ecospace results: Relative change in catches of all fleets (except sardine) in the Gulf of Nicoya for the different fishing closure settings (scenario 1–6), 2039 compared to the start of simulation (2013).

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6
	Closure	MC not allowed	entire gulf all fleets	4 months	new closure	no closure
Fleet name	Catch (end)	Catch (end)	Catch (end)	Catch (end)	Catch (end)	Catch (end)
Art. Gillnets	1,488233	1,488235	1,444152	1,459322	1,486502	1,5214
Art. Longline	0,321865	0,321854	0,299517	0,318206	0,322204	0,32202
Art. Handline	0,10825	0,108241	0,104201	0,104201	0,108244	0,117686
Art. Manual	0,057499	0,055261	0,046905	0,058711	0,055294	0,054813
SI Sardine	0,95866	0,972981	0,801319	0,966901	0,972993	0,984065
Total	2,934508	2,946571	2,696093	2,90734	2,945237	2,999984

Till date, the manual collection is not tackled by the fishing closure, but our modelling results revealed that the biomasses of all target species of this fleet are below the initial biomass level of 2013. Figure 5.7 displays the biomass development of the target species until 2039 provided that the fishing regulations remain intact and manual collection is not limited. Thus, we assessed the benefits of including the MC in the original closure and we see an addition 0.6% catch reduction over all.

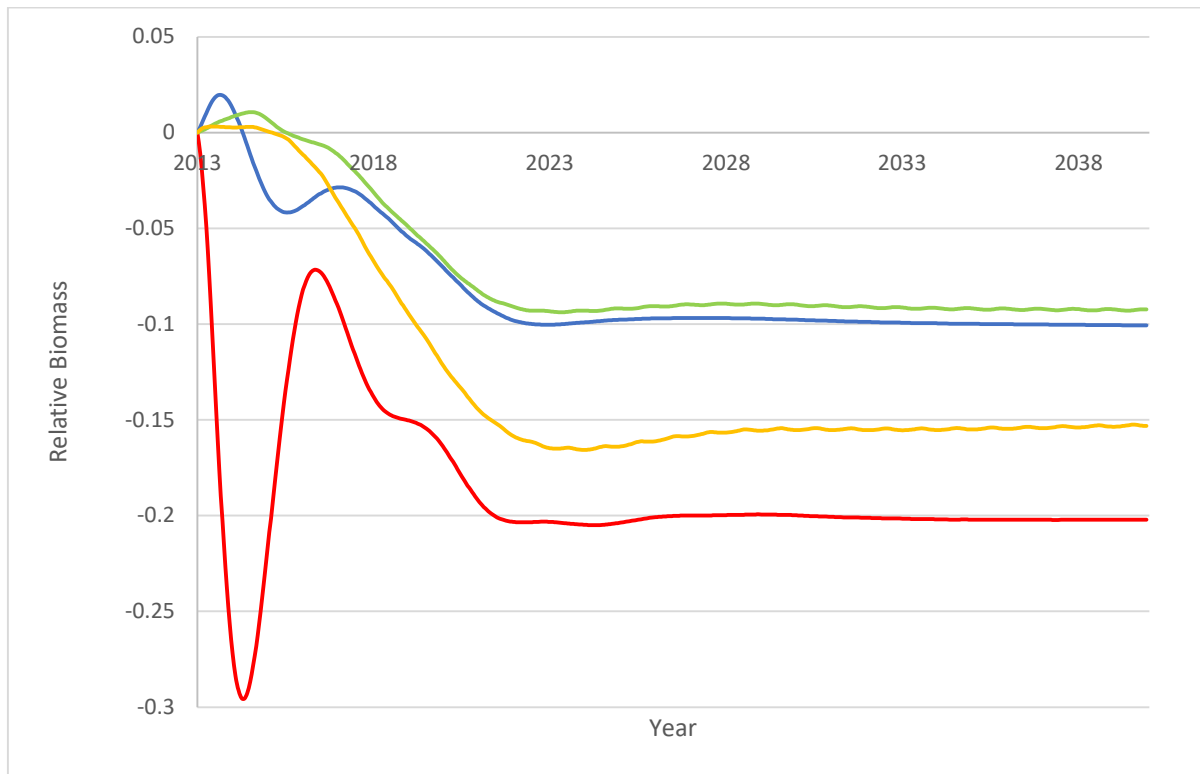


Figure 5.7 - Relative biomass of target species of the artisanal manual collection in the Gulf of Nicoya (Costa Rica) 2013-2039. Blue: Cephalopods, red: lobster, yellow: Epibenthos, green: Crabs.

As the manual collection is also practiced along the shoreline of zone 203, we evaluated the effect of closing the entire gulf for all fleets for 3 months. The fishing closure reduces the annual fishing effort by an average of around 8% compared to allowing all fleets to fish throughout the year. Catch reduction, mostly from longline and sardine fleet, favors the recovery of small pelagics and large predatory species, not further straining the artisanal fleet. The scenario increases biomass distribution in the outer gulf of predatory species, catfish in coastal areas. Small pelagics and small demersal fish biomass increases in the outer gulf (Figure 5.8).

A relative annual increase of the catches by all fleets running scenario 5 was later observed. Compared to 2013, our modeling results show an increase of the total catches by 2039. The annual catches increased for all fleets in 2039, the handline fleet increased its catches by 40% and the manual fleet even doubled its catches. This is the highest long-term increase in catches observed among all tested fishing closure settings (scenarios 1-6).

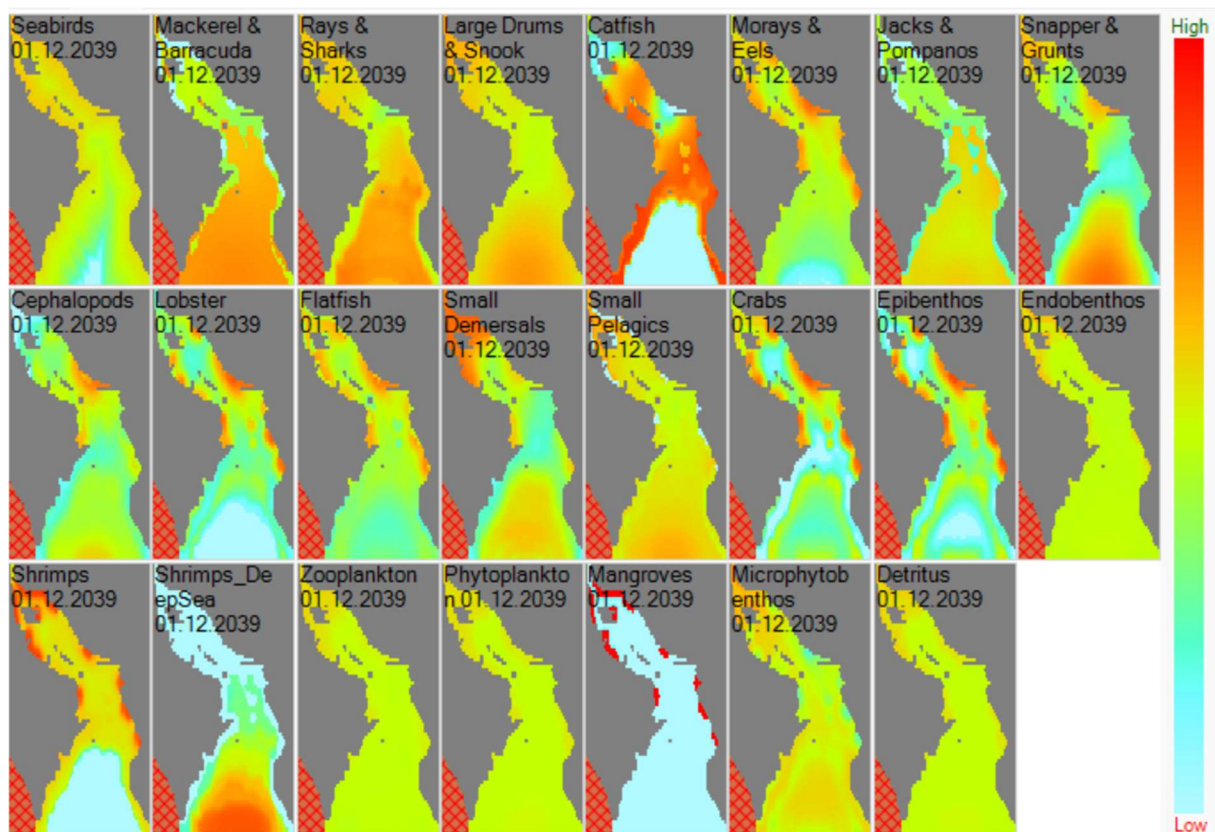


Figure 5.8 – Seasonal fishing closure. Ecospace output: Simulating biomass distribution of all functional groups after the implementation of a seasonal fishing closure covering the entire Gulf of Nicoya (Costa Rica) including restrictions for all fishing fleets. End year of simulation is 2039.

5.4 Discussion

Our results suggest the need for further conservation measures in the Gulf of Nicoya. Shrimp trawlers have been identified as a major cause of ecosystem degradation; a pattern observed in many regions worldwide. The ecological consequences depend on the extent and intensity of trawling, and impact multiple components of the ecosystem. Firstly, trawlers can cause physical disturbance and even damage habitat structures (Kaiser et al. 2002; Turner et al. 1999). Moreover, as stated above, trawlers exert pressure on the demersal community structure through the extraction of substantial amounts of shrimps and enormous quantities of bycatch species. The biomass of bycatch species strongly exceeds the biomass of shrimps landed, leading to shrimp-to-bycatch ratios of 1:9. In the Gulf of Nicoya, the shrimp-to-bycatch ratio was estimated to be at least 1:7 and the species landed and discarded are diverse.

A trawl ban provides a unique opportunity to investigate recoveries in ecosystem structure and function by removing the impacts from a major fishing gear operating in the ecosystem (Tao et al. 2021). The potential of a species to recover depends partly on its life history traits; smaller, short-lived species and fast-growing species, such as shrimps, tend to take a shorter time to recover from overfishing than larger species. However, we observed a slow recovery of shrimp biomass 20 years after the fishing closure, and the increase in relative biomass was not as significant as expected. This could be attributed to the status of the stock prior to the ban, which was strongly overexploited. Another reason for the slow recovery might be the influence of

predation mortality. As shown by our model, shrimps are the main prey for many species in the ecosystem. Thus, the reduction of fishing effort due to the ban on trawling and the associated increase in shrimp biomass availability, will lead to an increase in their predators, and thus in predation mortality. The predation mortality might be as high, that the shrimp stock biomass is not initially increasing. An additional factor, enhancing the predation effect, is that several predator species of shrimps have previously been caught as by-catch by the shrimp trawling fleet. After the ban the biomass of these groups increased, consequently the predation mortality on the shrimps further increased. As shown in other studies, middle and lower trophic levels may be sensitive to reductions in upper-level predator biomass (Hinke et al. 2004). The direct effects of reduced fishing mortality must be considered together with the indirect effects related to changes in predation and competition (Hinke et al. 2004). As in other studies, we can demonstrate that the speed of recovery significantly depends on the current status of the stock and it can vary among ecosystems and that recovery can be a challenging and prolonged process. For example, in the Indonesian shrimp fishery, the overfished shrimp stocks showed a recovery within a 7-year period after the ban (Chong et al. 1987). Another study in the Java Sea revealed a shift in shrimp species dominance after the ban, as well as an increase of shrimp biomass only in offshore waters, not in coastal areas (Tirtadanu and Suprato 2017). In Hong Kong waters, the shrimp biomass remained unchanged for the next 5 years after the ban (Mak et al. 2021). We observed a faster biomass increase of coastal than deep-sea shrimps in our model. This can be due to a more intense targeting of shrimp predators in coastal waters due to the operational radius of the artisanal fleets, compared to deeper waters, where the shrimp trawlers used to operate. Besides the effect of predation mortality, this can also be due to their lower initial biomass of deep-sea shrimps or lower production/biomass ratio.

Beside the recovery of shrimps as main target species, we expected the ban on trawling to have a positive effect on bycatch species, ocean-bottom flora and fauna as well as demersal bottom-associated species. In the Gulf of Nicoya, the bycatch species, such as catfish, morays and eels, and small demersals showed marked recovery initially. As shown in an embayment in Kenya (Munga et al. 2012), and in (estuarine) coastal waters of Hong Kong, China (Mak et al. 2021) and Indonesia (Bailey 1997) and the Mediterranean (Pipitone et al. 2023), the recovery of the demersal fish and shellfish assemblages, can be faster than the increase of shrimp biomass. The increasing abundance of lower trophic level species will most likely be followed by an increase in biomass of longer-lived predatory species, like drums and snook, mackerel, barracuda and sharks. Among these, demersal sharks and rays were particularly vulnerable bycatch species in the trawl fisheries of the Gulf of Nicoya (Clarke et al. 2016) that could benefit from the ban. The biomass distribution of predatory species, such as large mackerel & barracuda also increased in our model prediction by 2039. Rays and shark biomass increased slightly and the distribution of this group shifted more towards the inner Gulf of Nicoya, and the same can be observed for cephalopods. The predatory species follow the distribution of the prey species, such as small demersals, snappers and catfish. Some species do not exhibit an upward trend, which can be due to an increase in predation mortality caused by the increase in predator biomass. Notably, potential habitat destruction that might have been caused by the process of bottom trawling and its recovery after the ban is not part of the Ecospace model. Besides the changes in biomass distribution, the removal of the trawling effort also caused local shifts in the fishing efforts of the remaining fleets. These shifts can be related to the distribution of target

species. For instance, the handline fleet shifts its effort to the inner gulf, concentrating the effort on zone 201, following the distribution of its main target species, such as small demersal, snappers and catfish. The same process can be observed for the manual fleet. The semi-industrial sardine fleet is slightly affected by the change in fleet effort. This was expected, as these two fleets did not overlap in their target/landed species. The sardine fleet operates mainly in zone 203, and targets specifically small pelagic fish with a low bycatch ratio.

Overall, we see clear benefits of the ban on trawling for the ecosystem. However, the ecosystem has been strongly degraded over the last 30 years and this conservational measure might not be sufficient to achieve a recovery of the majority of functional groups across the ecosystem. We know from other studies, that a simultaneous recovery of all species in the system is unlikely to be achieved due to various trophic interactions and other complex external influences (Froese et al. 2016; Steven Mackinson et al. 2009) but we believe that for the Gulf of Nicoya more measures are necessary to turn the tide. Thus, we assessed fishing closures as a further conservational measure in the Gulf of Nicoya. The three-month closed fishing season in the inner and intermediate Gulf of Nicoya, known as ‘veda’ in Spanish has been applied for many years now. Seasonal closures have often been implemented to protect over-exploited spawning stocks in certain locations, but the effectiveness of these measures is rarely tested (Clarke et al., 2015). However, these measures, for example, in the case of Atlantic cod off the coast of Scotland, are easier to implement and control for single stocks. (Clarke et al., 2015). In multi-species fisheries, adjusting the closure to meet the needs of many different target species is challenging (Samy-Kamal et al., 2015).

The performance of the temporary marine protected area was compared with different levels of protection. The traditional seasonal fishing closure reduces the total annual catch by 2.2% compared to a situation without any restrictions. In recent years, due to the increasing economic challenges exacerbated by the COVID-19 pandemic, the regulations governing the seasonal closure have changed. The sardine purse seiners are partially allowed to fish during the closed season. Nowadays, sardine vessels are banned for only one of the three months, with the justification that sardines are sold as bait for the national fishing fleet. The total annual catch reduction of the closure with the new regulations is 1.8%, lower than during the traditional closure. On the one hand, our model does not indicate a significant negative effect of the small pelagic purse-seiners operating during the fishing closure, as the fleet operates selectively, exclusively catching its target species, and its main fishing ground is the outer Gulf, which is not part of the fishing closure. On the other hand, the exclusive right to fish during the seasonal closure could cause an unfavorable shift of effort into the Gulf. Furthermore, favoring the semi-industrial fleet against the artisanal fleet is not recommendable, as it can foster competition and decrease the compliance of the artisanal fishing communities. According to our modelling results, a temporal extension of the fishing closure by one month is of little benefit to the ecosystem and would cause a loss of income/ for the artisanal fleets.

From a conservation standpoint, our model results suggest that manual collection activities should be considered in the seasonal closure. Due to the downward trend of all its target species, we advise against allowing manual collection to operate during the closed season, as it operates above a sustainable threshold, especially concerning the bivalve extraction. Regarding declining benthic biomass, the impact of the artisanal fleet should not be disregarded. The heavy

exploitation of bivalves by coastal communities (yearly average = 20 metric tons in the last 5 years) has most likely caused the severe decline in epibenthos biomass. These species declined together with the loss of mangrove areas. A positive development was seen in other parts of Costa Rica. In the neighboring estuary, Golfo Dulce in Costa Rica INCOPESCA saw the urge to act and established a special fishing closure that permits the extraction of the bivalve species *Anadara tuberculosa* and *Anadara similis* in certain areas of the mangrove forest for the entire year 2023.

A fishing closure that includes all fleets increases efficiency but we see from our modelling results that the effect of the closure on the catches and biomass distribution of species could be enhanced. Thus, we tested a fishing closure covering the entire Gulf and including all fleets. Our results indicate that this fishing closure will lead to an annual catch reduction of approximately 10%. This is due to a substantial decrease of fishing effort, that benefits all species in the Gulf. We see increasing biomasses across almost all functional groups. Our biomass distribution maps indicate increasing biomasses of high trophic levels in the outer Gulf. Given that the fishing closure is only temporary, fishing effort remains high during the rest of the year, and the modelling results indicate benefits for almost all species, we recommend implementing a complete closure of the Gulf to all fleets. These closure settings, without exceptions, also simplify the enforcement and control of regulations. Regarding the benefits of the fishing closure, we should consider that the closed season is intended to support the hatching and reproduction of several target species in the Gulf of Nicoya. Our model does not assess the success of these endeavors and does not fully account for these additional closure benefits.

Besides the environmental benefits of these conservational measures, we also need to see the socioeconomic importance of the fisheries for many people. For example, manual collection significantly impacts the coastal communities, particularly for women engaged in this occupation. This is also applicable to sardine fisheries, as many people depend on the fishing sector. The seasonal fishing closure has an impact on the income and livelihood of the fishermen and leads to short-term disadvantages, especially for the fishermen who already live on the margins of subsistence (Brillo et al. 2019; Islam et al. 2021). Ideally, subsidies and alternative employment opportunities would be needed during fishing closure periods but in reality, there are often no alternative employment opportunities, subsidies are insufficient and alternative sources of income are rare (Colwell and Axelrod 2017; Owusu and Andriesse 2020). In some areas, artisanal fishermen who collect bivalves have been alternatively employed for the restoration of mangroves, providing a livelihood as well as ecosystem services to fish, shrimp and society. One also needs to consider the long-term benefits of the fishing closure. Catches can be higher after the closure and potentially reduce the travel distances from the port, leading to lower operating costs (Samy-Kamal et al., 2015). From an efficient fishing closure, we would expect increasing biomasses of the target stocks over the years, which would lead to higher catches of the fishing fleets. However, the short-term disadvantages often outweigh the benefits.

A permanent Marine Protected Areas (MPAs) could be an efficient measure of conservation in the Gulf of Nicoya, which has not yet been implemented. MPAs are a tool to safeguard marine natural systems, yet their effectiveness depends on how well they are integrated into the existing socioeconomic context (Horta e Costa et al. 2022). The problem of implementing a permanent

MPA in the Gulf of Nicoya is the socioeconomic structure of the fishing sector. The artisanal fishing fleet fishes from small ports close to their communities and the manual collection has an even smaller operational radius. Closing one certain area permanently puts high constraints on certain communities close to the area, cutting them out from their fishing grounds. The high level of subsistence fishing in the Gulf of Nicoya would make it even more difficult to enforce and control an MPA in certain areas. A permanent MPA in deeper waters further away from the coast would be more feasible from a socioeconomic point of view but mostly likely not the most efficient from a conservation point of view. The advanced artisanal fleet and the semi-industrial fleet cover larger distances from the port with advanced boats and gears and are impacted less severely by a stationary permanent MPA. Further ecological research to identify the definition of optimal trade-offs between alternative potential areas would be needed together with a socioeconomic evaluation of the negative impacts on the artisanal fishing communities.

Ecospace is a spatial modelling tool that allows for considering the impact of MPAs in an ecosystem context. It models the distribution of species biomass due to changes in effort distribution of the different fishing fleets and other possible factors. Ecospace has the ability to account for spatial effects in the ecosystem that are of importance for assessing the potential benefits of MPAs, such as the concentration of fishing effort at the edge of the MPAs and high movement rates of species. Cascade effects may generate a high concentration of prey outside the MPA, and attract predators out of the protected areas (Walters et al. 1999). Furthermore, these effects cannot be captured by non-spatial models. Ecospace can also be used to find policy alternatives that can afterwards be validated by more detailed analysis and experimental field testing. In this study, we utilize some of the capabilities of the Ecospace model to study the ecological effects of alternative MPAs settings, which could not be done with a non-spatial model. Ecospace was able to capture the effects of the interaction between spatial effort and species distribution and the predator-prey dynamics inside and outside the closure. These elements help to guide informed decisions which are based on ecosystem considerations.

5.5 Conclusion

We conclude that banning all shrimp trawlers from the Gulf of Nicoya is an important step towards conservation. This action is expected to benefit many species, improve the condition of bottom habitats and support the entire health of the ecosystem. However, with artisanal fleet likely to increase its catch in the future due to the loss of competition from the shrimp trawlers, some additional restrictions might be necessary, as the fishing closure with its recent settings might be insufficient. Our modelling results indicate that a closed season comprising the entire gulf including all fleets could be a promising measure, limiting also the extraction of mollusks and crustaceans from the gulf. A permanent MPA in the Gulf of Nicoya could be a good measure for conservation. In the next step, these questions can be assessed using the Ecospace model. The model's spatial optimization tool for protected area placement, which is an automated tool, can be used to find the appropriate areas for the placement of MPAs. The tool automatically detects suitable areas (cells) for protection by maximizing ecological, social and economic criteria (Christensen et al. 2009). This approach aligns with the policy search routine applied by (Alms, Romagnoni, and Wolff 2022).

Chapter 6

GENERAL DISCUSSION

6.1 Main Findings

We used the full scope of the EwE software package to assess the Gulf of Nicoya ecosystem, its past and present state, and its potential future development. To some extent, what we found is alarming. Thus, we also presented some possible future management scenarios to respond to the negative trend.

Our general analysis of the characteristics of the Gulf of Nicoya presented a productive bottom-up controlled system with few species at the intermediate level. The importance of phytoplankton and zooplankton is highly relative, driving the extremely productive system. At the same time, we highlight the importance of links to higher trophic levels (benthos, shrimps, and small pelagics), all species heavily targeted by the fisheries. These results are consistent with findings in other coastal or semi-enclosed ecosystems. Primary producers, zooplankton, small pelagics, and benthic organisms have been frequently observed to be among the most vital groups, e.g., in Chesapeake Bay (Baird and Ulanowicz 1989), the Gulf of Maine (Heymans 2001), and the Gulf of Thailand (Christensen 1998). Small pelagic fish, also called “wasp-waist” species, have been shown to exert top-down control on their prey species and bottom-up control on their predators (Shannon et al. 2007).

Our main finding concerning the fisheries’ impact on the Gulf of Nicoya ecosystem was a steady overexploitation since 1995 and a continuous degradation of the system, affecting the trophic level structure of the whole ecosystem and reducing the biomasses of many high trophic level species. These unfavorable developments are accompanied by severe economic losses for the semi-industrial and artisanal fleet, threatening subsistence fishing. In our attempt to analyze the reasons for the downward trend of the ecosystem, we found overfishing to be the main, but not the sole, reason/driver. Besides the fishing pressure, climate variations could be identified as an additional driver for ecosystem change. Our time-series models thus showed that the variations in the fisheries catch could not solely be explained by fishing pressure variations as shown in other systems (Shannon et al., 2008). In our study, the Chlorophyll_a concentration appeared to be the most relevant of tested environmental drivers for influencing the stocks of several functional groups (large drums and snook, crabs and morays, and eels). The environmental changes in the Gulf of Nicoya are influenced by the large-scale ecological conditions associated with the ENSO cycle.

We found some important potential countermeasures in our attempt to propose a way out of the negative trend. The ban on trawling, or at least a severe reduction of trawling, is an essential measure to allow for shrimp population recovery. Shrimp recovery is crucial for the entire ecosystem as shrimps are a fundamental link between trophic levels in the system. A trawling ban would also allow bycatch species regeneration, favorably impacting the ecosystem. However, a trawling ban alone is not sufficient for the full recovery of the ecosystem. These findings and the social and economic consequences of eliminating shrimp trawling need to be considered.

We also learned from our study that participatory management and the engagement of stakeholders in the process of policymaking can be successful and go well, along with the use of the Ecopath with Ecosim software.

6.1.1 Ecosystem Shifts

Compared to the status of the ecosystem in the early 90s (baseline model, (Mathias Wolff et al. 1998)) we detected a further degradation of the overall ecosystem health. (1) We see a decline in mangrove coverage by 10% due to land use changes. (2) A decrease in coastal shrimp species caused a shift in fishing efforts towards capturing deep-water shrimp species in the outer gulf. (3) A decline in bycatch species such as eel and catfish (4) and an overall decrease in the biomass of benthic invertebrates due to shrimp trawling, the excessive manual collection of bivalves, and the loss of the mangrove area. (5) A parallel decline in small pelagic fish biomass could be observed due to the immense fishing pressure of the semi-industrial fleet, leaving a hole in the intermediate trophic level, a vital link between the productive lower levels and the higher trophic levels, causing trophic cascades throughout the food web. (6) Thus, biomasses and catches of long-lived, high-trophic-level fish (rays and sharks) decreased. Our analysis highlights the importance of the mid-trophic level species, which are heavily exploited by semi-industrial fisheries (small pelagics) and artisanal fleets (small demersal). The observed decline of these species did not only have ecological, economic, and social consequences (Alms and Wolff 2019; Queiros et al. 2019) but may also lead to unfavorable shifts in the ecosystem.

6.1.2 Economic Loss

Although fishing efforts and overall catches increased between 1993 and 2013, these catches' total value and profit decreased substantially over the two decades of interest, causing a tremendous economic loss of approximately 50%. This loss of profits was due to changes in the catches toward low-value species, such as small pelagic, snappers, and small demersal species. In 2013, shrimps accounted for over 28% of the catch's total value and constituted only half as much as that in 1993.

These developments caused an expanding fishery for low-trophic-level species with low economic value while valuable shrimp resources and higher-trophic-level species were depleted. Both trends - fishing more heavily on low-trophic-level species and the expansion of the spatial scale to areas where large fish are still more abundant - are common phenomena in tropical fisheries (Pauly et al. 1998), along with advancements in the gears and modernization of the semi-industrial fleet (Vega-Corrales and Marín 2014). In the early 2000s, several illegal gillnets with smaller mesh sizes were introduced into the Gulf of Nicoya, potentially increasing the yield of smaller and juvenile fish and increasing the success of catching small, low-value coastal fish.

6.1.3 Drivers of Ecosystem Change

Our results indicate that the fisheries' exploitation rates mainly drive variations in the catch of some target species (e.g., cephalopods and shrimps). This is an expected result, especially for shrimps which are heavily fished. However, climate variations also substantially affect other species (e.g., corvina, snook, small pelagics, and crabs). The Chlorophyll-a concentration appeared to be the most relevant among tested environmental drivers. It is a biomass proxy for phytoplankton in the water column, which directly affects the bottom of the food chain. Increasing phytoplankton availability enhances zooplankton growth, resulting in higher prey availability for low trophic level species foraging on zooplankton. The associated increased production will then cascade through the entire food web.

Furthermore, the main finding of our study is that the Gulf of Nicoya is influenced by large-scale environmental conditions associated with the ENSO cycle. We observed variations in the catches of some trophic groups (large drums and snook, crabs, morays, and eels) related to the ENSO cycle. El Niño periods are associated with high Sea Surface Temperature (SST) extremes and increased precipitation. During these periods, phytoplankton productivity and zooplankton biomass decrease with bottom-up effects on the entire food chain.

We propose considering the system's environmental indicators and fishing effort regulations to form and develop appropriate management strategies. Thus, a stringent effort control that incorporates a buffer for external effects will help the ecosystem recover and increase its resilience towards climate change in the future.

6.2 Recommended Policy Plans/Management Implications

The results of this study confirm the urgency for the enforcement of regulations to protect the marine coastal resources in the Gulf of Nicoya. From our modelling results, we can state that a severe reduction of shrimp trawling is essential for recovering the Gulf of Nicoya ecosystem. All tested management scenarios that include the goals of increasing ecosystem health and biodiversity suggest a reduction of trawling by 50–100%. Despite the benefits of the ban on trawling, we see from our modelling results that it is insufficient to fully recover biomass levels of overall functional groups, especially of the key species in the system. Thus, two possible alternative management scenarios are proposed. Both call for additional reductions in fishing efforts, mainly by the semi-industrial purse seine fleet and the artisanal longline fleet. We also present the measures necessary to restore functional group biomass levels to the known state in 1993 (baseline model by Alms and Wolff 2019; Wolff et al. 1998), disregarding the feasibility and socio-economic aspects. In this scenario, the extent of ecosystem degradation is highlighted. Necessary measures would involve a ban on trawling and a minimum reduction of gillnet, handline, and purse seine fishing. The scenario would only allow the longline fleet to operate in the Gulf of Nicoya, together with manual collection, leading to an 80% loss of profit and social value.

Before implementing new policy measures, it can be beneficial to strengthen the enforcement and control mechanisms of already-existing regulations. One example of this would be the protection plan in the frame of the “Marine Area of Multiple Use” (Area Marina de Uso Multiple [AMUM]) implemented in the Gulf of Nicoya by the Costa Rican government in 1995 to protect and conserve marine and coastal resources through integrated management. To protect the marine resources in the AMUM, a “Veda” (an annual 3-month period) is implemented in which the Gulf of Nicoya is closed to fisheries. The temporary closure is supposed to overlap with the reproduction peak of commercially important species to support their recovery and aims to reduce fishing efforts in general. We suggest keeping and potentially extending the closure of seasonal spatial fisheries in the Gulf of Nicoya.

One could also tackle the severe problem of illegal mesh sizes in the Gulf of Nicoya. A minimum mesh size of 7.62 cm (3 in) for gillnets was implemented with little success. However, studies show that small meshed gillnets are not necessarily more destructive, as they target a smaller size range of fish, and may instead promote sustained production by allowing

a higher proportion of the spawning biomass to remain in the stock (Wolff et al. 2015). However, the specimens are removed from the gulf at a small size and generate low market values for fishermen. Our assessment results highlight the discrepancy between socioeconomic and conservational goals and the challenge of finding the right balance for policy-making. Increasing profits comes with improving the semi-industrial fishing effort at the expense of the ecosystem health. Efforts reduction of the artisanal fleet in favor of ecosystem health entail the loss of jobs for the fishermen in the communities around the Gulf of Nicoya. Ideally, policies that facilitate alternative income opportunities can be provided, some of which have already been implemented in other coastal regions. A good example of an alternative occupation in Costa Rica is the mangrove restoration. Artisanal fishermen who used to collect bivalves from the mangrove sediments have been alternatively employed to restore mangroves, planting mangrove trees in coastal areas. This occupation provides a livelihood for the fishermen and ecosystem services for fish, shrimp, and society. The same actions have been taken in other marine areas like India, Malaysia, Nigeria, and Panama (BDGP 2007; Porter and Orams 2014; Suman 2019; Woods et al., 2022). However, first of all, the burden of financing the Mangrove replanting program has to be overcome. These projects have been carried out as community initiatives, government-sponsored projects, or large-scale international development assistance programs (Primavera and Esteban 2008).

This study exemplifies how holistic ecosystem models can be used for management advice and how stakeholders can be engaged in policymaking. In eight participatory management workshops developed by our partners (Sánchez-Jiménez et al. 2019; Sánchez Jiménez et al. 2021), stakeholders have been invited, and ecological information about the Gulf of Nicoya ecosystem have been presented using our EwE models. It is a promising approach of participatory management in the Gulf of Nicoya to listen to the advice/perception of the fishermen themselves in stakeholder workshops and interviews. The EwE food web flowcharts are a descriptive illustration of the ecosystem that can serve as a basis for discussion. The suggested effort reduction by the fishermen was thereafter used for our policy search. We believe cooperation with stakeholders is crucial for designing future management plans and benefitting all parties involved. Management authorities can learn from the fishermen's local ecological knowledge, and science has the right tools to develop reliable fisheries management plans that minimize these livelihood-threatening uncertainties. We suggest that stakeholder engagement can be particularly intensified in the entire EwE modelling processes, starting with data collection (here, the discussion with fishermen on relevant biological groups to be considered for the model construction may be a very relevant first step), continuing with model development and model validation, and ending with management simulations. As shown in several cases, stakeholder engagement can be used for management strategy evaluation (Fulton et al. 2011; Koenigstein et al. 2016) as a stakeholder-informed ecosystem modelling approach and is a key aspect in providing good governance, dissolving top-down hierarchies and improving compliance of all actors (Pita et al. 2010; Soma 2003).

6.3 Critical Assessment of the Methodological Approach

Multi-species models per se do not always provide better advice than single-species models, which are still necessary for the management of many fisheries (Hollowed et al. 2000). However, we believe that depending on the research questions and data availability, the Ecopath with Ecosim software offers excellent tools for a detailed analysis of the ecosystem (Christensen and Steenbeek 2024). We successfully integrated available data on trophic interactions and external drivers in the model and were able to identify and separate the role of fishery and the environment in driving ecosystemic changes over time. Thus, multi-species models can potentially offer a greater biological reality concerning the incorporation of trophic interactions and the role of environmental disturbance (Hollowed et al. 2000).

The integrated policy search routine implemented in Ecosim is a valuable tool for identifying optimum levels of fishing efforts for multiple fleets that may each exploit numerous species from an ecosystem. Moreover, this method allows exploring the effects of different management policies on the ecosystem and the socio-economic well-being of the resource users. The spatial modelling tool Ecospace is one choice among others to visualize the efficiency of the seasonal fishing closure, but one of few that allows monitoring ecosystem and fisheries components at once.

This dissertation's different analyses are based on one trophic reference model. The quality and robustness of that model are crucial for every step in the assessment. Therefore, the correctness/quality of the input data is fundamental for the accuracy of the assessment of the results. Thus, all input data need to be examined and data quality validated. Data inserted had to be checked manually for plausibility and were also compared to other models and literature (e.g., diet composition). The Ecopath software uses a pedigree system to calculate the overall data quality fed into the model to support data validation. The calculated EwE Pedigree Index of 0.29, based on the confidence intervals of all input parameters, indicated that the model's level of precision is in the intermediate to the upper range compared to other published Ecopath models worldwide (Morissette 2007). The Ecopath models used in our study were also carefully validated with the prescribed balancing routine of Link (2010), as described in Chapter 2. We adhered to the suggestions of best practices in EwE food web models for ecosystem-based management by Heymans et al. (2016).

In the next step, it is mandatory for good scientific practice to fit the model to time-series data to observe whether the model can predict observed/measured values in a correct manner. This step has been carefully fulfilled for our model, which was calibrated with historical fishing data sets, and comparing the model predictions to reference data allowed for the validation of the model performance (Heymans et al. 2016). We followed the recommendation to adapt the Akaike Information Criterion (AIC) for small sample sizes and used the bias-corrected version AICc instead (Burnham et al. 2011). We believe that the AICc is a good measure for validating the selected model.

We also carefully tried to validate our Ecospace model. According to Christensen and Walters (2024), the fishing effort distribution predicted by the model should be compared to observed spatial effort patterns (vessel monitoring system and/or AIS data) to validate and modify input parameters. Spatial data are rare in our study area, but we used limited resources to validate our

model performance. We used global fishing watch (GFW) data based on data from the vessel monitoring system (VMS) provided by INCOPECA to visually validate the model performance. In addition, we monitored data from fishermen (2013) for a plausibility check of fleet and species distribution. We took these steps to validate our model and ensure the best possible performance. However, we should remember that a model is no more than a good estimation of reality.

A critical point of our study was general data availability. We were working in a data-deficient region, and the data resolution was not comparable to the data-rich regions (Northeast US) or data-rich stocks (Bigeye Tuna) in some other areas of the world (Wiedenmann et al. 2019; Li et al. 2022). Unfortunately, we have been unable to procure fisheries data on effort and landings in recent years. While we reported a substantial reduction of the officially reported fishing effort in the gulf, it can be assumed that unreported catches, fishing without licenses, and illegal fishing during the closed season have occurred during the observational period. These unreported fishing activities could not be quantified, a problem common in many fishing regions worldwide. The availability of recent fisheries data on the Gulf of Nicoya is scarce; thus, simulations with data from the last few years were impossible. The good news is that there is progress. Nowadays, Costa Rica is introducing GPS tracking to some of the vessels to act against illegal fishing. Another great initiative is the fact that Costa Rican Fisheries and Aquaculture Institute has collaborated with Global Fishing Watch, an initiative that uses satellite imagery, vessel GPS data, and machine learning to monitor fishing activity at sea to ensure a fair and sustainable use of the ocean (<https://globalfishingwatch.org/>).

Our study gives insights into the past and present status of the Gulf of Nicoya and the drivers of the observed changes that occurred over time. We assessed the possibilities of combining the commercial interest of artisanal and commercial fisheries with the conservation of the ecosystem and the biomass recovery of different functional groups. We also simulated and visualized in a spatial model the distribution of species biomasses and fishing efforts related to the ban on trawling and marine protected areas. Our study contributes to the Ecosystem Approach to Fisheries and presents numerous analyses possible with the Ecopath with Ecosim software. We also provide an example of an interdisciplinary work that aims to take the first step in fostering participatory management in the Gulf of Nicoya.

BIBLIOGRAPHY

- Abrantes, K. G., A. Barnett, R. Baker, and M. Sheaves. 2015. 'Habitat-Specific Food Webs and Trophic Interactions Supporting Coastal-Dependent Fishery Species: An Australian Case Study'. *Reviews in Fish Biology and Fisheries* 25 (2): 337–63.
- Alam, S., M. Rahman, and A. Al Arif. 2021. 'Challenges and Opportunities in Artisanal Fisheries (Sonadia Island, Bangladesh): The Role of Legislative, Policy and Institutional Frameworks'. *Ocean & Coastal Management* 201 (February):105424. <https://doi.org/10.1016/j.ocecoaman.2020.105424>.
- Allen, K. R. 1971. 'Relation Between Production and Biomass'. *Journal of the Fisheries Research Board of Canada* 28 (10): 1573–81. <https://doi.org/10.1139/f71-236>.
- Allison, E. H, and F. Ellis. 2001. 'The Livelihoods Approach and Management of Small-Scale Fisheries'. *Marine Policy* 25 (5): 377–88. [https://doi.org/10.1016/S0308-597X\(01\)00023-9](https://doi.org/10.1016/S0308-597X(01)00023-9).
- Alms, V., G. Romagnoni, and M. Wolff. 2022. 'Exploration of Fisheries Management Policies in the Gulf of Nicoya (Costa Rica) Using Ecosystem Modelling'. *Ocean & Coastal Management* 230:106349.
- Alms, V., and M. Wolff. 2019. 'The Gulf of Nicoya (Costa Rica) Fisheries System: Two Decades of Change'. *Marine and Coastal Fisheries* 11 (2): 139–61. <https://doi.org/10.1002/mcf2.10050>.
- Alms, V., and M. Wolff. 2020. 'Identification of Drivers of Change of the Gulf of Nicoya Ecosystem (Costa Rica)'. *Frontiers in Marine Science*. <https://doi.org/10.3389/fmars.2020.00707>.
- Alpízar, B. M. 2011. 'La Zona de Crianza Del Golfo de Nicoya'. Documento Técnico 5. Puntarenas, Costa Rica: Instituto Costarricense de Pesca y Acuicultura.
- Alpízar, B. M. 2014. 'Estudio Linea Base de Las Pesquerías Del Golfo de Nicoya, Año 1797'. Documento Técnico 19. Puntarenas, Costa Rica: Instituto Costarricense de Pesca y Acuicultura.
- Alpízar, B. M. 2015. 'Evaluación de Los Recursos Pesqueros y de La Veda 2014 En La Zona Interior Del Golfo de Nicoya'. Documento Técnico 20. Puntarenas, Costa Rica: Instituto Costarricense de Pesca y Acuicultura.
- Alpízar, B. M., H. A. Umaña, Ana R. Vásquez, and Juan Bautista Chavarría. 2013. 'Informe Del Resultado de La Veda 2012, Basado En Muestras Biologicas Realizados En 7 Comunidades Del Golfo de Nicoya'. Documento Técnico 15. Puntarenas, Costa Rica: Instituto Costarricense de Pesca y Acuicultura.
- Alpízar, B. M., and A. R. Vásquez. 2010. 'Evaluación de Los Recursos Pesqueros Post-Veda 2009 En El Interior Del Golfo de Nicoya'. Documento Técnico 2. Puntarenas, Costa Rica: Instituto Costarricense de Pesca y Acuicultura.

- Alpízar, B. M. 2012. 'Estimación de La Talla de Primera Madurez Sexual Criterio L50% TPM de La Corvina Reina (*Cynoscion Albus*, Perciformes: Sciaenidae) Bajo Condiciones de Sobreexplotación de Su Población En El Golfo de Nicoya, Costa Rica'. Documento Técnico 11. Puntarenas, Costa Rica: Instituto Costarricense de Pesca y Acuicultura.
- Andersen, K. H., and M. Pedersen. 2010. 'Damped Trophic Cascades Driven by Fishing in Model Marine Ecosystems'. *Proceedings of the Royal Society B: Biological Sciences* 277 (1682): 795–802. <https://doi.org/10.1098/rspb.2009.1512>.
- Anning, A., I. S. Egyir, G. T. M. Kwadzo, and J. K. M. Kuwornu. 2012. 'Willingness of Marine Artisanal Fishermen to Integrate Aquaculture in Enterprise Mix: Evidence from Ghana'. *Journal of Economics and Sustainable Development* 3 (1): 56–67.
- Arauz, R. 2000. 'Impact of High Seas Longline Fishery Operations on Shark and Sea Turtle Populations in the Exclusive Economic Zone of Costa Rica'. Forest Knolls, California: Turtle Island Restoration Network, Sea Turtle Restoration Project.
- Armengol, L. d', M. P. Castillo, I. Ruiz-Mallén, and E. Corbera. 2018. 'A Systematic Review of Co-Managed Small-Scale Fisheries: Social Diversity and Adaptive Management Improve Outcomes'. *Global Environmental Change* 52 (September):212–25. <https://doi.org/10.1016/j.gloenvcha.2018.07.009>.
- Arreguín-Sánchez, F., and L. E. Calderón-Aguilera. 2002. 'Evaluating Harvesting Strategies for Fisheries in the Central Gulf of California Ecosystem'. *Fisheries Centre Research Reports* 10 (2): 135.
- Arthur, R. Ian. 2020. 'Small-Scale Fisheries Management and the Problem of Open Access'. *Marine Policy* 115:103867.
- Bailey, C. 1997. 'Lessons from Indonesia's 1980 Trawler Ban'. *Marine Policy* 21 (3): 225–35.
- Bakun, A. 1999. 'The Pacific Central American Coastal LME'. Large Marine Ecosystems of the Pacific Rim, Assessment, Sustainability and Management.
- Batista, V. S., N. N. Fabr  , A. C. M. Malhado, and R. J. Ladle. 2014. 'Tropical Artisanal Coastal Fisheries: Challenges and Future Directions'. *Reviews in Fisheries Science & Aquaculture* 22 (1): 1–15. <https://doi.org/10.1080/10641262.2013.822463>.
- Baum, J. K., and B. Worm. 2009. 'Cascading Top-down Effects of Changing Oceanic Predator Abundances'. *Journal of Animal Ecology* 78 (4): 699–714. <https://doi.org/10.1111/j.1365-2656.2009.01531.x>.
- BDCP. 2007. 'Implementation of a Public Awareness and Public Participation Programme in Relation to Mangrove Depletion And Proposed Re-Forestation in Coastal Nigeria'. Feasibility Studies Report Mangrove and Nypa. Prepared for the United Nations Industrial Development Organization.

- Beddington, J. R., G. P. Kirkwood, J. F. Caddy, and J. C. Seijo. 2005. 'This Is More Difficult than We Thought! The Responsibility of Scientists, Managers and Stakeholders to Mitigate the Unsustainability of Marine Fisheries'. *Philosophical Transactions of the Royal Society B: Biological Sciences* 360 (1453): 59–75.
<https://doi.org/10.1098/rstb.2004.1567>.
- Bell, J. D., J. E. Johnson, and A. J. Hobday. 2011. 'Vulnerability of Tropical Pacific Fisheries and Aquaculture to Climate Change'. Nouméa, New Caledonia: Secretariat of the Pacific Community.
- Bennett, N. J., E. M. Finkbeiner, N. C. Ban, D. Belhabib, S. D. Jupiter, J. N. Kittinger, S. Mangubhai, J. Scholtens, D. Gill, and P. Christie. 2020. 'The COVID-19 Pandemic, Small-Scale Fisheries and Coastal Fishing Communities'. *Coastal Management* 48 (4): 336–47. <https://doi.org/10.1080/08920753.2020.1766937>.
- Bethel, M. B., L. F. Brien, M. M. Esposito, C. T. Miller, H. S. Buras, S. B. Laska, R. Philippe, K. J. Peterson, and C. P. Richards. 2014. 'Sci-TEK: A GIS-Based Multidisciplinary Method for Incorporating Traditional Ecological Knowledge into Louisiana's Coastal Restoration Decision-Making Processes'. *Journal of Coastal Research* 30 (5): 1081–99.
<https://doi.org/10.2112/JCOASTRES-D-13-00214.1>.
- Beverton, R. J. H., and S. J. Holt. 1957. On the Dynamics of Exploited Fish Populations. Reprint 1993. London: Chapman and Hall.
- Bianchi, G. 2008. 'The Concept of the Ecosystem Approach to Fisheries in FAO'. The *Ecosystem Approach to Fisheries*, 20–38.
- Bigelow, H. Bryant, and W. C. Schroeder. 2002. 'Bigelow and Schroeder's Fishes of the Gulf of Maine'.
- BIOMARCC-SINAC-GIZ. 2013a. Análisis de Vulnerabilidad de Las Zonas Oceánicas y Marino-Costas de Costa Rica Frente al Cambio Climático. San José, Costa Rica.
- BIOMARCC-SINAC-GIZ. 2013b. Evaluación de Las Pesquerías En La Zona Media y Externa Del Golfe de Nicoya En Costa Rica. Vol. 10. Serie Técnica. San José, Costa Rica.
- Blaber, S. J. M. 2002. '“Fish in Hot Water”: The Challenges Facing Fish and Fisheries Research in Tropical Estuaries'. *Journal of Fish Biology* 61:1–20.
- Blaber, S. J. M., D. P. Cyrus, J.-J. Albaret, C. V. Ching, J. W. Day, M. Elliott, M. S. Fonseca, et al. 2000. 'Effects of Fishing on the Structure and Functioning of Estuarine and Nearshore Ecosystems'. *ICES Journal of Marine Science* 57 (3): 590–602.
<https://doi.org/10.1006/jmsc.2000.0723>.
- Blaber, S. J. M. 2008. Tropical Estuarine Fishes: Ecology, Exploitation and Conservation. John Wiley & Sons.
- Bolaños, M. 2005. 'Characterization of the Costa Rican Semi-Industrial Coastal Shrimp Trawling Fishery and the Small-Scale Artisanal Shrimp Trawling Fishery'. San José, Costa Rica: Food and Agriculture Organization of the United Nations, Costa Rica Office.

- Brillo, B. B. C., H. C. Jalotjot, C. C. Cervantes, and A. C. Rola. 2019. 'Impact on Income and Livelihood of Fisheries Workers: Closed Fishing Season Policy for Sardines in Zamboanga Peninsula, Philippines'. *Journal of Coastal Conservation* 23:1057–67.
- Briones-Fourzán, P., and E. Lozano-Álvarez. 2013. 'Essential Habitats for Panulirus Spiny Lobsters'. *Lobsters: Biology, Management, Aquaculture and Fisheries*, 186–220.
- Brown, C. J., S. A., and P. J. Mumby. 2015. 'Minimizing the Short-term Impacts of Marine Reserves on Fisheries While Meeting Long-term Goals for Recovery'. *Conservation Letters* 8 (3): 180–89. <https://doi.org/10.1111/conl.12124>.
- Brugnoli-Olivera, E., E. Díaz-Ferguson, Mariana Delfino-Machin, Alvaro Morales-Ramírez, and Arturo Dominici Arosemena. 2004. 'Composition of the Zooplankton Community, with Emphasis in Copepods, in Punta Morales, Golfo de Nicoya, Costa Rica'. *Revista de Biología Tropical* 52 (4): 897–902.
- Brugnoli-Olivera, E., and A. Morales-Ramírez. 2008. 'Trophic Planktonic Dynamics in a Tropical Estuary, Gulf of Nicoya, Pacific Coast of Costa Rica during El Niño 1997 Event'. *Revista de Biología Marina y Oceanografía* 43 (1): 75–89.
- Burnham, K. P., and D. R. Anderson. 2004. 'Multimodel Inference: Understanding AIC and BIC in Model Selection'. *Sociological Methods & Research* 33 (2): 261–304.
- Burnham, K. P., D. R. Anderson, and K. P. Huyvaert. 2011. 'AIC Model Selection and Multimodel Inference in Behavioral Ecology: Some Background, Observations, and Comparisons'. *Behavioral Ecology and Sociobiology* 65:23–35.
- Cai, W., S. Borlace, M. Lengaigne, Pe. Van Rensch, M. Collins, G. Vecchi, A. Timmermann, A. Santoso, M. J. McPhaden, and L. Wu. 2014. 'Increasing Frequency of Extreme El Niño Events Due to Greenhouse Warming'. *Nature Climate Change* 4 (2): 111–16.
- Carlsson, L., and F. Berkes. 2005. 'Co-Management: Concepts and Methodological Implications'. *Journal of Environmental Management* 75 (1): 65–76.
- Carrillo, I. C., S. Partelow, R. Madrigal-Ballester, A. Schlüter, and I. Gutierrez-Montes. 2019. 'Do Responsible Fishing Areas Work? Comparing Collective Action Challenges in Three Small-Scale Fisheries in Costa Rica'. *International Journal of the Commons* 13 (1).
- Cavalli, R. O., W. Wasielesky Jr, S. Peixoto, L. Henrique da Silva Poersch, M. H. Silva Santos, and R. Borda Soares. 2008. 'Shrimp Farming as an Alternative to Artisanal Fishermen Communities: The Case of Patos Lagoon, Brazil'. *Brazilian Archives of Biology and Technology* 51 (5): 991–1001.
- Charles, A. T., and A. Herrera-Ulloa. 1994. 'Development and Diversification: Sustainability Strategies for a Costa Rican Fishing Cooperative'.
- Cheung, W. W. L., J. L. Sarmiento, J. Dunne, T. L. Frölicher, V. W. Y. Lam, M. L. Deng Palomares, R. Watson, and D. Pauly. 2013. 'Shrinking of Fishes Exacerbates Impacts of Global Ocean Changes on Marine Ecosystems'. *Nature Climate Change* 3 (3): 254–58.

- Cheung, W. W. L., and U. R. Sumaila. 2008. 'Trade-Offs between Conservation and Socio-Economic Objectives in Managing a Tropical Marine Ecosystem'. Special Section: Integrated Hydro-Economic Modelling for Effective and Sustainable Water Management 66 (1): 193–210. <https://doi.org/10.1016/j.ecolecon.2007.09.001>.
- Chong, K.-C., A. Dwiponggo, S. Ilyas, and P. Martosubroto. 1987. 'Some Experiences and Highlights of the Indonesian Trawl Ban: Bioeconomics and Socioeconomics'. In , 16–26.
- Christensen, V., Z. Ferdaña, and . Steenbeek. 2009. 'Spatial Optimization of Protected Area Placement Incorporating Ecological, Social and Economical Criteria'. *Ecological Modelling* 220 (19): 2583–93. <https://doi.org/10.1016/j.ecolmodel.2009.06.029>.
- Christensen, V., S. Lai, M. L. D. Palomares, D. Zeller, and D. Pauly. 2011. 'The State of Biodiversity and Fisheries in Regional Seas'.
- Christensen, V., and J. Steenbeek. 2024. User Guide for Ecopath with Ecosim (EwE). Vancouver, BC: The University of British Columbia.
- Christensen, V., and C. J Walters. 2004a. 'Ecopath with Ecosim: Methods, Capabilities and Limitations'. *Ecological Modelling* 172 (2–4): 109–39.
- Christensen, V., and C. J. Walters. 2004b. 'Trade-Offs in Ecosystem-Scale Optimization of Fisheries Management Policies'. *Bulletin of Marine Science* 74 (3): 549–62.
- Christensen, V., and C. J. Walters. 2024. 'Introduction to Ecospace'. Ecosystem Modelling with EwE.
- Christensen, V., C. J Walters, and Daniel Pauly. 2005. 'Ecopath with Ecosim: A User's Guide'. Fisheries Centre, University of British Columbia, Vancouver 154.
- Clark, D. A., S. Brown, D. W. Kicklighter, J. Q. Chambers, J. R. Thomlinson, J. Ni, and E. A. Holland. 2001. 'Net Primary Production in Tropical Forests: An Evaluation and Synthesis of Existing Field Data'. *Ecological Applications* 11 (2): 371–84.
- Clarke, T. J., D. M. Bailey, and P. J. Wright. 2015. 'Evaluating the Effectiveness of a Seasonal Spawning Area Closure'. *ICES Journal of Marine Science* 72 (9): 2627–37. <https://doi.org/10.1093/icesjms/fsv144>.
- Clarke, T. M., M. Espinoza, R. Ahrens, and I. S. Wehrtmann. 2016. 'Elasmobranch Bycatch Associated with the Shrimp Trawl Fishery off the Pacific Coast of Costa Rica, Central America.' *Fishery Bulletin* 114 (1).
- Cochrane, K.L. 2002. 'The Use of Ecosystem Models to Investigate Ecosystem-Based Management Strategies for Capture Fisheries: Introduction'. The Use of Ecosystem Models to Investigate Multispecies Management Strategies for Capture Fisheries 10:5–10.
- Coglan, L., and S. Pascoe. 2014. 'Fisheries Management and Fisher Discount Rates'.
- Coll, M., A. Bundy, and L. J. Shannon. 2009. 'Computers in Fisheries Research'. Comput. Fish. Res. Netherlands: Dordrecht: Springer, 37–49.

- Coll, M., J. Steenbeek, J. Sole, I. Palomera, and V. Christensen. 2016. 'Modelling the Cumulative Spatial–Temporal Effects of Environmental Drivers and Fishing in a NW Mediterranean Marine Ecosystem'. *Ecopath 30 Years – Modelling Ecosystem Dynamics: Beyond Boundaries with EwE 331* (July):100–114. <https://doi.org/10.1016/j.ecolmodel.2016.03.020>.
- Collins, M., S.-I. An, W. Cai, A. Ganachaud, E. Guilyardi, F.-F. Jin, M. Jochum, M. Lengaigne, S. Power, and A. Timmermann. 2010. 'The Impact of Global Warming on the Tropical Pacific Ocean and El Niño'. *Nature Geoscience* 3 (6): 391–97.
- Colwell, J. M. Novak, and M. Axelrod. 2017. 'Socio-Economic Impacts of a Closed Fishing Season on Resource-Dependent Stakeholders in Tamil Nadu, India: Differences in Income and Expenditure Effects by Occupational Group'. *Marine Policy* 77:182–90.
- Córdoba-Muñoz, R. 1998. 'Productividad Primaria En La Columna de Agua, Un Sistema de Manglar En El Golfo de Nicoya, Costa Rica'. *Revista de Biología Tropical* 46:257–62.
- Cornick, J., J. Jimenez, and M. Román. 2014. 'Public–Private Collaboration on Productive Development Policies in Costa Rica'. IDB Working Paper Series. Washington, D.C.: Inter-American Development Bank.
- Corrales, L. A. V., and C. M. Vindas. 2014. 'Evaluation of Escherichia Coli Concentrations in Crassostrea Gigas and Seawater in Two Oyster Growing Areas in the Gulf of Nicoya, Costa Rica'. *Revista Ciencias Marinas y Costeras* 6:155–66.
- Corredor Mariano del Pacífico Este Tropical - CMAR, ed. 2019. 'Plan de Acción Corredor Marino Del Pacífico Este Tropical (CMAR) 2019-2024'.
- Cota-Nieto J. J., B. Erisman, O. Aburto-Oropeza, M. Moreno-Báez, G. Hinojosa-Arango, and A. F. Johnson. 2018. 'Participatory Management in a -Scale Coastal Fishery—Punta Abreojos, Pacific Coast of Baja California Sur, Mexico'. *Regional Studies in Marine Science* 18 (February):68–79. <https://doi.org/10.1016/j.rsma.2017.12.014>.
- Côté, I. M., and E. S. Darling. 2010. 'Rethinking Ecosystem Resilience in the Face of Climate Change'. *PLoS Biology* 8 (7): e1000438.
- Coyle, V., R. Duca, C. Jones, and K. Roopcharan. 2015. 'Analyzing and Improving the Fishing Supply Chain in Costa Rica'. Fishing Supply Chain Cost Analysis. Worcester Polytechnic Institute, La Fundación MarViva, Worcester, Massachusetts.
- Crawford, R. J. M., and P. A. Shelton. 1978. 'Pelagic Fish and Seabird Interrelationships off the Coasts of South West and South Africa'. *Biological Conservation* 14 (2): 85–109.
- Cruz, I. A., M. Daniela, and P. G. Alvarado; 2022. 'Critical Routes: Women Fishers, Shrimp Peelers and Mollusk Workers in Costa Rica Identify Priority Areas for the Recognition and Formalisation of Their Work', no. Central America: Costa Rica.
- Cury, P., A. Bakun, R. J. M. Crawford, A. Jarre, R. A. Quinones, L. J. Shannon, and H. M. Verheye. 2000. 'Small Pelagics in Upwelling Systems: Patterns of Interaction and Structural Changes in "Wasp-Waist" Ecosystems'. *ICES Journal of Marine Science* 57 (3): 603–18.

- Cury, P. M., I. L. Boyd, S. Bonhommeau, T. Anker-Nilssen, R. J. M. Crawford, R. W. Furness, J. A. Mills, E. J. Murphy, H. Österblom, and M. Paleczny. 2011. 'Global Seabird Response to Forage Fish Depletion—One-Third for the Birds'. *Science* 334 (6063): 1703–6.
- Del Solar, A., G. Romagnoni, and M. Wolff. 2024. 'The Masked Influence of Remote ENSO-Related Drivers on the Trophic Dynamics of Independencia Bay, Peru: Insights for Management from Time-Series Analysis and Food-Web Modelling'. *Ocean & Coastal Management* 251:107047.
- Dimarchopoulou, D., I. Keramidas, K. Tsagarakis, V. Markantonatou, G. Halouani, and A. C. Tsikliras. 2024. 'Spatiotemporal Fishing Effort Simulations and Restriction Scenarios in Thermaikos Gulf, Greece (Northeastern Mediterranean Sea)'. *Ocean & Coastal Management* 247 (January):106914. <https://doi.org/10.1016/j.ocecoaman.2023.106914>.
- Duarte, L. O., and C. B. García. 2004. 'Trophic Role of Small Pelagic Fishes in a Tropical Upwelling Ecosystem'. *Ecological Modelling* 172 (2–4): 323–38. <https://doi.org/10.1016/j.ecolmodel.2003.09.014>.
- Enright, S. R., R. Meneses-Orellana, and I. Keith. 2021. 'The Eastern Tropical Pacific Marine Corridor (CMAR): The Emergence of a Voluntary Regional Cooperation Mechanism for the Conservation and Sustainable Use of Marine Biodiversity Within a Fragmented Regional Ocean Governance Landscape'. *Frontiers in Marine Science* 8:569. <https://doi.org/10.3389/fmars.2021.674825>.
- Evans, L. S. 2010. 'Ecological Knowledge Interactions in Marine Governance in Kenya'. *Ocean & Coastal Management* 53 (4): 180–91. <https://doi.org/10.1016/j.ocecoaman.2010.01.009>.
- FAO (Food and Agriculture Organization of the United Nations). 2002. 'Report of the Reykjavik Conference on Responsible Fisheries in the Marine Ecosystem, Reykjavik, Iceland, 1-4 October 2001'. In , FAO Fisheries Report No. 658:128.
- FAO (Food and Agriculture Organization of the United Nations). 2012. EAF Toolbox: The Ecosystem Approach to Fisheries. Food and Agriculture Organization of the United Nations.
- FAO (Food and Agriculture Organization of the United Nations). 2020. 'The State of World Fisheries and Aquaculture 2020. Sustainability in Action.', Rome, . <https://doi.org/10.4060/ca9229en>.
- FAO (Food and Agriculture Organization of the United Nations). 2007. 'Mangroves of North and Central America 1980–2005: Country Reports'. Forest Resources Assessment Working Paper 137.
- Feary, D.A., A.M. Fowler, and T.J. Ward. 2014. 'Developing a Rapid Method for Undertaking the World Ocean Assessment in Data-Poor Regions – A Case Study Using the South China Sea Large Marine Ecosystem'. *Ocean & Coastal Management* 95 (July):129–37. <https://doi.org/10.1016/j.ocecoaman.2014.04.006>.
- Fiedler, P. C. 2002. 'Environmental Change in the Eastern Tropical Pacific Ocean: Review of ENSO and Decadal Variability'. *Marine Ecology Progress Series* 244:265–83.

- Forrest, R. E., M. Savina, E. A. Fulton, and T. J. Pitcher. 2015. 'Do Marine Ecosystem Models Give Consistent Policy Evaluations? A Comparison of Atlantis and Ecosim'. *Fisheries Research* 167:293–312. <https://doi.org/10.1016/j.fishres.2015.03.010>.
- Frederiksen, M., M. Edwards, A. J. Richardson, N. C. Halliday, and S. Wanless. 2006. 'From Plankton to Top Predators: Bottom-up Control of a Marine Food Web across Four Trophic Levels'. *Journal of Animal Ecology* 75 (6): 1259–68.
- Froese, R., C. Garilao, H. Winker, G. Coro, N. Demirel, A. Tsikliras, D. Dimarchopoulou, G. Scarcella, and A. Sampang-Reyes. 2016. 'Exploitation and Status of European Stocks'.
- Froese, R., H. Winker, G. Coro, N. Demirel, A. C. Tsikliras, D. Dimarchopoulou, G. Scarcella, M. Quaas, and N. Matz-Lück. 2018. 'Status and Rebuilding of European Fisheries'. *Marine Policy* 93:159–70.
- Fulton, E. A., T. Jones, F. Boschetti, M. Sporcic, W. De la Mare, G. J. Syme, P. Dzidic, R. Gorton, L. R. Little, and G. Dambacher. 2011. 'A Multi-Model Approach to Stakeholder Engagement in Complex Environmental Problems'.
- Fulton, E. A., J. S. Link, I. C. Kaplan, M. Savina-Rolland, P. Johnson, C. Ainsworth, P. Horne, R. Gorton, R. J. Gamble, and A. D. M. Smith. 2011. 'Lessons in Modelling and Management of Marine Ecosystems: The Atlantis Experience'. *Fish and Fisheries* 12 (2): 171–88.
- García, C. B., and L. O. Duarte. 2002. 'Consumption to Biomass (Q/B) Ratio and Estimates of Q/B-Predictor Parameters for Caribbean Fishes'. *Naga* 25 (2): 19–31.
- García L. O., A. J. 2014. 'An Institutional, Socio-Economic, and Legal Analysis of Fisheries Co-Management and Regulation in the Gulf of Nicoya, Costa Rica'. Master's Thesis, Miami, USA: Florida International University.
- Gasalla, M. A., A. R. Rodrigues, L. F. A. Duarte, and U. R. Sumaila. 2010. 'A Comparative Multi-Fleet Analysis of Socio-Economic Indicators for Fishery Management in SE Brazil'. *Progress in Oceanography* 87 (1–4): 304–19.
- Gascuel, D., S. Guénette, and D. Pauly. 2011. 'The Trophic-Level-Based Ecosystem Modelling Approach: Theoretical Overview and Practical Uses'. *ICES Journal of Marine Science* 68 (7): 1403–16. <https://doi.org/10.1093/icesjms/fsr062>.
- Geers, T. M., E. K. Pikitch, and M. G. Frisk. 2016. 'An Original Model of the Northern Gulf of Mexico Using Ecopath with Ecosim and Its Implications for the Effects of Fishing on Ecosystem Structure and Maturity'. *The Gulf of Mexico Ecosystem - before, during and after the Macondo Blowout* 129 (July):319–31. <https://doi.org/10.1016/j.dsr2.2014.01.009>.
- Giarrizzo, T., D. Ferraz, and V. Isaac. 2013. 'Estimates of Annual Food Consumption/Biomass Ratio (Q/B) from the Fish Fauna of a Mangrove Estuary in North Brazil'. *Biota Amazonia* 3 (2): 149–54.
- Gocke, K., J. Cortés, and M. Murillo. 2002. 'Planktonic Primary Production in a Tidally Influenced Mangrove Forest on the Pacific Coast of Costa Rica'. *Revista de Biología Tropical* 49:279–88.

- Godinot, O., and V. Allain. 2003. 'A Preliminary Ecopath Model of the Warm Pool Pelagic Ecosystem'. In Working Paper BBRG5. Mooloolaba, Australia.
- Griffiths, S. P., R. J. Olson, and G. M. Watters. 2013. 'Complex Wasp-Waist Regulation of Pelagic Ecosystems in the Pacific Ocean'. *Reviews in Fish Biology and Fisheries* 23:459–75.
- Gu  nette, S., V. Christensen, and D. Pauly. 2001. 'Fisheries Impacts on North Atlantic Ecosystems: Models and Analyses'.
- Guti  rrez, M., G. Swartzman, A. Bertrand, and S. Bertrand. 2007. 'Anchovy (*Engraulis Ringens*) and Sardine (*Sardinops Sagax*) Spatial Dynamics and Aggregation Patterns in the Humboldt Current Ecosystem, Peru, from 1983–2003'. *Fisheries Oceanography* 16 (2): 155–68.
- Guzman, H. M., R. Cipriani, A. J. Vega, M. Lopez, and J. M. Mair. 2008. 'Population Assessment of the Pacific Green Spiny Lobster *Panulirus Gracilis* in Pacific Panama'. *Journal of Shellfish Research* 27 (4): 907–15.
- Guzm  n-Mora, A. G., and H. Molina-Ure  a. 2008. 'La Pesquer  a Artesanal de Peces En Golfo Dulce, Costa Rica'.
- Hare, J. A. 2020. 'Ten Lessons from the Frontlines of Science in Support of Fisheries Management'. *ICES Journal of Marine Science* 77 (3): 870–77.
- Hays, G. C., A. J. Richardson, and C. Robinson. 2005. 'Climate Change and Marine Plankton'. *Trends in Ecology & Evolution* 20 (6): 337–44.
- Hegde, M. R., V. P. Padate, and C. U. Rivonker. 2016. 'Seasonal Variations in Habitat Selection and Catch Trends of Sciaenids (Family: Sciaenidae) from the Tropical Waters off Goa, West Coast of India'.
- Heileman, S. 2012. 'XIV-48 Pacific Central-American Coastal: LME# 11'. In Large Marine Ecosystems of the World. Silver Spring, Maryland: National Oceanic and Atmospheric Administration.
- Hervann, P.-Y., D. Gascuel, A. Gr  ss, J.-N. Druon, D. Kopp, I. Perez, C. Piroddi, and M. Robert. 2020. 'The Celtic Sea Through Time and Space: Ecosystem Modeling to Unravel Fishing and Climate Change Impacts on Food-Web Structure and Dynamics'. *Frontiers in Marine Science* 7. <https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2020.578717>.
- Herrera, M., T. M. Clarke, B. Naranjo-Elizondo, M. Espinoza, and I. S. Wehrtmann. 2016. 'Size at Maturity of the Pacific Bearded Brotula (Ophidiidae: Brotula Clarkae): A Commercially Exploited Species in the Pacific of Costa Rica'. *Latin American Journal of Aquatic Research* 44 (3): 657–61. <https://doi.org/DOI: 10.3856/vol44-issue3-fulltext-25>.
- Heymans, J. J., A. Bundy, V. Christensen, M. Coll, K. De Mutsert, E. A. Fulton, Ch. Piroddi, Y.-J Shin, J. Steenbeek, and M. Travers-Trolet. 2020. 'The Ocean Decade: A True Ecosystem Modeling Challenge'. *Frontiers in Marine Science* 7:766. <https://doi.org/10.3389/fmars.2020.554573>.

- Heymans, J. J., M. Coll, S. Libralato, L. Morissette, and V. Christensen. 2014. 'Global Patterns in Ecological Indicators of Marine Food Webs: A Modelling Approach'. *PloS One* 9 (4): e95845.
- Heymans, J. J., M. Coll, J. S. Link, S. Mackinson, J. Steenbeek, Carl Walters, and Villy Christensen. 2016. 'Best Practice in Ecopath with Ecosim Food-Web Models for Ecosystem-Based Management'. *Ecological Modelling* 331:173–84.
- Hidalgo, M., T. Rouyer, J. C. Molinero, E. Massutí, J. Moranta, B. Guijarro, and N. C. Stenseth. 2011. 'Synergistic Effects of Fishing-Induced Demographic Changes and Climate Variation on Fish Population Dynamics'. *Marine Ecology Progress Series* 426:1–12.
- Hinke, J. T., I. C. Kaplan, K. Aydin, G. M. Watters, R. J. Olson, and J. F. Kitchell. 2004. 'Visualizing the Food-Web Effects of Fishing for Tunas in the Pacific Ocean'. *Ecology and Society* 9 (1). <http://www.jstor.org/stable/26267649>.
- Hollowed, A. B., N. Bax, R. Beamish, J. Collie, M. Fogarty, P. Livingston, J. Pope, and J. C. Rice. 2000. 'Are Multispecies Models an Improvement on Single-Species Models for Measuring Fishing Impacts on Marine Ecosystems?' *ICES Journal of Marine Science* 57 (3): 707–19. <https://doi.org/10.1006/jmsc.2000.0734>.
- Horta e Costa, B., M. H. Guimarães, M. Rangel, A. Ressurreição, P. Monteiro, F. liveira, L. Bentes, N. S. Henriques, I. Sousa, and S. Alexandre. 2022. 'Co-Design of a Marine Protected Area Zoning and the Lessons Learned from It'. *Frontiers in Marine Science* 9:969234.
- Hoyt, E., and M. Iñíguez. 2008. 'The State of Whale Watching in Latin America'. Chippenham, UK: Whale and Dolphin Conservation Society.
- Hunnam, K. 2021. 'The Biology and Ecology of Tropical Marine Sardines and Herrings in Indo-West Pacific Fisheries: A Review'. *Reviews in Fish Biology and Fisheries* 31 (3): 449–84.
- Hutchison, J., Mark S., and P. Zu Ermgassen. 2014. 'The Role of Mangroves in Fisheries Enhancement'. *The Nature Conservancy and Wetlands International* 54:434.
- Islam, M. M., A. Begum, S. M. A. Rahman, and H. Ullah. 2021. 'Seasonal Fishery Closure in the Northern Bay of Bengal Causes Immediate but Contrasting Ecological and Socioeconomic Impacts'. *Frontiers in Marine Science* 8:704056.
- Jacinto, E. R., and R. S. Pomeroy. 2011. 'Developing Markets for Small-Scale Fisheries: Utilizing the Value Chain Approach'. *Small-Scale Fisheries Management: Frameworks and Approaches for the Developing World*, 160–77.
- Jennings, S., and M. J. Kaiser. 1998. 'The Effects of Fishing on Marine Ecosystems'. In *Advances in Marine Biology*, 34:201–352. Elsevier.

- Steenbeek, J., and M. Coll. 2024. 'MPA Dynamics Plug-in - a Brief Users' Guide Ecopath with Ecosim Software Guides Version 1'. Ecopath International Initiative. Spain.
- Kaiser, M. J., J. S. Collie, S. J. Hall, S. Jennings, and I. R. Poiner. 2002. 'Modification of Marine Habitats by Trawling Activities: Prognosis and Solutions'. *Fish and Fisheries* 3 (2): 114–36.
- Kalikoski, D. C., S. Jentoft, P. McConney, and S. Siar. 2019. 'Empowering Small-Scale Fishers to Eradicate Rural Poverty'. *Maritime Studies* 18:121–25.
- Kelleher, K., L. Westlund, E. Hoshino, D. Mills, R. Willmann, G. de Graaf, and R. Brummett. 2012. Hidden Harvest: The Global Contribution of Capture Fisheries. Worldbank; WorldFish.
- Kempton, R. A. 2002. 'Species Diversity'. In A. H. El-Shaarawi and W. W. Piegorsch. Eds. *Encyclopedia of Environmetrics*, 4th ed., 2086–89. Chichester.
- Kittinger, J. N., E. M. Finkbeiner, N. C. Ban, K. Broad, M. H. Carr, J. E. Cinner, S. Gelcich, M. L. Cornwell, J. Z. Koehn, and X. Basurto. 2013. 'Emerging Frontiers in Social-Ecological Systems Research for Sustainability of Small-Scale Fisheries'. *Current Opinion in Environmental Sustainability* 5 (3–4): 352–57.
- Kleisner, K., D. Pauly, V. Christensen, S. Lai, M. L. D. Palomares, and D. Zeller. 2011. 'The State of Biodiversity and Fisheries in Regional Seas'.
- Koenigstein, S., M. Ruth, and S. Gößling-Reisemann. 2016. 'Stakeholder-Informed Ecosystem Modeling of Ocean Warming and Acidification Impacts in the Barents Sea Region'. *Frontiers in Marine Science* 3:93.
- Lam, V. W. Y., E. H. Allison, J. D. Bell, J. Blythe, W. W. L. Cheung, T. L. Frölicher, M. A. Gasalla, and U. Rashid Sumaila. 2020. 'Climate Change, Tropical Fisheries and Prospects for Sustainable Development'. *Nature Reviews Earth & Environment* 1 (9): 440–54. <https://doi.org/10.1038/s43017-020-0071-9>.
- Last, P. R., W. T. White, D. C. Gledhill, A. J. Hobday, R. Brown, G. J. Edgar, and G. Pecl. 2011. 'Long-term Shifts in Abundance and Distribution of a Temperate Fish Fauna: A Response to Climate Change and Fishing Practices'. *Global Ecology and Biogeography* 20 (1): 58–72. <https://doi.org/10.1111/j.1466-8238.2010.00575.x>.
- Lea, B., P. Béarez, and J. McCosker. 2010. 'Brotula Clarkae'. In The IUCN Red List of Threatened Species. IUCN.
- Leontief, W. W. 1951. The Structure of American Economy, 1919–1939: An Empirical Application of Equilibrium Analysis - Enlarged. Oxford, UK: Oxford University Press.
- Lercari, D., and F. Arreguín-Sánchez. 2008. 'An Ecosystem Modelling Approach to Deriving Viable Harvest Strategies for Multispecies Management of the Northern Gulf Of...'
- Leroy, B., T. Peatman, T. Usu, S. Caillot, B. Moore, A. Williams, and S. Nicol. 2016. 'Interactions between Artisanal and Industrial Tuna Fisheries: Insights from a Decade of Tagging Experiments'. *Marine Policy* 65 (March):11–19. <https://doi.org/10.1016/j.marpol.2015.12.001>.

- Lery, J.-M., J. Prado, and U. Tietze. 1999. Economic Viability of Marine Capture Fisheries: Findings of a Global Study and an Interregional Workshop. Vol. 377. Food & Agriculture Org.
- Li, R. V., L. M. Barquero, and D. M. Fernández. 2013. 'Mapping Mangrove Species Composition with Rapideye Satellite Images in the Nicoya Gulf, Costa Rica: How Far Can We Go?'
- Li, Y., J. Zhu, X. Dai, D. Fu, and Y. Chen. 2022. 'Using Data-Limited Approaches to Assess Data-Rich Indian Ocean Bigeye Tuna: Data Quantity Evaluation and Critical Information for Management Implications'. *Acta Oceanologica Sinica* 41 (3): 11–23. <https://doi.org/10.1007/s13131-021-1933-9>.
- Libralato, S., V. Christensen, and D. Pauly. 2006. 'A Method for Identifying Keystone Species in Food Web Models'. *Ecological Modelling* 195 (3–4): 153–71. <https://doi.org/10.1016/j.ecolmodel.2005.11.029>.
- Link, J. S. 2010. 'Adding Rigor to Ecological Network Models by Evaluating a Set of Pre-Balance Diagnostics: A Plea for PREBAL'. *Ecological Modelling* 221 (12): 1580–91. <https://doi.org/10.1016/j.ecolmodel.2010.03.012>.
- Lloret, J., I. G. Cowx, H. Cabral, M. Castro, T. Font, J. M. S. Gonçalves, A. Gordo. 2018. 'Small-Scale Coastal Fisheries in European Seas Are Not What They Were: Ecological, Social and Economic Changes'. *Marine Policy* 98 (December): 176–86. <https://doi.org/10.1016/j.marpol.2016.11.007>.
- Lopez-Peralta, R. H., and C. A. T. Arcila. 2002. 'Diet Composition of Fish Species from the Southern Continental Shelf of Colombia'.
- Lozano, A., J. García, and J. T. Heinen. 2016a. 'Identifying Drivers of Collective Action for the Co-Management of Coastal Marine Fisheries in the Gulf of Nicoya, Costa Rica'. *Environmental Management* 57 (4): 759–69.
- Lozano, A., and J. T. Heinen. 2016b. 'Property Relations and the Co-Management of Small-Scale Fisheries in Costa Rica: Lessons from Marine Areas for Responsible Fishing in the Gulf of Nicoya'. *Marine Policy* 73:196–203. <https://doi.org/10.1016/j.marpol.2016.08.011>.
- Mackinson, S., G. Daskalov, J.J. Heymans, S. Neira, H. Arancibia, M. Zetina-Rejón, H. Jiang, et al. 2009. 'Which Forcing Factors Fit? Using Ecosystem Models to Investigate the Relative Influence of Fishing and Changes in Primary Productivity on the Dynamics of Marine Ecosystems'. Selected Papers from the Sixth European Conference on Ecological Modelling - ECEM '07, on Challenges for Ecological Modelling in a Changing World: Global Changes, Sustainability and Ecosystem Based Management, November 27-30, 2007, Trieste, Italy 220 (21): 2972–87. <https://doi.org/10.1016/j.ecolmodel.2008.10.021>.
- Mackinson, S., B. Deas, D. Beveridge, and J. Casey. 2009. 'Mixed-Fishery or Ecosystem Conundrum? Multispecies Considerations Inform Thinking on Long-Term Management of North Sea Demersal Stocks'. *Canadian Journal of Fisheries and Aquatic Sciences* 66 (7): 1107–29.

- Mak, Y. K. Y., L. S. R. Tao, V. Ho, D. Dudgeon, W. W. L. Cheung, and K. M. Y. Leung. 2021. 'Initial Recovery of Demersal Fish Communities in Coastal Waters of Hong Kong, South China, Following a Trawl Ban'. *Reviews in Fish Biology and Fisheries* 31 (4): 989–1007. <https://doi.org/10.1007/s11160-021-09685-5>.
- Maureaud, A., D. Gascuel, M. Colléter, M. L. D. Palomares, H. Du Pontavice, D. Pauly, and W. W. L. Cheung. 2017. 'Global Change in the Trophic Functioning of Marine Food Webs'. *PloS One* 12 (8): e0182826.
- Melissa, M. C. 2012. 'Identificación y Caracterización de Actores Institucionales y de La Sociedad Civil Claves En La Gestión de Las AMUM Golfo de Nicoya y Pacífico Sur Proyecto SINAC-BID-MarViva'. Sistema Nacional de Áreas de Conservación, Banco Interamericano de Desarrollo, MarViva, Puntarenas, Costa Rica.
- Morato, T., R. Watson, T. J. Pitcher, and D. Pauly. 2006. 'Fishing down the Deep'. *Fish and Fisheries* 7 (1): 24–34. <https://doi.org/10.1111/j.1467-2979.2006.00205.x>.
- Morissette, L. 2007. 'Complexity, Cost and Quality of Ecosystem Models and Their Impact on Resilience'. Vancouver, BC: University of British Columbia 278.
- Mug-Villanueva, M. 2002. 'Análisis de Las Tendencias Del Comportamiento y Desempeño Del Sector Pesquero En Costa Rica'. Octavo Informe Estado de La Nación. San José: Programa Estado de La Nación.
- Munga, C., S. Ndegwa, B. Fulanda, J. Manyala, E. Kimani, J. Ohtomi, and A. Vanreusel. 2012. 'Bottom Shrimp Trawling Impacts on Species Distribution and Fishery Dynamics; Ungwana Bay Fishery Kenya before and after the 2006 Trawl Ban'. *Fisheries Science* 78 (2): 209–19. <https://doi.org/10.1007/s12562-011-0458-0>.
- Mutsert, K. de, M. Coll, J. Steenbeek, C. Ainsworth, J. Buszowski, D. Chagaris, V. Christensen, S. J. J. Heymans, K. A. Lewis, and S. Libralato. 2024. 'Advances in Spatial-temporal Coastal and Marine Ecosystem Modeling Using Ecospace'.
- Navia, A. F., V. H. Cruz-Escalona, A. Giraldo, and A. Barausse. 2016. 'The Structure of a Marine Tropical Food Web, and Its Implications for Ecosystem-Based Fisheries Management'. *Ecological Modelling* 328 (May):23–33. <https://doi.org/10.1016/j.ecolmodel.2016.02.009>.
- Neira, S., H. Arancibia, and L. Cubillos. 2004. 'Comparative Analysis of Trophic Structure of Commercial Fishery Species off Central Chile in 1992 and 1998'. *Ecological Modelling* 172 (2–4): 233–48.
- Neira, S., C. Moloney, V. Christensen, P. Cury, L. Shannon, and H. Arancibia. 2014. 'Analysing Changes in the Southern Humboldt Ecosystem for the Period 1970–2004 by Means of Dynamic Food Web Modelling'. *Ecological Modelling* 274 (February):41–49. <https://doi.org/10.1016/j.ecolmodel.2013.09.022>.
- Neira, S., C. L. Moloney, P. Cury, C. Mullon, and V. Christensen. 2009. 'Mechanisms Affecting Recovery in an Upwelling Food Web: The Case of the Southern Humboldt'. *Progress in Oceanography* 83 (1–4): 404–16.

- Nielsen, J. R., and T. Vedsmand. 1999. 'User Participation and Institutional Change in Fisheries Management: A Viable Alternative to the Failures of 'Top-down' Driven Control?' *Ocean & Coastal Management* 42 (1): 19–37.
- NOAA (National Oceanic and Atmospheric Administration). 1999. 'Discounting and the Treatment of Uncertainty in Natural Resource Damage Assessment'. *Technical Paper* 99-1, Silver Spring, . <https://www.sfu.ca/>.
- NOAA (National Oceanic and Atmospheric Administration) 2020. El Niño/Southern Oscillation (ENSO) Technical Discussion. NOAA.gov. Available online at: <https://www.ncdc.noaa.gov/teleconnections/enso/enso-tech.php> (accessed February 2020).
- Odum, E. P. 1969. 'The Strategy of Ecosystem Development: An Understanding of Ecological Succession Provides a Basis for Resolving Man's Conflict with Nature.' *Science* 164 (3877): 262–70.
- Okafor-Yarwood, I., N. I. Kadagi, D. Belhabib, and E. H. Allison. 2022. 'Survival of the Richest, Not the Fittest: How Attempts to Improve Governance Impact African Small-Scale Marine Fisheries'. *Marine Policy* 135:104847.
- Okey, T. 2004. 'A Trophic Model of a Galápagos Subtidal Rocky Reef for Evaluating Fisheries and Conservation Strategies'. *Ecological Modelling* 172 (2–4): 383–401. <https://doi.org/10.1016/j.ecolmodel.2003.09.019>.
- Okey, T., and B. A. Wright. 2004. 'Toward Ecosystem-Based Extraction Policies for Prince William Sound, Alaska: Integrating Conflicting Objectives and Rebuilding Pinnipeds'. *Bulletin of Marine Science* 74 (3): 727–47.
- Olson, R. J., and G. M. Watters. 2003. 'A Model of the Pelagic Ecosystem in the Eastern Tropical Pacific Ocean'.
- Ovando, D., D. Dougherty, and J. R. Wilson. 2016. 'Market and Design Solutions to the Short-term Economic Impacts of Marine Reserves'. *Fish and Fisheries* 17 (4): 939–54. <https://doi.org/10.1111/faf.12153>.
- Owusu, V., and E. Andriesse. 2020. 'From Open Access Regime to Closed Fishing Season: Lessons from Small-Scale Coastal Fisheries in the Western Region of Ghana'. *Marine Policy* 121:104162.
- Pace, M. L., J. J. Cole, S. R. Carpenter, and J. F. Kitchell. 1999. 'Trophic Cascades Revealed in Diverse Ecosystems'. *Trends in Ecology & Evolution* 14 (12): 483–88. [https://doi.org/10.1016/S0169-5347\(99\)01723-1](https://doi.org/10.1016/S0169-5347(99)01723-1).
- Palomares, M.-L. D., and D. Pauly. 2019. 'Coastal Fisheries: The Past, Present, and Possible Futures'. In *Coasts and Estuaries*, 569–76. Elsevier.
- Parsons, T. R., M. Takahashi, and B. Hargrave. 2013. *Biological Oceanographic Processes*. Elsevier.
- Pauly, D. 1983. *Some Simple Methods for the Assessment of Tropical Fish Stocks*. Food & Agriculture Org.

- Pauly, D., V. Christensen, and C. Walters. 2000. 'Ecopath, Ecosim, and Ecospace as Tools for Evaluating Ecosystem Impact of Fisheries'. *ICES Journal of Marine Science* 57 (3): 697–706.
- Pichegru, L., P. G. Ryan, R. J. M. Crawford, C. D. van der Lingen, and D. Grémillet. 2010. 'Behavioural Inertia Places a Top Marine Predator at Risk from Environmental Change in the Benguela Upwelling System'. *Marine Biology* 157:537–44.
- Pinnegar, J.K., N.V.C. Polunin, P. Francour, F. Badalamenti, R. Chemello, M.-L. Harmelin-Vivien, B. Hereu, et al. 2000. 'Trophic Cascades in Benthic Marine Ecosystems: Lessons for Fisheries and Protected-Area Management'. *Environmental Conservation* 27 (2): 179–200. <https://doi.org/10.1017/S0376892900000205>.
- Pipitone, C., D. Agnetta, A. Zenone, V. Maximiliano Giacalone, F. Badalamenti, F. Fiorentino, P. Rinelli, M. Sinopoli, T. V. Fernández, and G. D'Anna. 2023. 'When the Trawl Ban Is a Good Option: Opportunities to Restore Fish Biomass and Size Structure in a Mediterranean Fisheries Restricted Area'. *Sustainability* 15 (3): 2425.
- Pita, C., G. J. Pierce, and I. Theodossiou. 2010. 'Stakeholders' Participation in the Fisheries Management Decision-Making Process: Fishers' Perceptions of Participation'. *Marine Policy* 34 (5): 1093–1102. <https://doi.org/10.1016/j.marpol.2010.03.009>.
- Plagányi, É E, and D. S. Butterworth. 2004. 'A Critical Look at the Potential of Ecopath with Ecosim to Assist in Practical Fisheries Management'. *African Journal of Marine Science* 26:261–87.
- Porter, B. A., and M. B. Orams. 2014. 'Exploring Tourism as a Potential Development Strategy for an Artisanal Fishing Community in the Philippines: The Case of Barangay Victory in Bolinao'. *Tourism in Marine Environments* 10 (1–2): 49–70.
- Primavera, J. H., and J. Morvenna, and A. Esteban. 2008. 'A Review of Mangrove Rehabilitation in the Philippines: Successes, Failures and Future Prospects'. *Wetlands Ecology and Management* 16:345–58.
- Prince W., and N. Fantom. 2014. 'World Development Indicators 2014'. The World Bank.
- Proyecto Golfos. 2012a. 'Diagnóstico de Los Grupos Asociativos de Pescadores y Piangueros y Alternativas Productivas En Las AMUM Pacífico Sur y Golfo de Nicoya (Costa Rica)'. Sistema Nacional de Áreas de Conservación, Instituto Nacional de Biodiversidad. Golfo de Nicoya, Costa Rica.
- Proyecto Golfos. 2012b. 'Identificación y Caracterización de Actores Institucionales y de La Sociedad Civil Claves En a Gestión de Las AMUN Golfo de Nicoya y Pacífico Sur'. In , 94. Costa Rica.
- Proyecto Golfos. 2013. 'Estado Del Área Marina de Uso Múltiple Pacífico Sur'. In , 83. Costa Rica.
- Punt, A. E., M. G. Dalton, G. D. Adams, S. J. Barbeaux, W. Cheng, A. J. Hermann, K. K. Holsman, P.-J. F. Hulson, T. P. Hurst, and A. Rovellini. 2024. 'Capturing Uncertainty When Modelling Environmental Drivers of Fish Populations, with an Illustrative Application to Pacific Cod in the Eastern Bering Sea'. *Fisheries Research* 272 (April):106951. <https://doi.org/10.1016/j.fishres.2024.106951>.

- Queiros, Q., J.-M. Fromentin, E. Gasset, G. Dutto, C. Huiban, L. Metral, L. Leclerc, Q. Schull, D. J. McKenzie, and C. Saraux. 2019. 'Food in the Sea: Size Also Matters for Pelagic Fish'. *Frontiers in Marine Science* 6:385.
- Quesada-Alpízar, M. A., and A. Morales-Ramírez. 2004. 'Comportamiento de Las Masas de Agua En El Golfo Dulce, Costa Rica Durante El Niño (1997-1998)'. *Revista de Biología Tropical* 52:95–103.
- Aizpurúa, Q. A. 2014. 'Análisis Del Estado Actual de La Pesquería de Barracuda Sphyræna Ensis (Sphyrænidae), En El Golfo de Nicoya, Costa Rica'. Master's Thesis, Puntarenas, Costa Rica: Universidad Nacional Costa Rica.
- Requejo, I.S. 2011. 'El Sistema Pelágico Del Golfo de Nicoya y Río Tempisque (Costa Rica)'. University of Cadiz, Cadiz, Spain.
- Robertson, D. R. 2009. 'Shorefishes of the Tropical Eastern Pacific Online Information System'. In . Smithsonian Institution Scholarly Press.
- Roessig, J. M., Christa M. Woodley, Joseph J. Cech, and Lara J. Hansen. 2004. 'Effects of Global Climate Change on Marine and Estuarine Fishes and Fisheries'. *Reviews in Fish Biology and Fisheries* 14 (2): 251–75. <https://doi.org/10.1007/s11160-004-6749-0>.
- Romagnoni, G., S. Mackinson, J. Hong, and A. M. Eikeset. 2015. 'The Ecospace Model Applied to the North Sea: Evaluating Spatial Predictions with Fish Biomass and Fishing Effort Data'. *Ecological Modelling* 300 (March):50–60. <https://doi.org/10.1016/j.ecolmodel.2014.12.016>.
- Roux, J.-P., C. D. van der Lingen, M. J. Gibbons, N. E. Moroff, L. J. Shannon, A. D. M. Smith, and Ph. M. Cury. 2013. 'Jellyfication of Marine Ecosystems as a Likely Consequence of Overfishing Small Pelagic Fishes: Lessons from the Benguela'. *Bulletin of Marine Science* 89 (1): 249–84.
- Runge, J. A. 1988. 'Should We Expect a Relationship between Primary Production and Fisheries? The Role of Copepod Dynamics as a Filter of Trophic Variability'. In , 61–71. Springer.
- Ryabinin, V., J. Barbière, P. Haugan, G. Kullenberg, N. Smith, C. McLean, A. Troisi, A. Fischer, S. Aricò, and T. Aarup. 2019. 'The UN Decade of Ocean Science for Sustainable Development'. *Frontiers in Marine Science* 6:470. <https://doi.org/10.3389/fmars.2019.00470>.
- Sabau, G. 2017. 'Costa Rica: A Champion of the Small-Scale Fisheries Guidelines'. The Small-Scale Fisheries Guidelines: Global Implementation, 355–78.
- Salas, S., R. Chuenpagdee, A. T. Charles, and J. C. Seijo. 2011. Coastal Fisheries of Latin America and the Caribbean. Vol. 544. Food and Agriculture Organization of the United Nations ^ eRome Rome.
- Samy-Kamal, M., A. Forcada, and J. L. Sánchez Lizaso. 2015. 'Effects of Seasonal Closures in a Multi-Specific Fishery'. *Fisheries Research* 172 (December):303–17. <https://doi.org/10.1016/j.fishres.2015.07.027>.

- Sánchez Jiménez, A., D. MacMillan, M. Wolff, A. Schlüter, and M. Fujitani. 2021. 'The Importance of Values in Predicting and Encouraging Environmental Behavior: Reflections From a Costa Rican Small-Scale Fishery'. *Frontiers in Marine Science* 8:137. <https://doi.org/doi:10.3389/fmars.2021.543075>.
- Sánchez-Jiménez, A., M. Fujitani, D. MacMillan, A. Schlüter, and M. Wolff. 2019. 'Connecting a Trophic Model and Local Ecological Knowledge to Improve Fisheries Management: The Case of Gulf of Nicoya, Costa Rica'. *Frontiers in Marine Science* 6:126. <https://doi.org/10.3389/fmars.2019.00126>.
- Sathyendranath, S., V. Stuart, A. Nair, K. Oka, T. Nakane, H. Bouman, Mh. Forget, H. Maass, and T. Platt. 2009. 'Carbon-to-Chlorophyll Ratio and Growth Rate of Phytoplankton in the Sea'. *Marine Ecology Progress Series* 383 (May):73–84. <https://doi.org/10.3354/meps07998>.
- Schmidt, T. C. dos Santos, and J. Ferraz Dias. 2012. 'Pattern of Distribution and Environmental Influences on the Scienidae Community of the Southeastern Brazilian Coast'. *Brazilian Journal of Oceanography* 60:233–43.
- Scott, E., N. Serpetti, J. Steenbeek, and J. J. Heymans. 2016. 'A Stepwise Fitting Procedure for Automated Fitting of Ecopath with Ecosim Models'. *SoftwareX* 5:25–30. <https://doi.org/10.1016/j.softx.2016.02.002>.
- 'Sea Around Us'. 2024. <https://www.seaaroundus.org/data/#/eez>.
- Sea Around Us Project. 2014. 'Stock Status.' University of British Columbia, Fisheries Centre, Vancouver. www.seaaroundus.org.
- Shannon, L. J., M. Coll, S. Neira, P. M. Cury, and J.-P. Roux. 2007. 'Role of Small Pelagic Fish on the Ecosystem'. *Climate Change and Small Pelagic Fish*. Cambridge University Press, Cambridge.
- Shannon, L. J., S. Neira, and M. Taylor. 2008. 'Comparing Internal and External Drivers in the Southern Benguela and the Southern and Northern Humboldt Upwelling Ecosystems'. *African Journal of Marine Science* 30 (1): 63–84.
- SINAC. 2019. 'Estrategia Regional Para El Manejo y Conservación de Los Manglares En El Golfo de Nicoya-Costa Rica 2019-2030'.
- Smith, H., and X. Basurto. 2019. 'Defining Small-Scale Fisheries and Examining the Role of Science in Shaping Perceptions of Who and What Counts: A Systematic Review'. *Frontiers in Marine Science* 6:236.
- Soma, K. 2003. 'How to Involve Stakeholders in Fisheries Management—a Country Case Study in Trinidad and Tobago'. *Marine Policy* 27 (1): 47–58. [https://doi.org/10.1016/S0308-597X\(02\)00050-7](https://doi.org/10.1016/S0308-597X(02)00050-7).
- Steenbeek, J., X. Corrales, M. Platts, and M. Coll. 2018. 'Ecosampler: A New Approach to Assessing Parameter Uncertainty in Ecopath with Ecosim'. *SoftwareX* 7:198–204.
- Stiles, M. L., J. Stockbridge, M. Lande, and Mi. F. Hirshfield. 2010. 'Impacts of Bottom Trawling'. Oceana, Washington DC.

- Sumaila, U. R., A. S. Khan, A. J. Dyck, R. Watson, G. Munro, P. Tydemers, and D. Pauly. 2010. 'A Bottom-up Re-Estimation of Global Fisheries Subsidies'. *Journal of Bioeconomics* 12 (3): 201–25.
- Suman, D. O. 2019. 'Chapter 31 - Mangrove Management: Challenges and Guidelines'. In *Coastal Wetlands (Second Edition)*, edited by Gerardo M.E. Perillo, Eric Wolanski, Donald R. Cahoon, and Charles S. Hopkins, 1055–79. Elsevier.
<https://doi.org/10.1016/B978-0-444-63893-9.00031-9>.
- Tam, J., M. H. Taylor, V. Blaskovic, P. Espinoza, R. M. Ballón, E. Díaz, C. Wosnitza-Mendo, J. Argüelles, S. Purca, and P. Ayón. 2008. 'Trophic Modeling of the Northern Humboldt Current Ecosystem, Part I: Comparing Trophic Linkages under La Niña and El Niño Conditions'. *Progress in Oceanography* 79 (2–4): 352–65.
- Tao, L. SR., G. Lui, K. JH. Wong, T. TY. Hui, Y. KY. Mak, R. C. Sham, J. KC. Yau, W. WL. Cheung, and K. MY. Leung. 2021. 'Does a Trawl Ban Benefit Commercially Important Decapoda and Stomatopoda in Hong Kong?' *Ecosystems* 24 (5): 1157–70.
<https://doi.org/10.1007/s10021-020-00574-9>.
- Taylor, M.H., J. Tam, V. Blaskovic, P. Espinoza, R. M. Ballón, C. Wosnitza-Mendo, J. Argüelles, E. Díaz, S. Purca, and N. Ochoa. 2008. 'Trophic Modeling of the Northern Humboldt Current Ecosystem, Part II: Elucidating Ecosystem Dynamics from 1995 to 2004 with a Focus on the Impact of ENSO'. *Progress in Oceanography* 79 (2–4): 366–78.
- Taylor, M. H., M. Wolff, F. Vadas, and C. Yamashiro. 2008. 'Trophic and Environmental Drivers of the Sechura Bay Ecosystem (Peru) over an ENSO Cycle'. *Helgoland Marine Research* 62:15–32.
- Thrane, J.-E., D. O. Hessen, and T. Andersen. 2014. 'The Absorption of Light in Lakes: Negative Impact of Dissolved Organic Carbon on Primary Productivity'. *Ecosystems* 17:1040–52.
- Thrush, S. F., and P. K. Dayton. 2002. 'Disturbance to Marine Benthic Habitats by Trawling and Dredging: Implications for Marine Biodiversity'. *Annual Review of Ecology and Systematics* 33 (1): 449–73. <https://doi.org/10.1146/annurev.ecolsys.33.010802.150515>.
- Tietze, U. 2001. Report of the Regional Workshop on the Effects of Globalization and Deregulation on Fisheries in the Caribbean, Castries, Saint Lucia, 4-8 December 2000. Food & Agriculture Org.
- Tirtadanu, T., and S. Suprato. 2017. 'Trawling Ban Impact on the Stock Density of Shrimps in the Java Sea, Indonesia'. *Indo Pacific Journal of Ocean Life* 1 (2): 49–54.
- Travers, M., K. Watermeyer, L.J. Shannon, and Y.-J. Shin. 2010. 'Changes in Food Web Structure under Scenarios of Overfishing in the Southern Benguela: Comparison of the Ecosim and OSMOSE Modelling Approaches'. *Journal of Marine Systems* 79 (1–2): 101–11.
- Trégarot, E., J. P. D'Olivo, A. Z. Botelho, A. Cabrito, G. O. Cardoso, G. Casal, C. C. Cornet, S. M. Cragg, A. K. Degia, and S. Fredriksen. 2024. 'Effects of Climate Change on Marine Coastal Ecosystems—A Review to Guide Research and Management'. *Biological Conservation* 289:110394.

- Trimble, M., and F. Berkes. 2013. 'Participatory Research towards Co-Management: Lessons from Artisanal Fisheries in Coastal Uruguay'. *Journal of Environmental Management* 128 (October):768–78. <https://doi.org/10.1016/j.jenvman.2013.06.032>.
- Trujillo, P., A. M. Cisneros-Montemayor, S. Harper, and D. Zeller. 2012. 'Reconstruction of Costa Rica's Marine Fisheries Catches (1950-2008)'.
- Turner, S. J., S.F. Thrush, J.E. Hewitt, V.J. Cummings, and G. Funnell. 1999. 'Fishing Impacts and the Degradation or Loss of Habitat Structure'. *Fisheries Management and Ecology* 6 (5): 401–20.
- Ulanowicz, R. E., and C. J. Puccia. 1990. 'Mixed Trophic Impacts in Ecosystems'. *Coenos* 5 (1): 7–16.
- Ullah, Md. H., Md. Rashed-Un-Nabi, and Md. Abdulla Al-Mamun. 2012. 'Trophic Model of the Coastal Ecosystem of the Bay of Bengal Using Mass Balance Ecopath Model'. *Ecological Modelling* 225:82–94. <https://doi.org/10.1016/j.ecolmodel.2011.11.013>.
- UNEP (United Nations Environment Program). 2004. 'The Regional Seas Programme, Large Marine Ecosystems: North East Pacific Region'. Nairobi, Kenya.
- Valette-Silver, N. J., and D. Scavia. 2003. 'Ecological Forecasting: New Tools for Coastal and Marine Ecosystem Management'.
- Vargas, E. C., J. Boza Abarca, and K. Berrocal Artavia. 2014. 'Efectos de El Niño y La Niña Sobre El Comportamiento Del Microfitoplancton Marino y Las Variables Fisicoquímicas Durante El 2008 a 2010 En El Golfo de Nicoya, Costa Rica'. *Revista Ciencias Marinas y Costeras* 6:115–33.
- Vargas, J. A. 1995. 'The Gulf of Nicoya Estuary, Costa Rica: Past, Present, and Future Cooperative Research'. *Helgoländer Meeresuntersuchungen* 49 (1–4): 821–28. <https://doi.org/10.1007/BF02368405>.
- Vargas, J. A. 2016. 'The Gulf of Nicoya: Estuarine Ecosystem'. *Costa Rican Ecosystems*, 139–61.
- Vargas-Montero, M., and E. Freer. 2004. 'Harmful Blooms of Cyanobacteria (Oscillatoriaceae) and Dinoflagellates (Gymnodiniaceae) in the Golfo de Nicoya, Costa Rica'. *Revista de Biología Tropical* 52:121–25.
- Christensen, V., and C.J. Walters. 2024a. *Ecosystem Modelling with EwE*. UBC.
- Walters, C., V. Christensen, and D. Pauly. 1997. 'Structuring Dynamic Models of Exploited Ecosystems from Trophic Mass-Balance Assessments'. *Reviews in Fish Biology and Fisheries* 7:139–72.
- Walters, C. J., V. Christensen, and D. Pauly. 2002. 'Searching for Optimum Fishing Strategies for Fishery Development, Recovery and Sustainability'. *The Use of Ecosystem Models to Investigate Multispecies Management Strategies for Capture Fisheries* Fisheries Centre Research Reports 10 (2): 11–15.

- Walters, C., D. Pauly, and V. Christensen. 1999. 'Ecospace: Prediction of Mesoscale Spatial Patterns in Trophic Relationships of Exploited Ecosystems, with Emphasis on the Impacts of Marine Protected Areas'. *Ecosystems* 2:539–54.
- Ward, P., and R. A. Myers. 2005a. 'Shifts in Open-Ocean Fish Communities Coinciding with the Commencement of Commercial Fishing'. *Ecology* 86 (4): 835–47.
<https://doi.org/10.1890/03-0746>.
- Ward, P., and R. A. Myers. 2005b. 'Shifts in Open-Ocean Fish Communities Coinciding with the Commencement of Commercial Fishing'. *Ecology* 86 (4): 835–47.
<https://doi.org/10.1890/03-0746>.
- Ward, R. D., D. A. Friess, R. H. Day, and R. A. Mackenzie. 2016. 'Impacts of Climate Change on Mangrove Ecosystems: A Region by Region Overview'. *Ecosystem Health and Sustainability* 2 (4): e01211.
- Wehrtmann, I. S., and J. Cortés. 2008. Marine Biodiversity of Costa Rica, Central America. Vol. 86. Springer Science & Business Media.
- Wehrtmann, I. S., and V. Nielsen-Muñoz. 2009. 'The Deepwater Fishery along the Pacific Coast of Costa Rica, Central America'. *Latin American Journal of Aquatic Research* 37 (3): 543–54.
- Wiedenmann, J., C. M. Free, and O. P. Jensen. 2019. 'Evaluating the Performance of Data-Limited Methods for Setting Catch Targets through Application to Data-Rich Stocks: A Case Study Using Northeast US Fish Stocks'. *Fisheries Research* 209:129–42.
- Wilson, D. C., J. Raakjær, and P. Degnbol. 2006. 'Local Ecological Knowledge and Practical Fisheries Management in the Tropics: A Policy Brief'. *Marine Policy* 30 (6): 794–801.
<https://doi.org/10.1016/j.marpol.2006.02.004>.
- Wolf, R., and J. Heuschele. 2018. 'Water Browning Influences the Behavioral Effects of Ultraviolet Radiation on Zooplankton'. *Frontiers in Ecology and Evolution* 6:26.
- Wolff, M., C. Wosnitza-Mendo, and J. Mendo. 2003. 'The Humboldt Current—Trends in Exploitation, Protection and Research'. *Large Marine Ecosystems of the World*. Elsevier, 279–309.
- Wolff, M., V. Koch, J. B. Chavarría, and J. A. Vargas. 1998. 'A Trophic Flow Model of the Golfo de Nicoya, Costa Rica'. *Revista de Biología Tropical*, 63–79.
- Wolff, M., 2006. 'Biomass Flow Structure and Resource Potential of Two Mangrove Estuaries: Insights from Comparative Modelling in Costa Rica and Brazil'. *Revista de Biología Tropical* 54 (S1): 69–86.
- Wolff, M., 2010. 'Galapagos Does Not Show Recent Warming but Increased Seasonality', Galapagos Research 67, Charles Darwin Foundation, Puerto Ayora, Galapagos, Ecuador
- Wolff, M., D. J. Ruiz, and M. Taylor. 2012. 'El Niño Induced Changes to the Bolivar Channel Ecosystem (Galapagos): Comparing Model Simulations with Historical Biomass Time Series'. *Marine Ecology Progress Series* 448:7–22.

- Wolff, M., M. Taylor, and G. Tesfaye. 2015. 'Implications of Using Small Meshed Gillnets for the Sustainability of Fish Populations: A Theoretical Exploration Based on Three Case Studies'. *Fisheries Management and Ecology* 22 (October):379–87. <https://doi.org/10.1111/fme.12137>.
- Woods, P. J., J. I. Macdonald, H. Bárðarson, S. Bonanomi, W. J. Boonstra, G. Cornell, G. Cripps, R. Danielsen, L. Färber, and A.S. Ferreira. 2022. 'A Review of Adaptation Options in Fisheries Management to Support Resilience and Transition under Socio-Ecological Change'. *ICES Journal of Marine Science* 79 (2): 463–79.

APPENDIX

Table A.1 – Taxonomic composition of the functional groups included in the models.

Functional group	Main taxa of group
1. Seabirds	Anhinga <i>Anhinga anhinga</i> , magnificent frigatebird <i>Fregata magnificens</i> , great frigatebird <i>Fregata minor</i> , black storm-petrel <i>Hydrobates melania</i> , American white pelican <i>Pelecanus erythrorhynchos</i> , brown pelican <i>Pelecanus occidentalis</i> , neotropic cormorant <i>Phalacrocorax brasilianus</i> , boobies <i>Sula</i> spp.
2. Mackerels and barracudas	Pacific Sierra <i>Scomberomorus sierra</i> , Mexican Barracuda <i>Sphyrna ensis</i>
3. Rays and sharks	Spotted Eagle Ray <i>Aetobatus narinari</i> , Longtail Stingray <i>Hypanus (Dasyatis) longus</i> , Rasptail Skate <i>Raja velezi</i> , Peruvian Torpedo <i>Torpedo peruana</i> , stingrays <i>Urotrygon</i> spp., Witch Guitarfish <i>Zapteryx xyster</i> ; Bigeye Thresher <i>Alopias superciliosus</i> , Shortfin Mako <i>Isurus oxyrinchus</i> , Brown Smoothhound <i>Mustelus henlei</i> , Sicklefins Smoothhound <i>Mustelus lunulatus</i> , Scalloped Hammerhead <i>Sphyrna lewini</i>
4. Large drums and snooks	Queen Corvina <i>Cynoscion albus</i> , Yellowtail Corvina <i>Cynoscion stolzmanni</i> ; Black Snook <i>Centropomus nigrescens</i> , White Snook <i>Centropomus viridis</i>
5. Catfishes	<i>Arius</i> spp., Chihuil <i>Bagre panamensis</i> , Long-barbeled Sea Catfish <i>Bagre pinnimaculatus</i> , Sculptured Sea Catfish <i>Notarius kessleri</i> , Chili Sea Catfish <i>Notarius troschelii</i>
6. Morays and eels	Pacific Bearded Brotula <i>Brotula clarkae</i> , Conehead Eel <i>Cynoponticus coniceps</i> , morays <i>Gymnothorax</i> spp., Yellow Snake Eel <i>Ophichthus zophochir</i> , and others
7. Carangids (jacks & pompanos)	Green Jack <i>Caranx caballus</i> , jacks <i>Caranx</i> spp., Golden Trevally <i>Gnathanodon speciosus</i> , leatherjacks <i>Oligoplites</i> spp., Pacific Moonfish <i>Selene peruviana</i> , Almaco Jack <i>Seriola rivoliana</i>
8. Snappers and grunts	Colorado Snapper <i>Lutjanus colorado</i> , Spotted Rose Snapper <i>Lutjanus guttatus</i> , Pacific Dog Snapper <i>Lutjanus novemfasciatus</i> , <i>Lutjanus</i> spp.; Yellowstripe Grunt <i>Haemulopsis axillaris</i> , Elongate Grunt <i>Haemulopsis elongatus</i> , Mojarra Grunt <i>Haemulon scudderii</i> , <i>Haemulon</i> spp., <i>Pomadasy</i> sp.
9. Cephalopods	Crescent octopus <i>Euaxoctopus panamensis</i> , squids <i>Lolliguncula</i> spp., common octopus <i>Octopus vulgaris</i>
10. Lobster	Green spiny lobster <i>Panulirus gracilis</i>
11. Flatfishes	Toothed Flounder <i>Cyclopsetta querna</i> , Dappled Flounder <i>Paralichthys woolmani</i> , Beach Flounder <i>Syacium latifrons</i> , Oval Flounder <i>Syacium ovale</i> , tonguefishes <i>Symphurus</i> spp., soles <i>Trinectes</i> sp., and others
12. Small demersals	Small and juvenile Sciaenidae and Centropomidae, Mullidae, Sparidae, Blenniidae, Gerreidae, Gobiidae, Scorpaenidae, Serranidae, and others
13. Small pelagics	Anchovies <i>Anchoa</i> spp., Anchoveta <i>Cetengraulis mysticetus</i> , thread herrings <i>Opisthonema</i> spp.
14. Crabs and stomatopods	Arched box crab <i>Calappa convexa</i> , <i>Callinectes arcuatus</i> , <i>Callinectes</i> spp., <i>Pleuroncodes</i> sp., mantis shrimp <i>Squilla</i> sp., Gecarcinidae, Grapsidae, Xanthidae
15. Epibenthos	Ark clams <i>Anadara</i> spp., Mytilidae, Ostreidae, and other bivalves; Strombidae and other gastropods; Diogenidae, Echinodermata, Isopoda, Paguridae
16. Endobenthos	Polychaeta: Capitellida, Orbiniida, Spionida, Branchiostomidae, and others
17. Shrimps	Penaeidae: Litopenaeus spp., Heterocarpus spp., Farfantepenaeus spp., Xiphopenaeus riverti, titi shrimp <i>Protrachypene precipua</i> , Rimapenaeus byrdi; Solenoceridae: kolobri shrimp <i>Solenocera agassizii</i>
18. Zooplankton	Copepoda: Calanoida and others; Chaetognatha, Crustacea larvae, ichthyoplankton, Bivalvia and Brachiopoda larvae, Ostracoda, Cladocera, Foraminifera, Appendicularia
19. Phytoplankton	Centric and pennate diatoms, dinoflagellates, and others
20. Mangroves	True mangroves <i>Rhizophora</i> sp., tea mangroves <i>Pelliciera</i> sp., <i>Avicennia</i> sp.
21. Microphytes	Cyanobacteria, benthic diatoms
22. Detritus	

Table A.2 – Diet matrix of all functional groups in the Gulf of Nicoya (Costa Rica) model for 2013. The numbers in parentheses indicate the initial input parameters prior to the model balancing routine.

Group number	Prey	Predator																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	Seabirds																		
2	Mackerels and barracudas																		
3	Rays and sharks			0.005															
4	Large drums and snooks																		
5	Catfishes			0.05			0.02												
6	Morays and eels		0.05	0.01	0.02														
7	Carangids		0.15				0.01	0.01	0.06										
8	Snappers and grunts	0.07			0.05	0.05													
		(0.08)																	
9	Cephalopods	0.02			0.08	0.05	0.05	0.1	0.03	0.02	0.15								
10	Lobster	0.01		0.14	0.02														
11	Flatfishes		0.05	0.095	0.03			0.04											
12	Small demersals	0.05	0.15	0.1	0.3	0.1	0.12		0.02	0.02		0.2							
			(0.14)																
13	Small pelagics	0.4	0.5	0.05				0.35	0.22	0.2									
		(0.39)																	
14	Crabs	0.05	0.05	0.25	0.15	0.1	0.05	0.1	0.06	0.08	0.15								
			(0.06)																
15	Epibenthos	0.3	0.05	0.25	0.1	0.35	0.35	0.2	0.4	0.3	0.55	0.45	0.3	0.3		0.05			
													(0.29)						
16	Endobenthos	0.1				0.2	0.2			0.1	0.15	0.2	0.2		0.2			0.05	
17	Shrimps			0.05	0.2	0.1	0.15	0.14	0.12	0.1		0.15	0.1		0.1				
													(0.11)						
18	Zooplankton							0.1		0.18			0.2	0.5	0.1	0.05	0.05	0.2	0.1
19	Phytoplankton												0.06	0.4		0.6	0.18		0.8
20	Mangroves															0.01	0.05	0.2	
21	Microphytobenthos												0.04	0.1		0.14	0.05	0.2	
22	Detritus				0.05	0.05	0.05		0.05				0.1		0.3	0.2	0.67	0.3	0.1

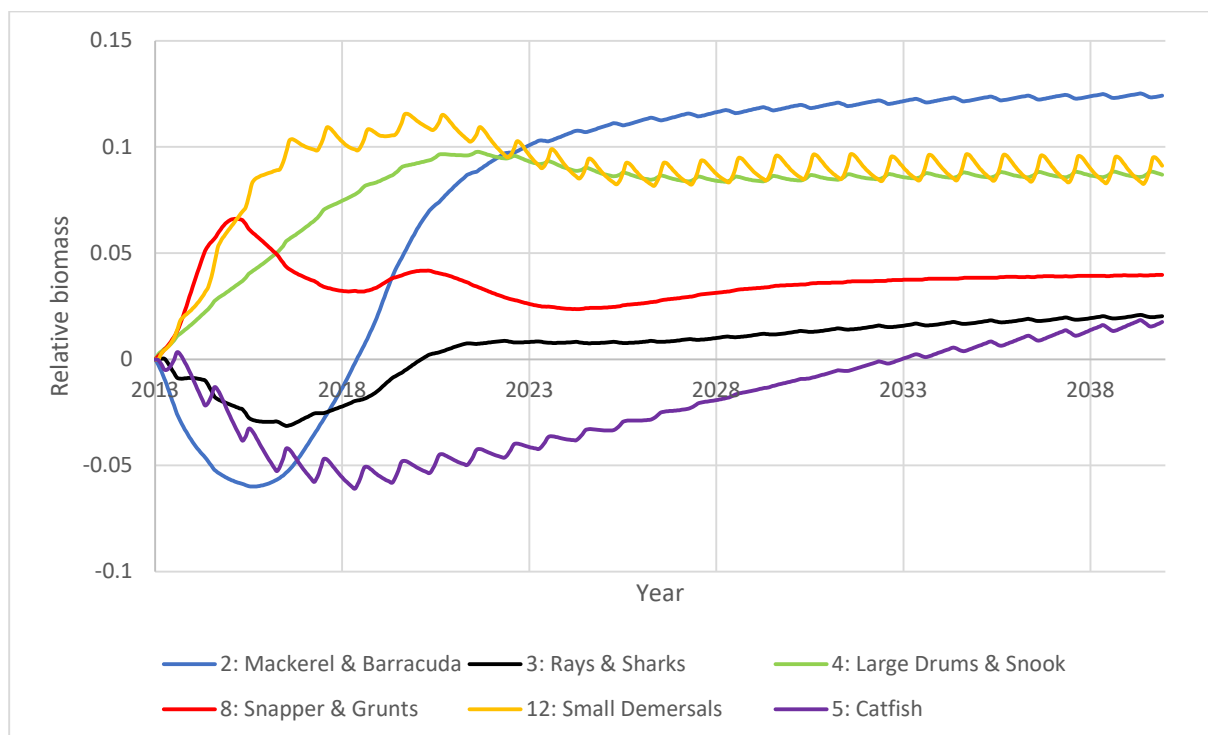
Table A.3 – Detailed references for the input parameters (biomass [B_i], production-to-biomass [P_i/B_i], consumption-to-biomass ratio [Q_i/B_i], and catch) of each functional group in the updated Ecopath model GN₊₂₀ (2013) for the Gulf of Nicoya, Costa Rica (INCOPESCA = Instituto Costarricense de Pesca y Acicultura; BIOMARCC = Biodiversidad Marino Costera y Adaptación al Cambio Climático).

Group name	B_i	P_i/B_i	Q_i/B_i	Catch
Seabirds	Wolff et al. (1998), Orias (2004)	Opitz (1996), Wolff et al. (1998), Olson and Watters (2003)	Opitz (1996), Wolff et al. (1998), Olson and Watters (2003)	
Mackerels and barracudas	INCOPESCA	Mendoza (1993), Olson and Watters (2003)	Mendoza (1993), Wolff et al. (1998), García and Duarte (2002), Olson and Watters (2003)	INCOPESCA
Rays and sharks	Clarke et al. (2011, 2014), Aires-da-Silva et al. (2014), INCOPESCA	Arreguín-Sánchez et al. (2002), Kitchell et al. (2002), Clarke et al. (2014)	Zanella et al. (2011)	INCOPESCA
Large drums and snooks	Alpízar (2011, 2015), Alpízar and Vásquez (2012, 2013), BIOMARCC (2013a, 2013b), INCOPESCA	INCOPESCA	García and Duarte (2002)	INCOPESCA
Catfishes	INCOPESCA	Mendoza (1993)	Mendoza (1993), Wolff et al. (1998), García and Duarte (2002)	INCOPESCA
Morays and eels	INCOPESCA	Wolff et al. (1998)	García and Duarte (2002)	INCOPESCA
Carangids	INCOPESCA	Arreguín-Sánchez et al. (2002)	García and Duarte (2002), Giarrizzo et al. (2013)	INCOPESCA
Snappers and grunts	INCOPESCA	Arreguín-Sánchez et al. (2002), Mendoza (1993)	Wolff et al. (1998), García and Duarte (2002), Arreguín-Sánchez et al. (2002), Giarrizzo et al. (2013)	INCOPESCA
Cephalopods	INCOPESCA	Opitz (1996), Arreguín-Sánchez et al. (2002)	Giarrizzo et al. (2013)	INCOPESCA
Lobster	Wehrmann and Nielsen-Muñoz (2009), INCOPESCA	Hernández and Wehrmann (2014)	Opitz (1996), Wolff et al. (1998)	INCOPESCA
Flatfishes	INCOPESCA	Wolff et al. (1998)	Wolff et al. (1998), Giarrizzo et al. (2013)	INCOPESCA
Small demersals	Alpízar (2011, 2015), Alpízar and Vásquez (2012, 2013)	Opitz (1996), Arreguín-Sánchez et al. (2002)	García and Duarte (2002), Giarrizzo et al. (2013)	INCOPESCA
Small pelagics	Chacón et al. (2007), Alpízar (2011, 2015), Alpízar and Vásquez (2012, 2013), BIOMARCC (2013a, 2013b)	INCOPESCA	Wolff et al. (1998), García and Duarte (2002), Giarrizzo et al. (2013)	INCOPESCA
Crabs	Fischer and Wolff (2006)	Dittel and Epifanio (1990), Arreguín-Sánchez et al. (2002)	Fischer and Wolff (2006)	INCOPESCA

Table A.3 – Continued

Group name	B_i	P_i/B_i	Q_i/B_i	Catch
Epibenthos	Vargas and Cortés (1999), Willis and Cortés (2001), Stern-Pirlot and Wolff (2006), Corrales (2010), INCOPESCA	Opitz (1996), Stern-Pirlot and Wolff (2006)	Opitz (1996), Wolff et al. (1998)	INCOPESCA
Endobenthos	Dean (2004)	Opitz (1996)	Opitz (1996), Wolff et al. (1998)	INCOPESCA
Shrimps	Tabash Blanco and Palacios (1996), Tabash Blanco (2007), Wehrmann and Nielsen-Muñoz (2009), Trujillo et al. (2012), BIOMARCC (2013a, 2013b), INCOPESCA	INCOPESCA	Arreguín-Sánchez et al. (2002), Wolff et al. (1998)	INCOPESCA
Zooplankton	Brugnoli-Olivera et al. (2004)	Morales-Ramírez (2001), Brugnoli-Olivera et al. (2004)	Brugnoli-Olivera et al. (2004)	
Phytoplankton	Brugnoli-Olivera and Morales-Ramírez (2008), Soto Rojas (2013)	Córdoba-Muñoz (1998), Gocke et al. (2002), Brugnoli-Olivera and Morales-Ramírez (2008), Soto Rojas (2013)	Wolff et al. (1998)	
Mangroves	Wolff et al. (1998), FAO (2007), Cortés et al. (2010), Venegas-Li et al., unpublished paper	Wolff et al. (1998), FAO (2007)		
Microphytobenthos	Wolff et al. (1998)	Opitz (1996), Wolff et al. (1998)		

Figure A.4 - Changes in the relative biomass of several functional groups 20 years after the in the Gulf of Nicoya over time (2013 until 20 years after the ban on trawling), Mackerel & Barracuda (blue), Rays & Sharks (black), Large Drums & Snook (green), Snapper & Grunts (red), Small Demersals (yellow) and Catfish (violet).



EIDESSTATTLICHE ERKLÄRUNG

Name **Viola Alms**

Hiermit erkläre ich, dass ich die Doktorarbeit mit dem Titel „*Modelling Two Decades of Change and Future Developments in the Gulf of Nicoya (Costa Rica) Ecosystem Using the Ecopath with Ecosim Toolbox*“ selbstständig verfasst und geschrieben habe und außer den angegebenen Quellen keine weiteren Hilfsmittel verwendet habe.

Ebenfalls erkläre ich hiermit, dass es sich bei den mir abgegebenen Arbeiten um drei identische Exemplare handelt.

Hamburg, den 19.12.2024